



Internal Use Only

MOBILE PHONE SERVICE MANUAL

CAUTION

BEFORE SERVICING THE UNIT, READ THE "SAFETY PRECAUTIONS" IN THIS MANUAL

MODEL : LG-M250

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1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

C. Attention

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.

Following information is ESD handling:

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

2.1 Band Specification

Support Band	TX Freq (MHz)	RX Freq (MHz)
WCDMA(FDD1)	1920 – 1980	2110 – 2170
WCDMA(FDD2)	1850 – 1910	1930 – 1990
WCDMA(FDD5)	824 – 849	869 – 894
WCDMA(FDD8)	880 – 915	925 – 960
LTE(FDD1)	1920 - 1980	2110 – 2170
LTE(FDD3)	1710 – 1785	1805 – 1880
LTE(FDD7)	2500 – 2570	2620 – 2690
LTE(FDD8)	880 - 915	925 – 960
LTE(FDD20)	832 - 862	791 - 821
EGSM	880 - 915	925 – 960
GSM850	824 - 849	869 – 894
DCS1800	1710 - 1785	1805 – 1880
PCS	1850 - 1910	1930 - 1990

2.2 HW Features

List	Type / Spec.	
1. Phone Type	DOP Type	
2. Size	148.7mm x 75.3mm x 7.9mm	
3. Weight	142 g (with Battery)	
4. Battery	2,800mAh(Typ) (Li-Ion) , 2,700mAh(Min.)	
5. Chipset	MT6750 1.5GHz Octa core	
6. Memory	16GB(EMMC) + 2GB(LPDDR3) External Memory(SD Card) : Up to 2TB	
7. LCD	Size	5.3 inch
	Display Type	Active matrix TFT, Transmissive Type
	Color	16.7M colors
	Resolution	1280(H) X 720(V), 277ppi
8. Touch	Type	5.3 inch Capacitive type
9. Main Camera (13M)	Type	CMOS image sensor
	Resolution	4224(H) X 3136(V) pixels.
	Image Scaling Down	13M(4160x3120), 9.7M(4160x2340), 4.9M(2560x1920), 3.7M(2560x1440), 2.1M(1440x1440)
	Format	Image : JPG, Video : MP4

2.2 HW Features

10. Audio	Receiver	12 X 06 X 2T Receiver
	Speaker	16 X 12 X 2.6T Speaker
	Format	MP3, AAC, MIDI, EAAC+, OGG, AMR
11. Bluetooth	Standard	Bluetooth 4.2
	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
12. WLAN	Standard	IEEE 802.11 b/g/n
	Throughput	Max 40Mbps (SDIO Driver performance)
	Depend on environment	0 ~ 50m (depend on environment)
13. GPS	type	A-GPS
14. FM	type	FM Radio, 3.5pi Ear-jack

2.3 RSSI Display

RSSI BAR	GSM	WCDMA	LTE	Comment
BAR 5->4	-91dBm $\pm$ 3	-85dBm $\pm$ 4	-85dBm $\pm$ 4	1. Call Connected & CIPPH Level=-3.3 2. LTE: RSRP
BAR 4->3	-95dBm $\pm$ 3	-90dBm $\pm$ 4	-95dBm $\pm$ 4	
BAR 3->2	-98dBm $\pm$ 3	-96dBm $\pm$ 4	-105dBm $\pm$ 4	
BAR 2->1	-102dBm $\pm$ 3	-100dBm $\pm$ 4	-115dBm $\pm$ 4	
BAR 1->0	-104dBm $\pm$ 3	-105dBm $\pm$ 4	-128dBm $\pm$ 4	

2.4 Current consumption

	Stand by		Voice Call (LTE Data connection)	Measurement Condition
	Bluetooth Off	Bluetooth Connected		
WCDMA Only	10 mA under (DRX=7)	11 mA under (DRX=7)	250 mA under (Tx=10dBm)	LCD off
GSM Only	10 mA under (PP=5)	MultiPoint지원 : Stanby 대비 Δ 3.0mA $\downarrow$ (PP=5)	400 mA under (Lvl=5)	LCD off
LTE	10 mA under (LTE 2.56S)	11 mA under (LTE 2.56s)	450 mA (@TX 10 dBm)	LCD off

2.5 Battery bar

Battery Bar icon	Specification(%)
Level 20	98 ~ 100
Level 19	93 ~ 97
Level 18	88 ~ 92
Level 17	83 ~ 87
Level 16	78 ~ 82
Level 15	73 ~ 77
Level 14	68 ~ 72
Level 13	63 ~ 67
Level 12	58 ~ 62
Level 11	53 ~ 57
Level 10	48 ~ 52
Level 9	43 ~ 47
Level 8	38 ~ 42
Level 7	33 ~ 37
Level 6	28 ~ 32
Level 5	23 ~ 27
Level 4	16 ~ 22
Level 3	13 ~ 15
Level 2	8 ~ 12
Level 1	3 ~ 7
Level 0	0 ~ 2
Low Battery Pop-up	16% → 15% : One Time popup (No call) There is no pop-up until 5% if you delete pop-up.
Critical Low battery pop-up	6% → 5% : One Time popup (No call) There is no pop-up until 3% if you delete pop-up.
Power off	2% → >1%

2.6 SW Specification

Function	Detail Item	O / X	Specification	Etc.
OS	OS	O	Android N-OS	
Data	Circuit	X		
	Packet	O		
Connectivity	Infrared (IrDA)	X		
	Bluetooth	O	Ver. 4.2	
	Wi-Fi	O	802.11 b/g/n	
	USB	O	Ver. 2.0	
	USB Mass storage	X	MTP support	
	RS232	X		
	DLNA1.5	X		
Voice Function	Voice Recording	O		
	Voice Command	X	Digit Dial	
		X	Name Dial	
		X	Menu	
		X	Phonebook Searching	
	Answering machine	X	Driving, Meeting, user define(absent)	
Memory	User Memory	O	Total	16GB eMMC/ 2GB DDR + external MicroSD
		O	Total User Memory	8.4GB (TBD) (w/o MicroSD)
		O	MMS	
		O	Pictures (Still Image & Moving Image)	
		O	MP3 (Music Contents)	
		X	Java Contents	
		O	Wallpaper	
		O	Ringtone	
		O	external memory (microSD)	up to 32GB
Camera (Still Capture)	Memory	O	MicroSD	
	Camera Module	O	13M AF	
	Auto Focus	O	Still Mode AF	
	Digital Zoom	O	4x	
	Capture Resolution	O	Rear 4160x3120, 4160x2340, 3120x3120 Front 2560x1920, 2560x1440, 1920x1920	
	Continuous Shot	O		
	Quality	X		
	Preview Image	O	960x720(Normal), 1280x720(Wide))	
	TV-Out	X		
	Flash	O	LED	

2.6 SW Specification

Camera (Movie Capture)	Movie Resolution	O	Rear 1920x1080, 1280x720 Front 1920x1080, 1280x720	
	Filter	X		
	Quality	X		
	MPEG Visual Bit Rate	X		Default code H.264 And video Bit rate is 17Mbps We can support MPEG-4, but the bit rate is depend on Application setting
	Audio Encode	O	AAC, , AMR-NB	
	Audio Decode	O	AMR	
	Movie Recode	O	Recode, Stop, Pause, Resume	
	Movie Playback	O	Play, Stop, Pause	
	Stream Format (Mux)	O	mp4, 3gp	mp4 – normal 3gp – mms
	Stream Format (Demux)	O	mp4, 3gp	mp4 – normal 3gp – mms
	Recoding Time	O	Real Time Recoding	
	TV-Out	X		
Audio Multimedia	Voice Codec	O	AMR/ 3gpp	
	AAC /AAC+/ EAAC+	O		
	MP3, WAV	O		
	WMA	O		
	Midi	O	- Type 0 and 1 (.mid, .xmf, .mxmf) - RTTTL/RTX (.rtttl, .rtx) - OTA (.ota) - iMelody (.imy)	
	Music ringtone	O	MP3, AAC, AAC+, eAAC+, AMR, OGG, FLAC, ALAC, PCM, WMA, OPUS	
	AMR-NB	O		
	Music player skin	O	Music Player	
	FM Radio	O		
	Speakerphone	O		

2.6 SW Specification

Multimedia	MPEG4/H.263	O	MPEG4 - Decoding : 1080p/30fps/20Mbps H.263 - Decoding : 704*576/30fps/15Mbps	
	H.264	O	Decoding : 1080p/30fps/20Mbps	
	WMV	X		
	Normal View	O	Moving Image Capture & Movie Contents	contents size
	Wide View	O	Full Screen mode	
	Custom DTV Zoom	X		
Other Features	File Viewer	X		Android App can be downloaded in Play Store
	Macromedia Flash Lite	X	Not Available	
	e-Book/e-Comic Book	X	*.doc, *.jpg, *.gif	E-book reader program is not included in ROM. But user can download in internet..
	Flight Mode	O		
Internet Browser	WAP (Windows Mobile 6.0 IE Mobile)	X	version 2.0	
		X	WML1.3	
		O	XHTML1.1	
		O	HTML4.01	Partially support - HTML 5
Java	MIDP	X		
	J2ME support	X		
	Game & Contents	X		
	CLDC	X		
Flash	Flash Game	X		
Streaming Service	H.263+ AAC/AMR-NB	O	H.263 - Decoding : 704*576/30fps/15Mbps	
	MPEG4+ AAC/AMR-NB	O	MPEG4 - Decoding : 1080p/30fps/20Mbps	
	WMV9	X		
DRM	OMA DRM	O	Forward Lock	
		X	Separate Delivery	
		X	Combined Delivery - Count-only	
OTA	OTA	O	OTA 1.0	
Display	RSSI	O	6 level (Include No service)	
	Battery Level	O	with percentage	Yes
	RTC	O		
	Multi Language	O	depending on build language.	
	Quick Access Mode	X		
	PLMN/Service Indicator	O		
	Dimming Clock	X		
			B a t t e r y p o w e r 15s, 30s, 1min, 2min, 5min 10min, 15min E x t e r n a l p o w e r 15s, 30s, 1min, 2min, 10min, 15min	Screen timeout supports

2.6 SW Specification

Normal Features	Last Dialed Number	O		
	Last Received Number	O		
	Last Missed Number	O		
	Scratch Pad Memory	X		
Call Management Managemen	Call Waiting	O		
	Call Swap	O		
	Call Retrieve	O		
	Auto Answer	O		
	Automatic Redial	X		
	Calling Line Identification	O		
	Full Call divert	O		
	Speed Dialing	O		
	Last Number Redial	X		
	Multiparty Call (Conference Call)	O		
	Explicit Call Transfer	O		
Network	Automatic Network Selection	O		
	Manual Network Selection	O		
	Network Service Status	O		
DTMF	DTMF Signaling	O		
	DTMF Enable & Disable	X		
Audio	Key Tone Volume	O	16 Level (Include Mute)	
	Ring Tone Volume	O	16 Level (Include Mute)	
	Ring Tone Pattern	O	ogg	
	Ring Type Silent	O	Vibrator & Ring (Indicator)	
	Earpiece Volume	O	16 Level (Include Mute)	
	Mute	O		
Cell Broadcast	Read Cell Broadcast	O		
	Cell Broadcast Categories	X		
	Cell Broadcast Message Language	O	Supports filtering with language IDs (German, Italian, French, Spanish, Dutch, Swedish, Danish, Portuguese, Finnish, Norwegian, Greek, Turkish, Hungarian, Polish, Czech)	

2.6 SW Specification

Phone Book	Entry	O	Name, Phone, Email, Groups, Postal address, Ringtone, Notification Sound, Phonetic name, Organization, IM, Notes, Nickname, Website, Event	
	Field	O	Phone(Mobile,Work,Home,Main, Work Fax ,Home Fax,Pager, Other, Custom) Email(Home,Work,Other,Custom) Groups(Coworkers,Family,Friends) Postal address(Home,Work,Other,Custom) Organization(Company,Title) IM(AIM,Windows Live,Yahoo,Skype,QQ,Hangouts,ICQ,Jabber,Custom) Event(Birthday,Anniversary,Other, Custom)	
	Numeric Store and Recall	O		
	Alphabetic Store	O		
	Alphabetic Recall	X		
	Scroll by alphabetic or numeric order	O		
	Last Number Dialed	O		
	Last Number Missed	O		
	Last Number Received	O		
	Copy & Move	X	Move not support	
	Fixed Dial Number	O		
	Barred Dial Number	X		
	Service Dial Number	X	It depends on SIM card	
	Email Entry	O		
	Picture ID	O	Relating Photo caller ID	
Call Cost	My Name card	O	Bluetooth transmission (OPP)	
	Last Call Timer	X		
	Last Call Charge Units	X		
	Total Call Timer	X		
	Total Charge Units	X		
Supp. Services	Visible, Audible Call Time /Cost Indication	X		
	Call Forwarding	O	All Incoming Calls,	
		O	No Reply	

2.6 SW Specification

SIM	Plug In Type	O		
	SIM Lock	O		
	SIM Toolkit	O		
	Prepaid SIM Operation	O		
Short Messaging	Read Message	O	EMS Not Support	
	Write and Edit Message	O		
	Send and Receive Message	O		
	Reply to Message	O		
	Forward Message	O		
	Extract Number from Message	O		
	Message Status	O		
	Message Unread Indicator	O		
	Settable Message Center Number,	O		
	Reply Path and Validity	X		
	Visible and Audible Message Receive Alerting	O		
	Voice Mail	O		
	Settable Voice Mail Center Number	X		
	Message Protocol	X		
	Message Overflow Indicator	O		
	Message Center Number	O		
	Help Menu	X		
Multimedia Messaging	Picture Attachable	O		Max bytes is it, According to inserted usim
	Moving Picture Attachable	O	Video MMS	Max bytes is it, According to inserted usim
	Voice Attachable	O	AMR	
	Help Menu	X		

2.6 SW Specification

Instant Messaging Service	IM	X		Android App can be downloaded in Play Store
	PC Compatable	X		
	Menu Option	X		
E-Mail	Protocol	O	IMAP4	
		O	SMTP	
		O	POP3	
		X	Help Menu	
		X	Full Duplex	
Voice Instant Messaging Service	PTT	X	Half Duplex	
		X	Grouping	
		X	Status List	
		X	Help	
		X	Others	
Sound Contents	Wallpaper	X		
	Screen Saver	X		
	Ringtones	O		
	Karaoke	X		
	Stutter Sound	O		
	Flip tone	X		
	Button tone	O	can be enable or disable button press sound.	
	Others	X		
Calendar	New Event	O		
	Edit Event	O		
	Agenda	O		
	Reminder	O		
	Google sync	O		
Miscellaneous Function	Development & Test Facility	O		
	Field Test Facility	X		
	Display Software Version	O		
	IMEI	O	*#06#	
Text Input	Language	O		
	Predictive word input	O	T9 predictive supported.	
Scheduler	Schedule	O		
	To Do List	O		
	Memo	O		
	D-day counter	X		
	Send via Bluetooth	O		
World Time	Setting Local Time	O		
	Display Two Number of Cities Time	X		

2.6 SW Specification

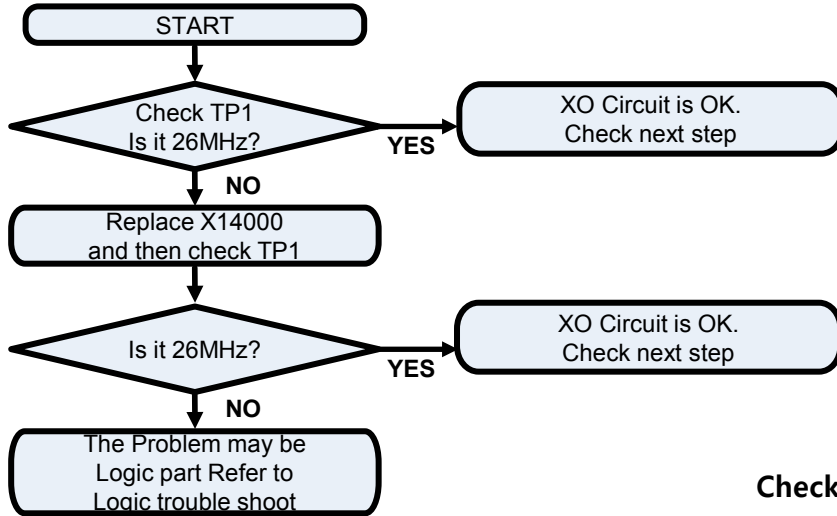
Unit converter		X		
Stop watch		O		
Calculator	Calculation	O		
PC Sync	Phone Book Sync	X	PC Bridge supports backup and restore (not sync)	
	Message Sync	x	PC Bridge supports backup and restore (not sync)	
	Multimedia Contents Sync	x	PC Bridge supports backup and restore (not sync)	
	Scheduler Sync	X	PC Bridge supports backup and restore (not sync)	
Sync ML	DS	X		
	DM	X		
Game		X	TBD	
Menu	Quick Access Mode (Profile)	O		
External Interface	Electrical Man Machine Interface	X		
	Development and Test Facility	X		
Handset	Restore Factory Setting	O		
	Read Software Version	O		
	Battery Charging Mode	O	USB/ TA Charging	USB/ TA Charging
Security	Emergency Call	O		
	Handset Lock	O		
	Security Code	O		
	SIM Lock	O		
	Key guard	X		
Real Time Clock	12/24 hour	O		
	Calendar	O		
	Time Zone	O		
	Alarm Manager	O		
	Dimming Clock	X	B a t t e r y p o w e r 15s, 30s, 1min, 2min, 10min, 15min E x t e r n a l p o w e r 15s, 30s, 1min, 2min, 10min, 15min	Screen timeout supports
	Power-off Alarm	X		
	On Alarm Event	O	Display	

3. TROUBLE SHOOTING

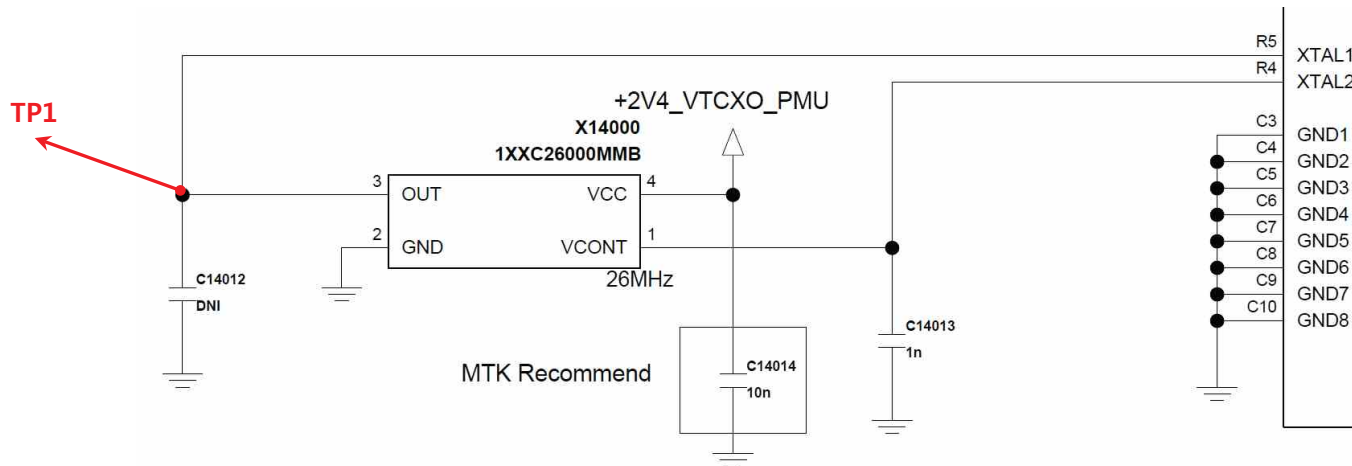
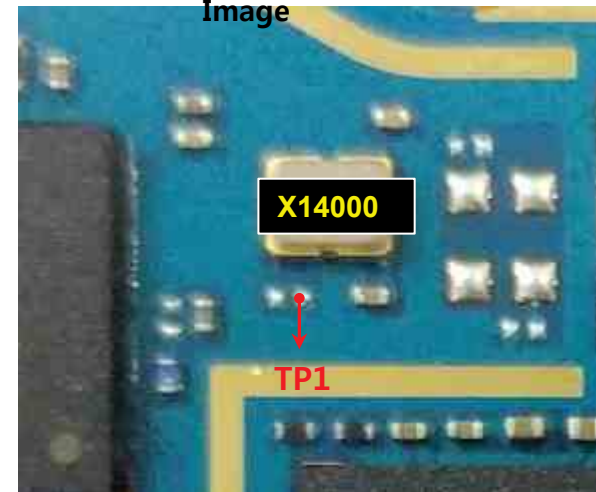
3.1 Checking XO Block

The out put frequency(26MHz) of VTCXO(X14000) is used as the reference one of MT6169

Checking Flow



Checking Flow

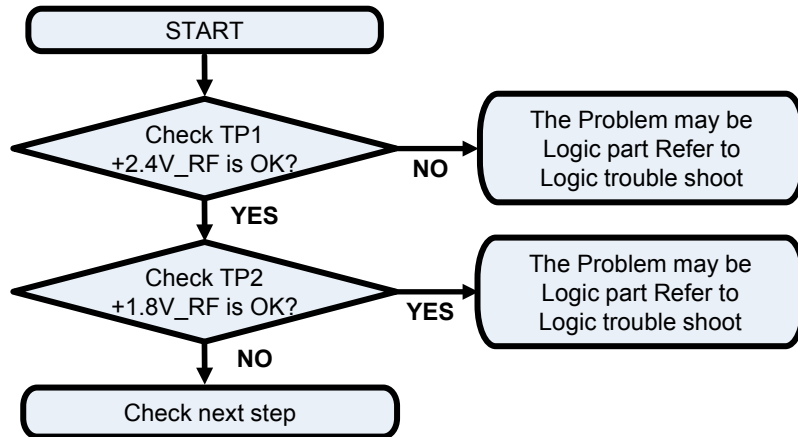


3. TROUBLE SHOOTING

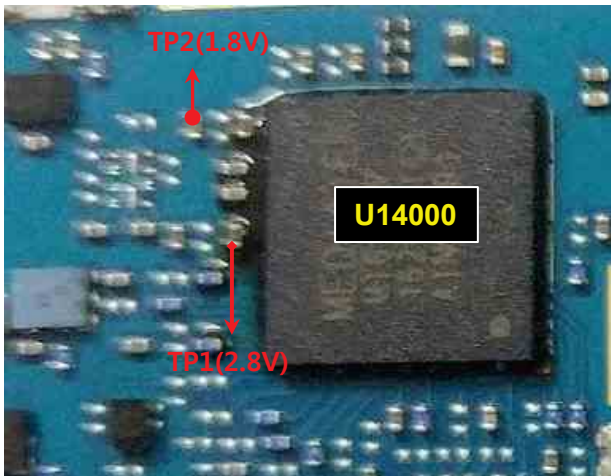
3.2 Checking Transceiver DC Power Supply Circuit Block

The MT6169 operating voltages used two voltage sources 1.8V and 2.4V

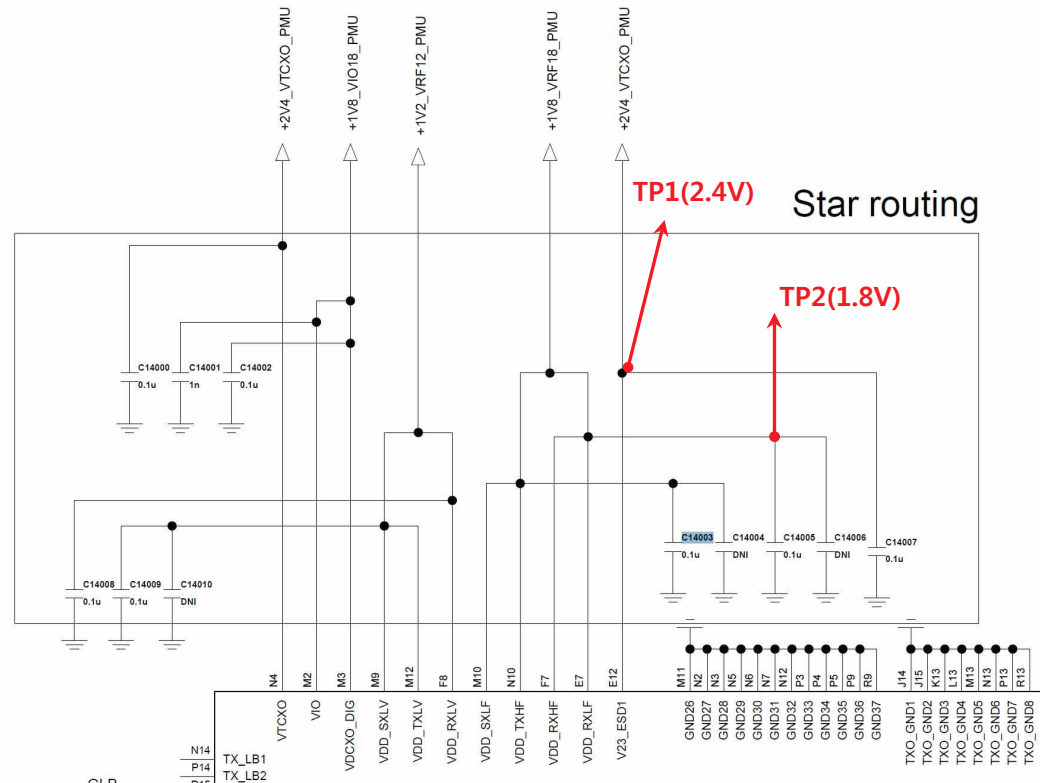
Checking Flow



Image



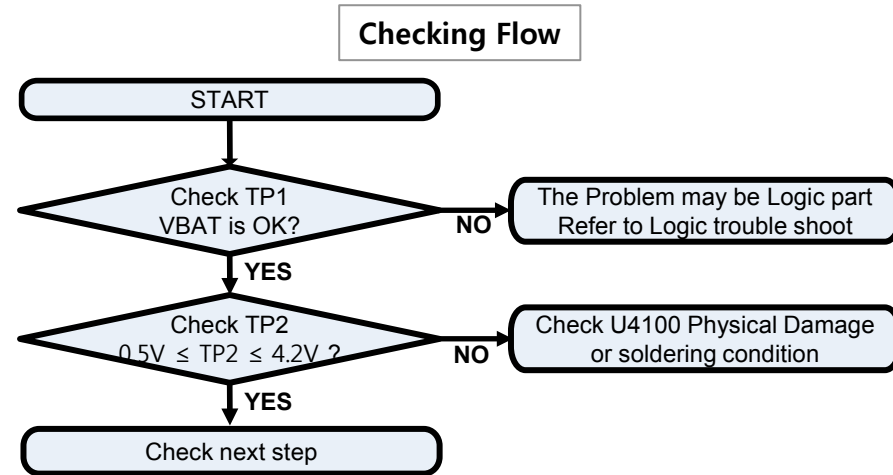
Circuit Diagram



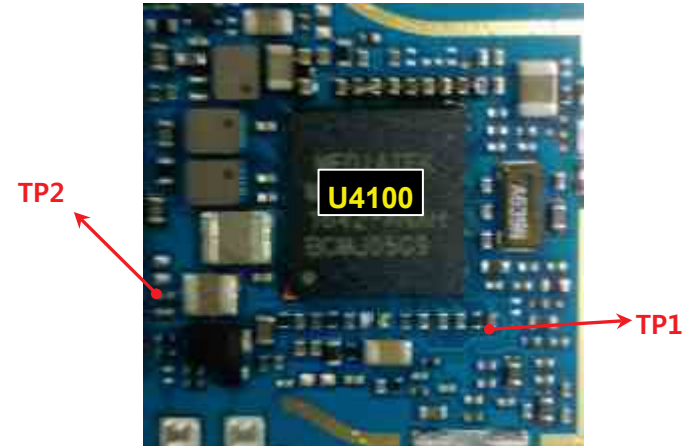
3. TROUBLE SHOOTING

3.3 Checking DC-DC Block

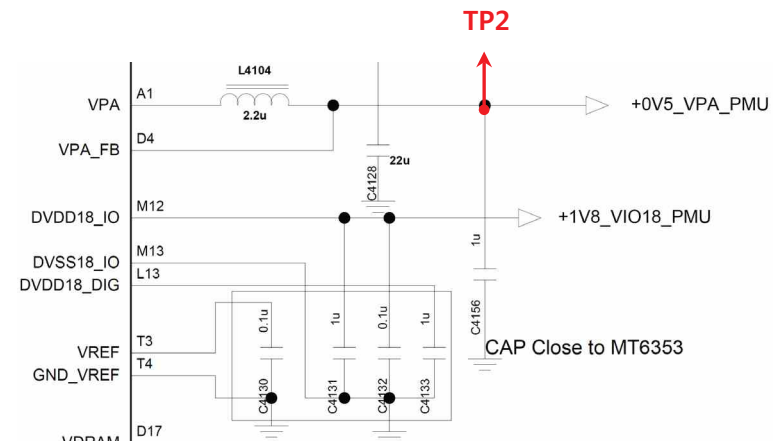
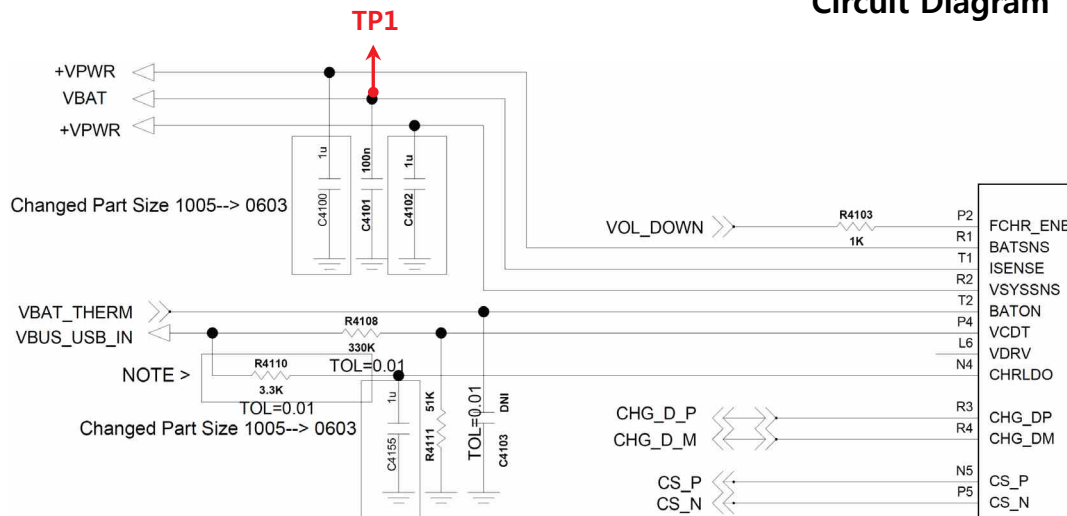
The DC-DC(MT6328, U4100) output voltages is used as the reference one of SKY77643-31



Image



Circuit Diagram

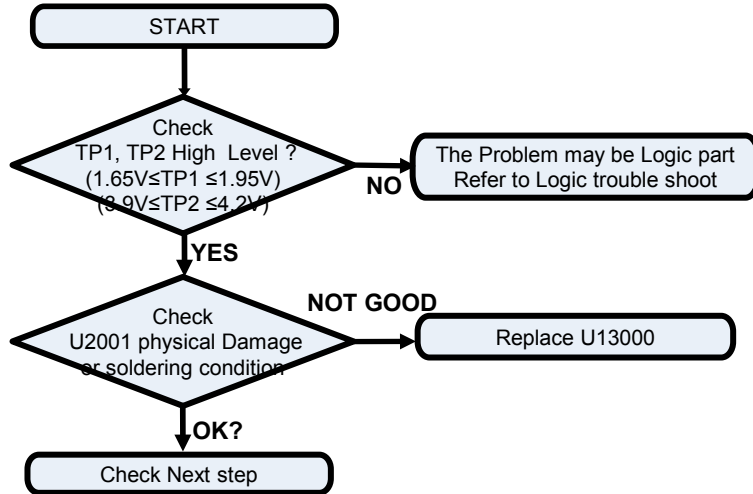


3. TROUBLE SHOOTING

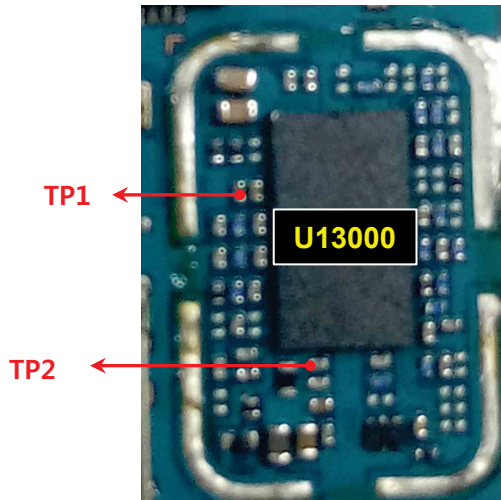
3.4 ASM(Antenna Switch Module) Block

3.4.1 Checking ANT #1 ASM (GSM 850/900, W B5/8, LTE B7/28)

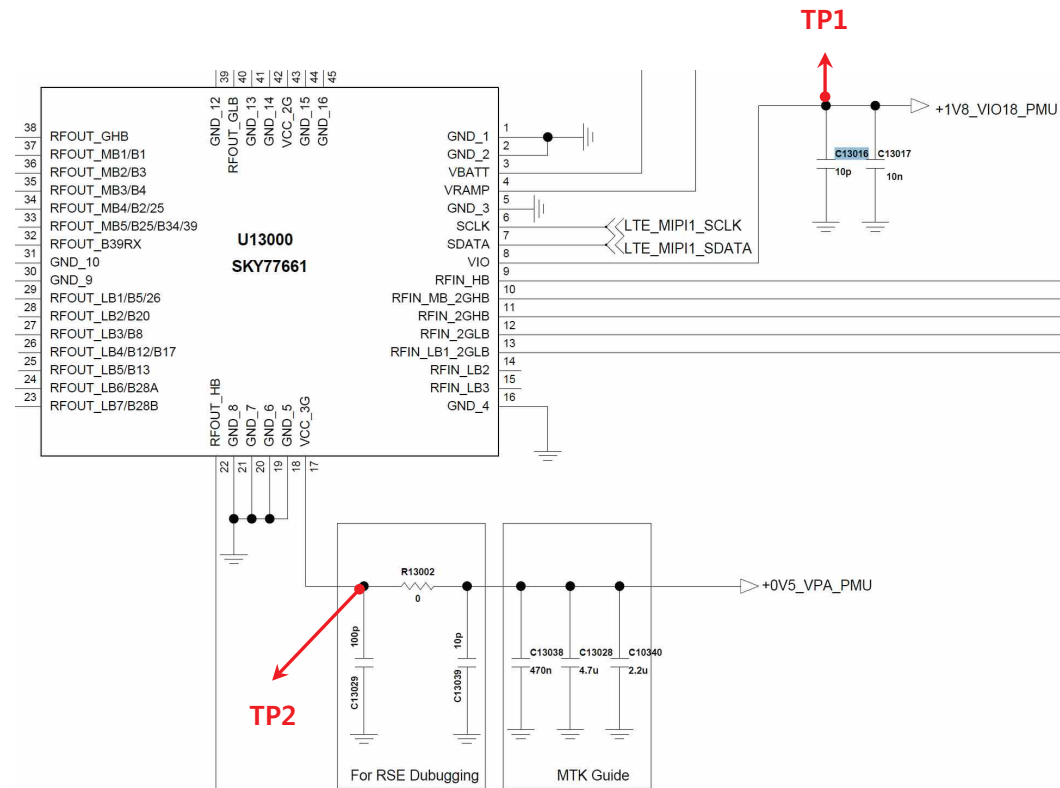
Checking Flow



Image



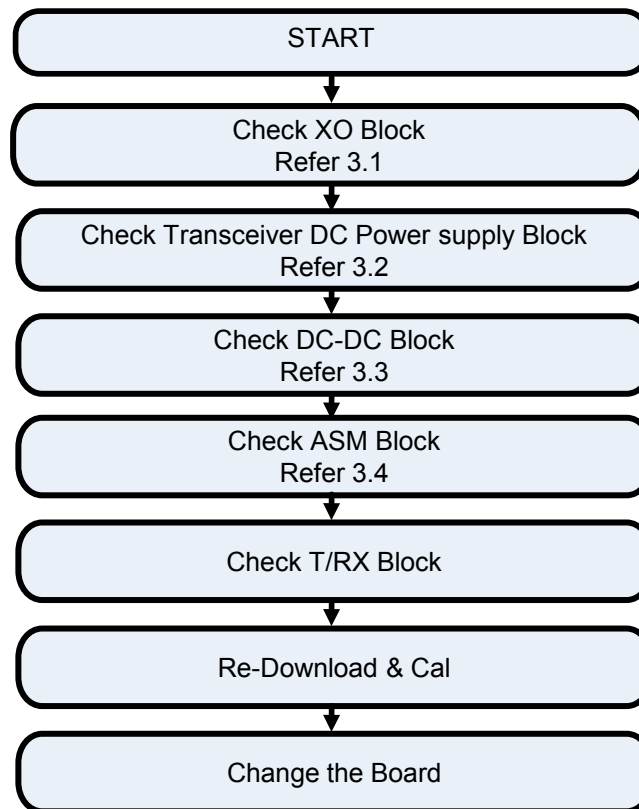
Circuit Diagram



3.5 GSM RF PART

GSM RF Part support GSM850/900/1800/1900 with ASM, PAM, Transceiver component

Checking Flow

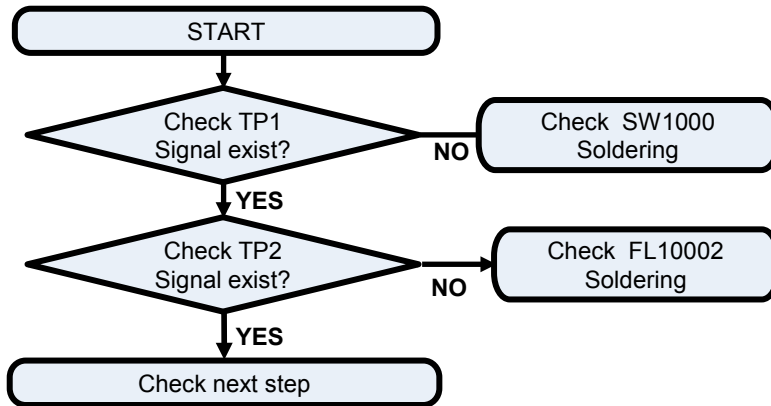


3. TROUBLE SHOOTING

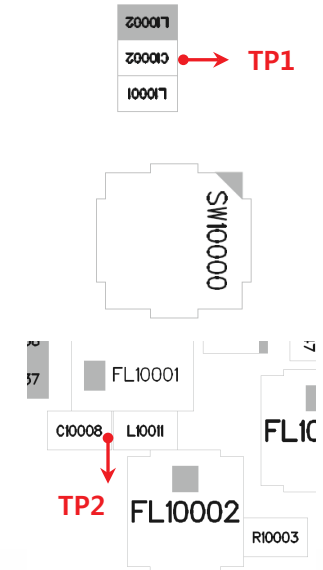
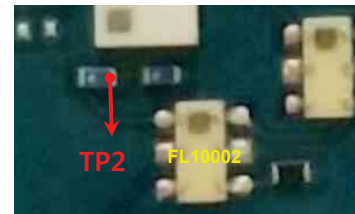
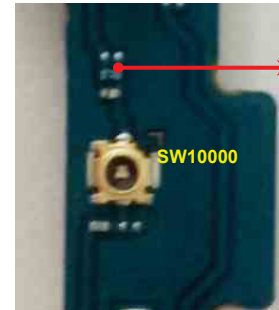
3.5 GSM RF PART

3.5.2 Checking RF Signal RX path(SW, GSM850/900)

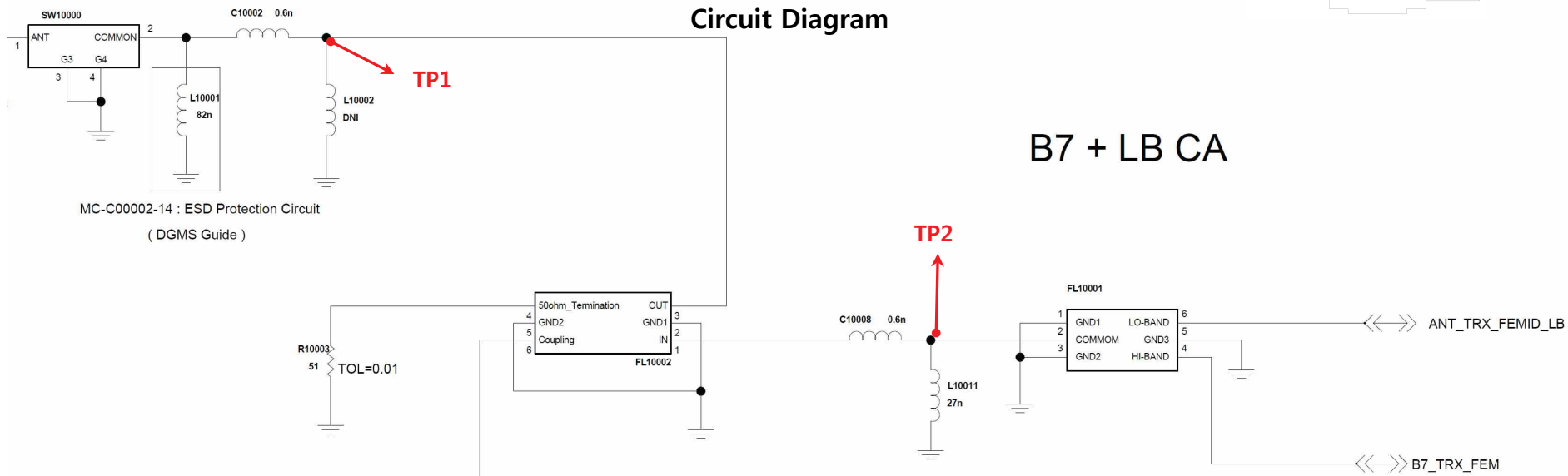
Checking Flow



Image



Circuit Diagram

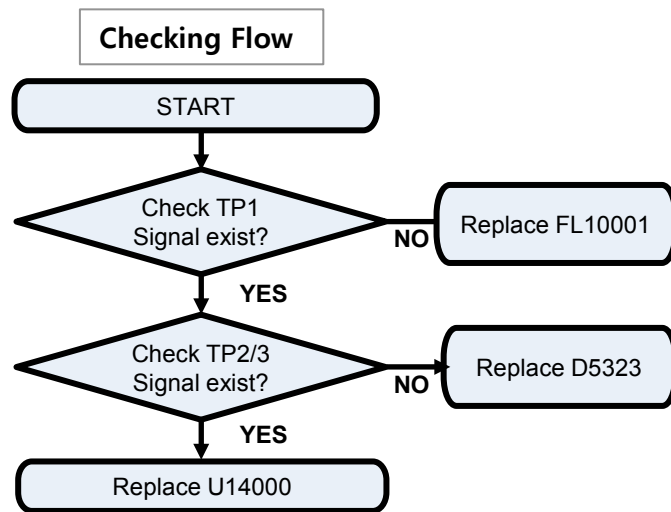


B7 + LB CA

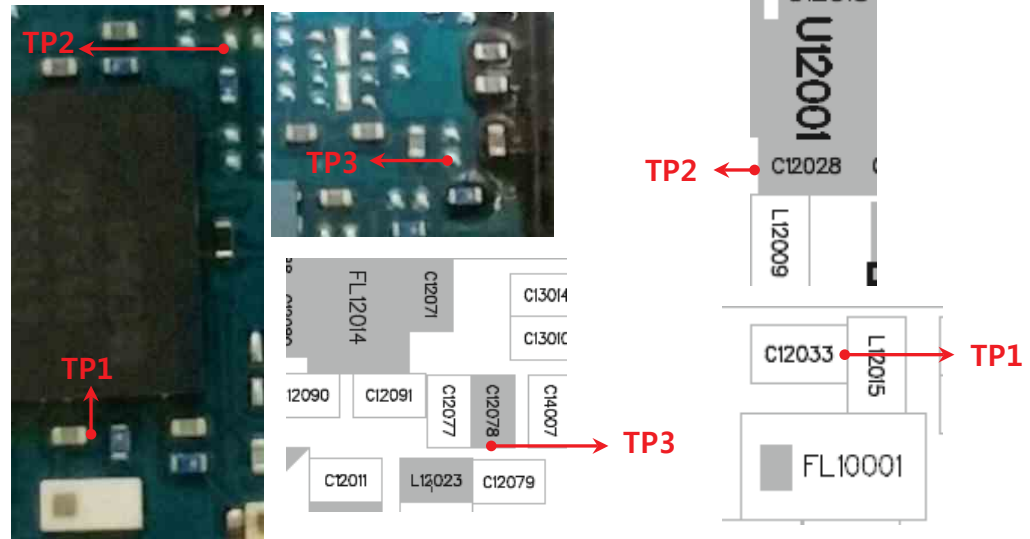
3. TROUBLE SHOOTING

3.5 GSM RF PART

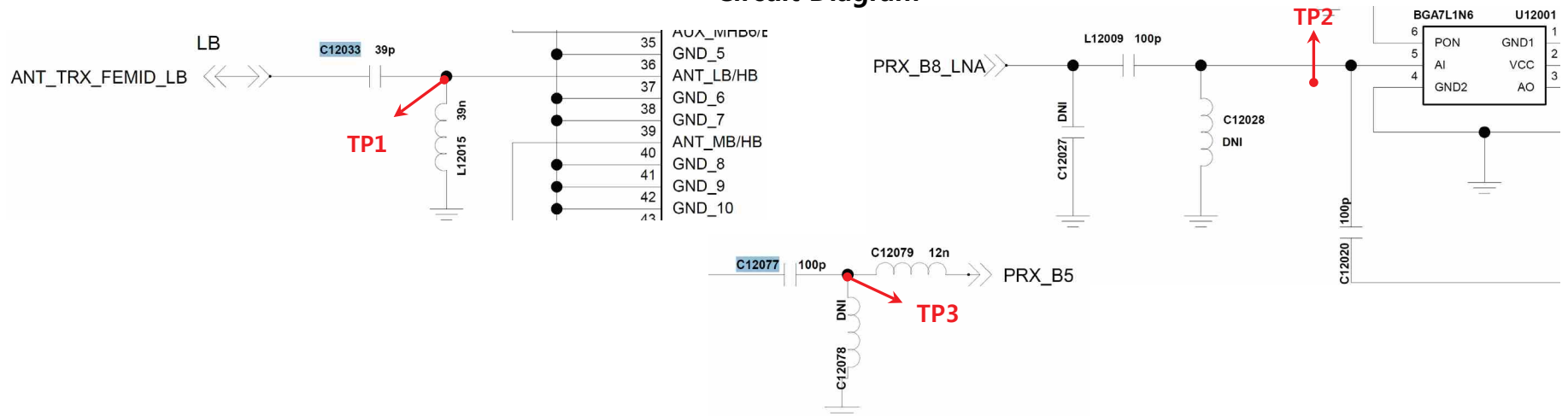
3.5.3 Checking RF Signal RX path(GSM850/900)



Image



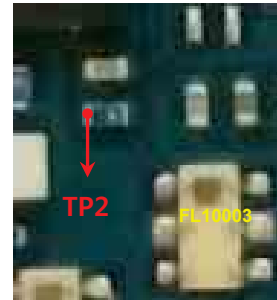
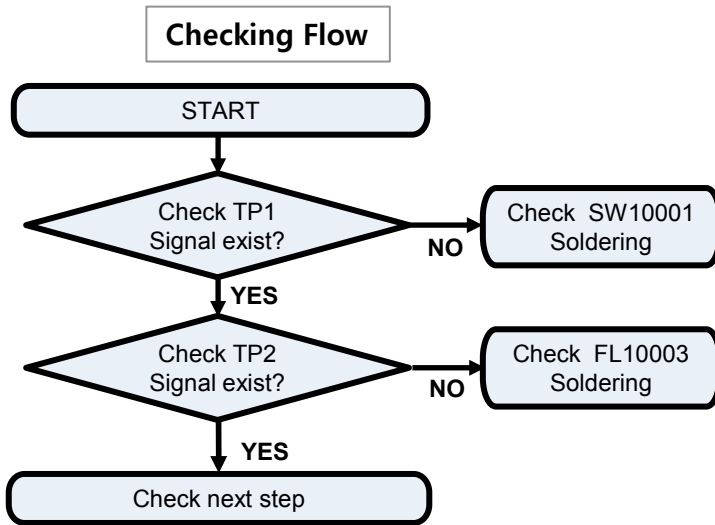
Circuit Diagram



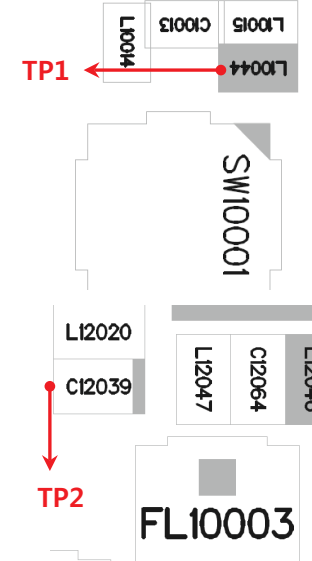
3. TROUBLE SHOOTING

3.5 GSM RF PART

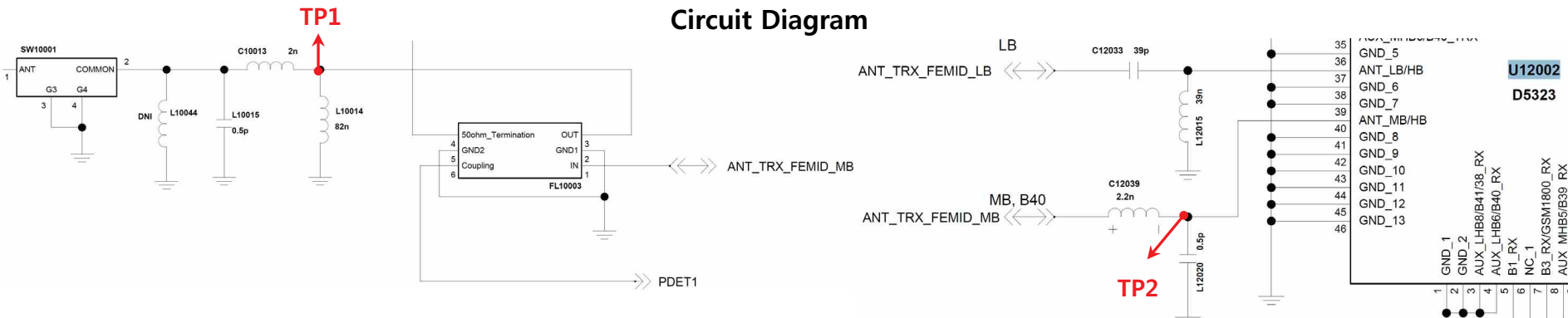
3.5.4 Checking RF Signal RX path(GSM1800/1900)



Image



Circuit Diagram

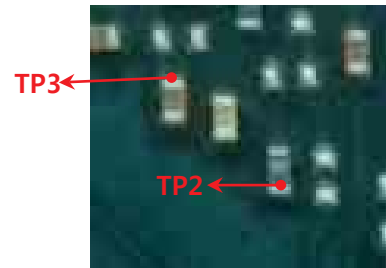
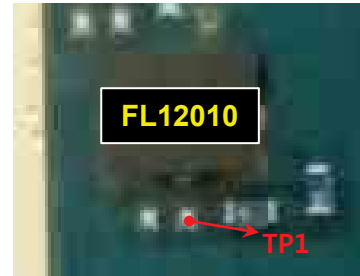
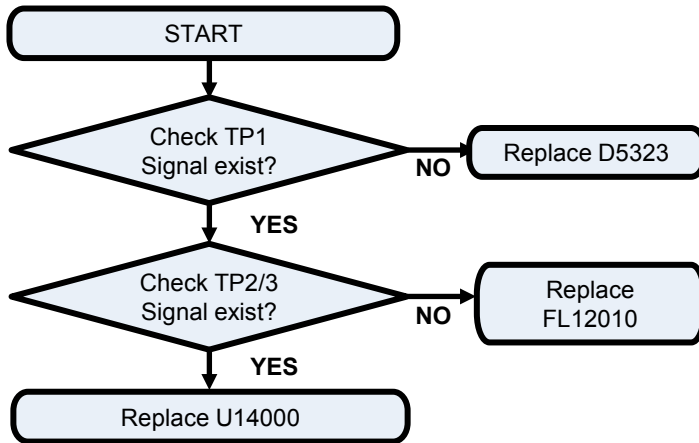


3. TROUBLE SHOOTING

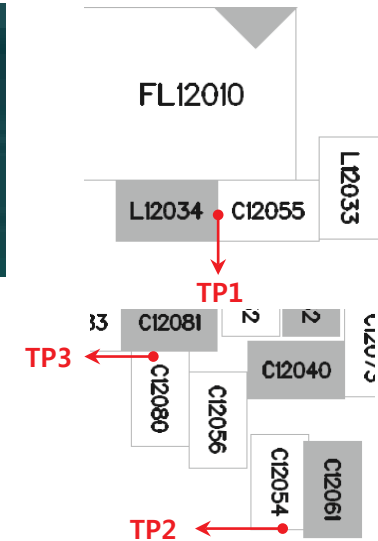
3.5 GSM RF PART

3.5.5 Checking RF Signal RX path(GSM1800/1900)

Checking Flow

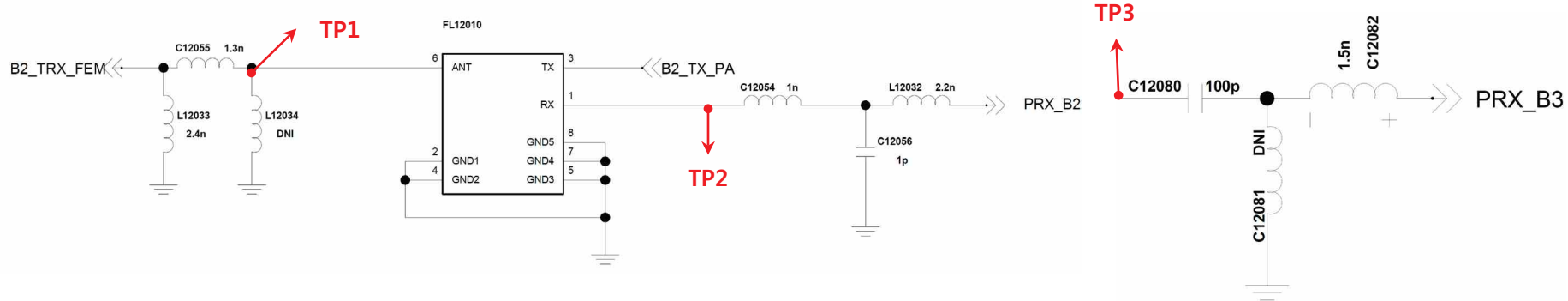


Image



Circuit Diagram

B2 / W1900 / GSM1900

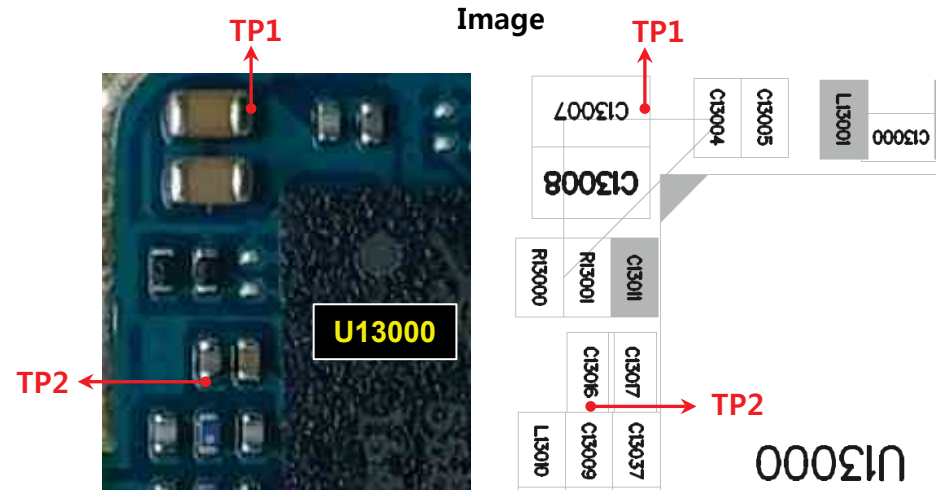
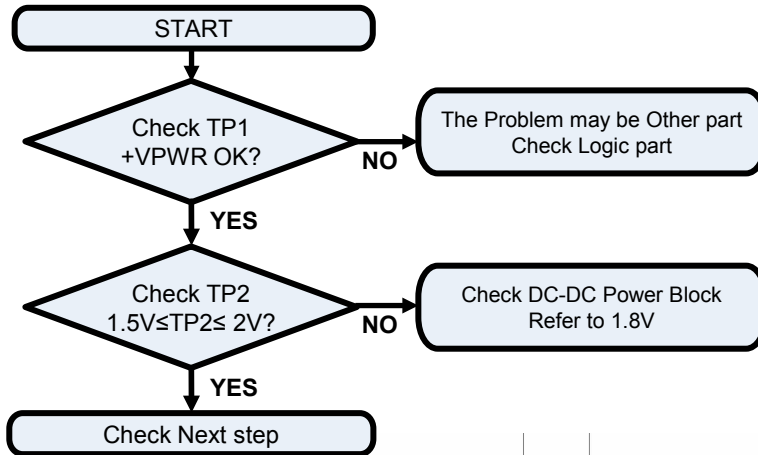


3. TROUBLE SHOOTING

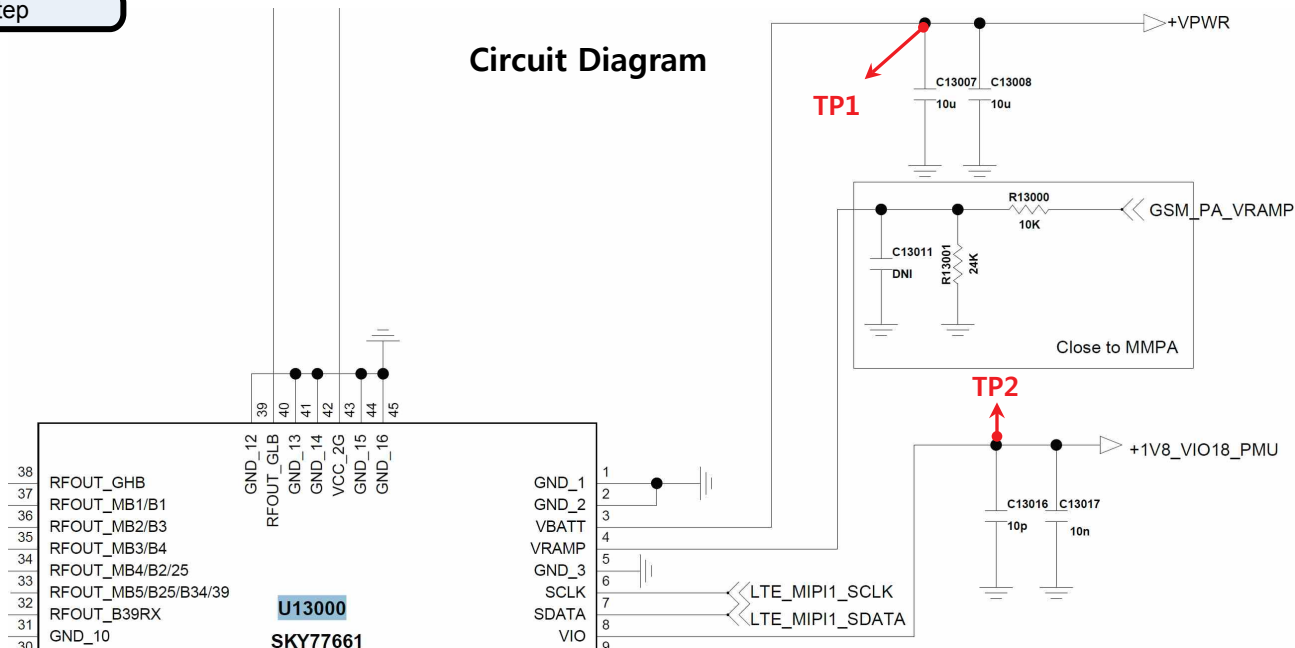
3.5 GSM RF PART

3.5.7 Checking GSM PAM DC Power Circuit

Checking Flow



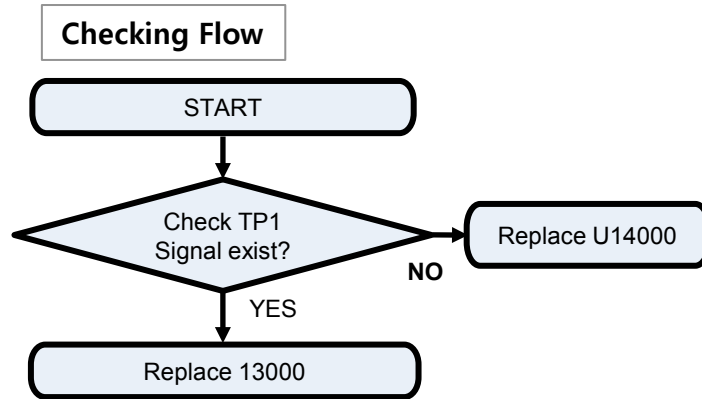
Circuit Diagram



3. TROUBLE SHOOTING

3.5 GSM RF PART

3.5.8 Checking RF Signal TX path(GSM850/900)



Circuit Diagram

Image

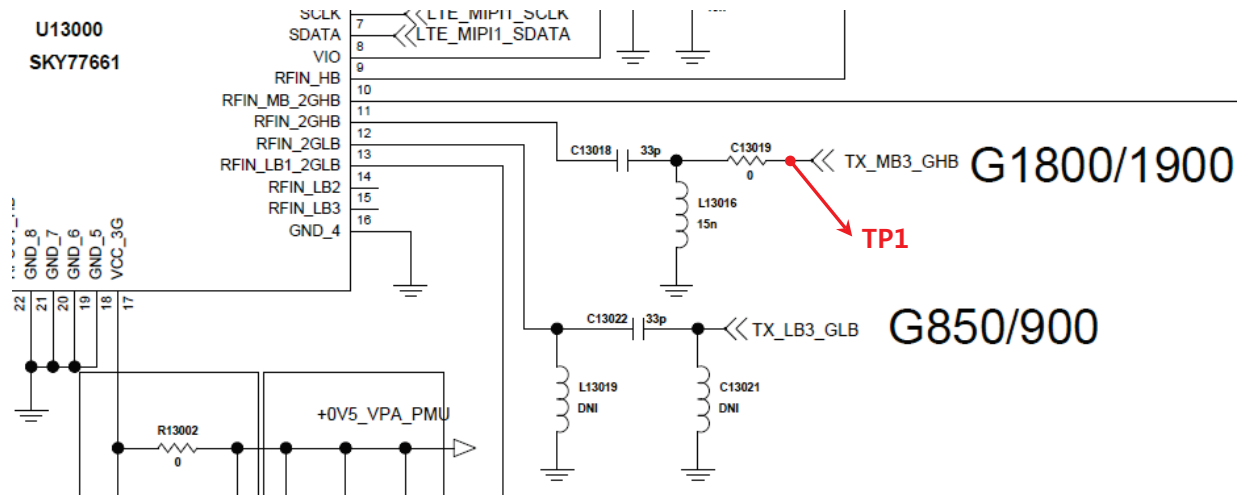
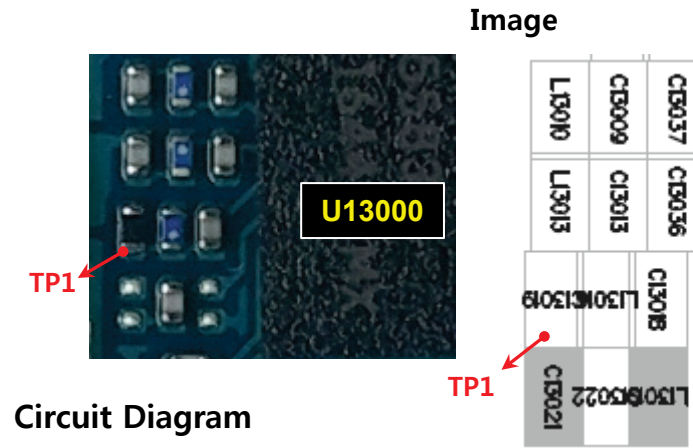
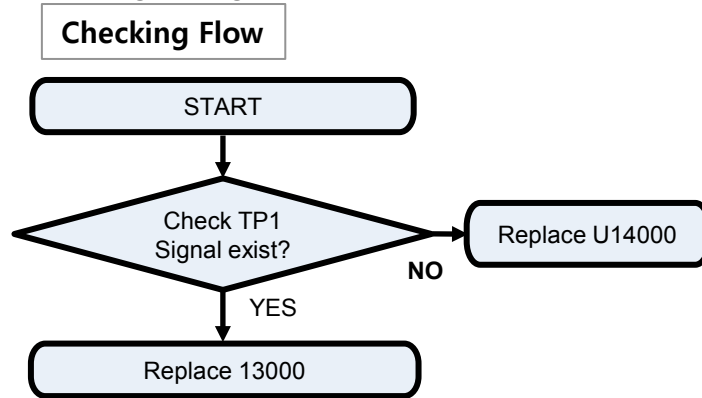


000510

3. TROUBLE SHOOTING

3.5 GSM RF PART

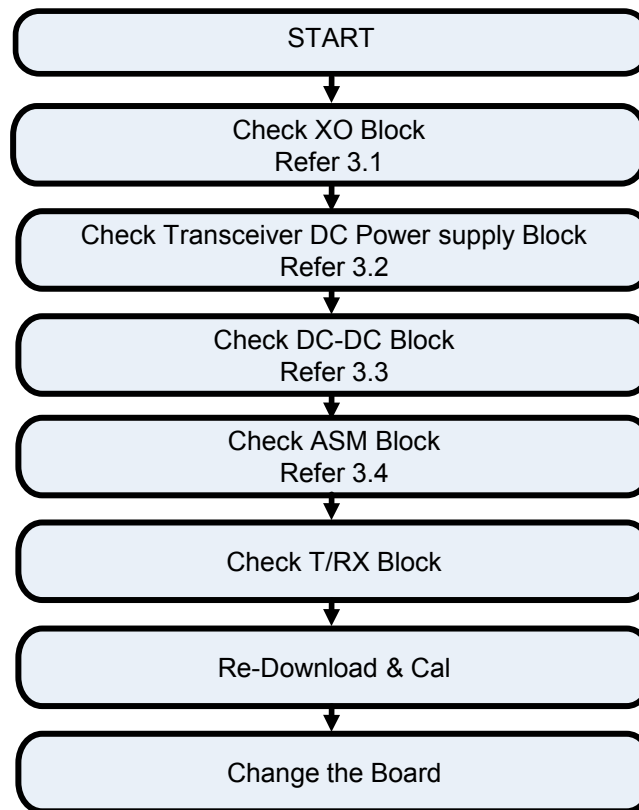
3.5.9 Checking RF Signal TX path(GSM1800/1900)



3.6 WCDMA RF PART

WCDMA RF Part support WCDMA B1/2/5/8 with ASM, PAM, Transceiver component

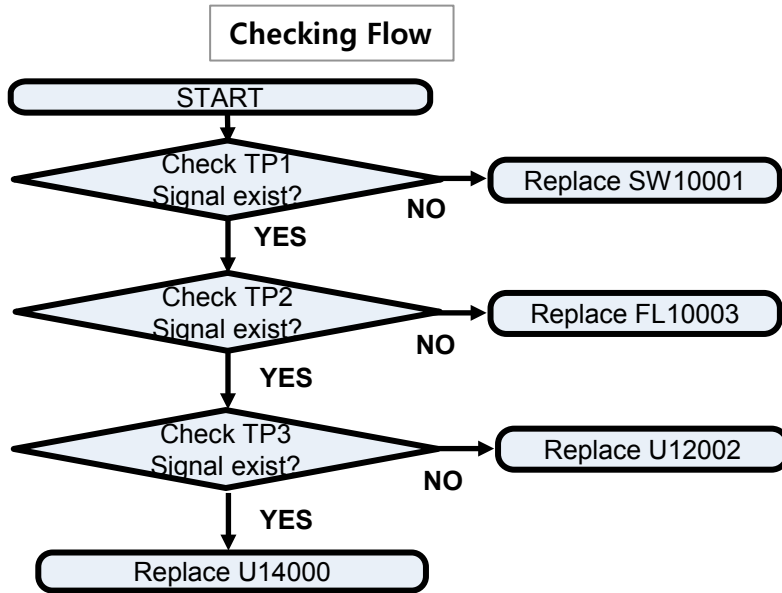
Checking Flow



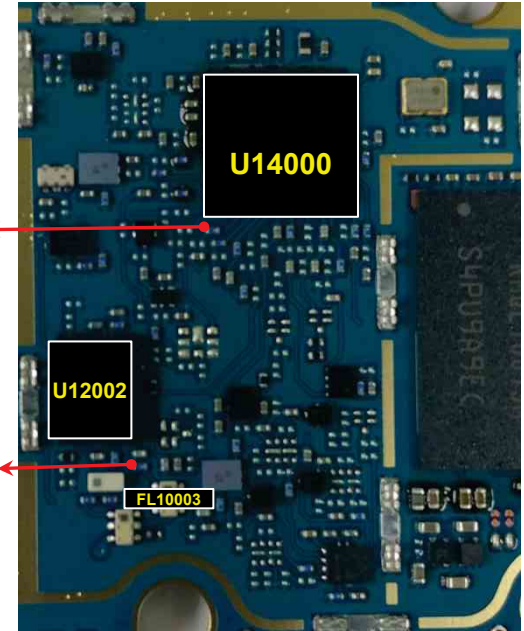
3. TROUBLE SHOOTING

3.6 WCDMA RF PART

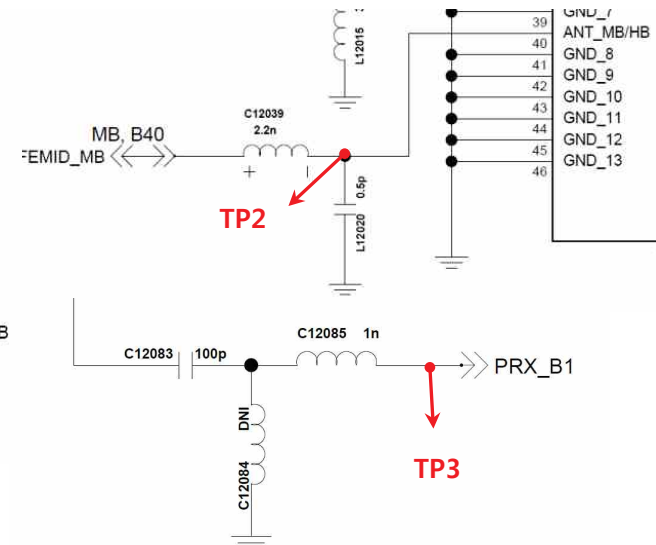
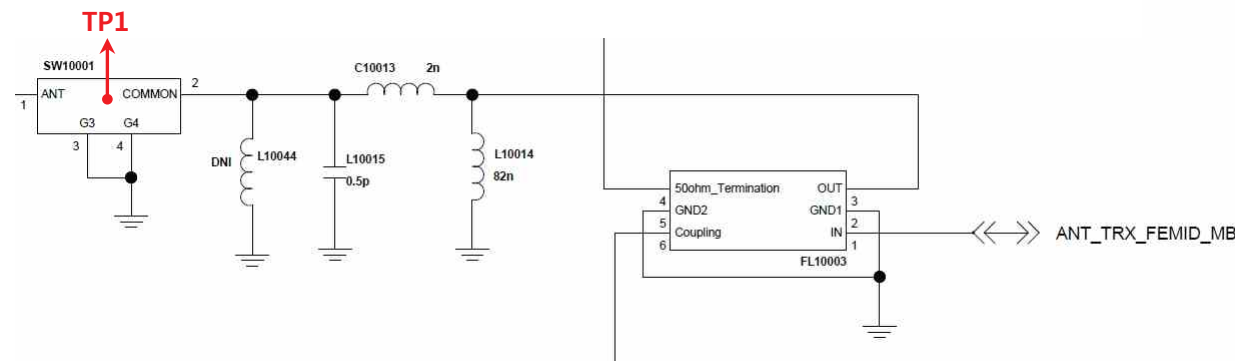
3.6.2 Checking RF Signal RX path(WCDMA B1)



Image



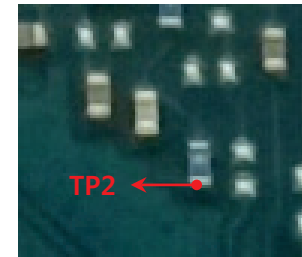
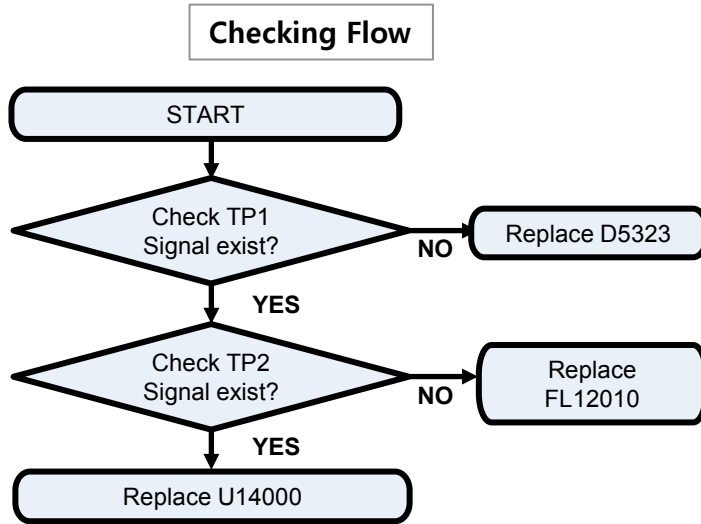
Circuit Diagram



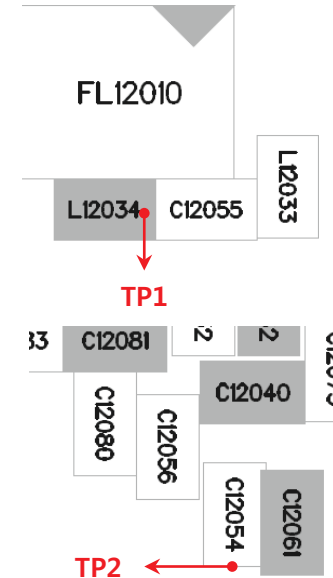
3. TROUBLE SHOOTING

3.6 WCDMA RF PART

3.6.2 Checking RF Signal RX path(WCDMA B2)

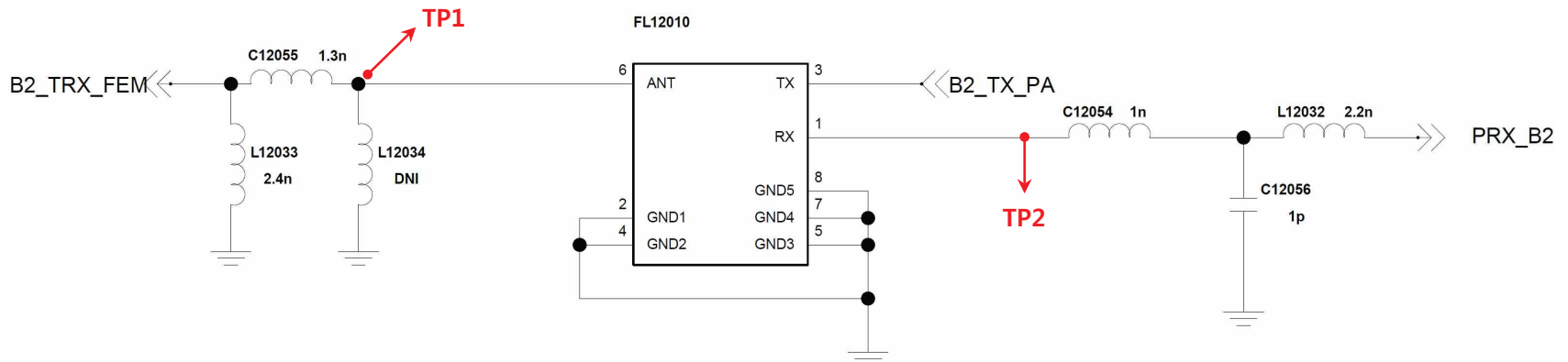


Image



B2 / W1900 / GSM1900

Circuit Diagram

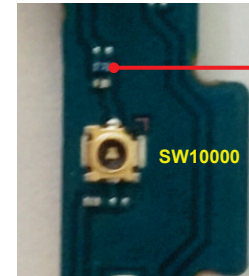
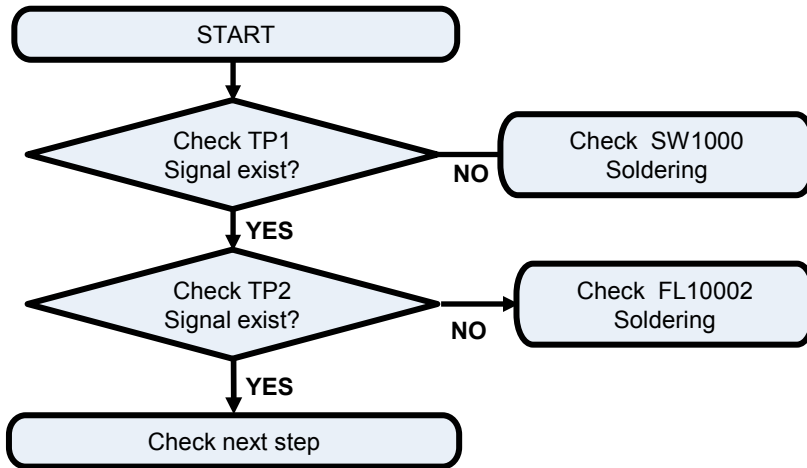


3. TROUBLE SHOOTING

3.6 WCDMA RF PART

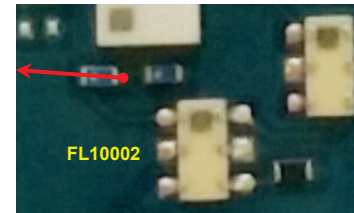
3.6.2 Checking RF Signal RX path(SW, WCDMA B5/B8)

Checking Flow

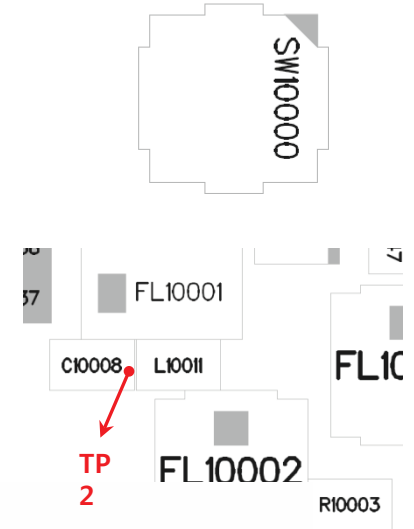


Image

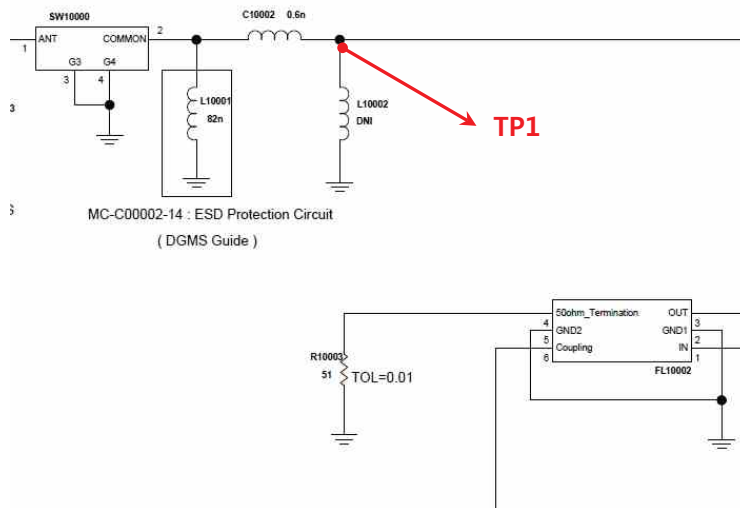
TP1



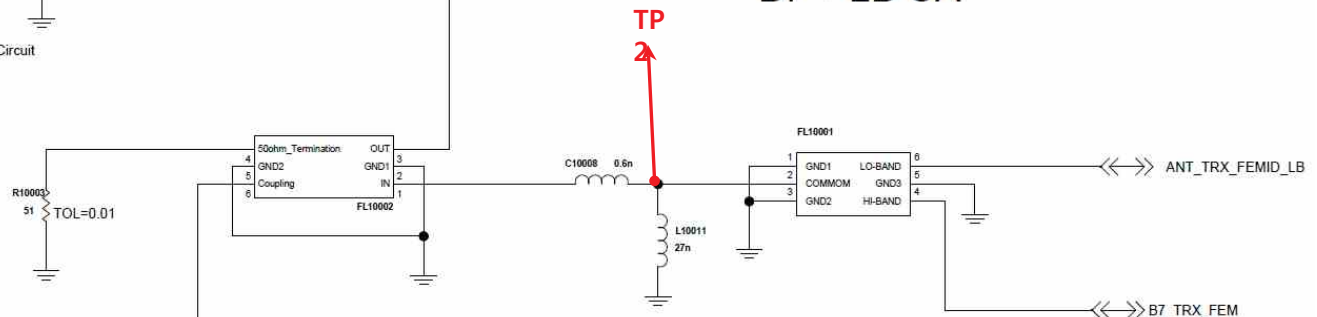
TP2



Circuit Diagram



B7 + LB CA

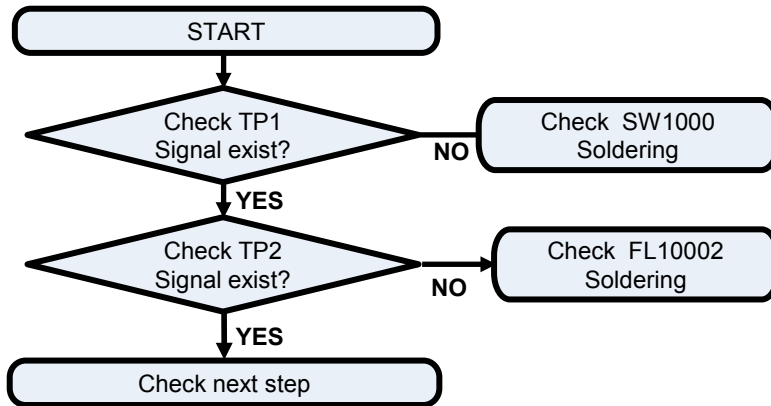


3. TROUBLE SHOOTING

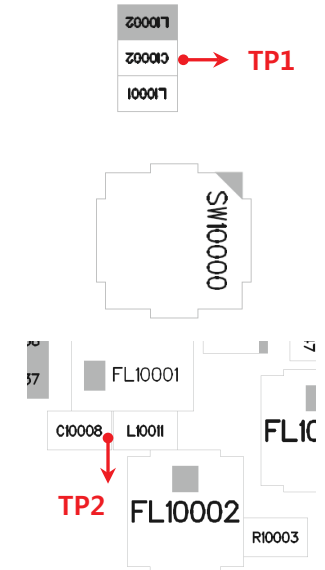
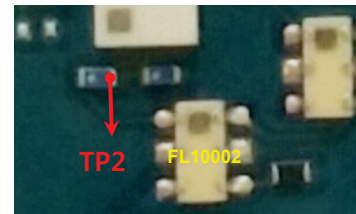
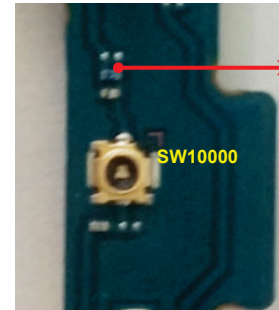
3.6 WCDMA RF PART

3.6.2 Checking RF Signal RX path(SW, WCDMA B5/B8)

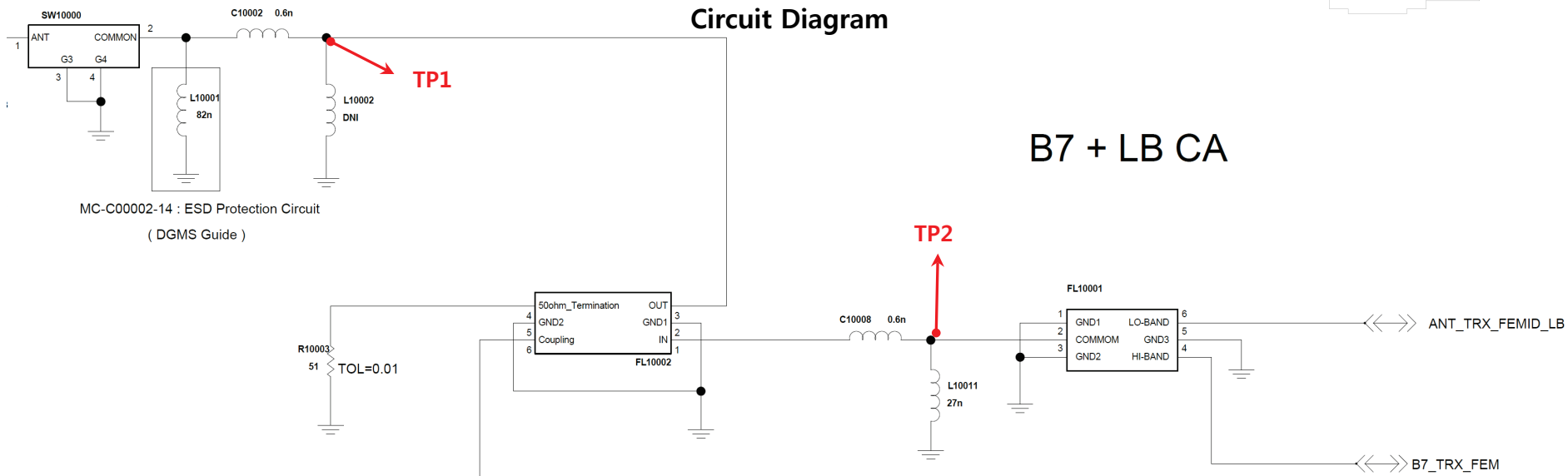
Checking Flow



Image



Circuit Diagram

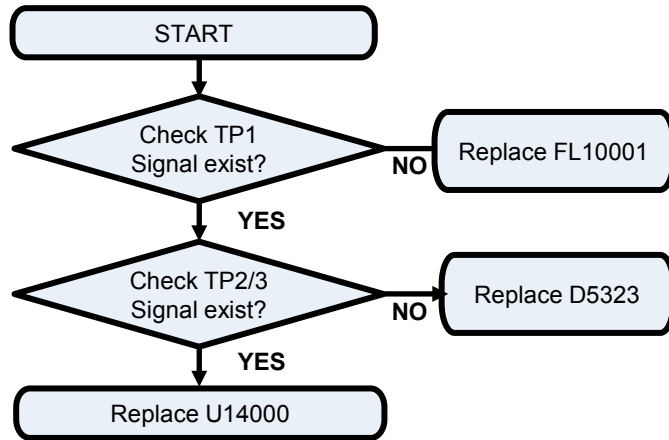


3. TROUBLE SHOOTING

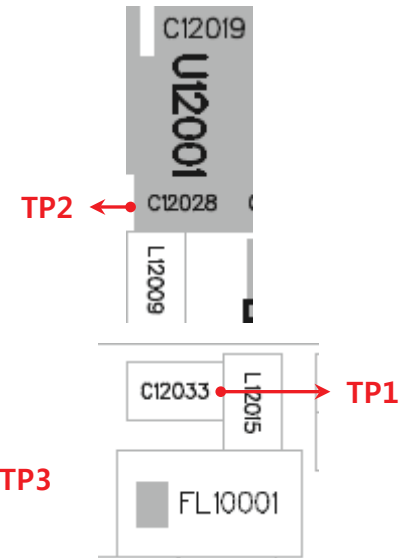
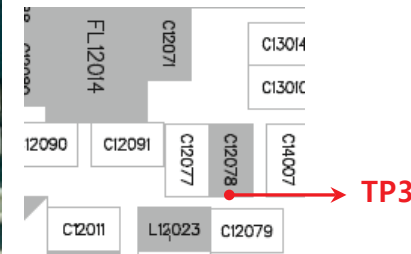
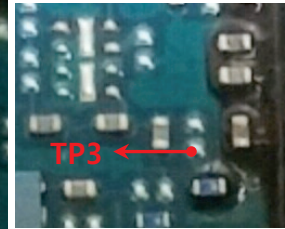
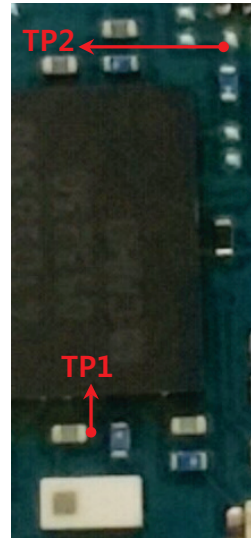
3.6 WCDMA RF PART

3.6.2 Checking RF Signal RX path(WCDMA B5/B8)

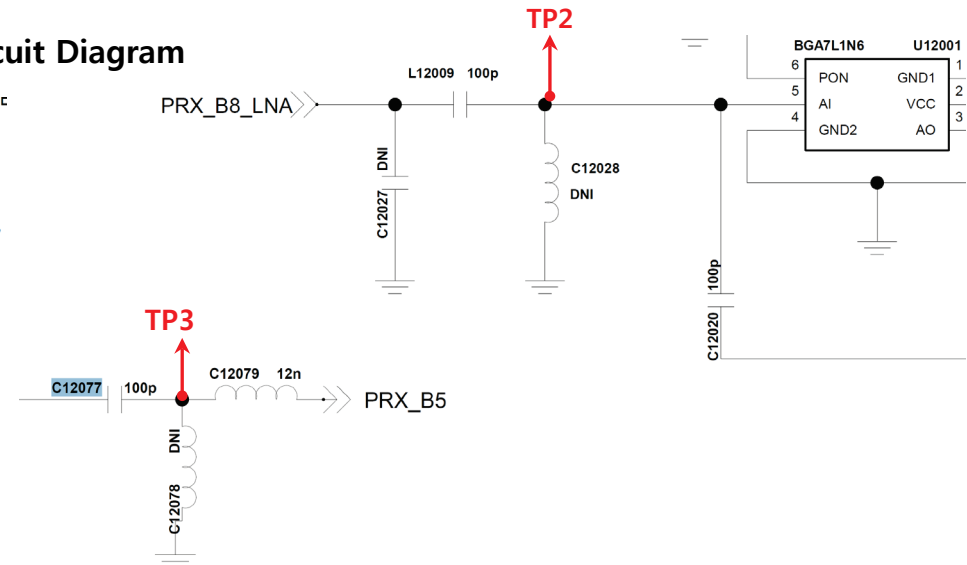
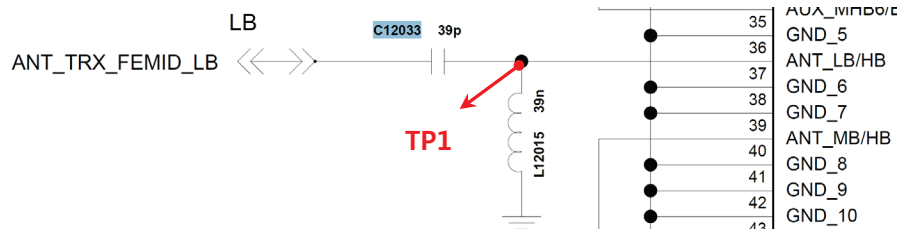
Checking Flow



Image



Circuit Diagram

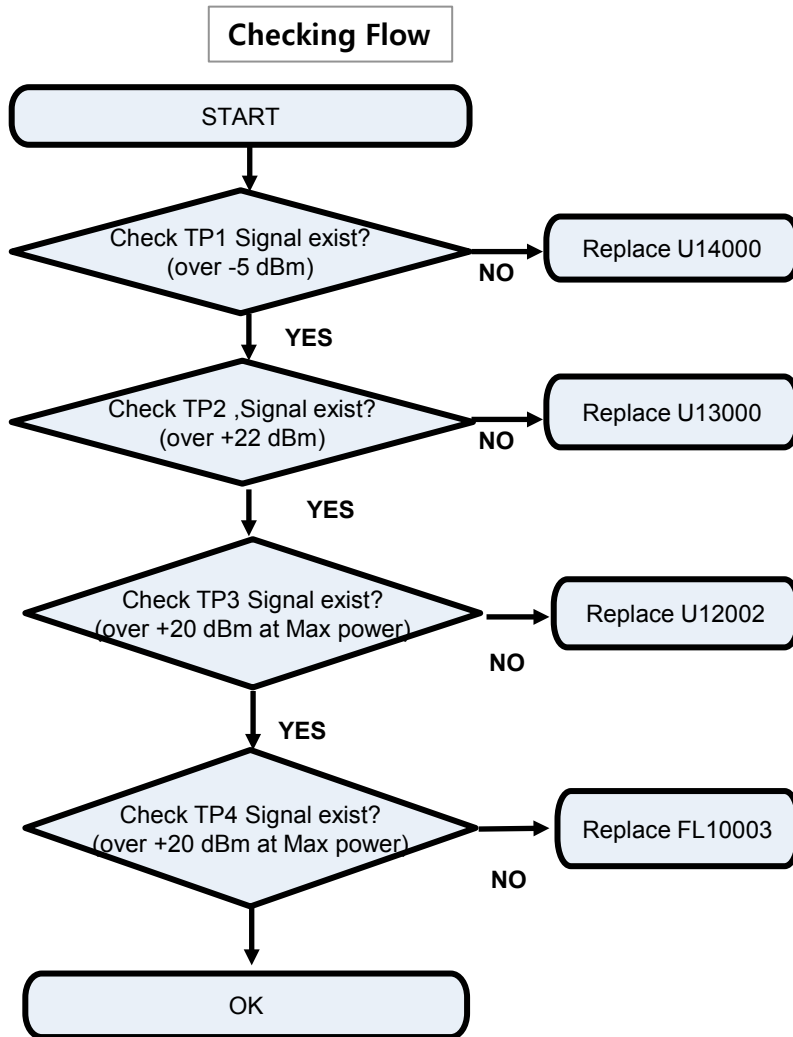


3. TROUBLE SHOOTING

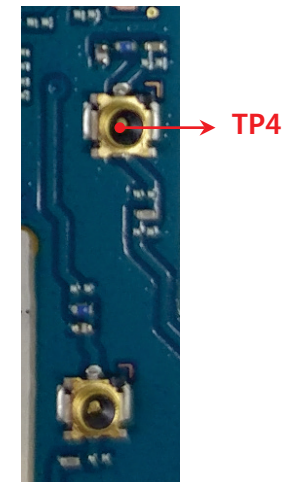
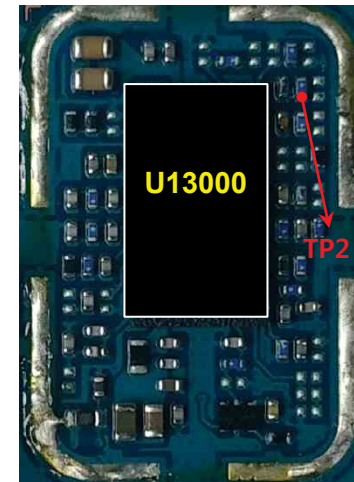
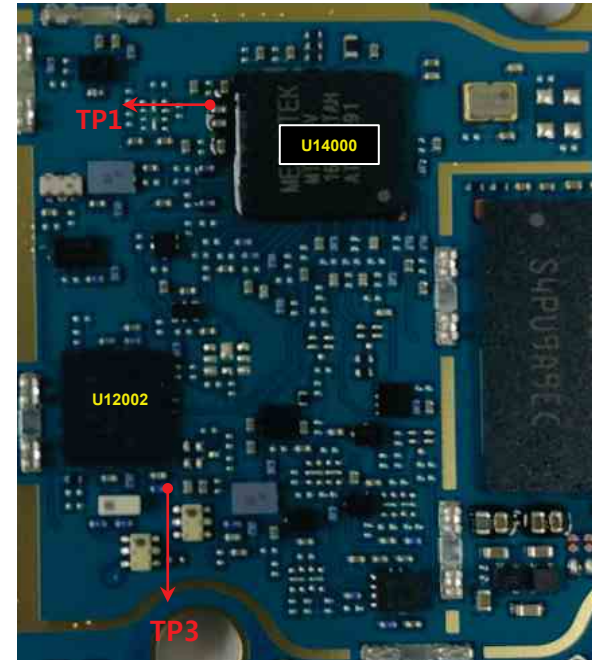
3.6 WCDMA RF PART

3.6.5 Checking RF Signal TX path(WCDMA B1/B2)

Checking Flow



Image

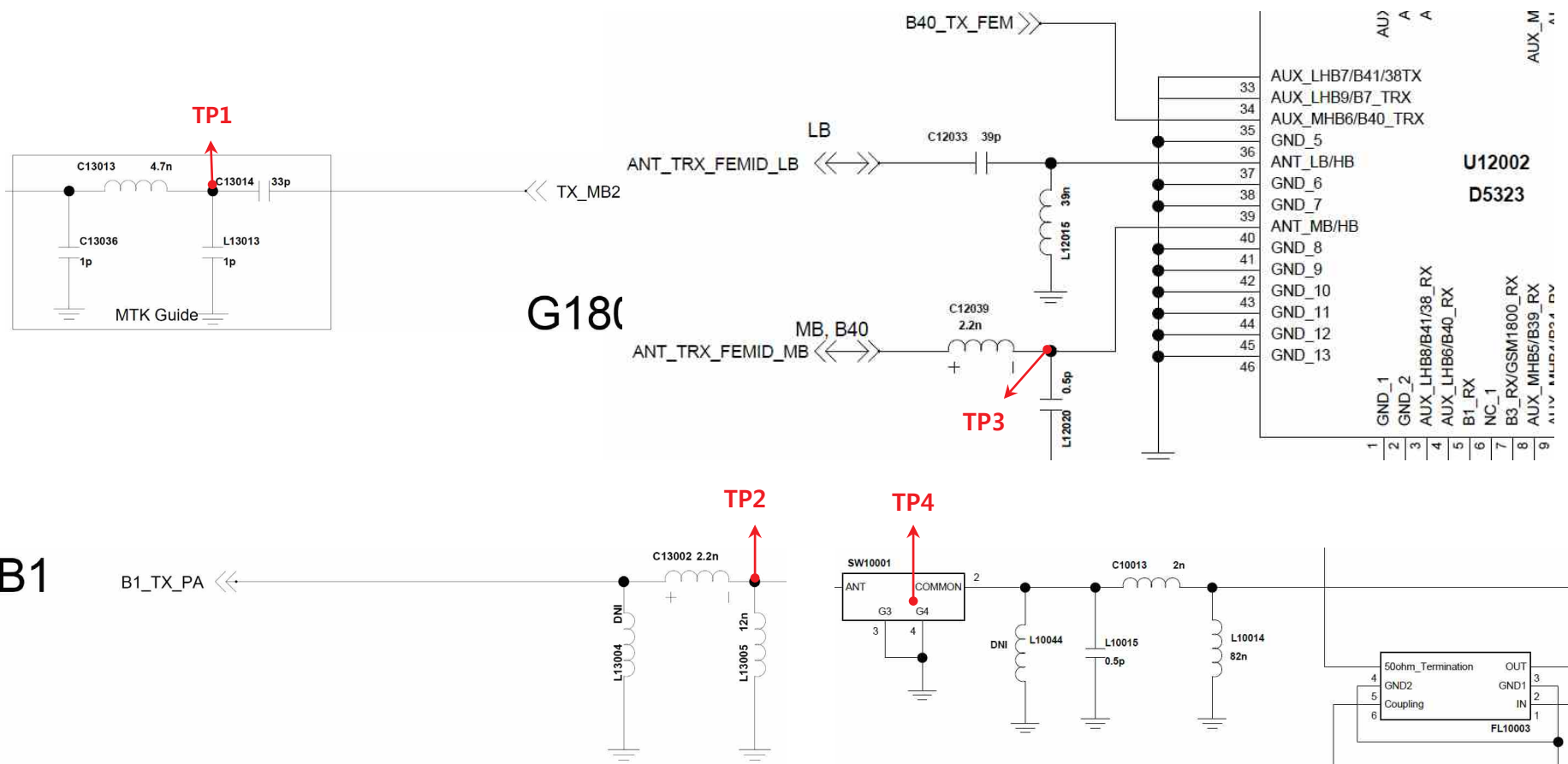


3.6 WCDMA RF PART

3.6.5 Checking RF Signal TX path(WCDMA B1/B2)

3. TROUBLE SHOOTING

Circuit Diagram

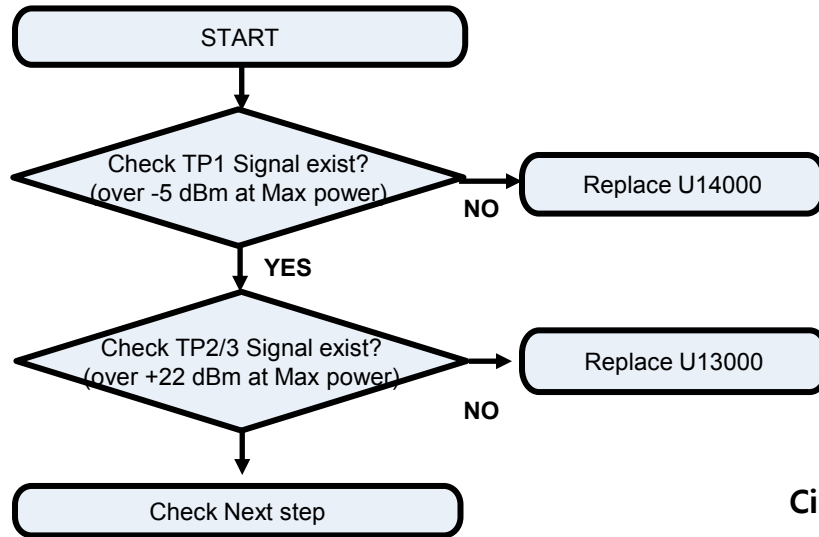


3. TROUBLE SHOOTING

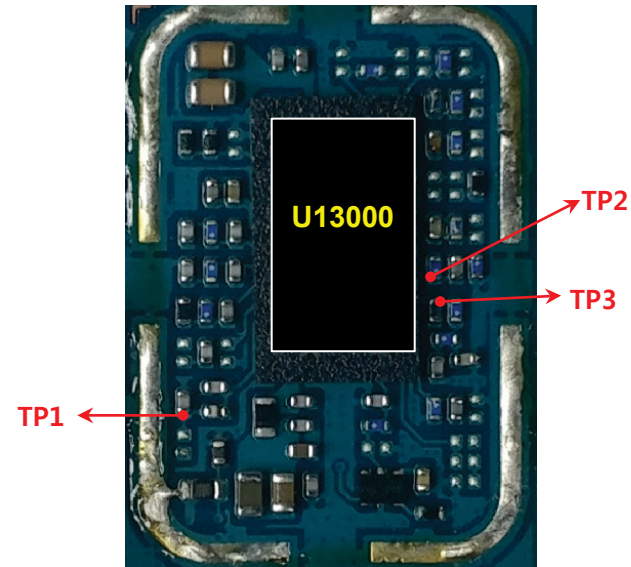
3.6 WCDMA RF PART

3.6.5 Checking RF Signal TX path(WCDMA B5/B8)

Checking Flow

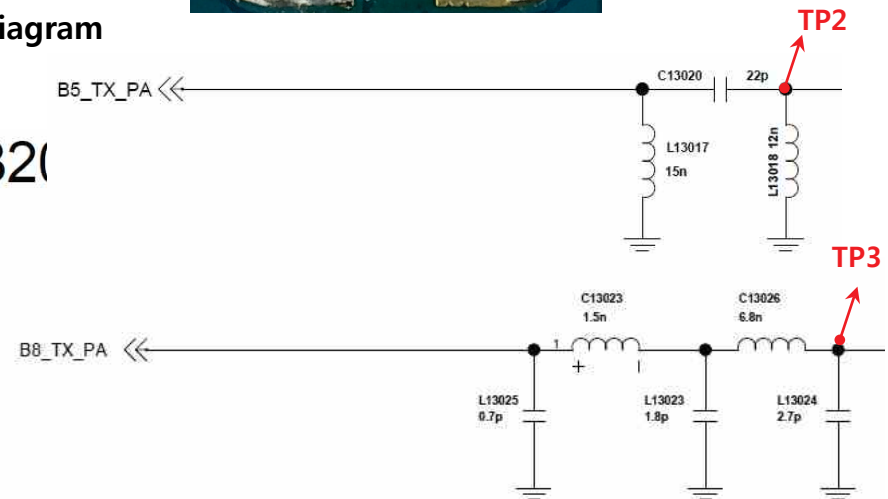
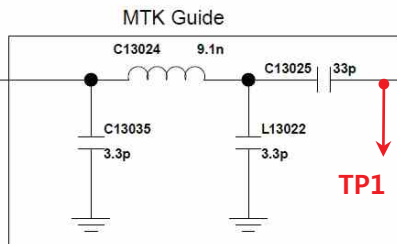


Image



Circuit Diagram

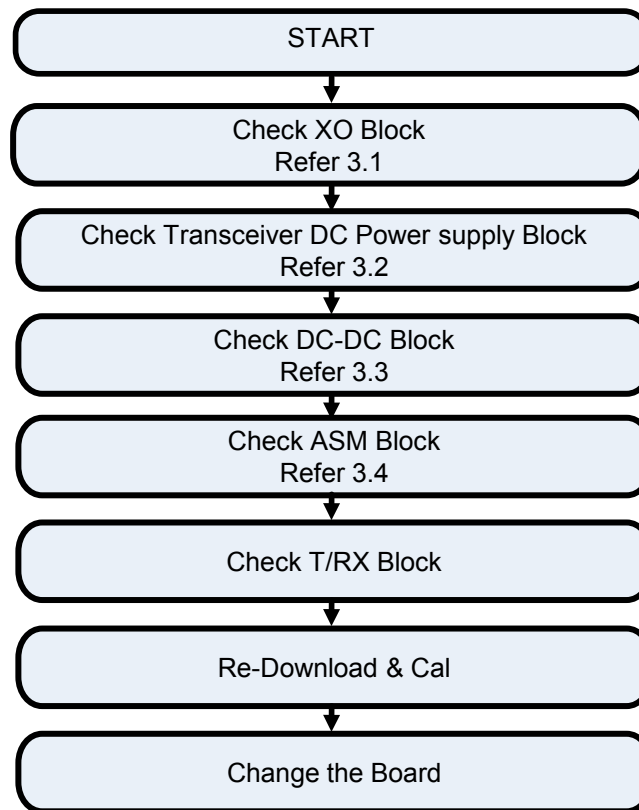
B5/B8/B17/B20



3.7 LTE RF PART

LTE RF Part support LTE B1/3/7/8/20 with ASM, PAM, Transceiver component

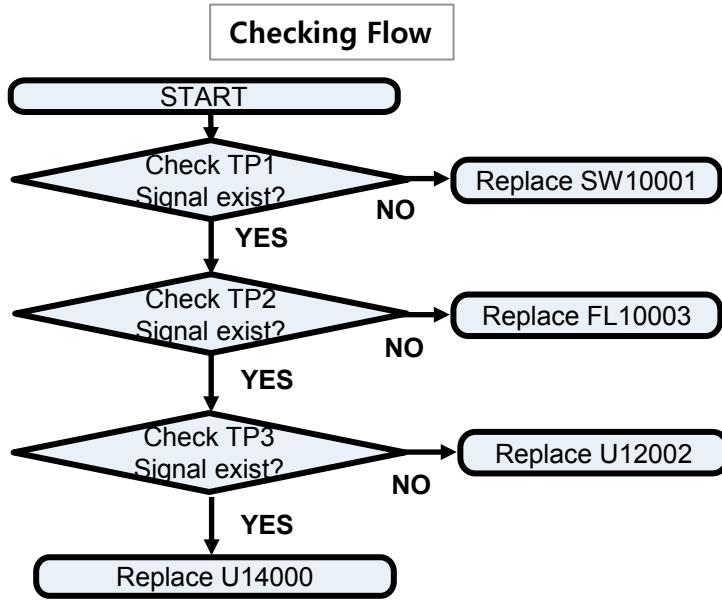
Checking Flow



3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.2 Checking RF Signal RX path(LTE B1)



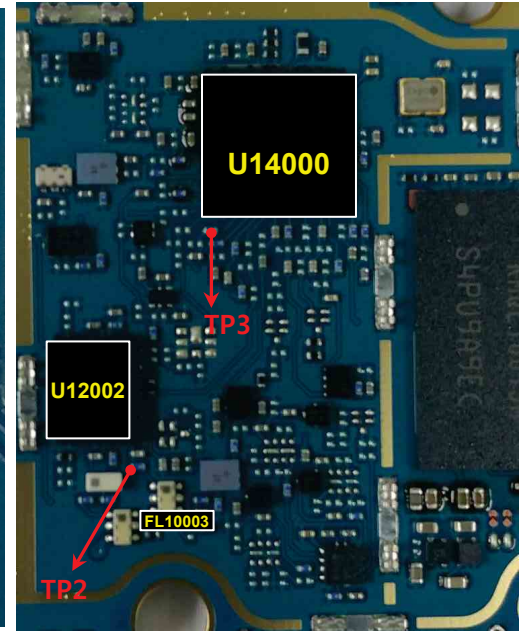
Image

TP1

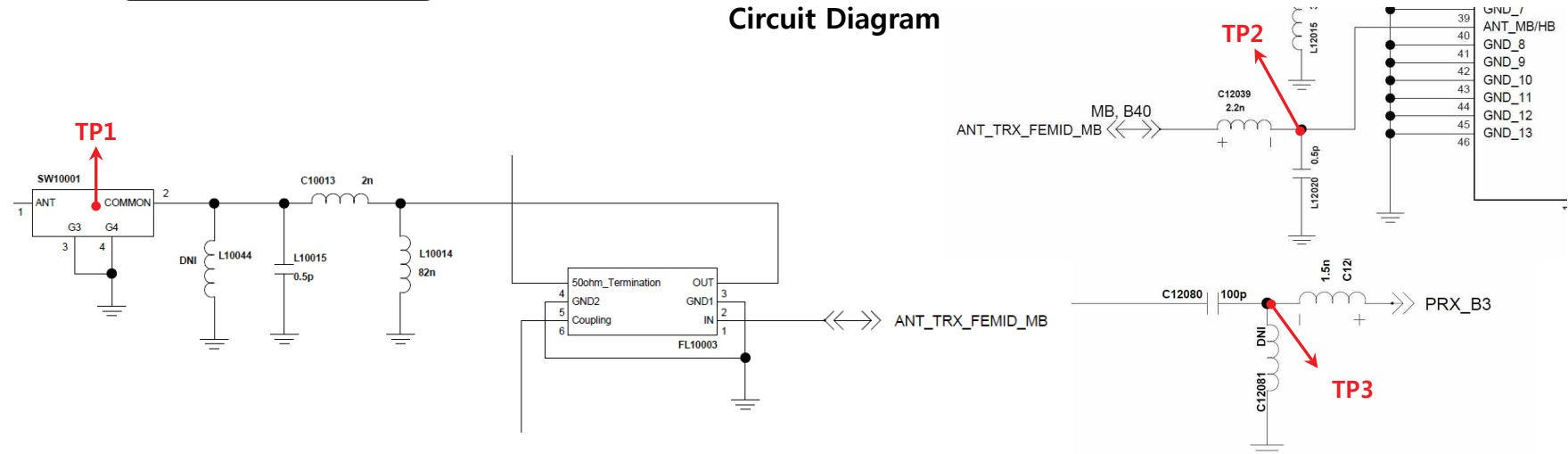


U14000

TP3



Circuit Diagram

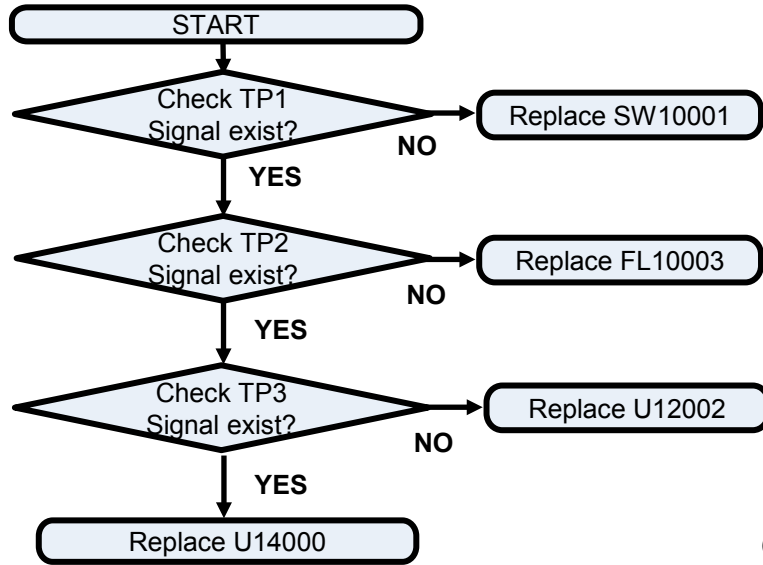


3. TROUBLE SHOOTING

3.7 LTE RF PART

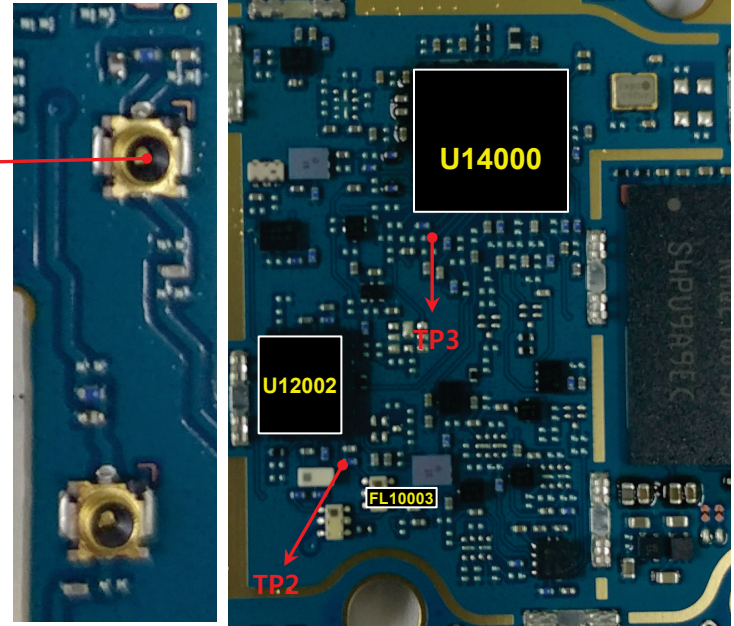
3.7.2 Checking RF Signal RX path(LTE B3)

Checking Flow

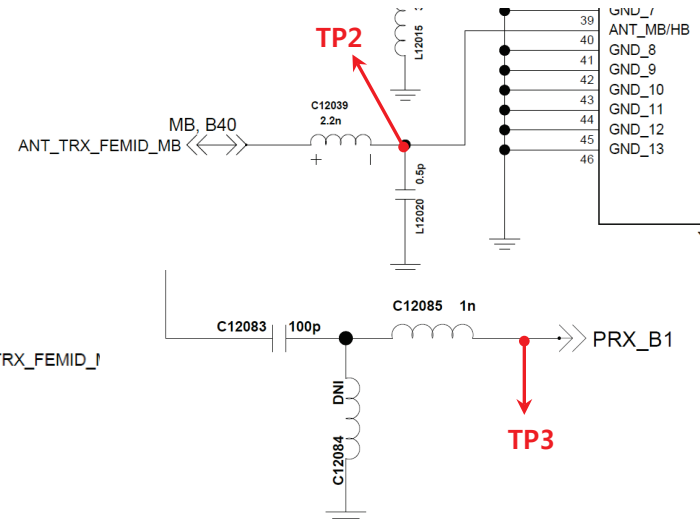
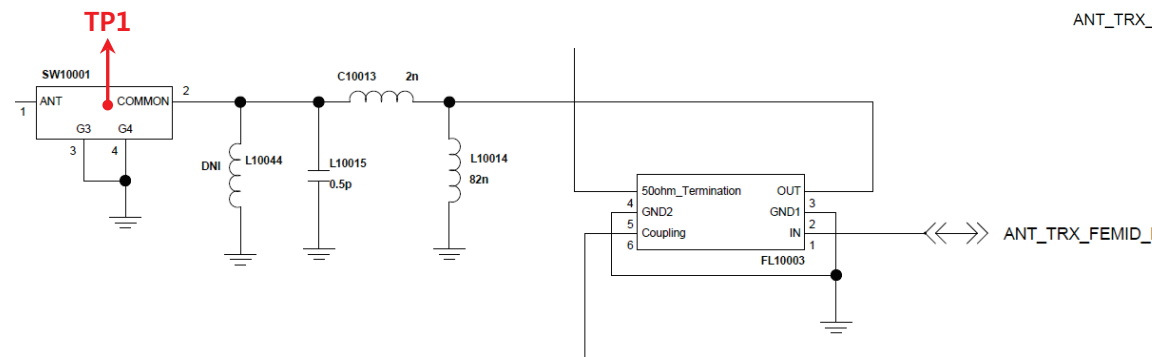


Image

TP1



Circuit Diagram

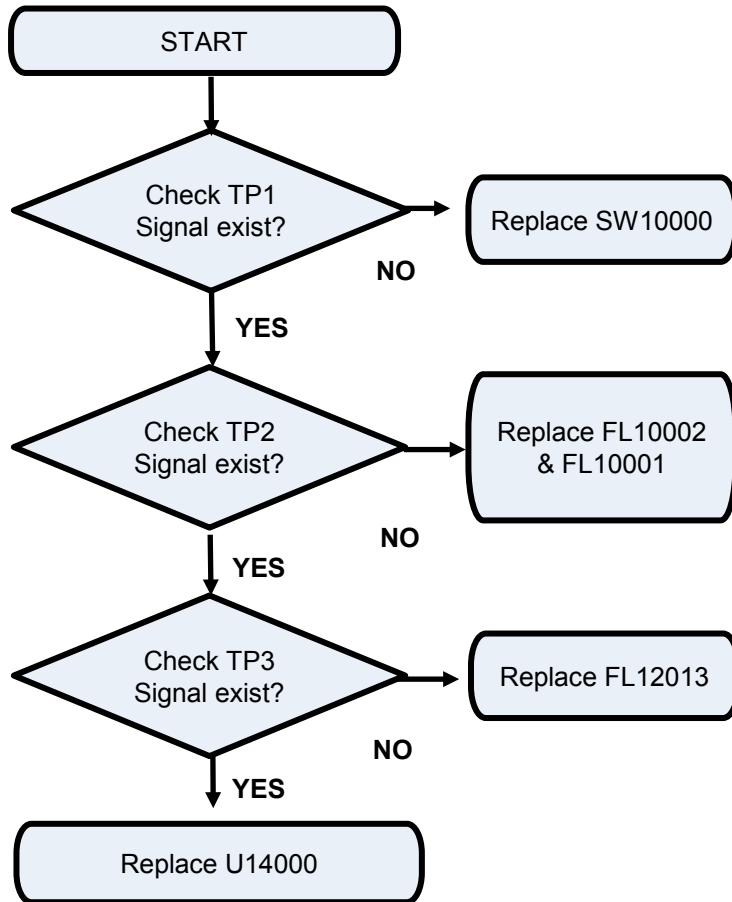


3. TROUBLE SHOOTING

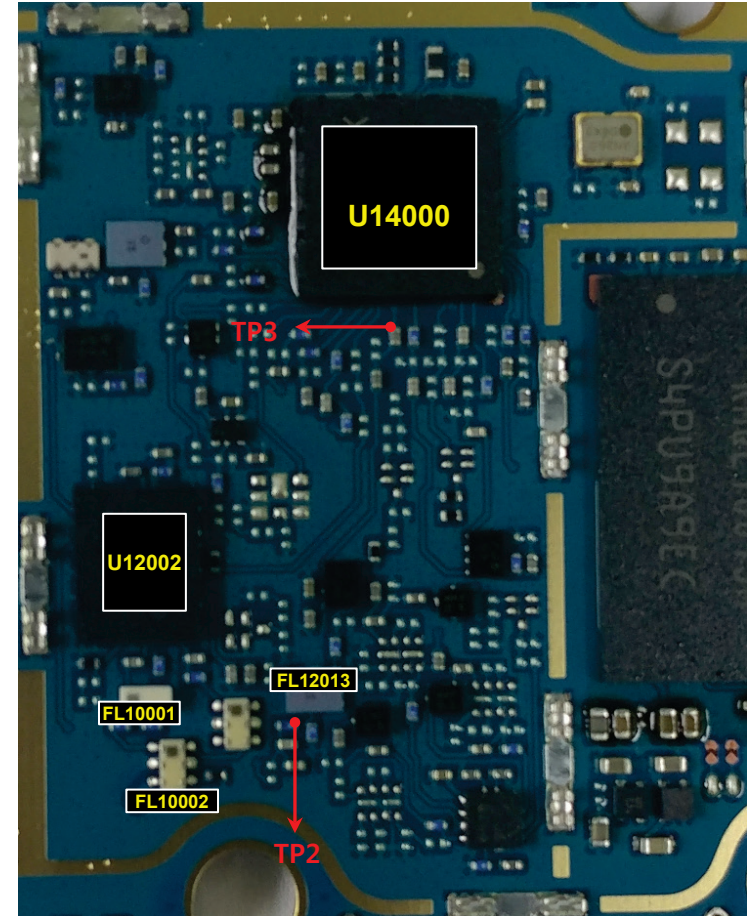
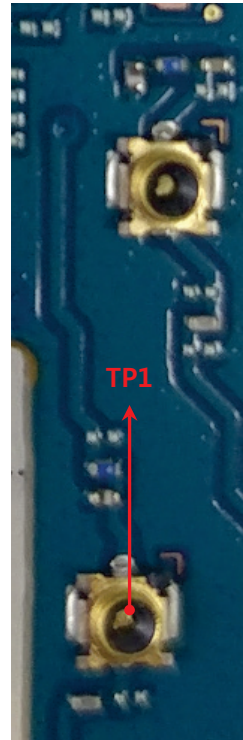
3.7 LTE RF PART

3.7.2 Checking RF Signal RX path(LTE B7)

Checking Flow



Image

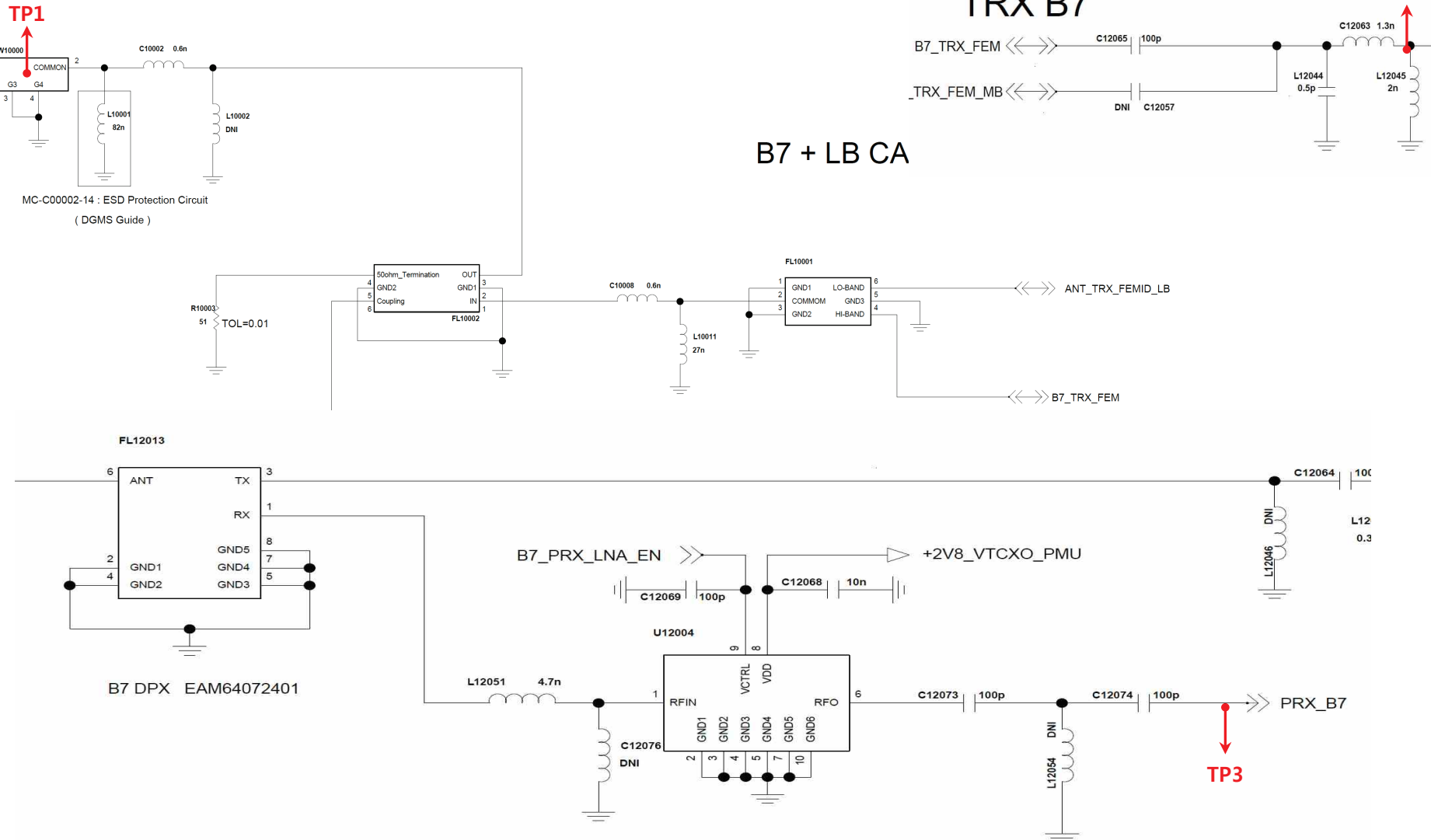


3. TROUBLE SHOOTING

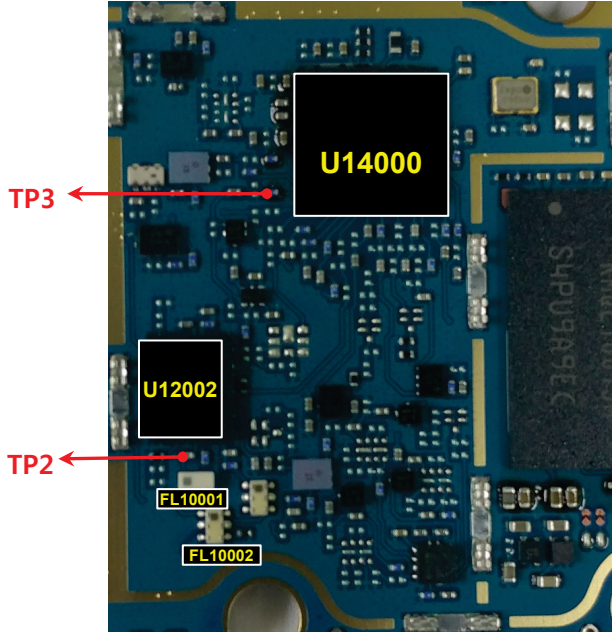
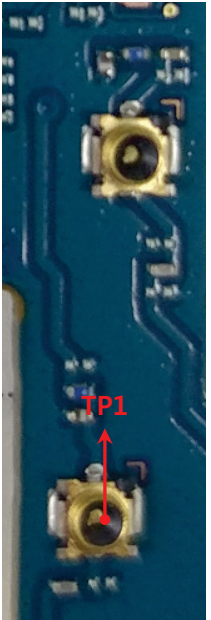
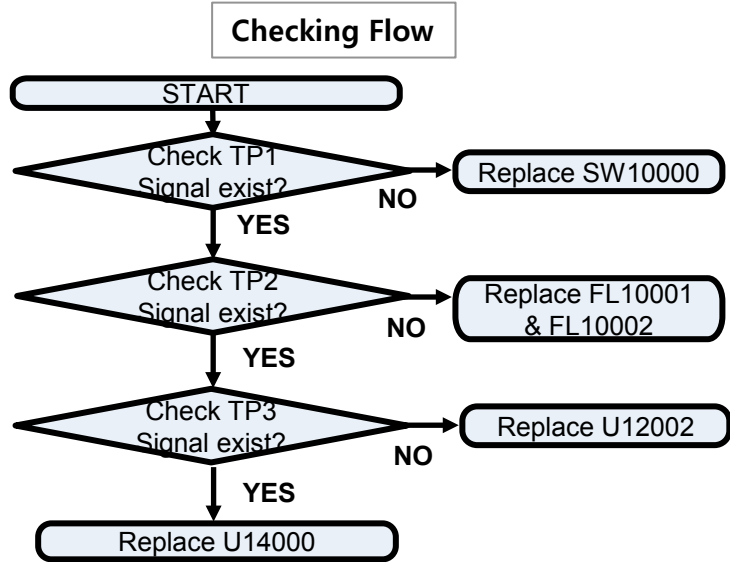
3.7 LTE RF PART

3.7.2 Checking RF Signal RX path(LTE B7)

Circuit Diagram

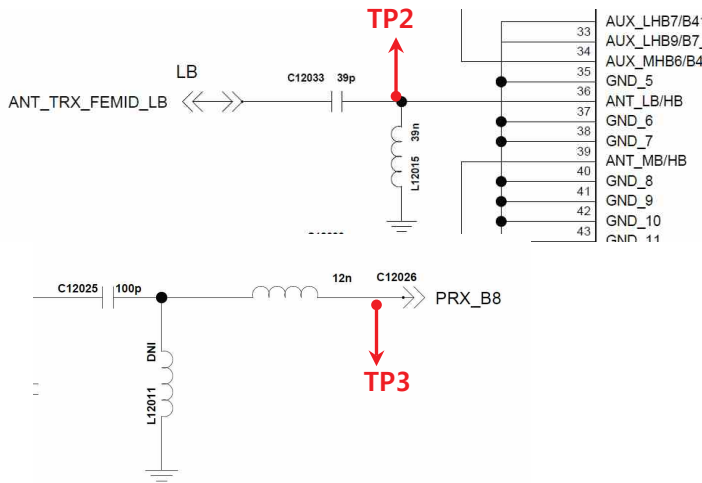
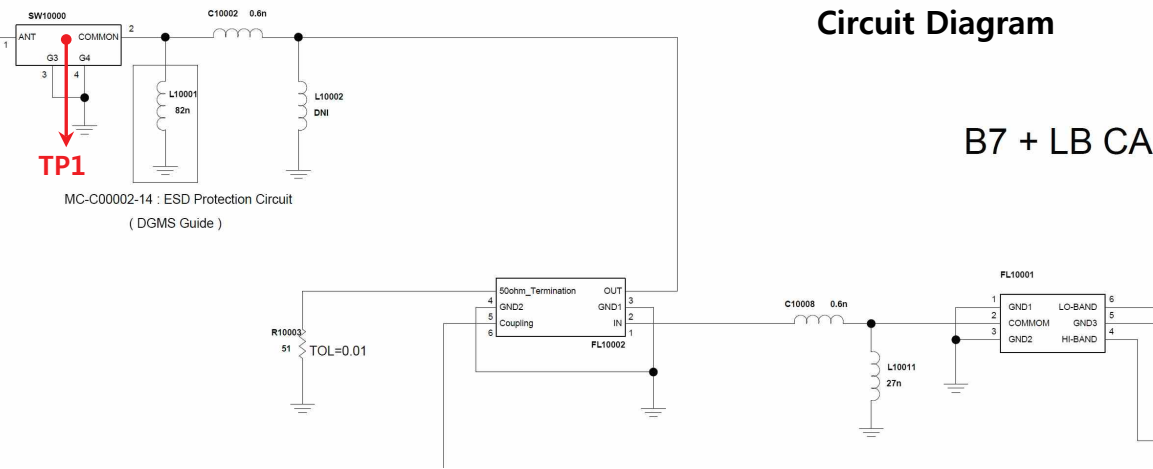


3.7 LTE RF PART
3.7.2 Checking RF Signal RX path(LTE B8)



Circuit Diagram

B7 + LB CA

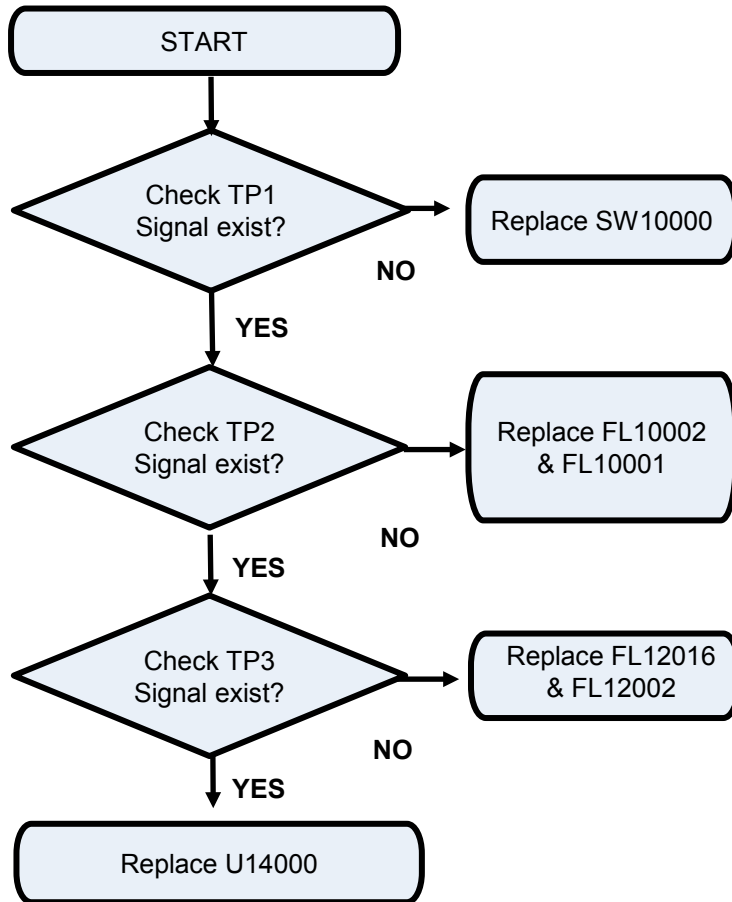


3. TROUBLE SHOOTING

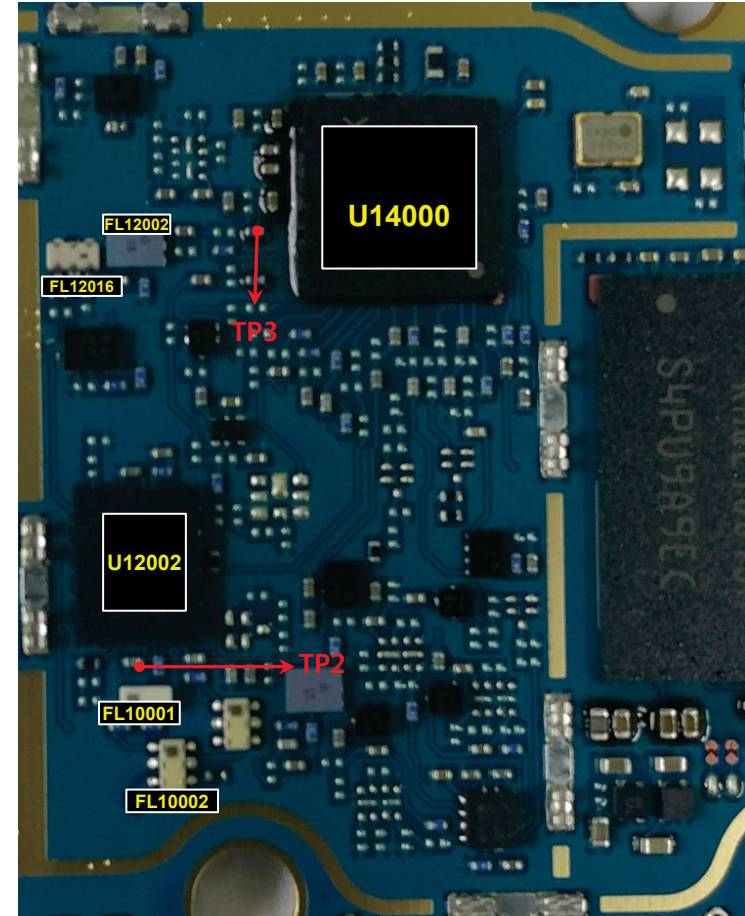
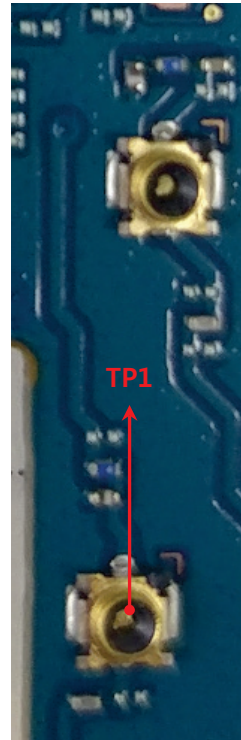
3.7 LTE RF PART

3.7.2 Checking RF Signal RX path(LTE B20)

Checking Flow



Image

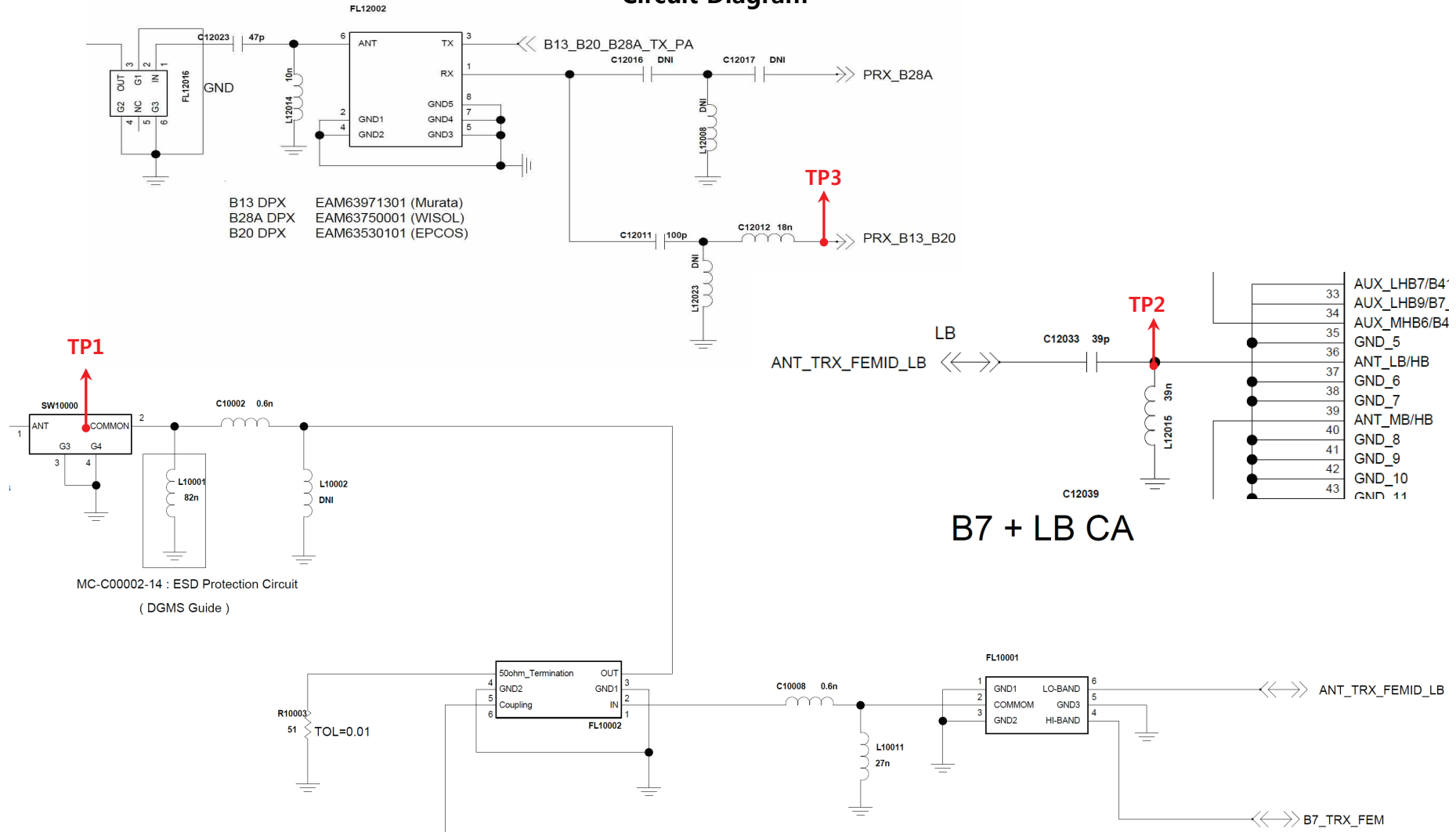


3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.2 Checking RF Signal RX path(LTE B20)

Circuit Diagram

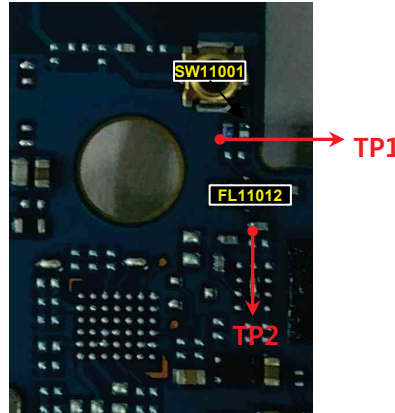
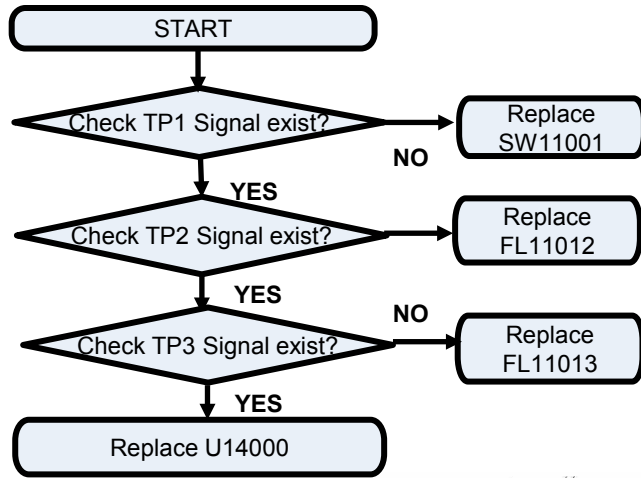


3. TROUBLE SHOOTING

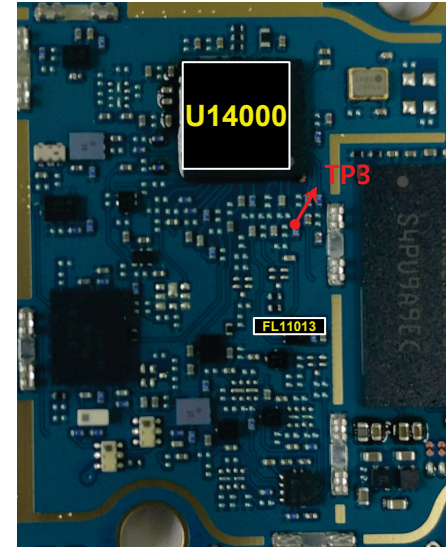
3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B1)

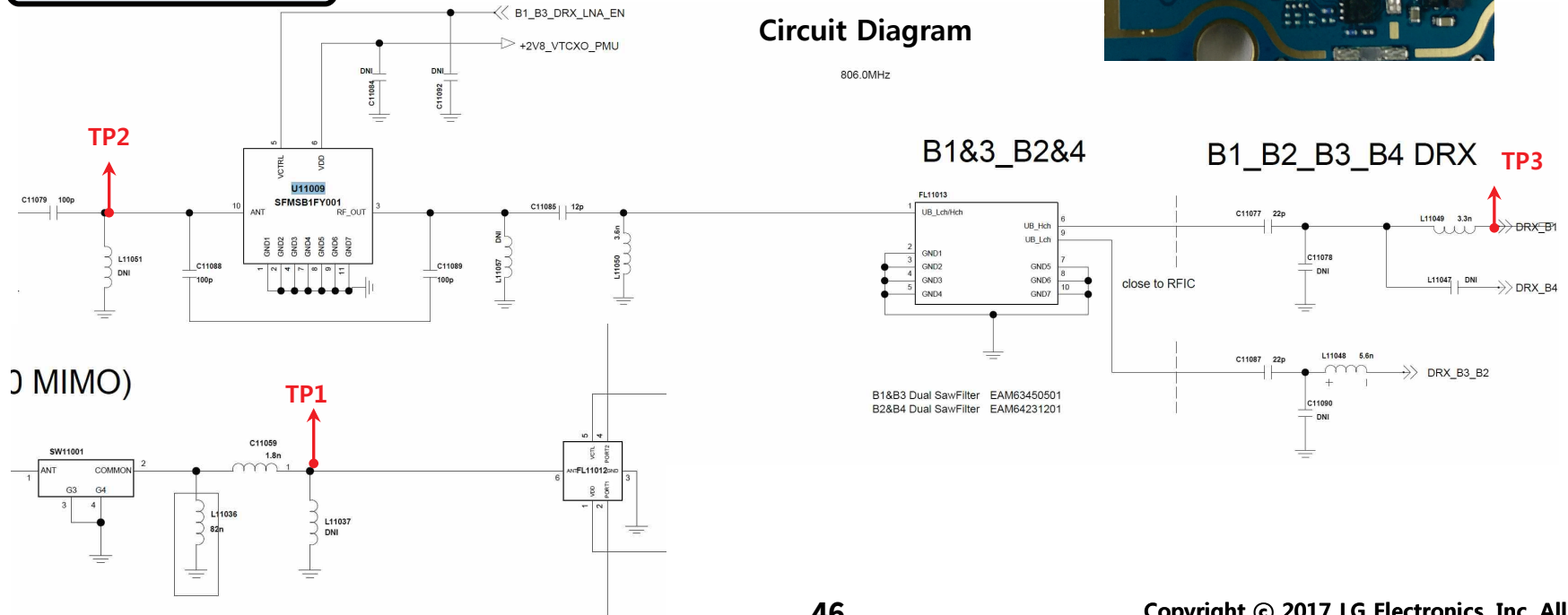
Checking Flow



Image



Circuit Diagram



3 MIMO)

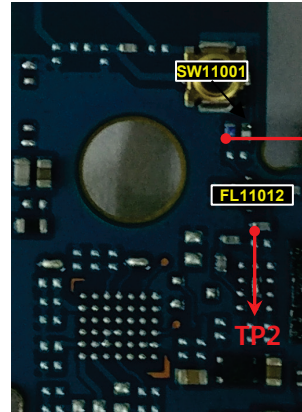
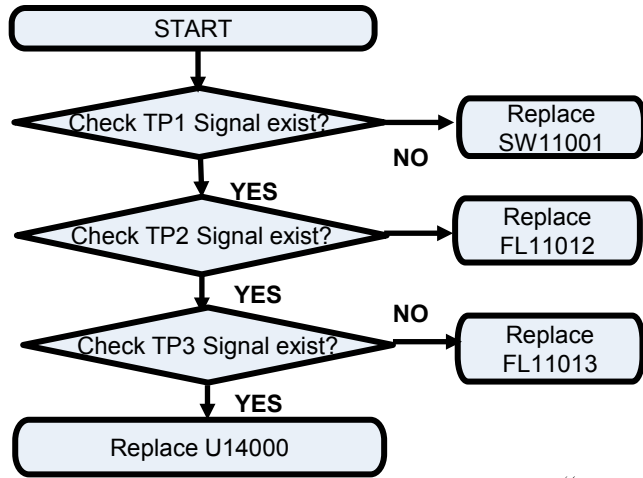
LGE Internal Use Only

3. TROUBLE SHOOTING

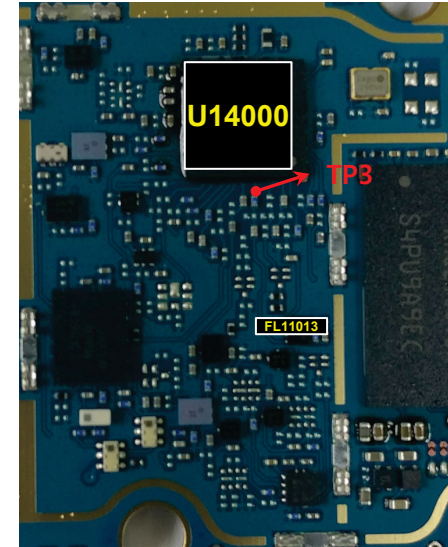
3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B3)

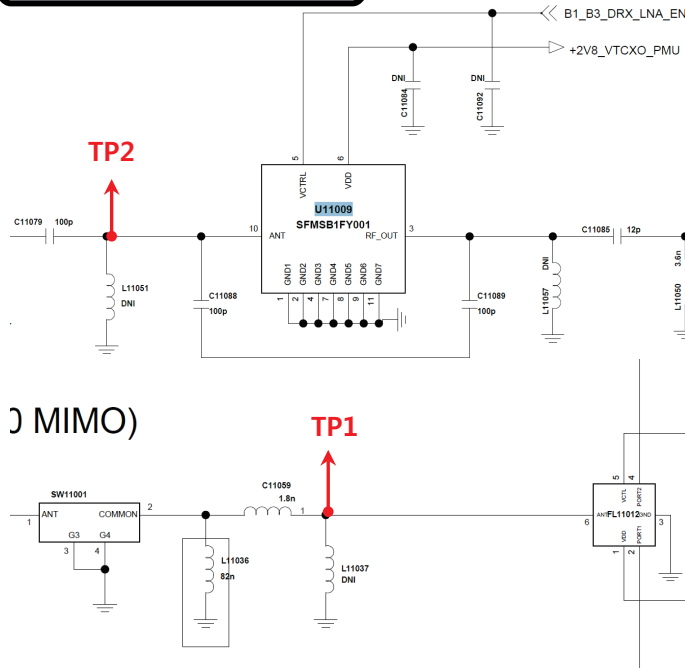
Checking Flow



Image



Circuit Diagram



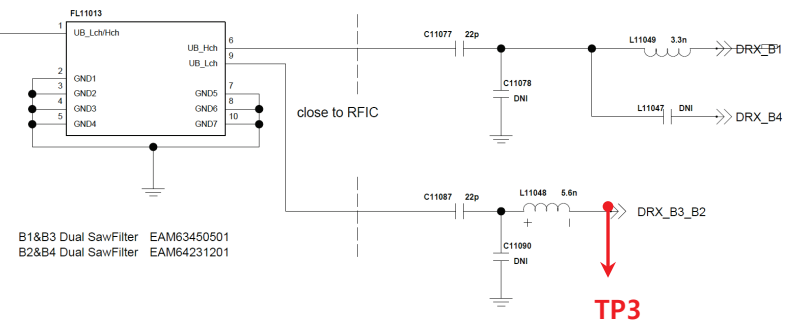
3 MIMO)

TP1

806.0MHz

B1&3_B2&4

B1_B2_B3_B4 DRX



B1&B3 Dual SawFilter EAM63450501
B2&B4 Dual SawFilter EAM64231201

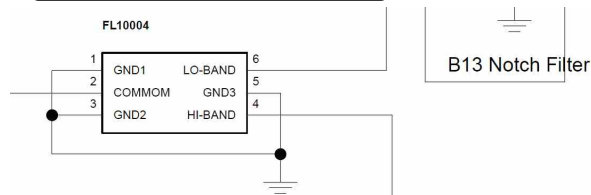
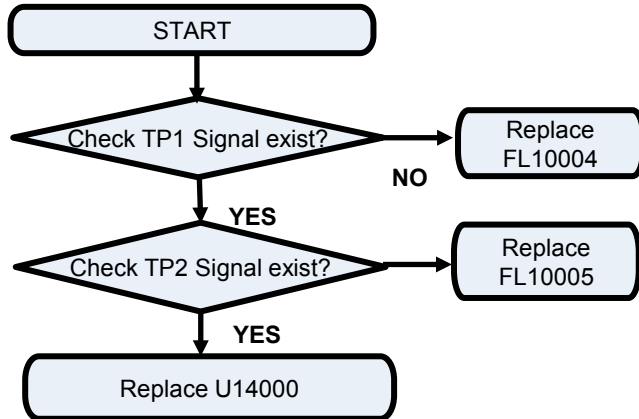
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B7)

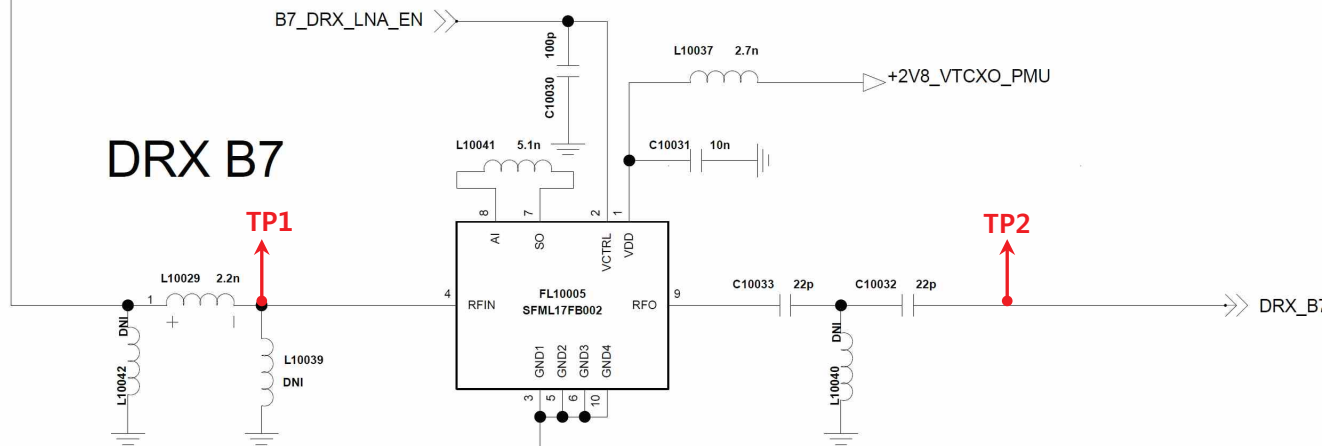
Image

Checking Flow



tion Circuit

DRX B7



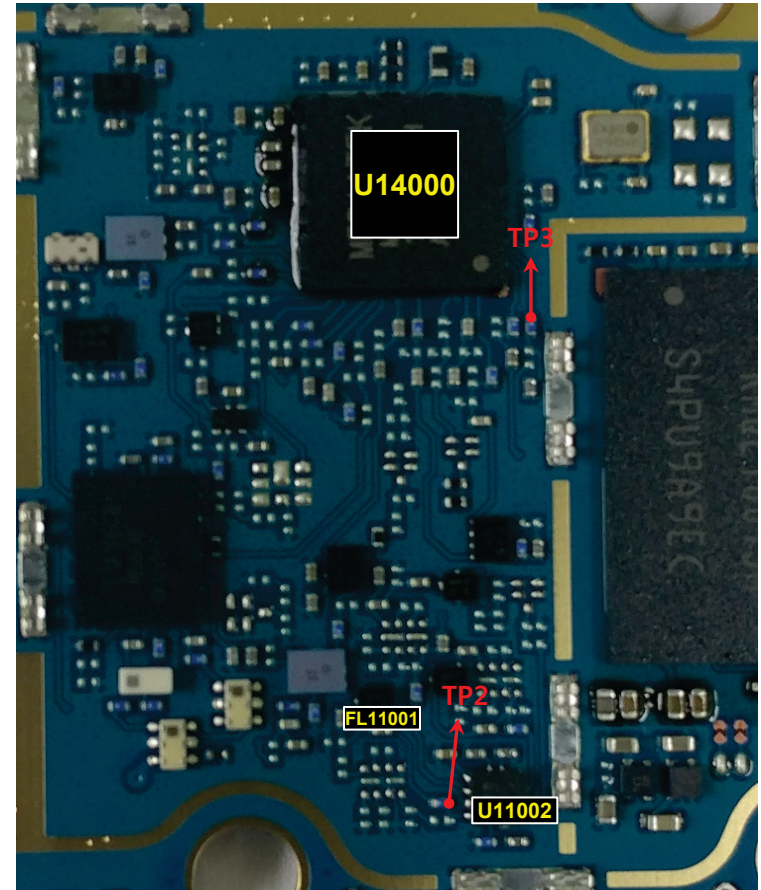
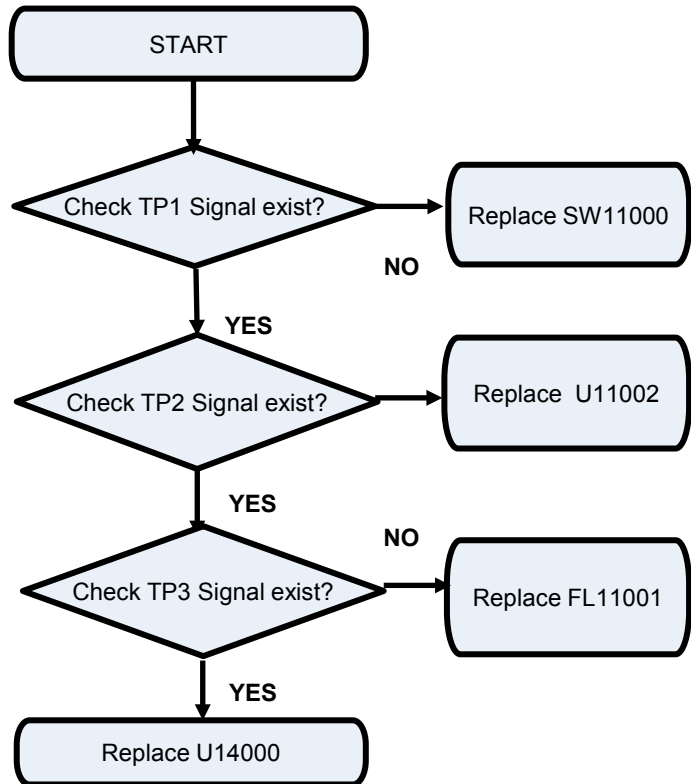
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B8)

Image

Checking Flow

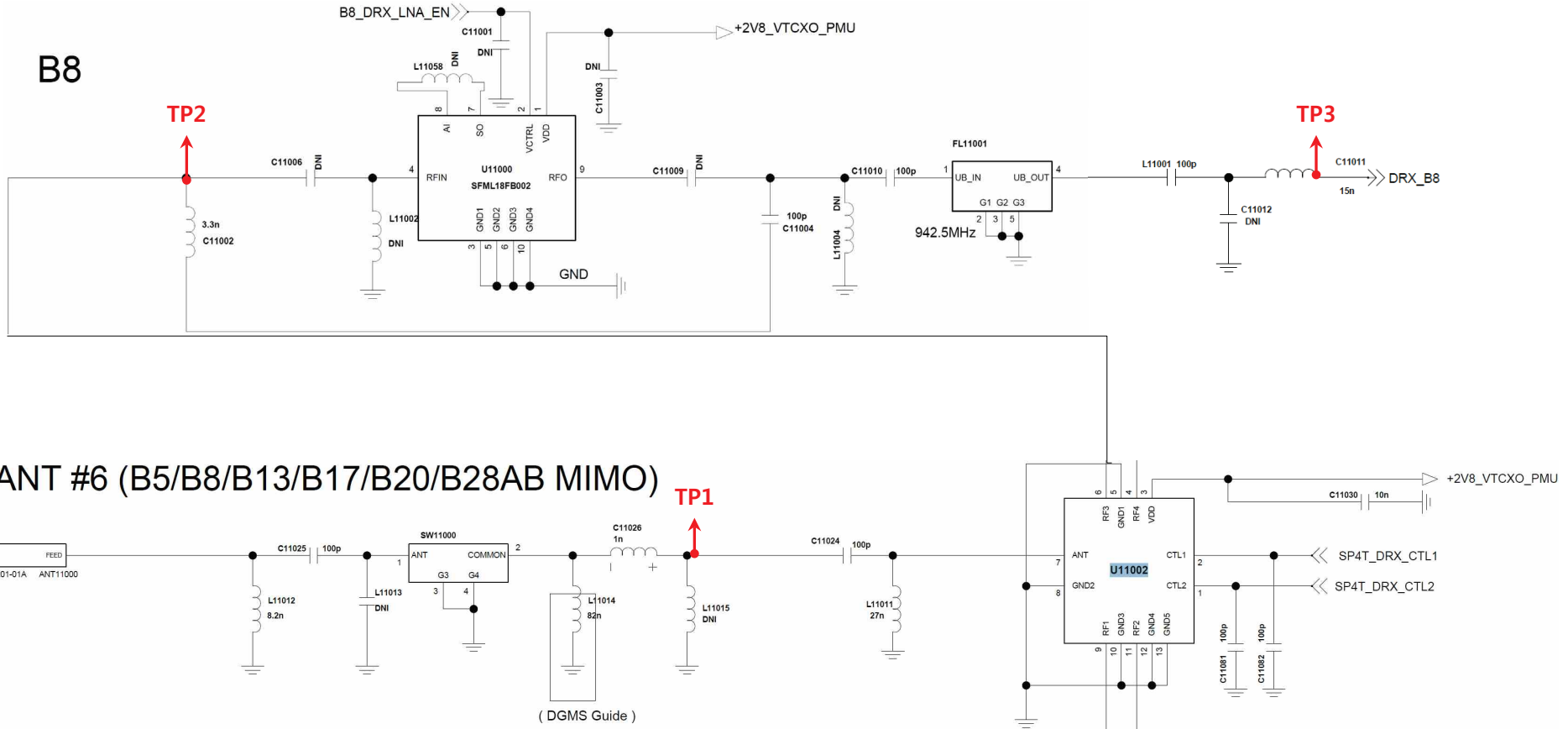


3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B8)

Circuit Diagram

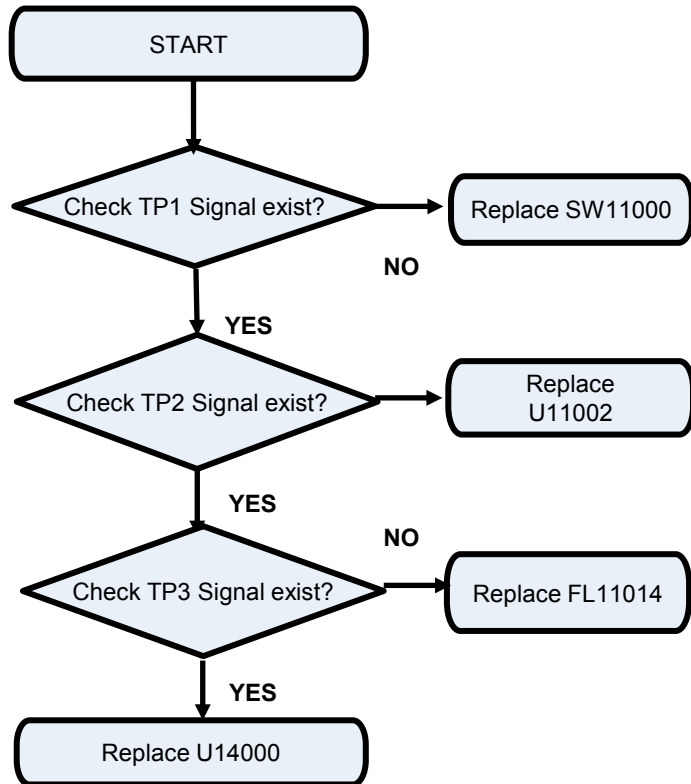


3. TROUBLE SHOOTING

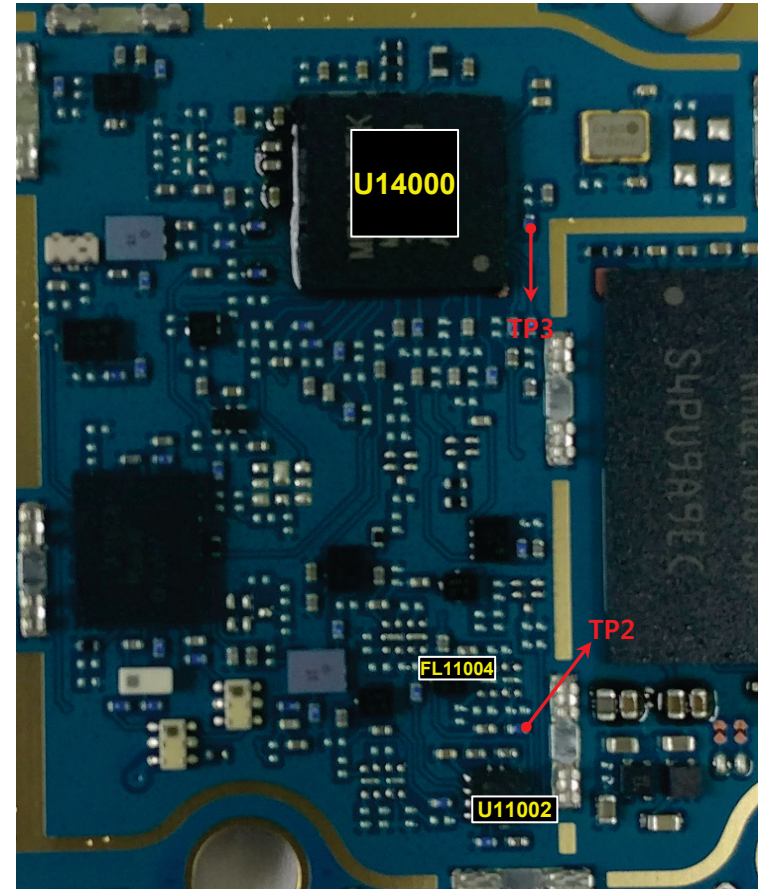
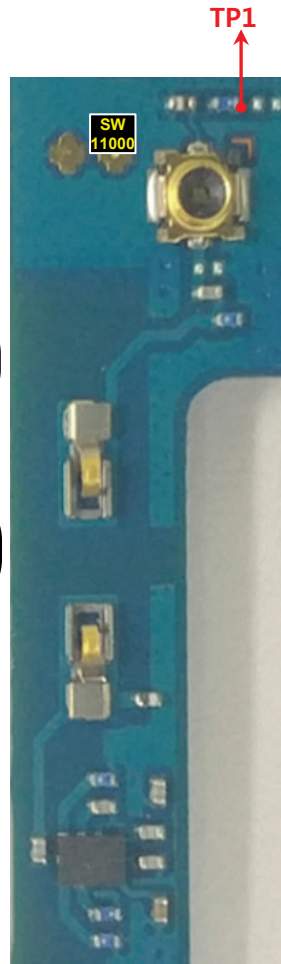
3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B20)

Checking Flow



Image



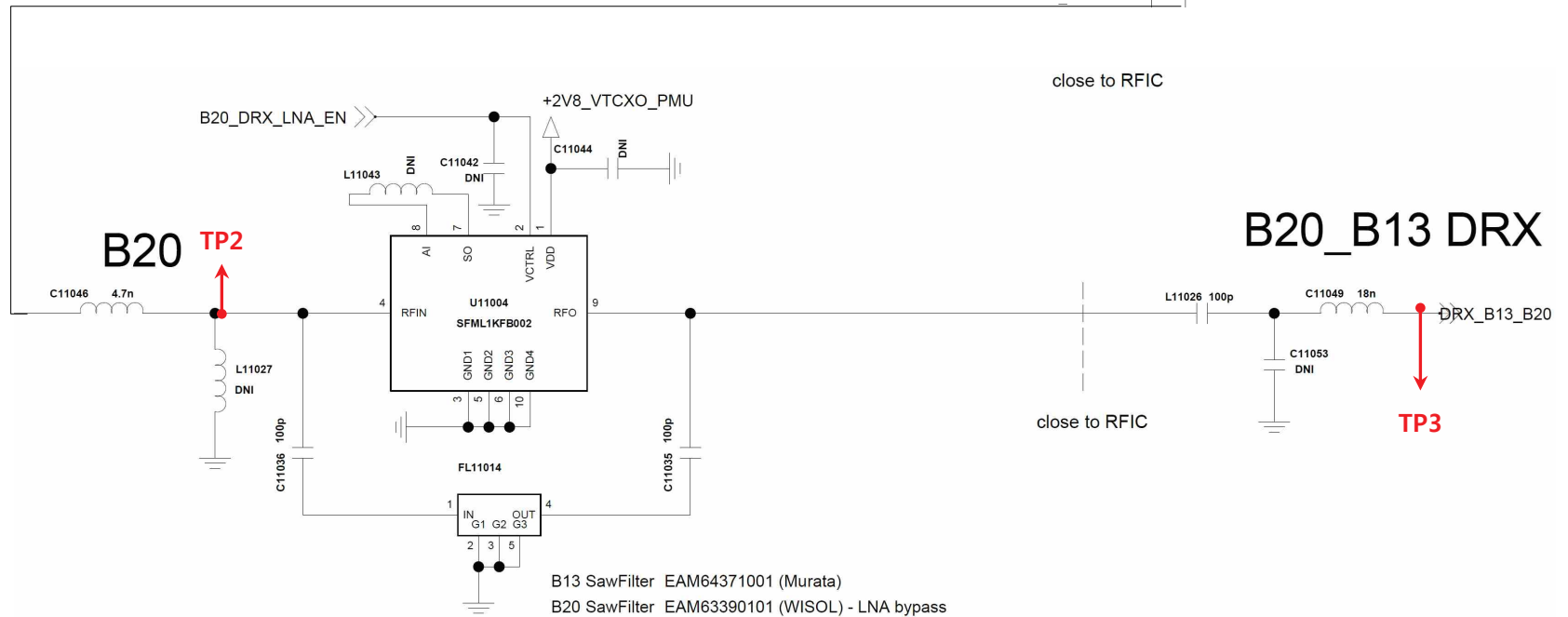
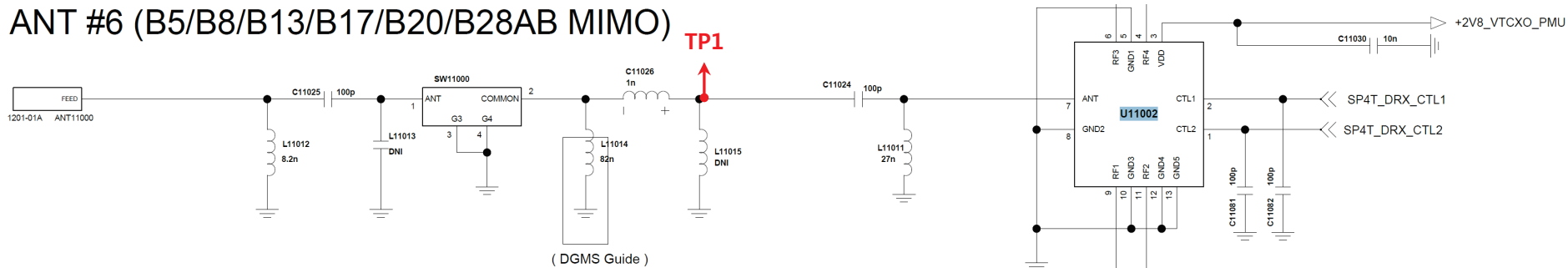
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTE B20)

Circuit Diagram

ANT #6 (B5/B8/B13/B17/B20/B28AB MIMO)

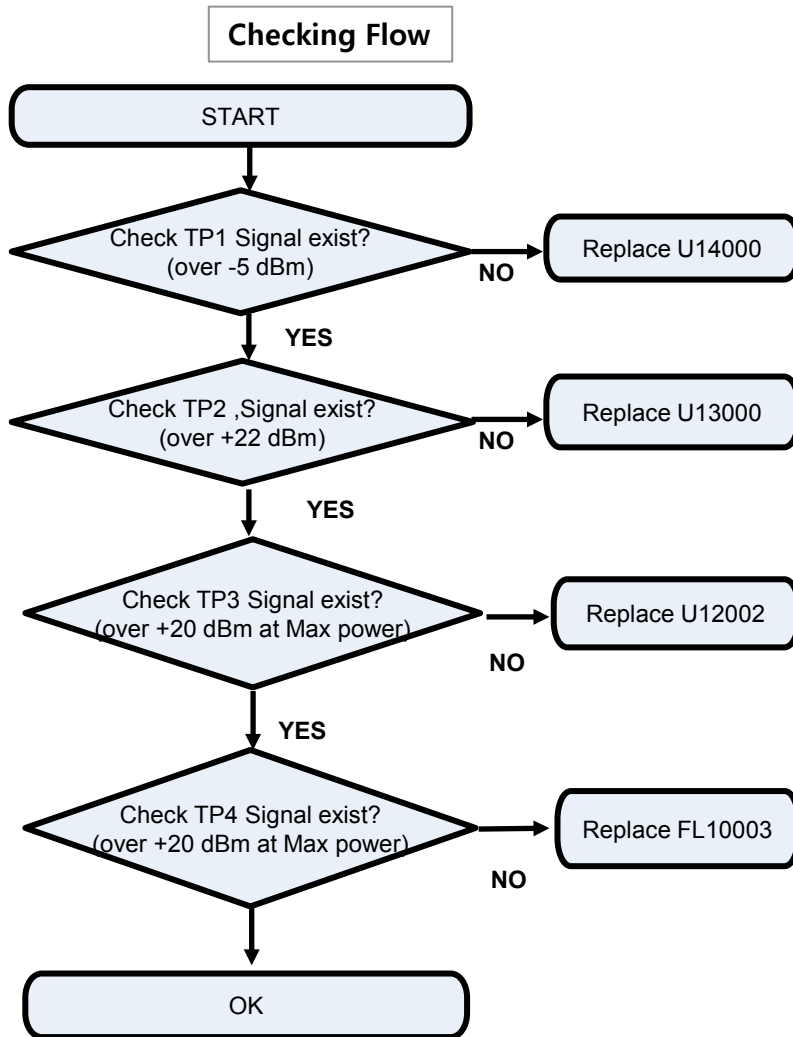


3. TROUBLE SHOOTING

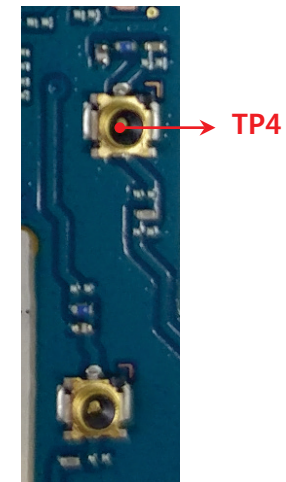
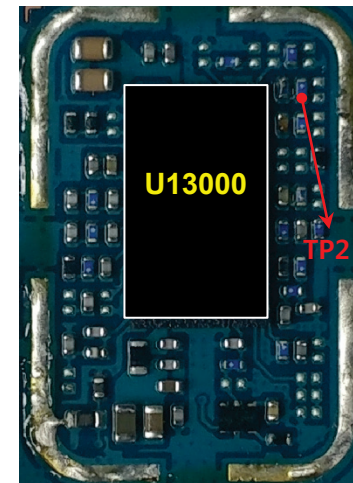
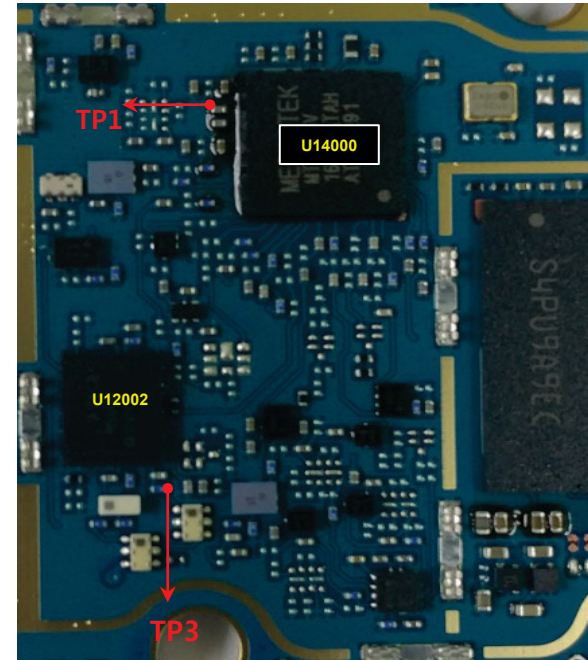
3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B1)

Checking Flow



Image

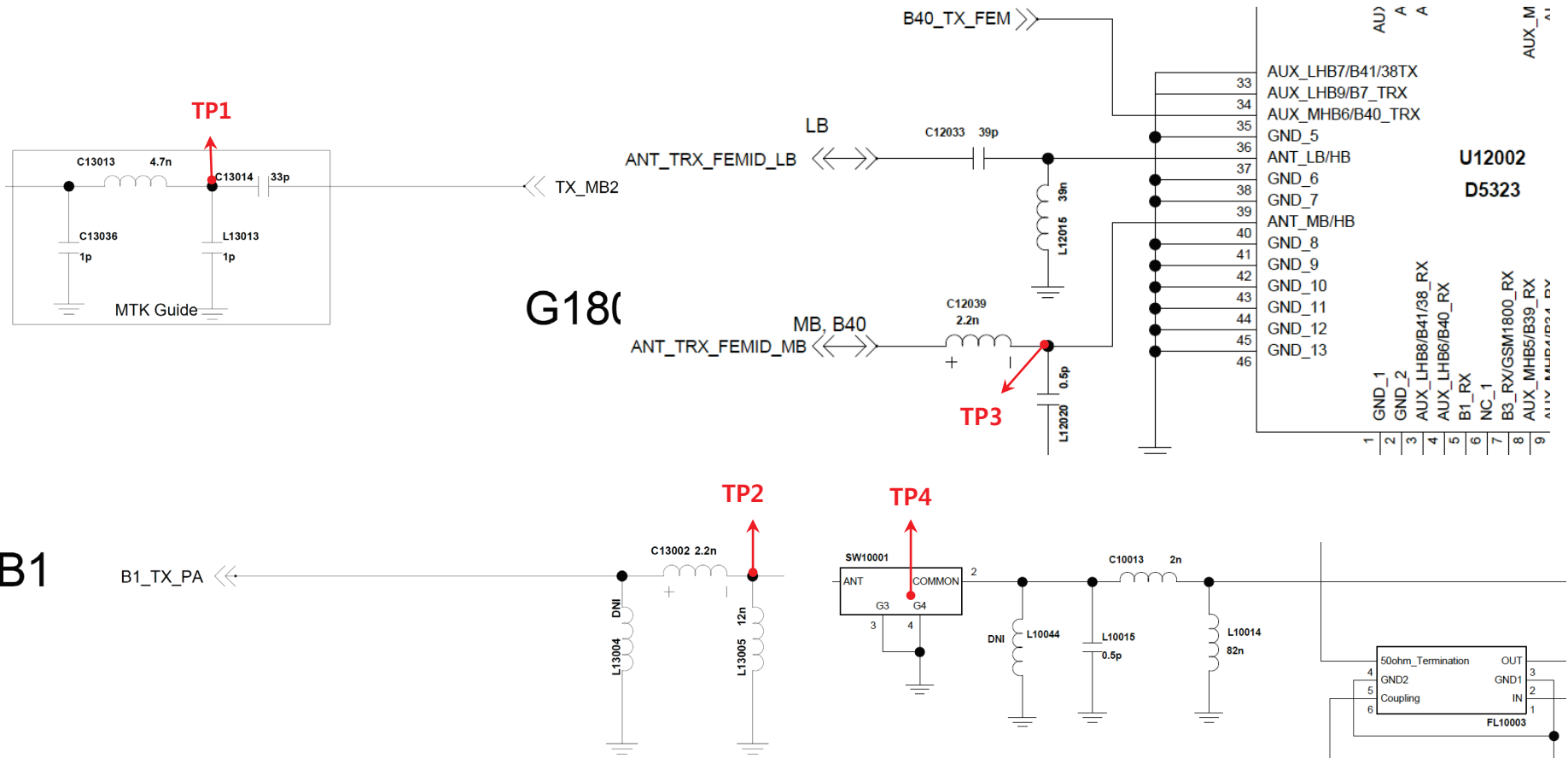


3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B1)

Circuit Diagram

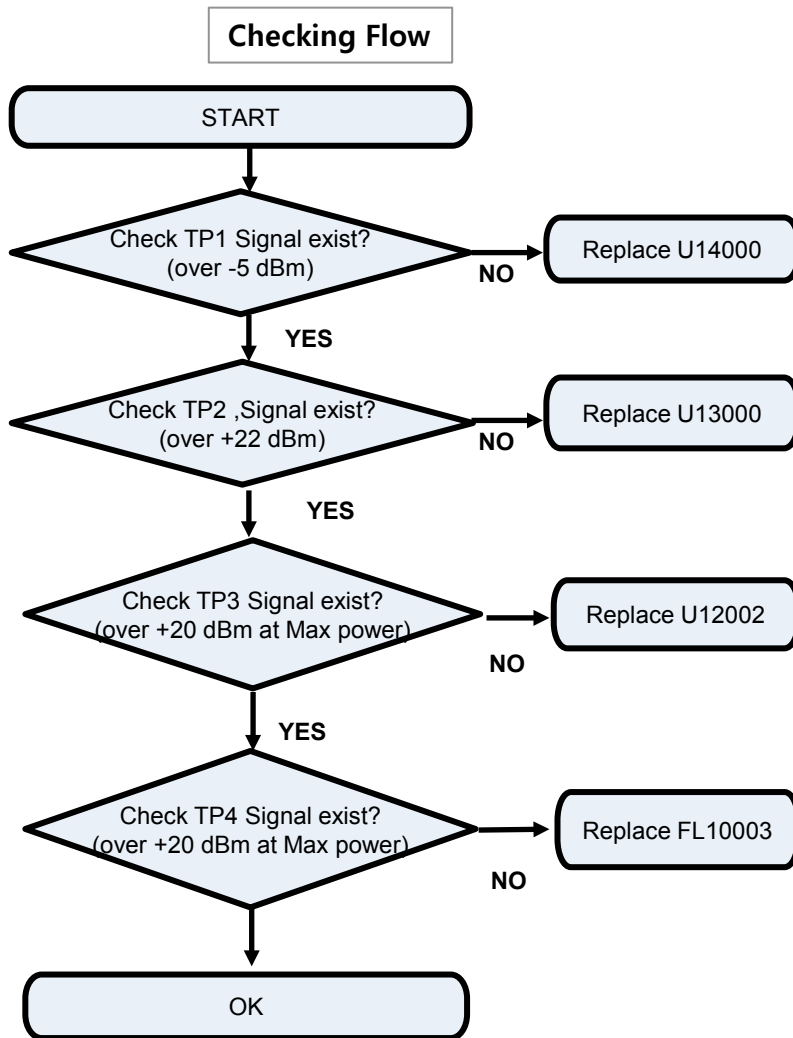


3. TROUBLE SHOOTING

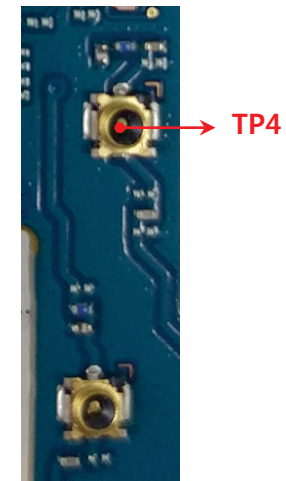
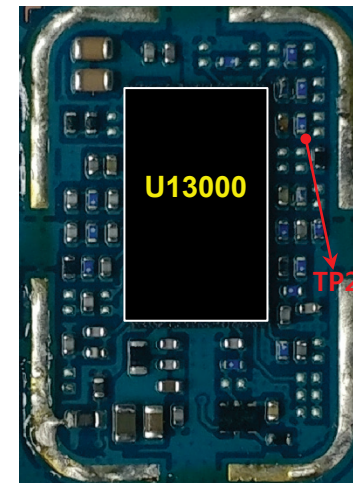
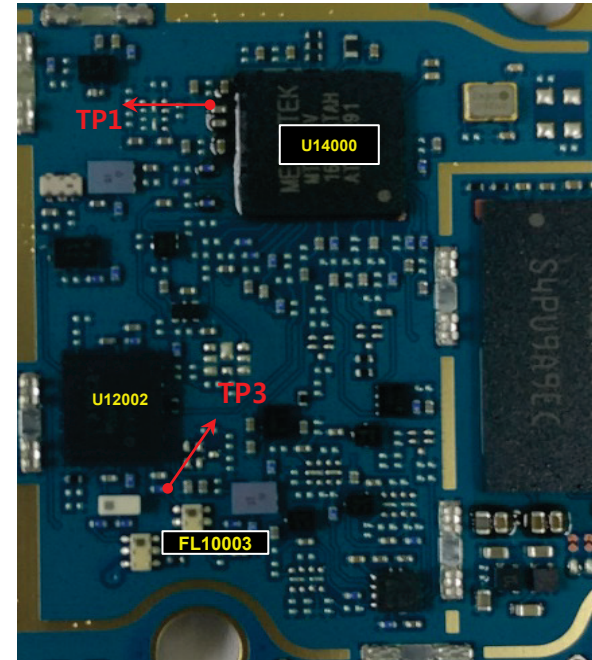
3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B3)

Checking Flow



Image

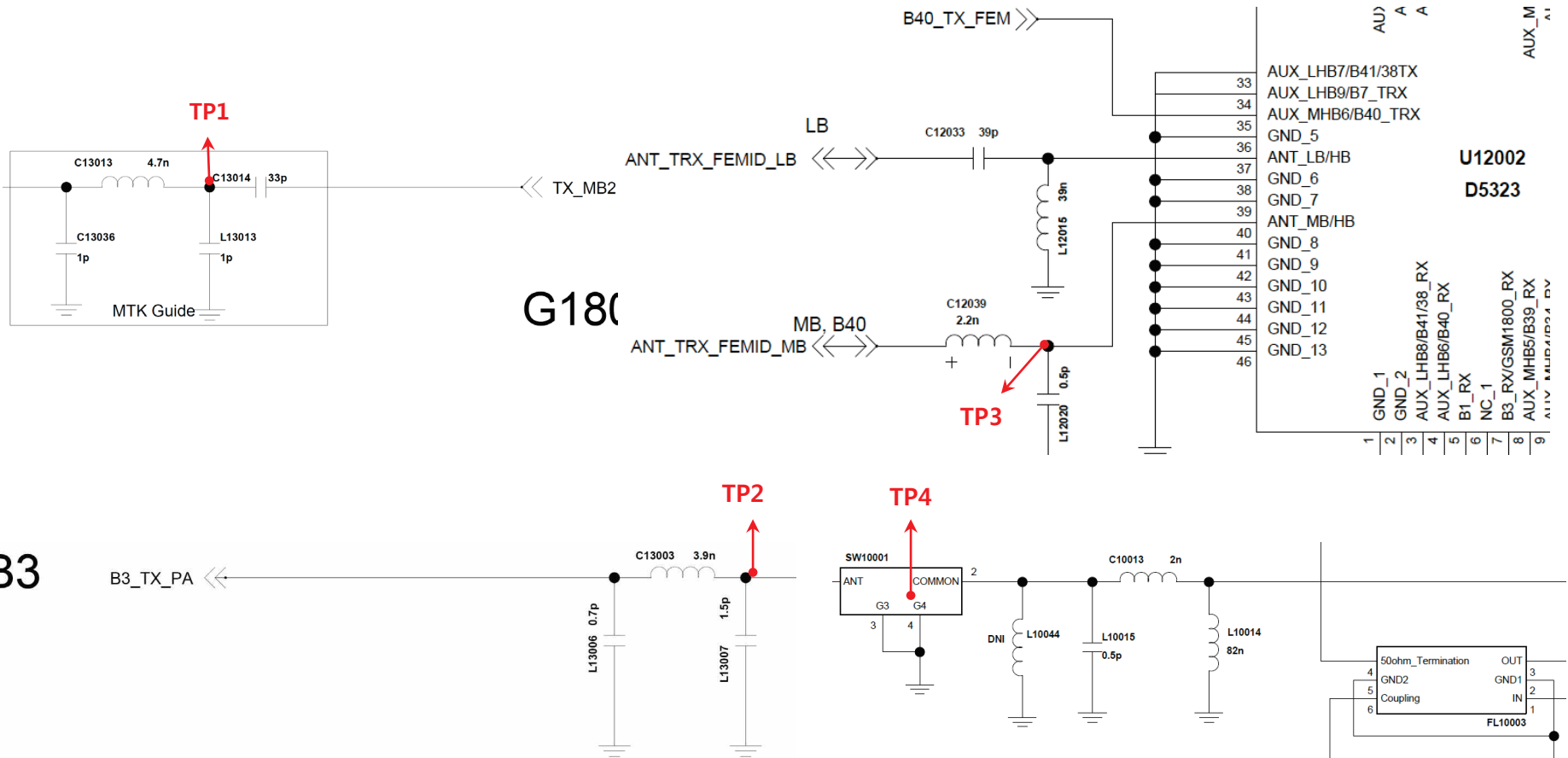


3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B3)

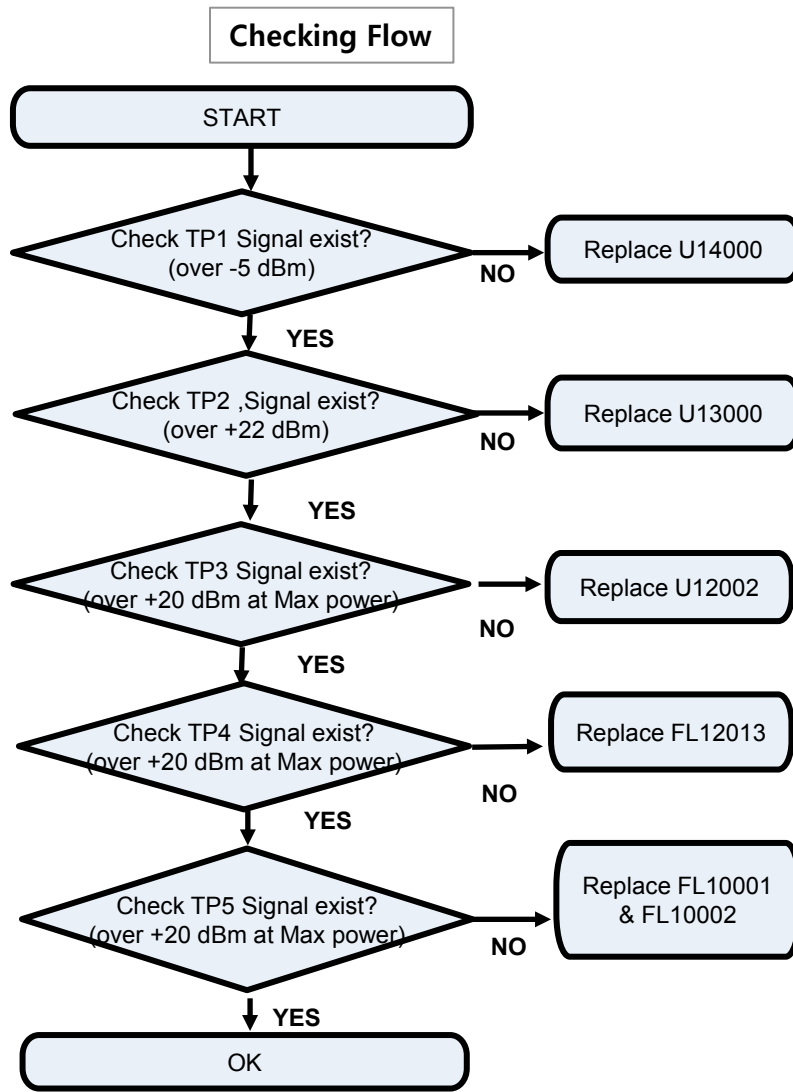
Circuit Diagram



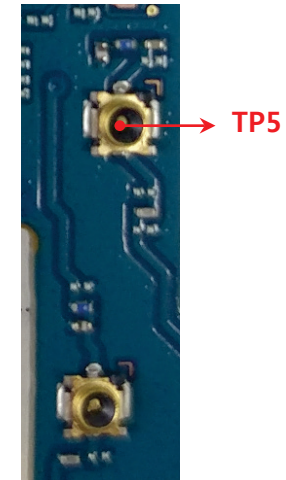
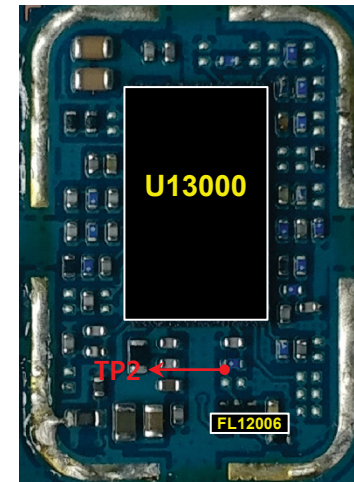
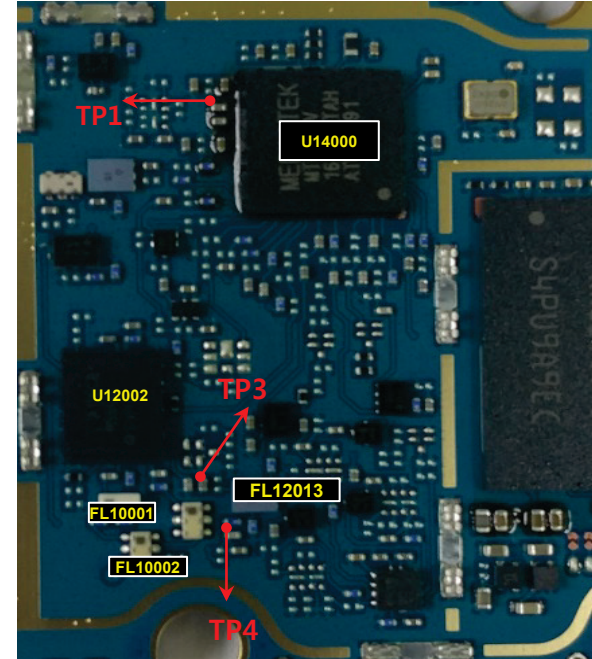
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B7)



Image

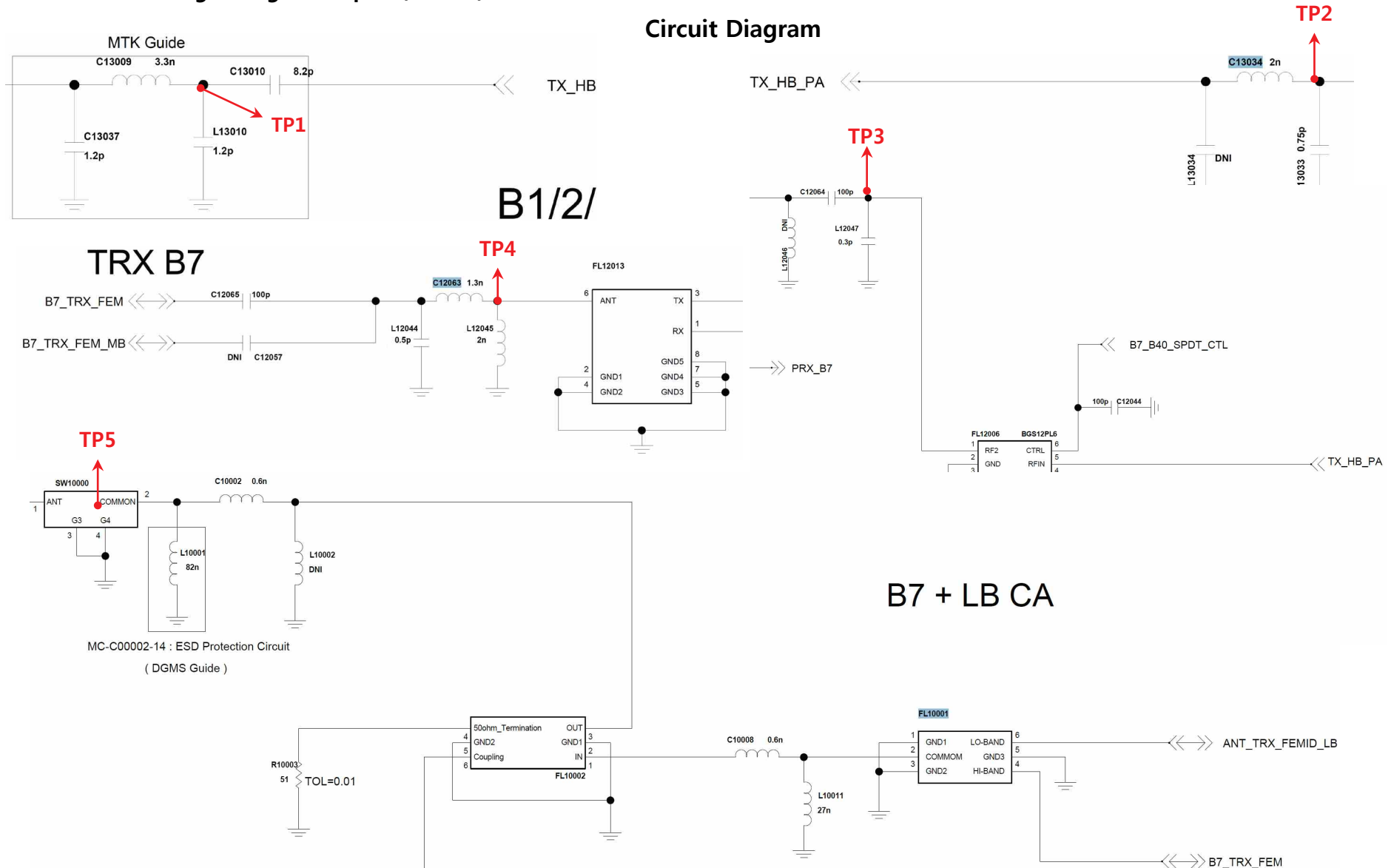


3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B7)

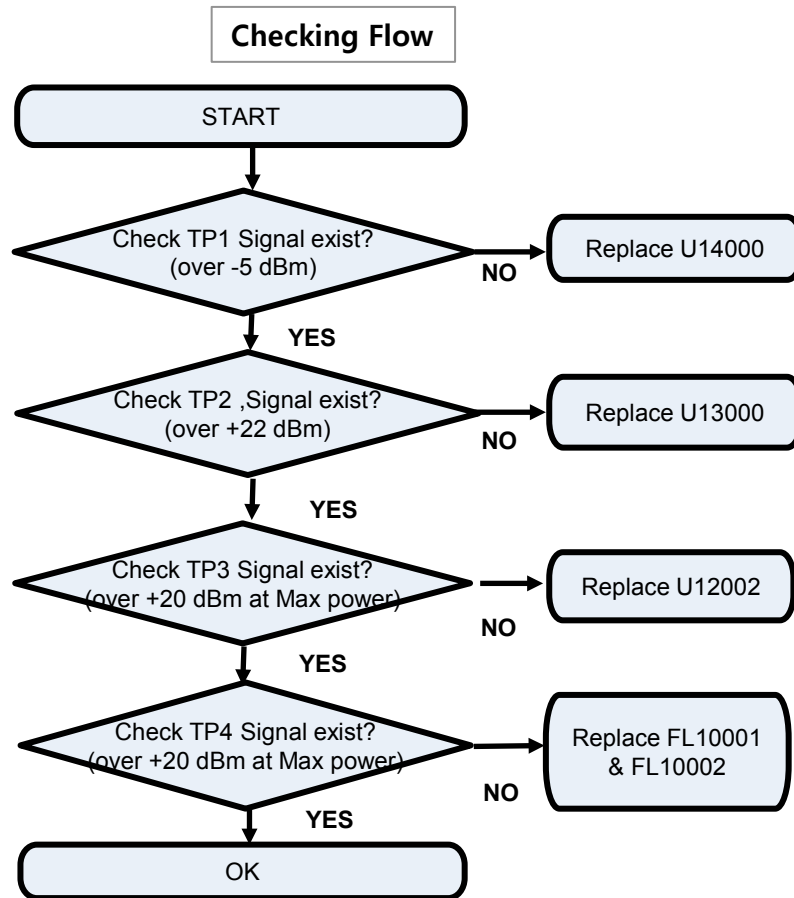
Circuit Diagram



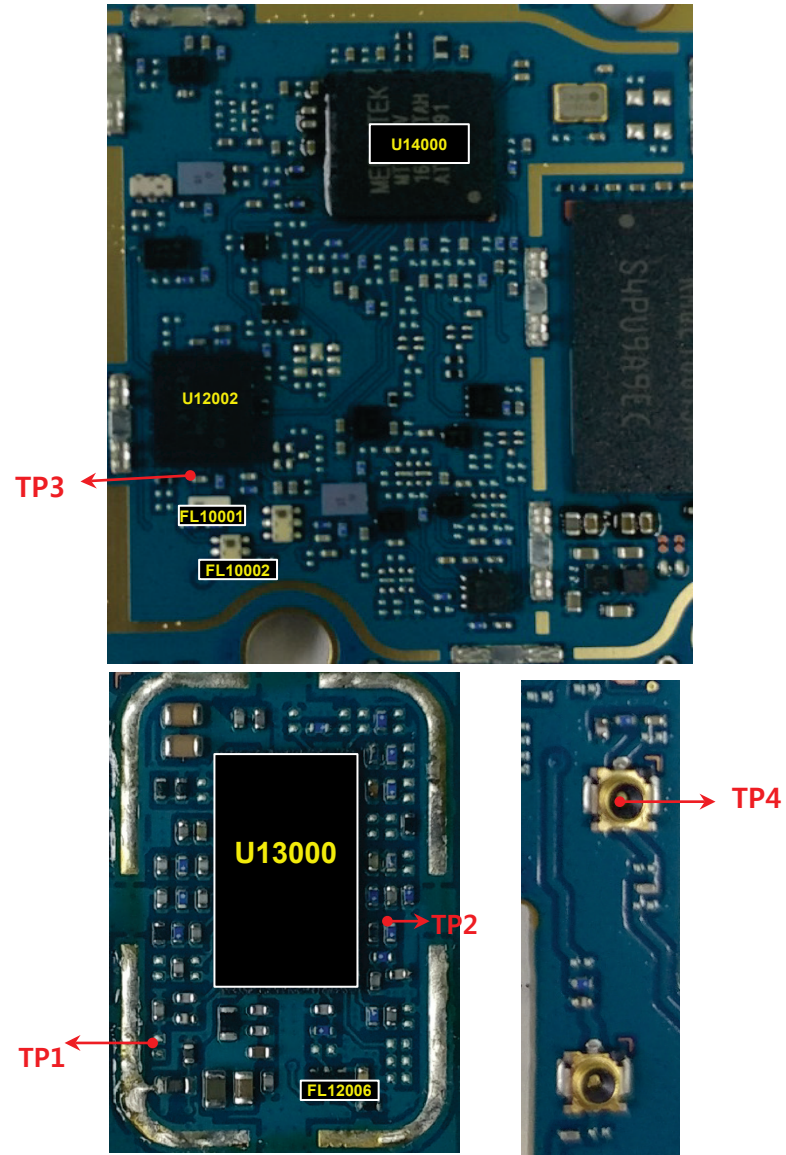
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B8)



Image



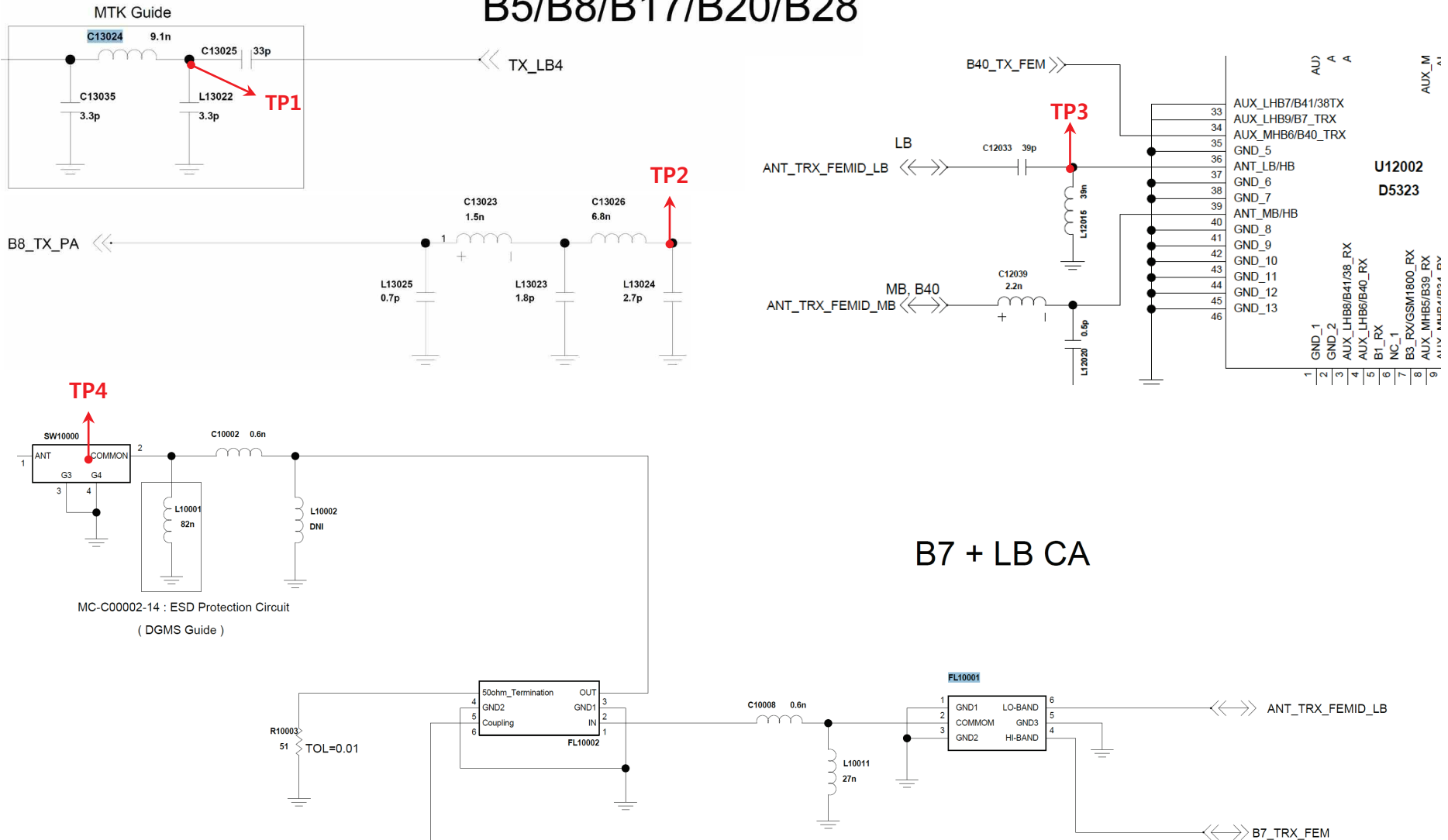
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B8)

Circuit Diagram

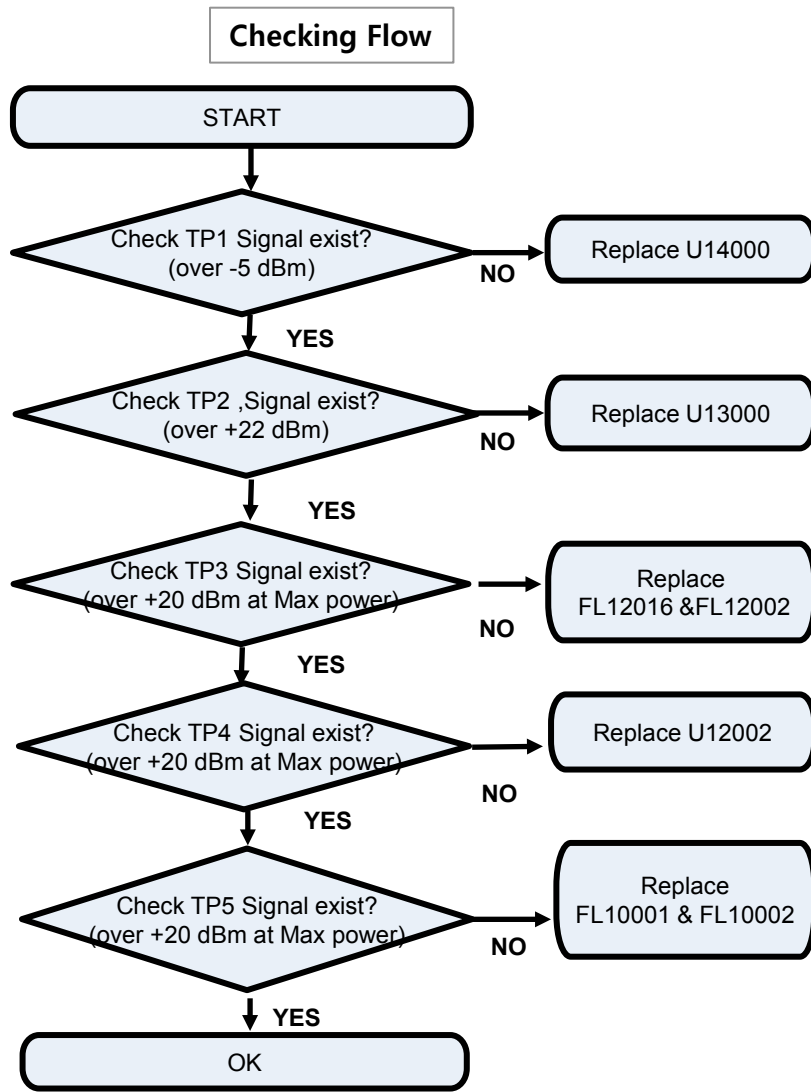
B5/B8/B17/B20/B28



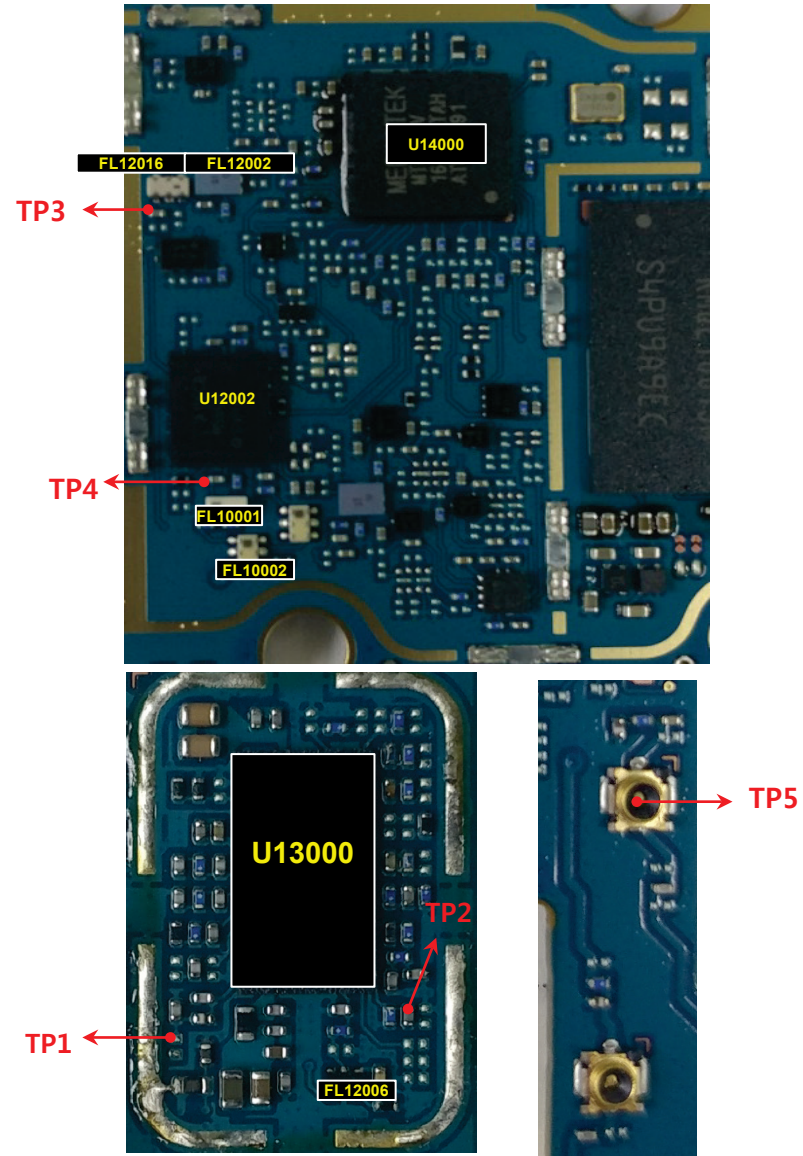
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B20)



Image



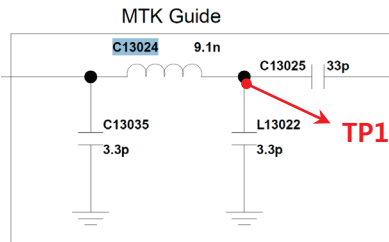
3. TROUBLE SHOOTING

3.7 LTE RF PART

3.7.12 Checking RF Signal TX path(LTE B20)

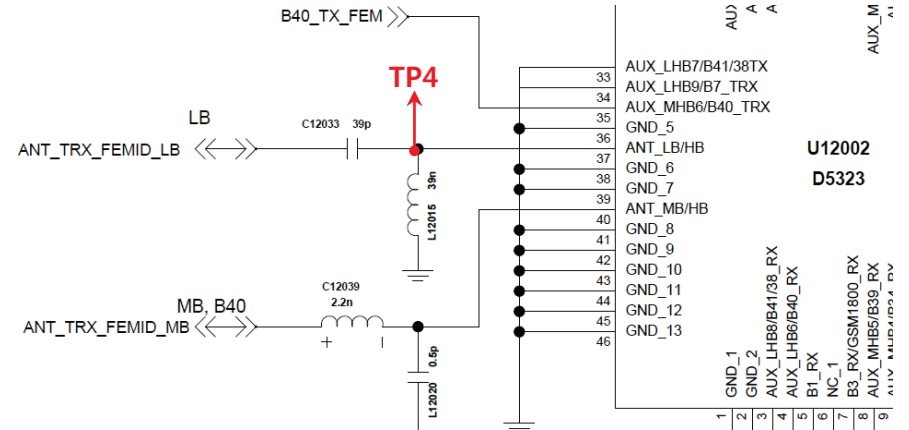
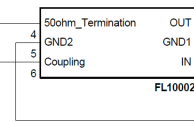
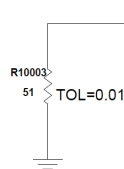
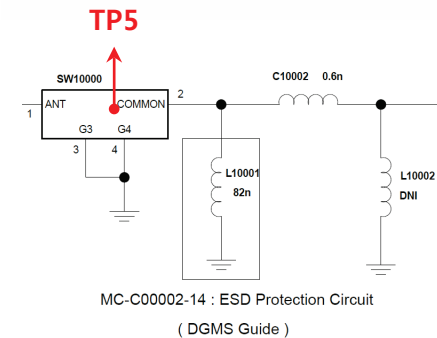
Circuit Diagram

B5/B8/B17/B20/B28



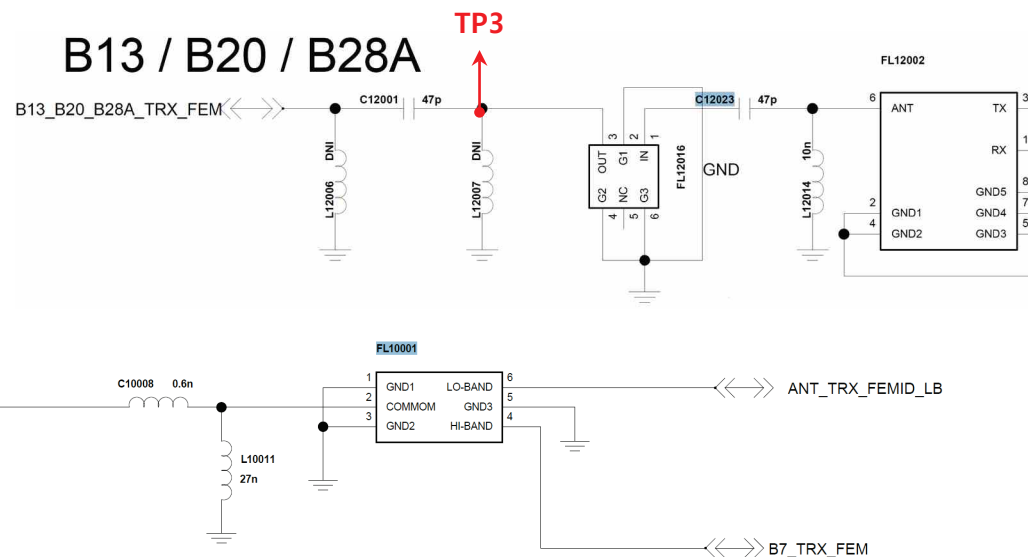
B13 / B20 / B28A

B13_B20_B28A_TX_PA



B13 / B20 / B28A

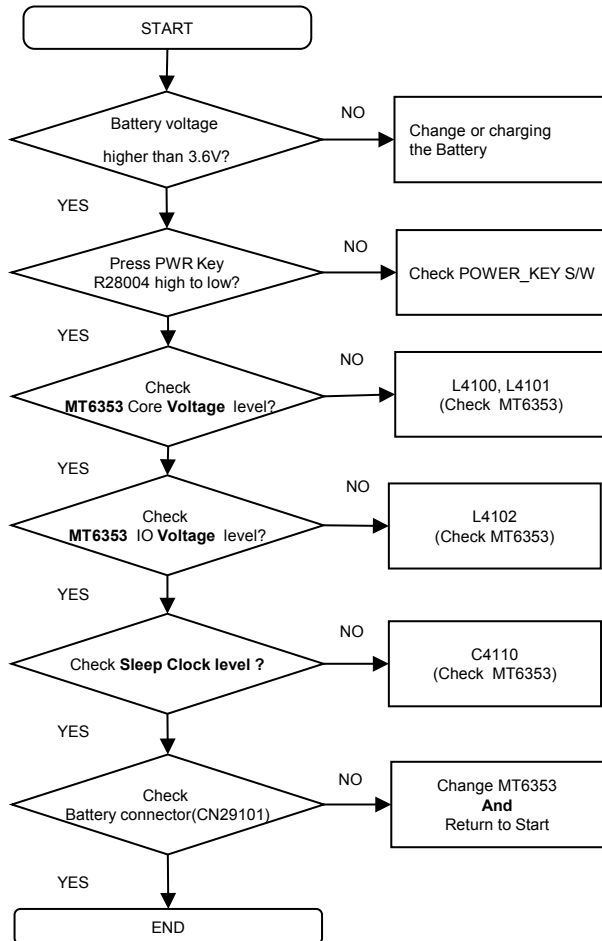
B13_B20_B28A_TRX_FEM



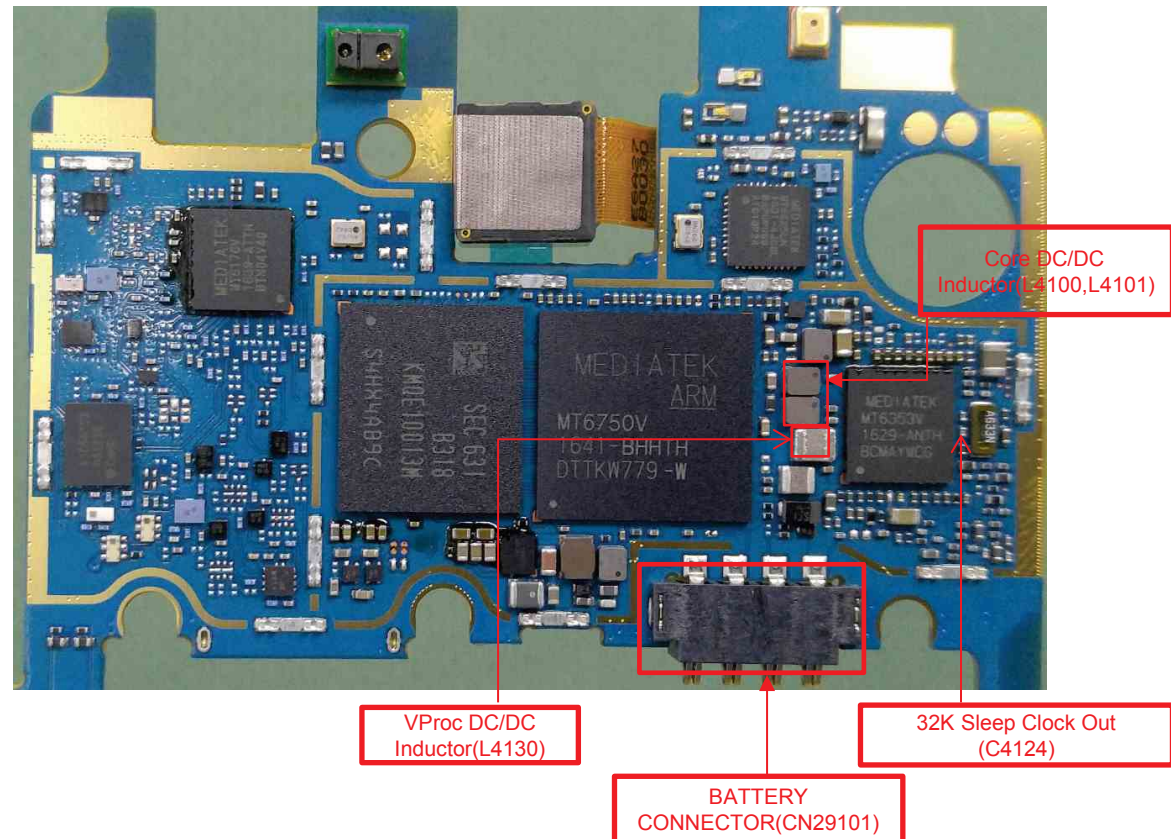
3.8 Power

Checking Power signal (Battery connector, Power Key, PMIC Regulator)

Checking Flow



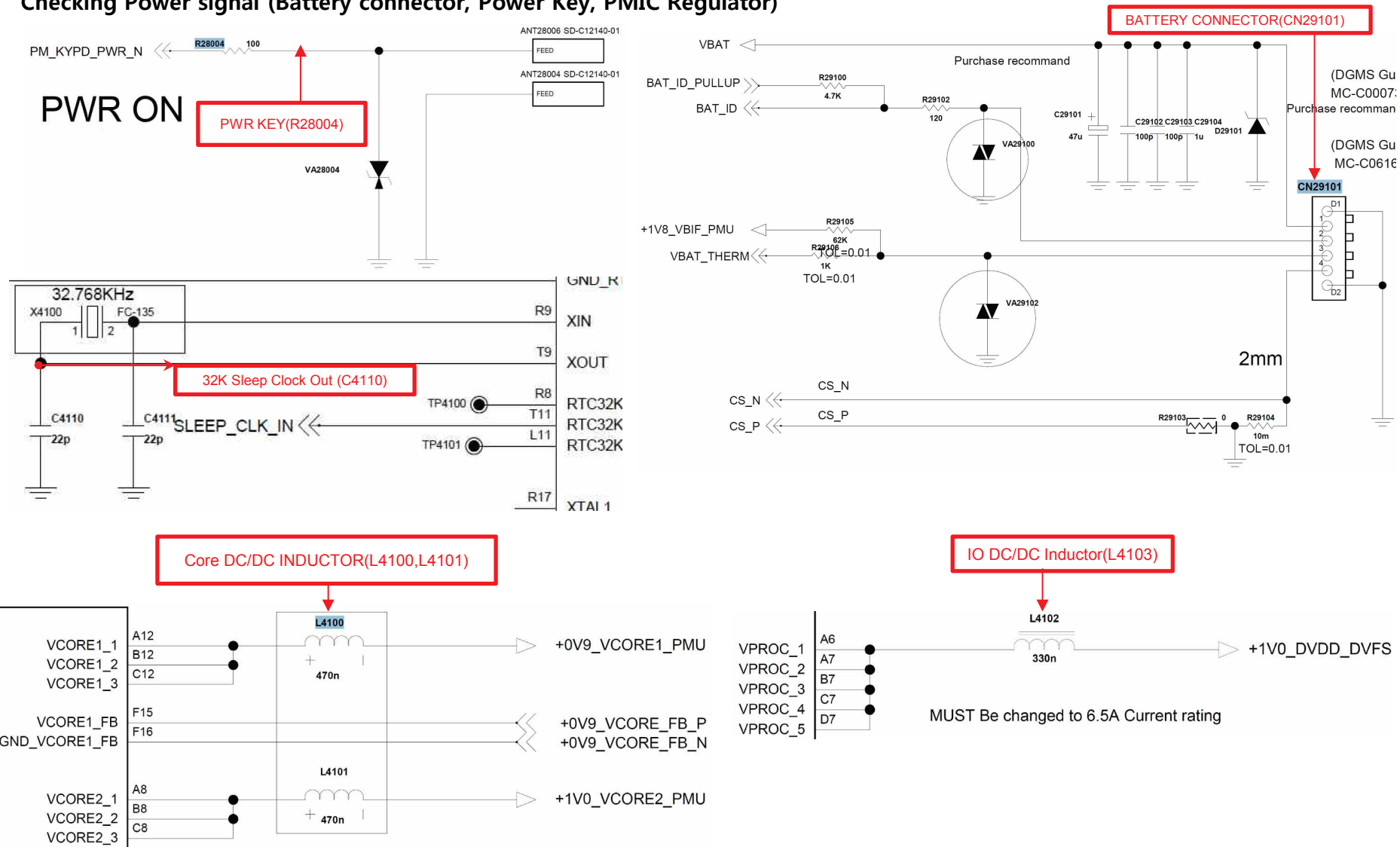
Image



3. TROUBLE SHOOTING

3.8 Power

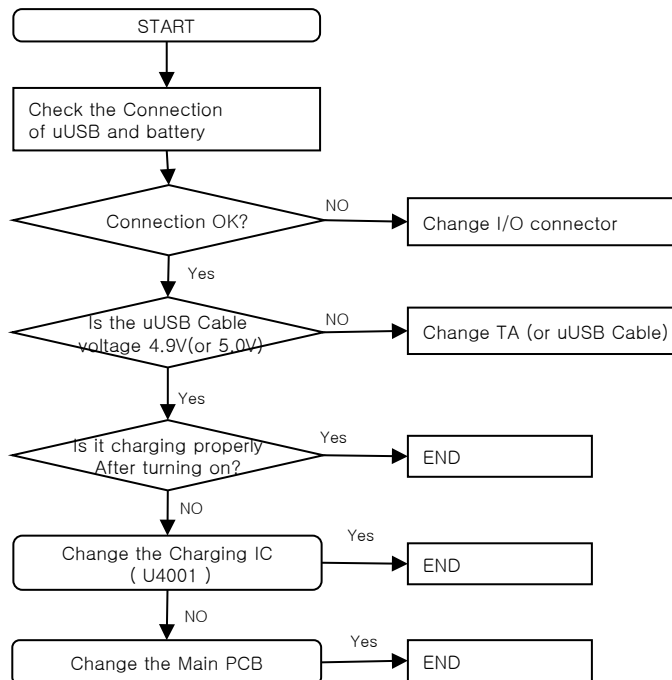
Checking Power signal (Battery connector, Power Key, PMIC Regulator)



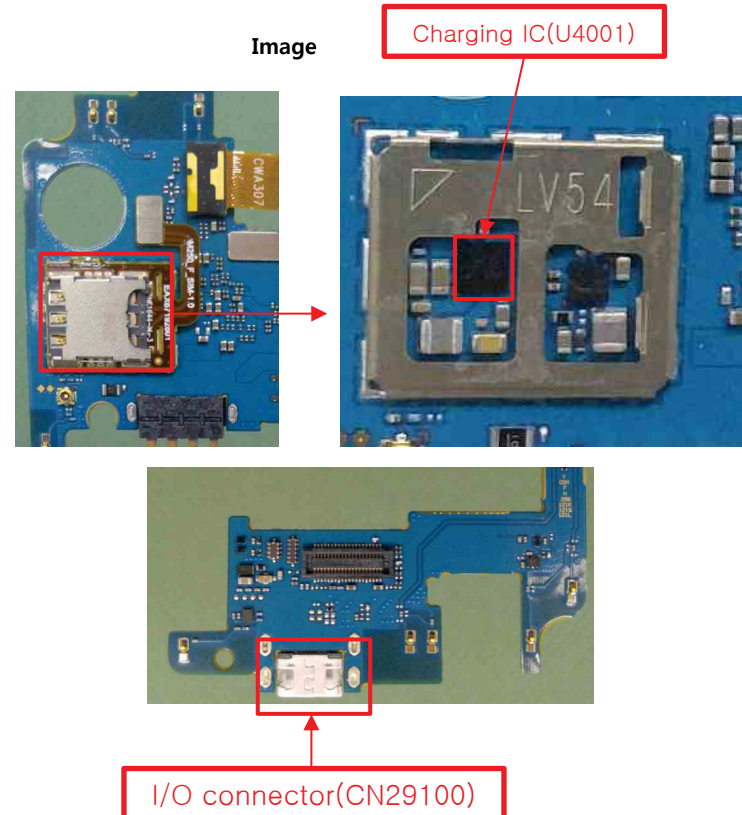
3.9 Charger

The I/O connector and uUSB cable voltage(5.0V) is used as the reference one of PMIC for charging.

Checking Flow



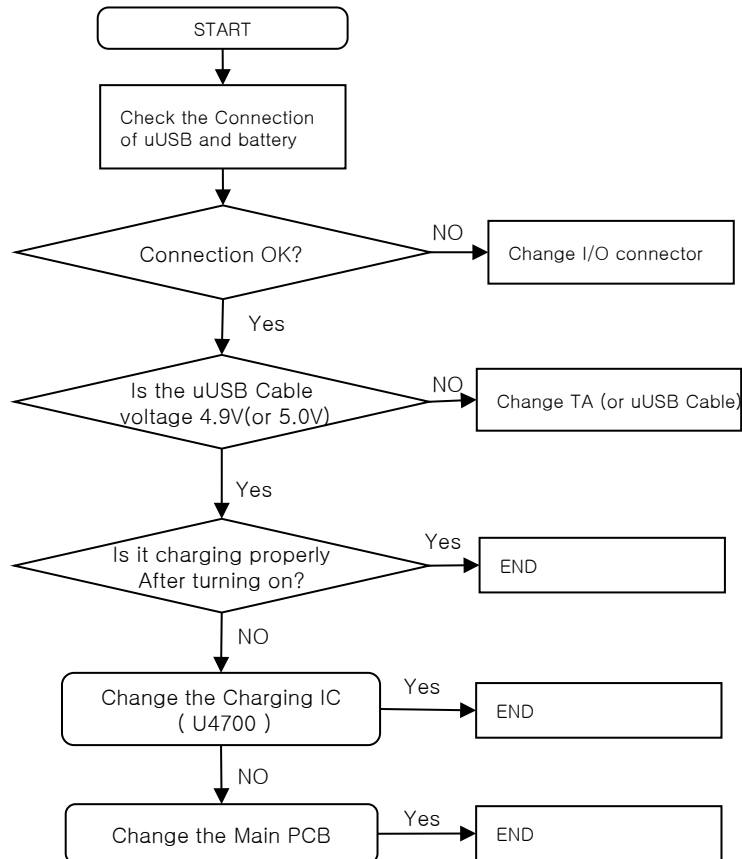
Image



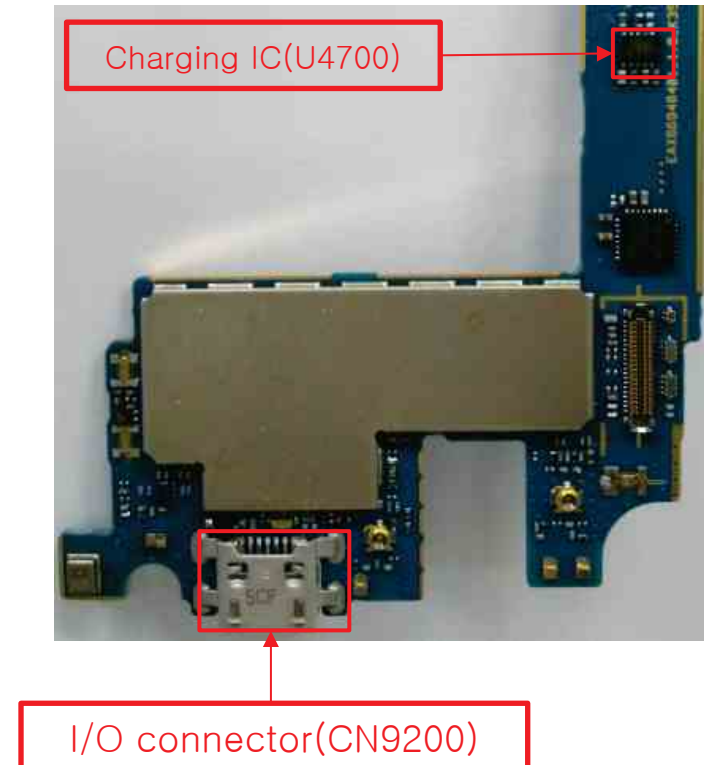
3.9 Charger

The I/O connector and uUSB cable voltage(5.0V) is used as the reference one of PMIC for charging.

Checking Flow



Image

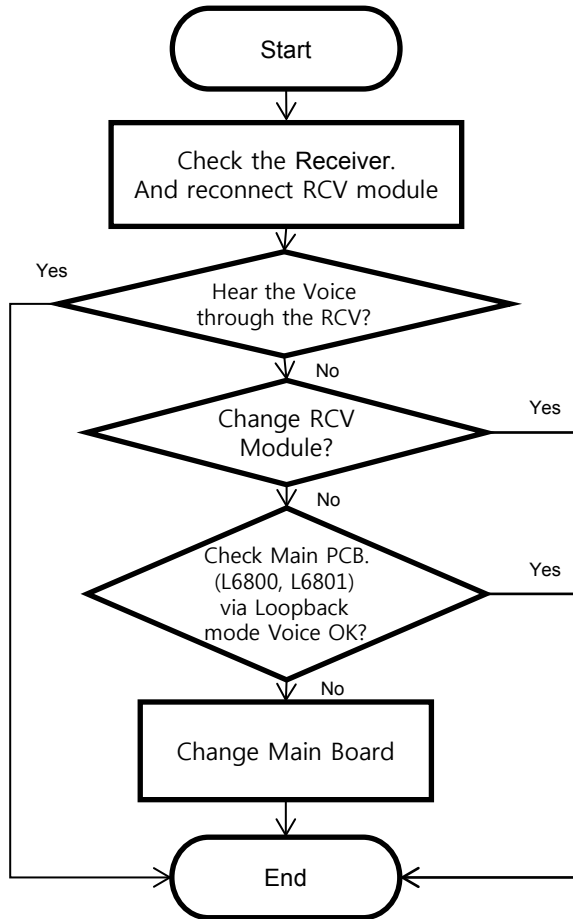


3. TROUBLE SHOOTING

3.10 Audio Block(3.10.1 Audio receiver)

The receiver control signals are generated by MT6353(U4100), the MT6353 chip and the receiver are to be checked out.

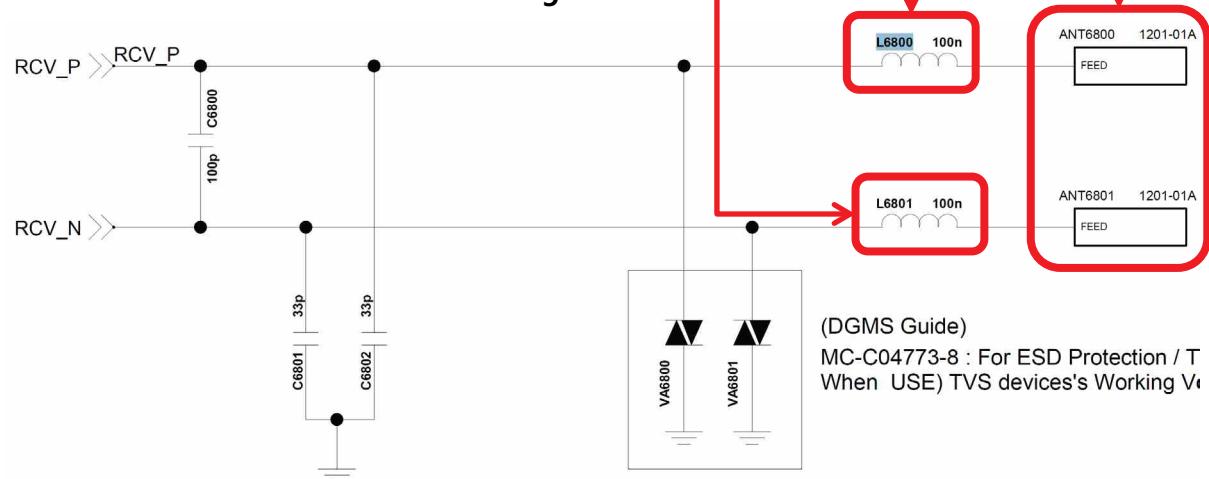
Checking Flow



Image



Circuit Diagram

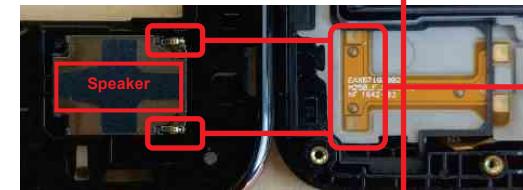
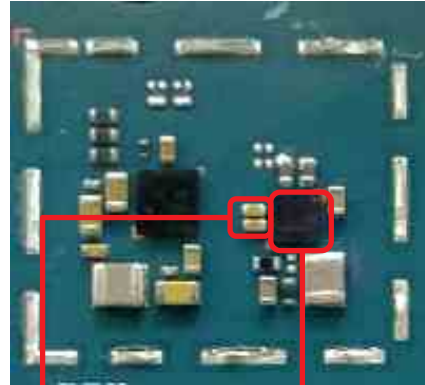


3. TROUBLE SHOOTING

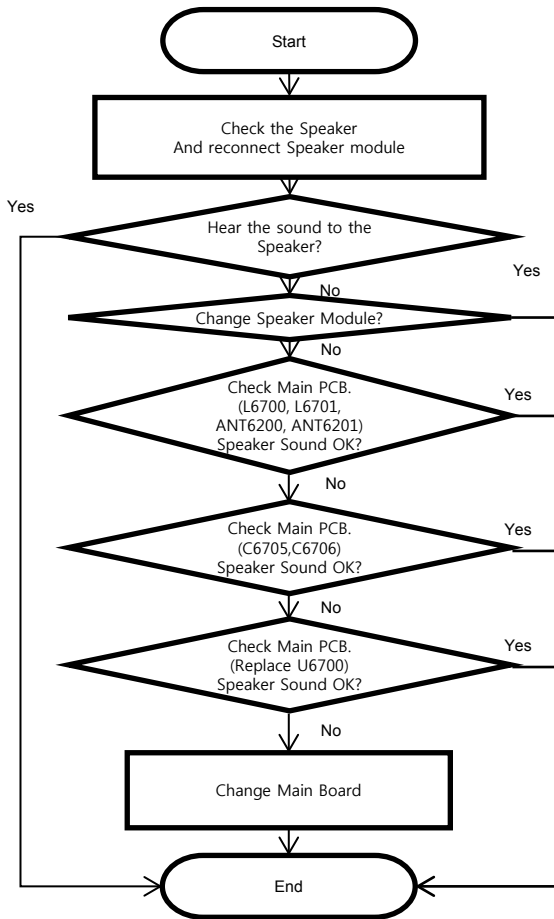
3.10.1 Audio_speaker

The Speaker control signals are generated by MT6353(U4100), the MT6353 chip and the speaker are to be checked out.

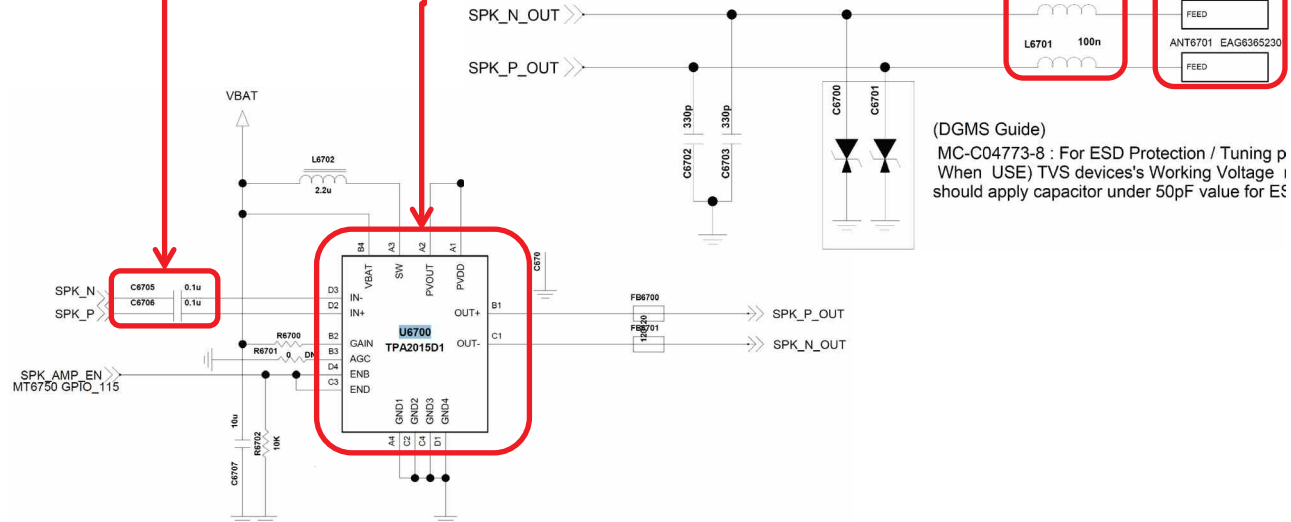
Image



Checking Flow



Circuit Diagram

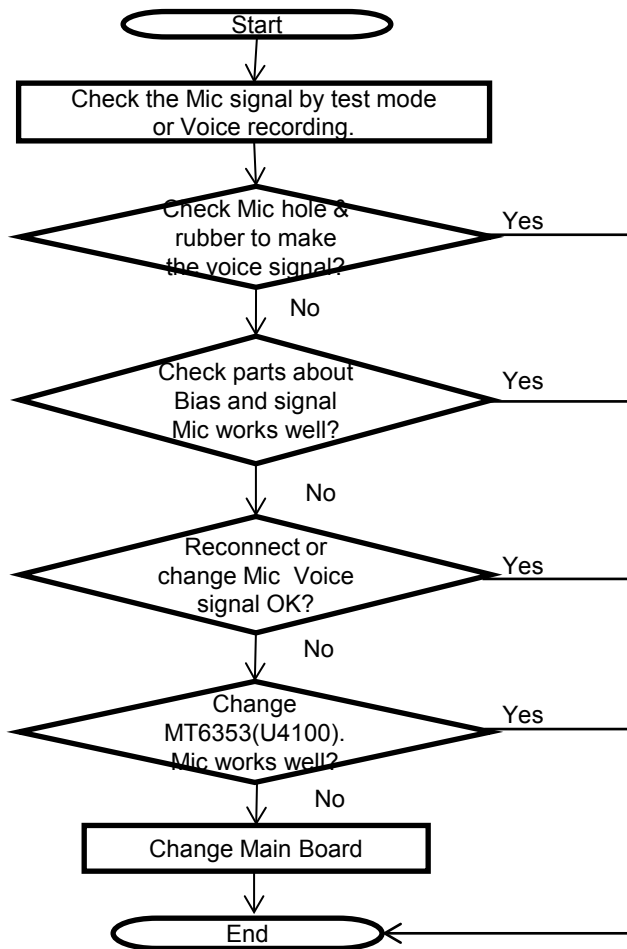


3. TROUBLE SHOOTING

3.10.2 Audio_Main MIC

It's operating voice call(except speakerphone), voice recording, camcorder recording.

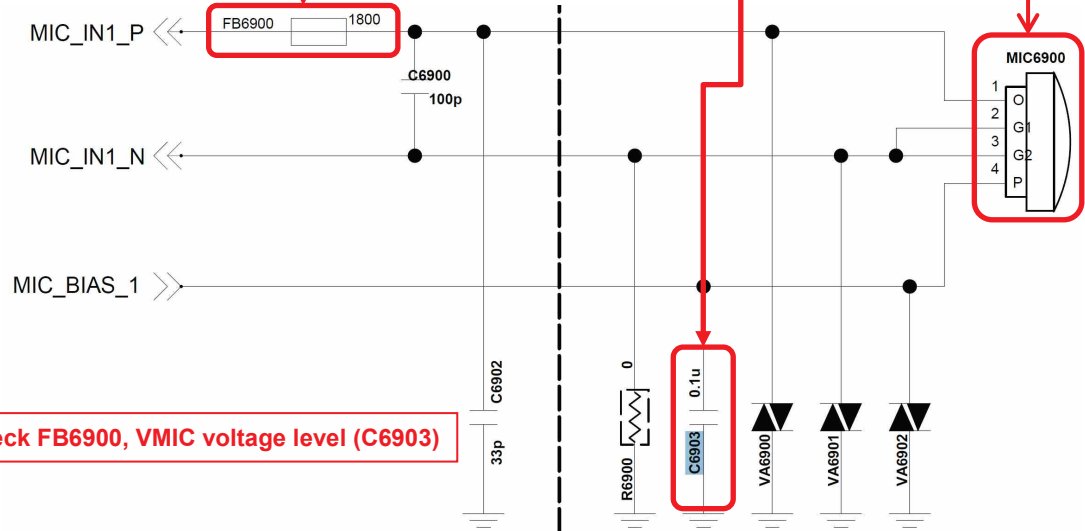
Checking Flow



Image



Circuit Diagram



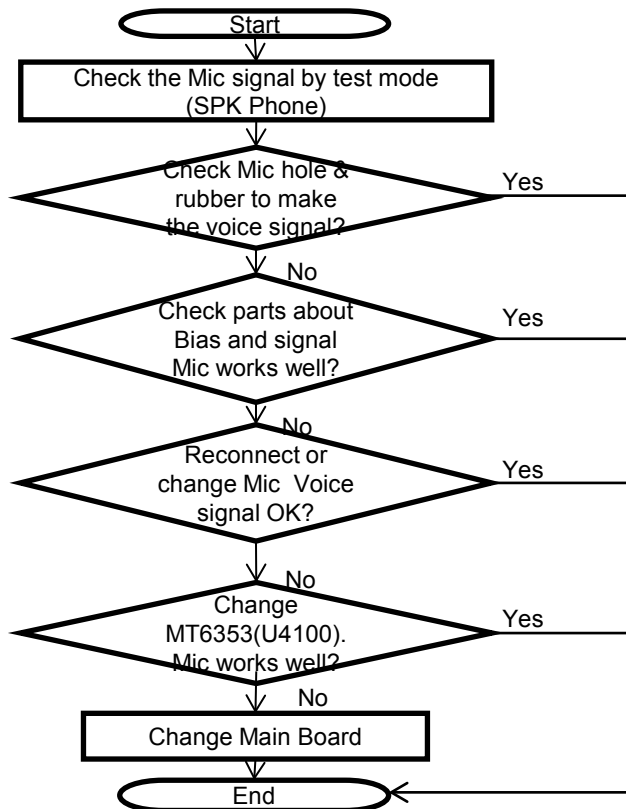
Check FB6900, VMIC voltage level (C6903)

3. TROUBLE SHOOTING

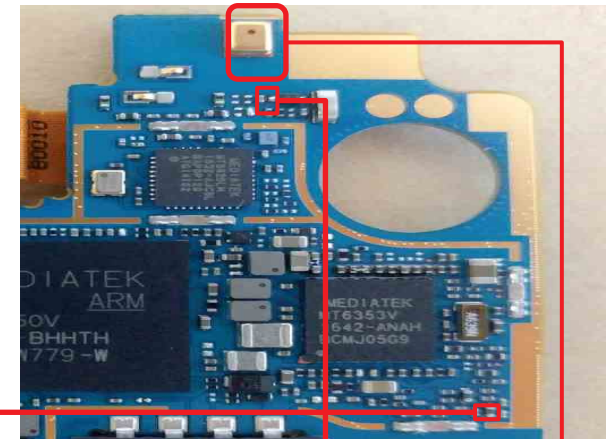
3.10.3 Audio_sub MIC

It's operating Speakerphone call.

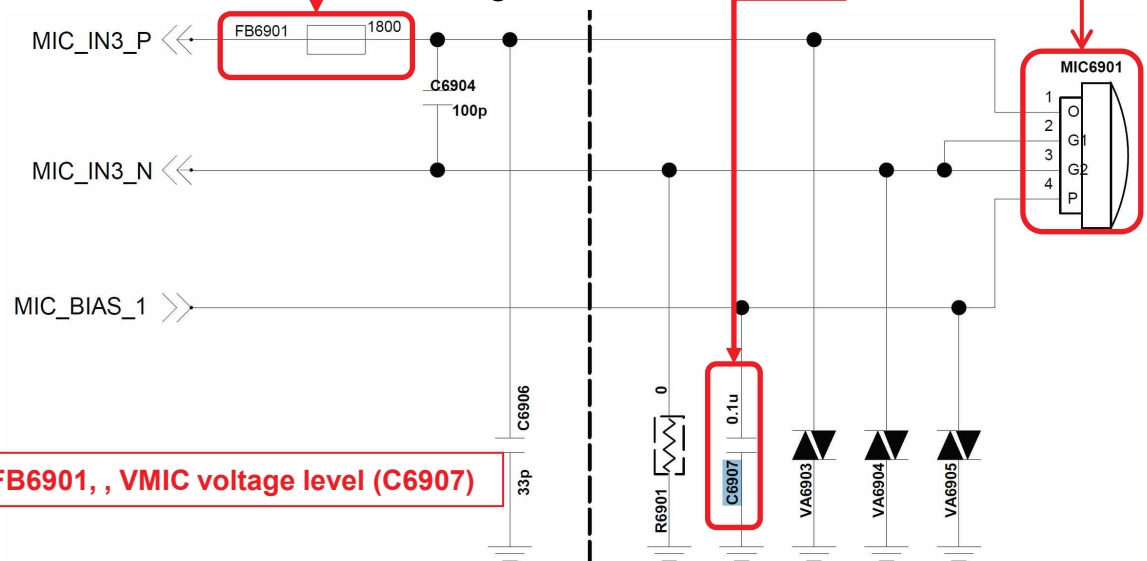
Checking Flow



Image



Circuit Diagram



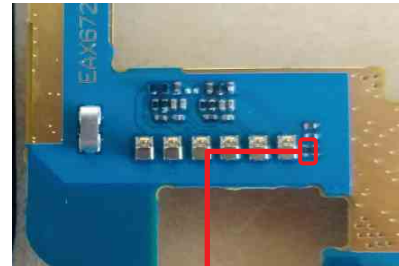
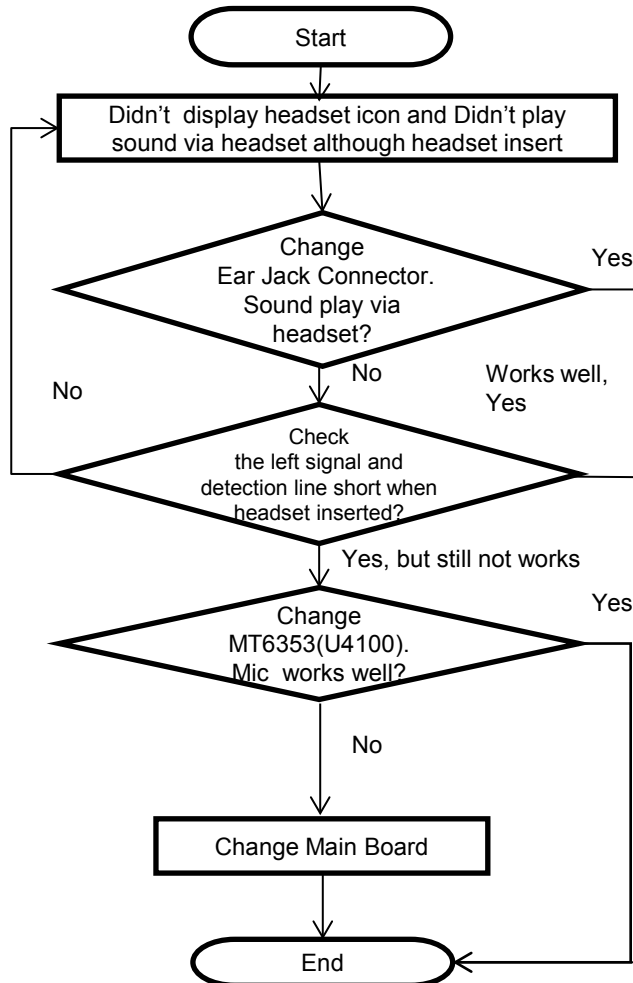
Check FB6901, , VMIC voltage level (C6907)

3. TROUBLE SHOOTING

3.10.4 Audio_Ear MIC Jack

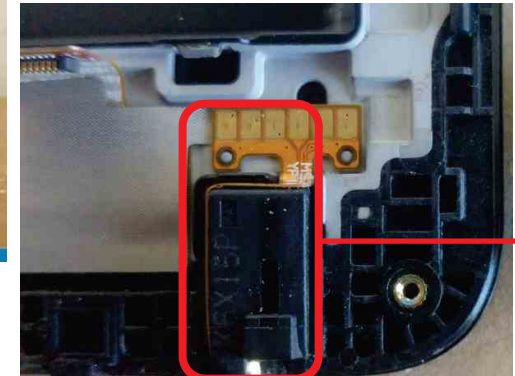
Disable detecting headset insert or No sound from Earphone, Check the Ear Mic and MT6353

Checking Flow



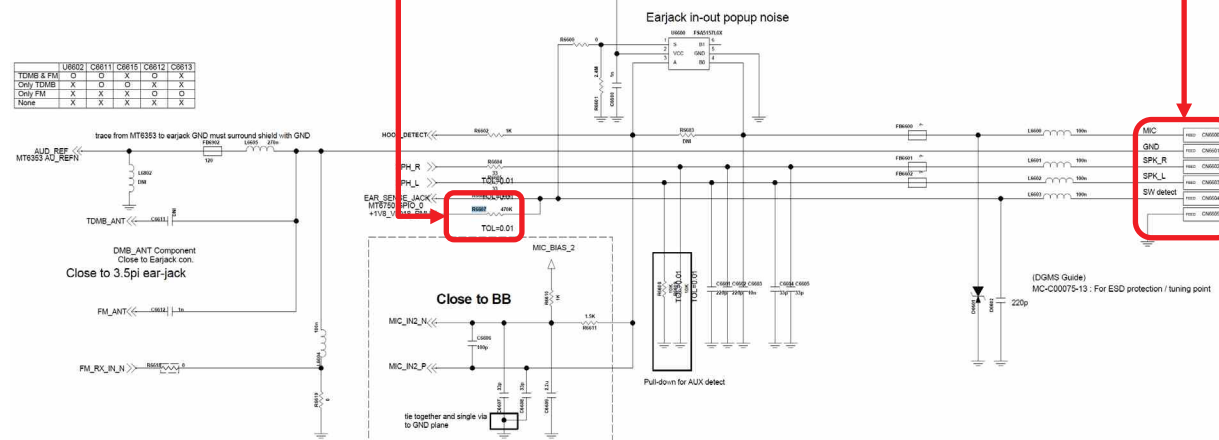
Jack detect Voltage Level Check at R6607
 - 1.8V(before Jack)
 - low(after Jack)

Image



< 6-6-1_Earjack >

Circuit Diagram

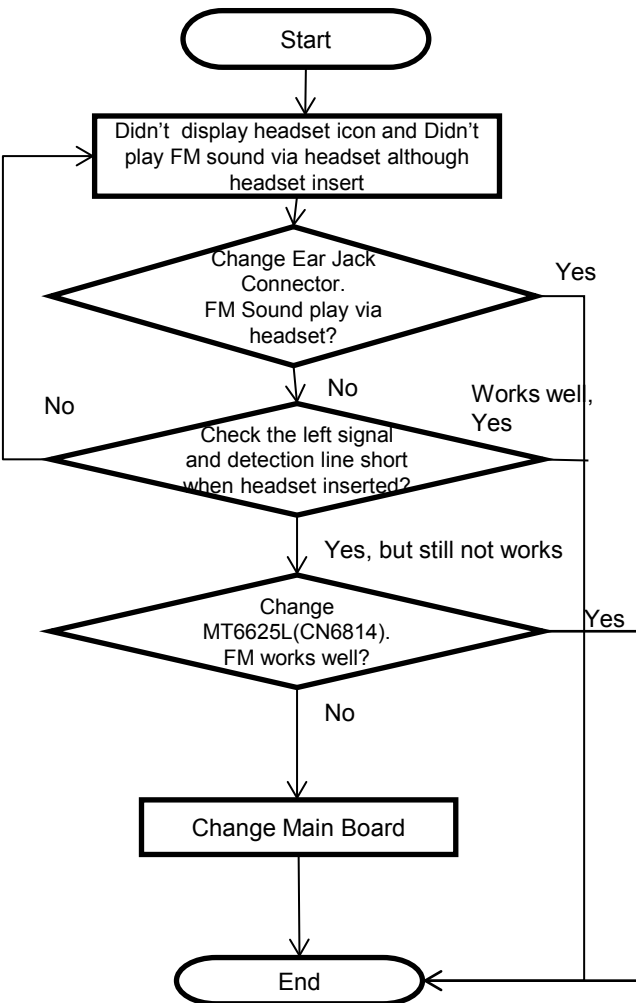


3. TROUBLE SHOOTING

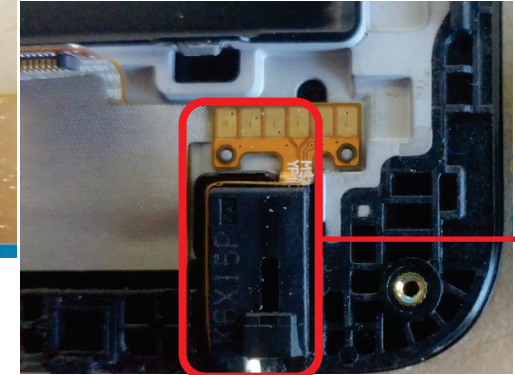
3.10.5 FM_Radio

Disable detecting headset insert or No FM sound from Earphone, Check the Ear mic and MT6625L

Checking Flow

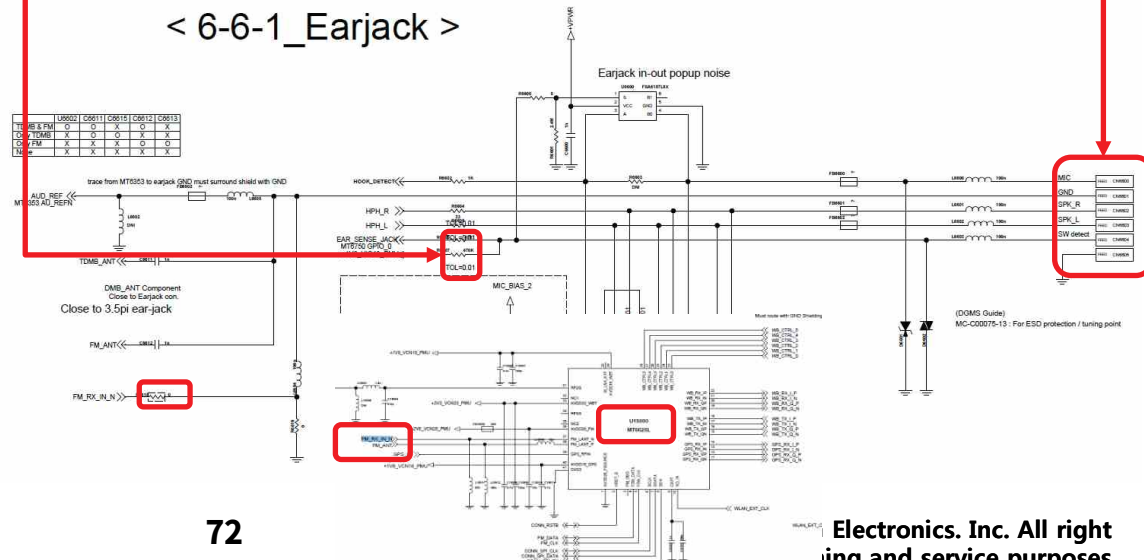


Image



Jack detect
Voltage Level Check at R6607
 - 1.8V(before Jack)
 - low(after Jack)

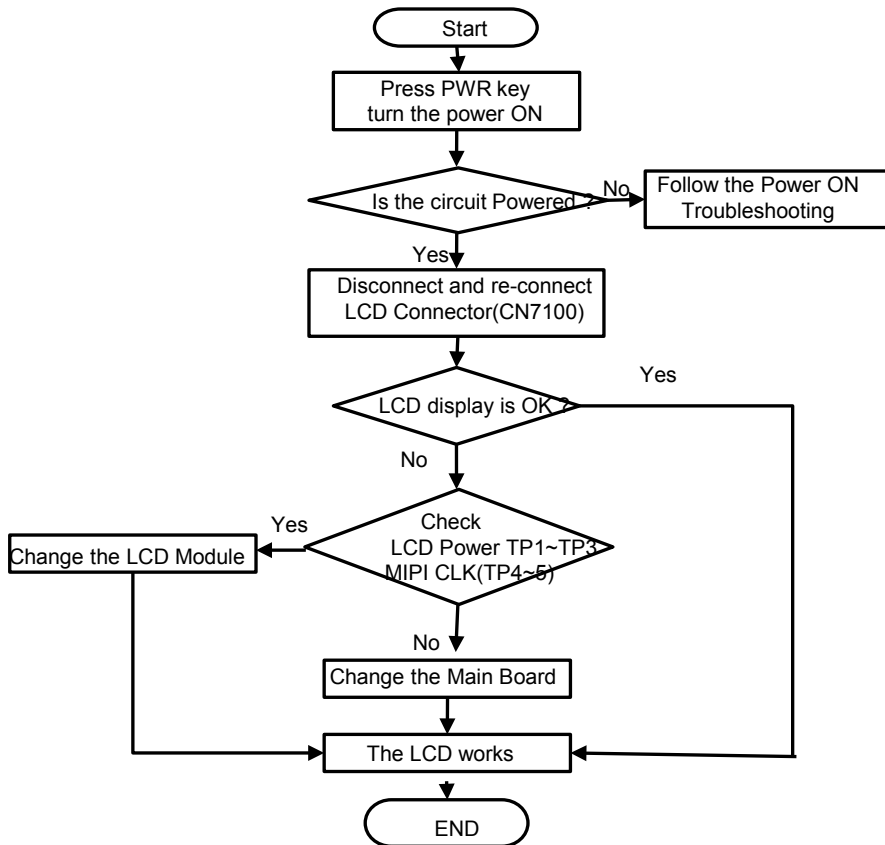
Circuit Diagram



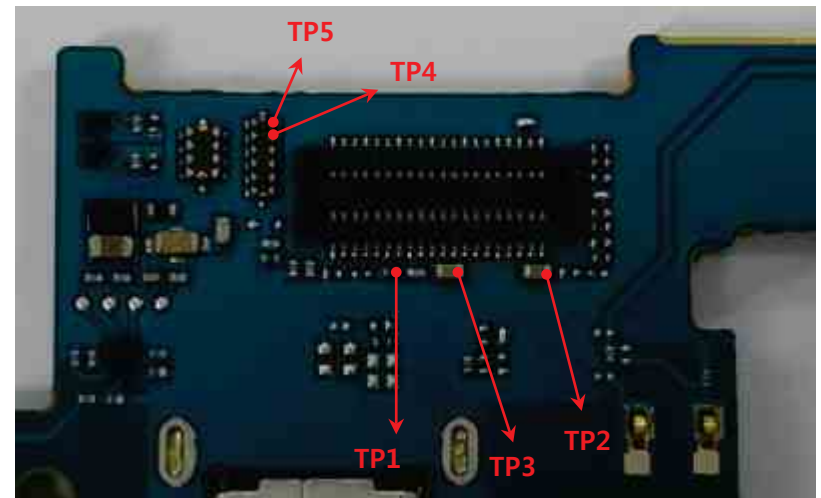
3.11 Checking LCD Block

The LCD control signals are generated by MT6750. Its interface is MIPI having four data lanes and one clock lane.

Checking Flow



Image

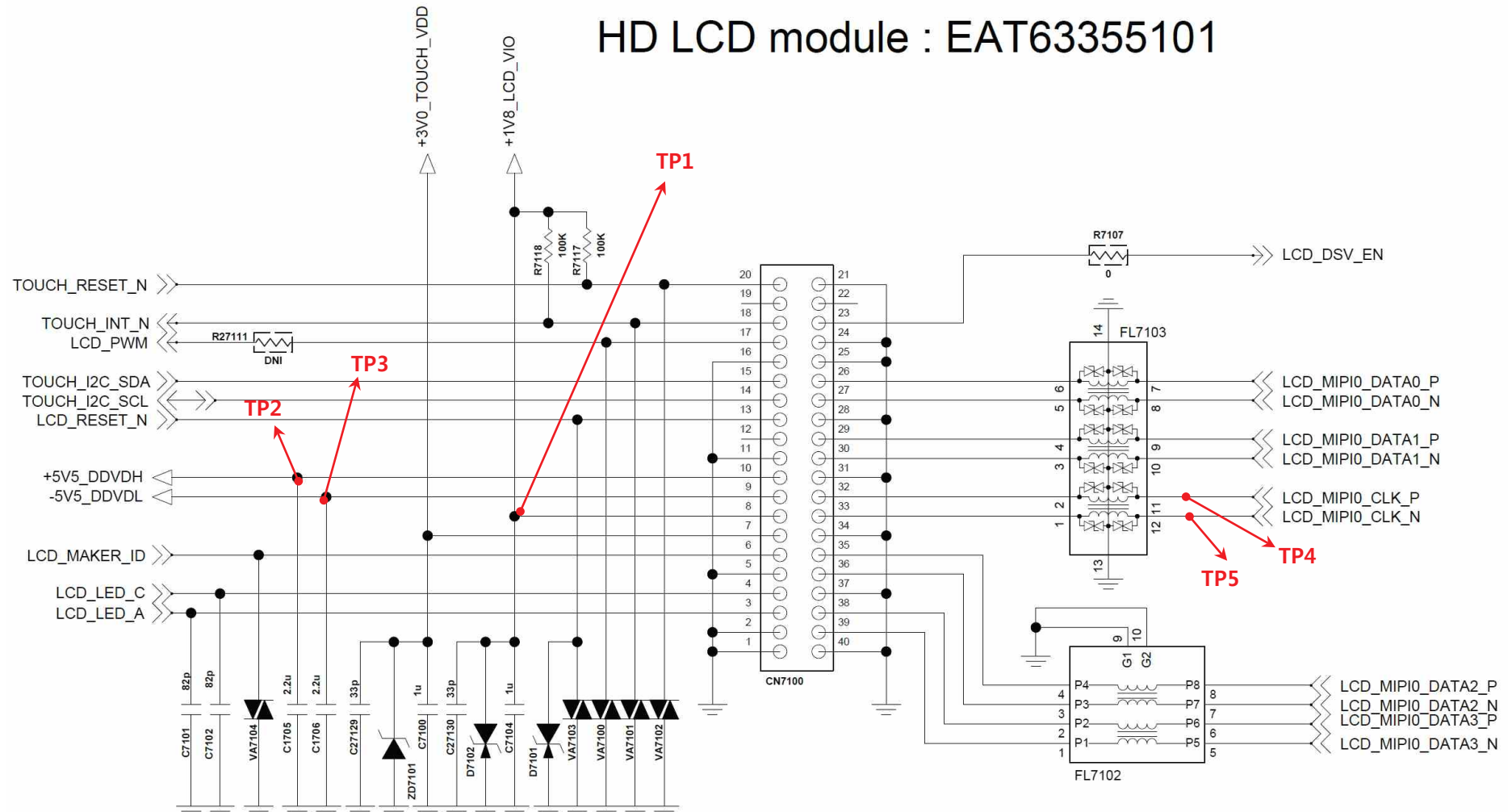


3.11 Checking LCD Block

The LCD control signals are generated by MT6750. Its interface is MIPI having four data lanes and one clock lane.

Circuit Diagram

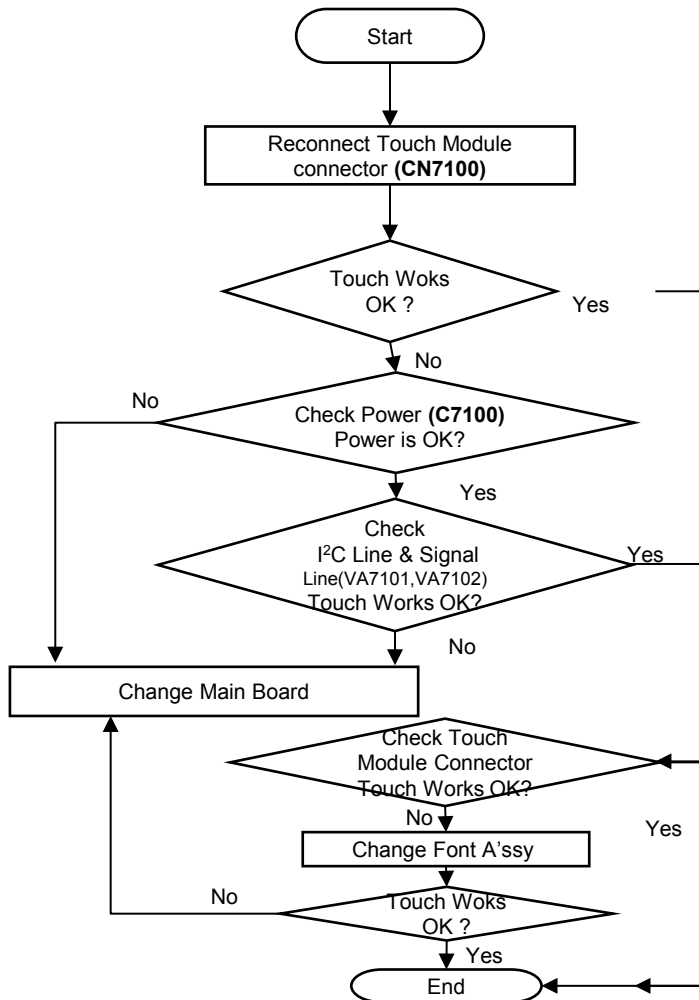
HD LCD module : EAT63355101



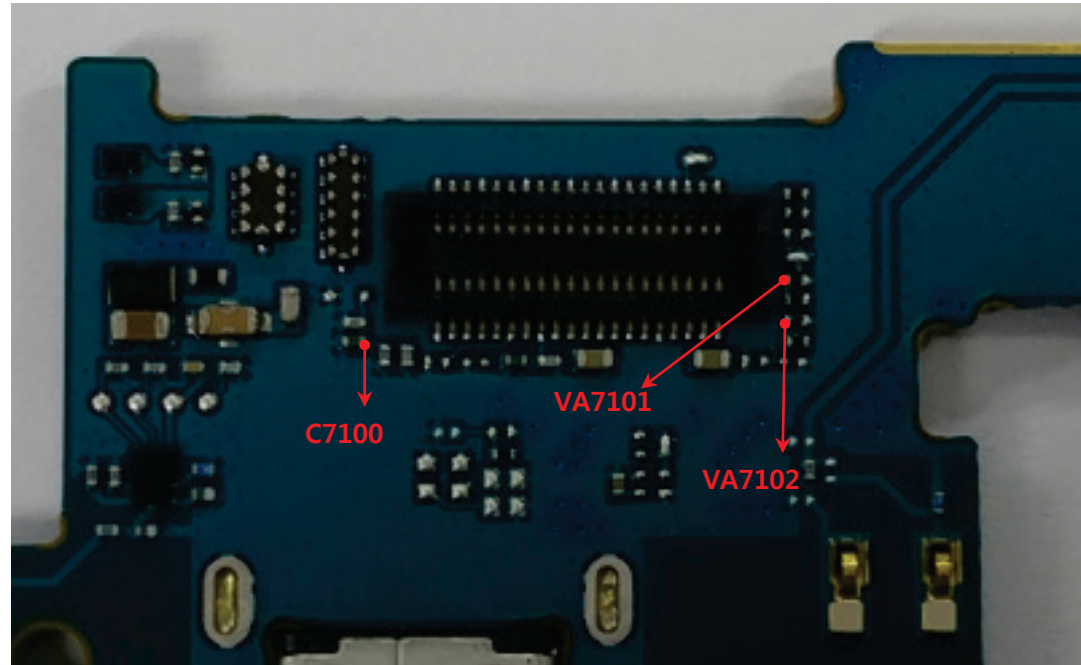
3.12 Checking Touch Block

The Touch control signals are generated by MT6750. It is assembled with LCD.

Checking Flow



Image

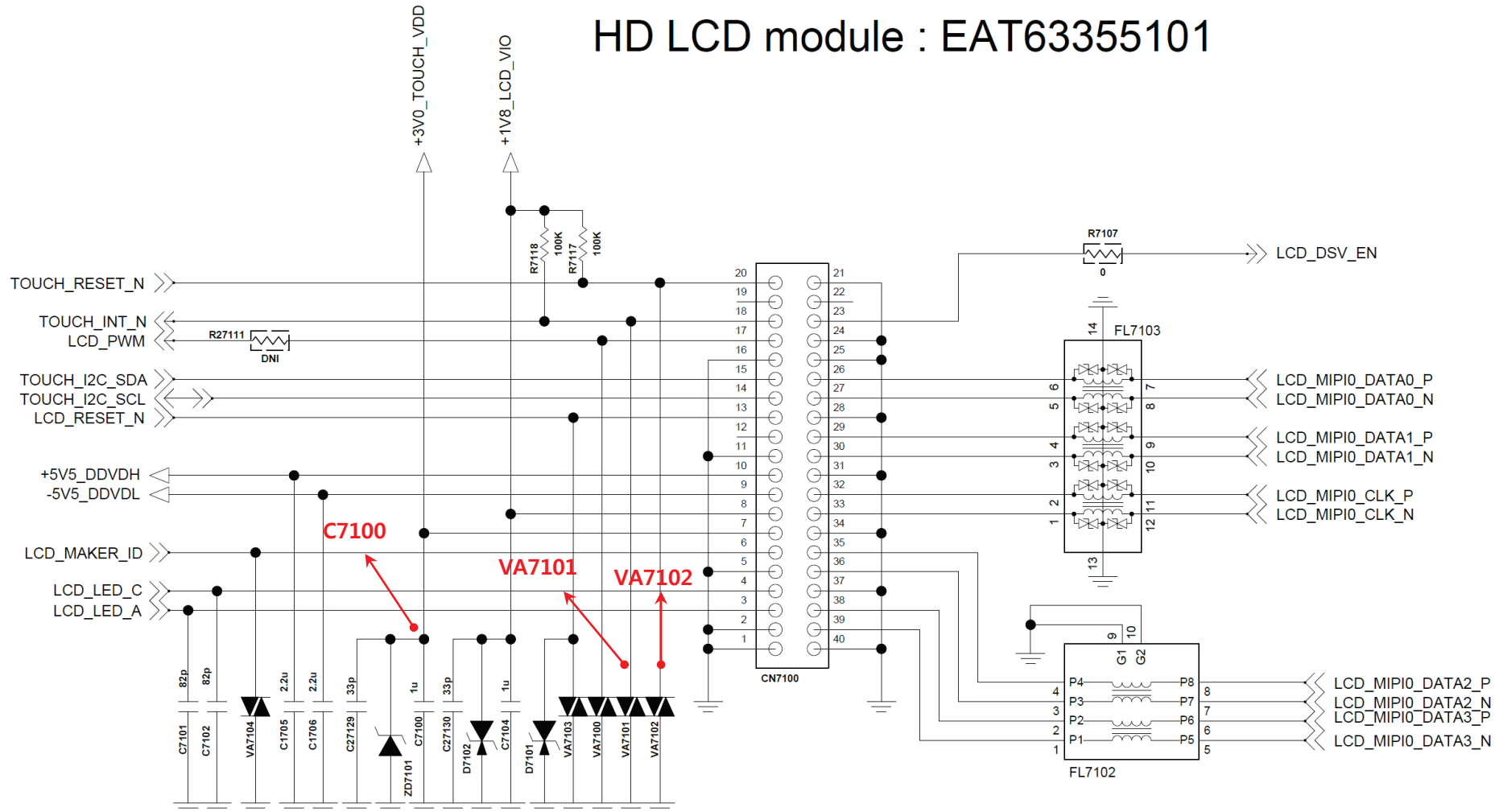


3.12 Checking Touch Block

The Touch control signals are generated by MT6750. It is assembled with LCD.

Circuit Diagram

HD LCD module : EAT63355101

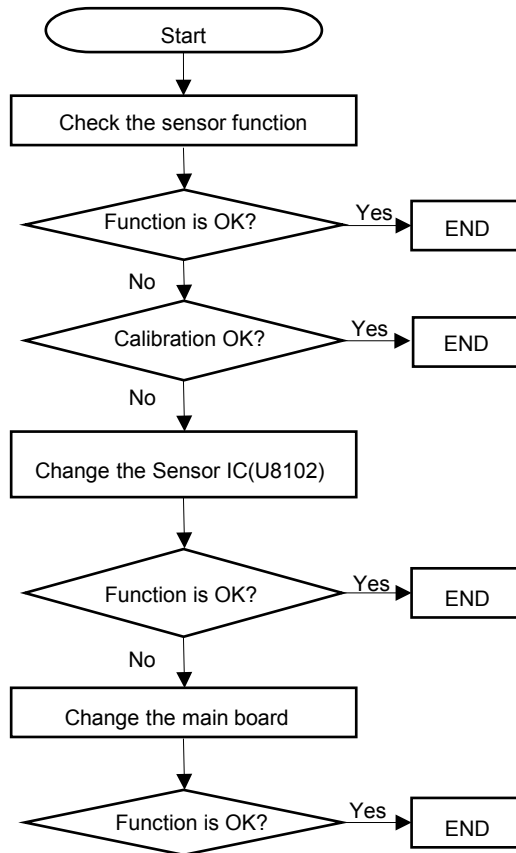


3. TROUBLE SHOOTING

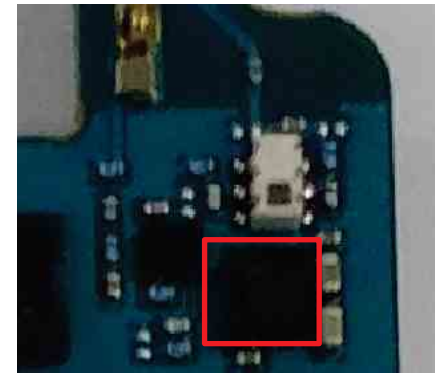
3.13 Checking accelmeter+compass sensor Block

The Accl. & compass sensors are calibrated by using SW algorithm.

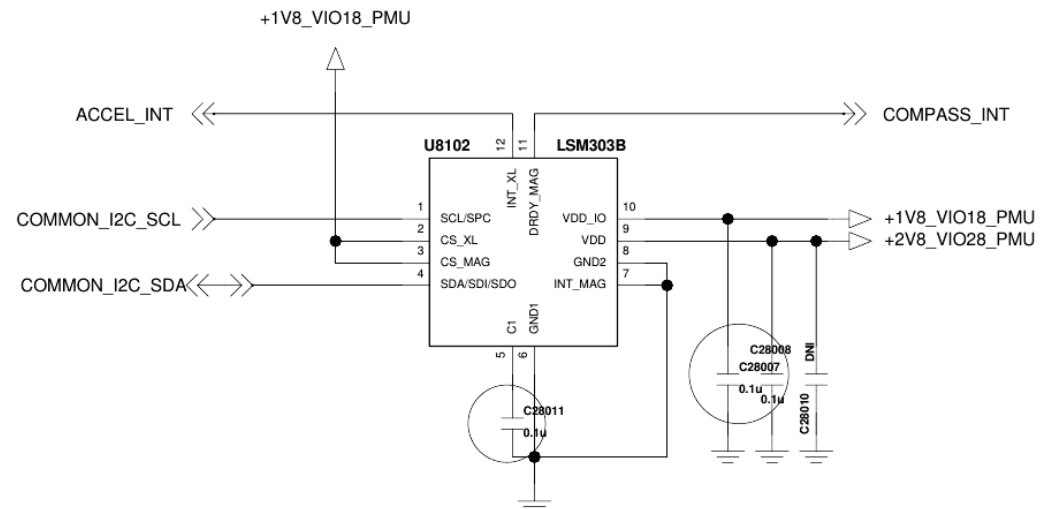
Checking Flow



Image



Circuit Diagram

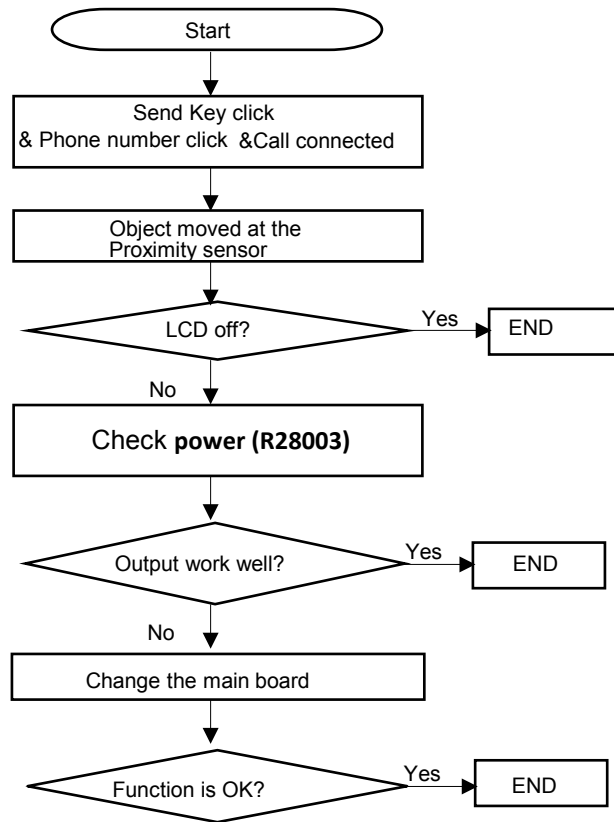


3.13 Checking proximity sensor Block

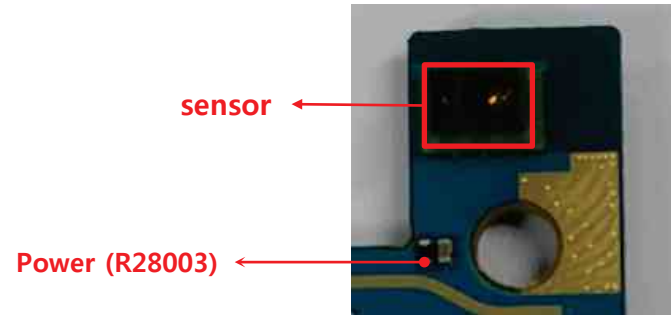
Proximity Sensor is worked as below: Send Key click → Phone number click → Call connected

→ Object moved at the sensor → Control the screen's on/off operation automatically

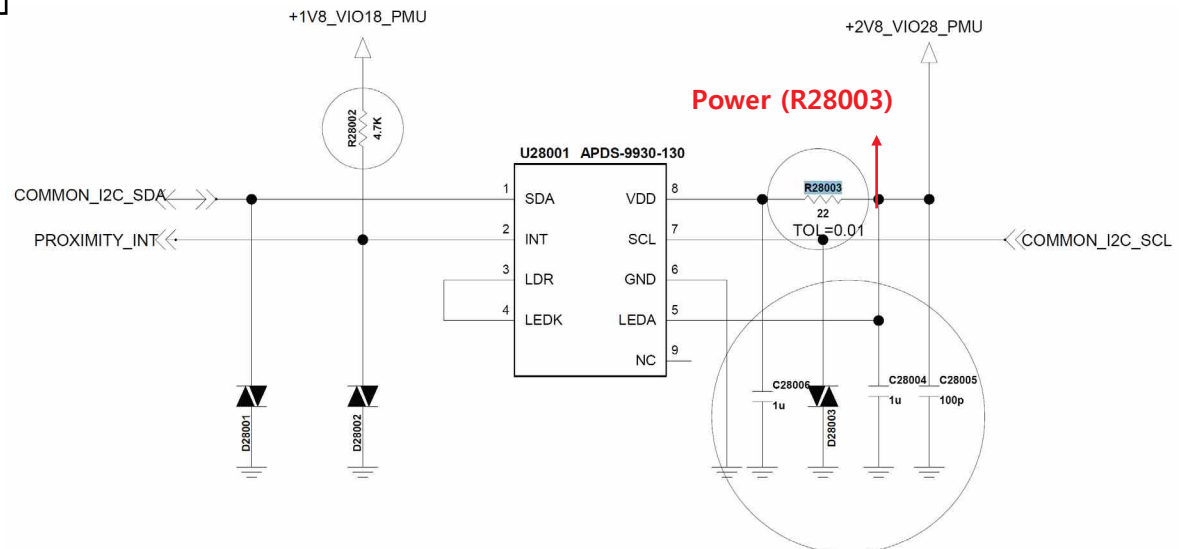
Checking Flow



Image



Circuit Diagram

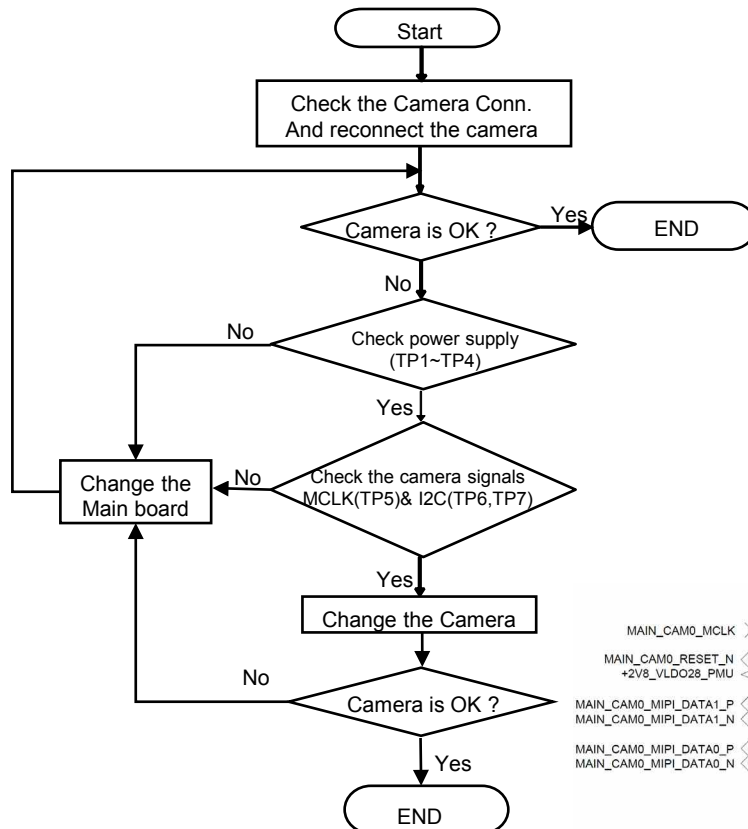


3. TROUBLE SHOOTING

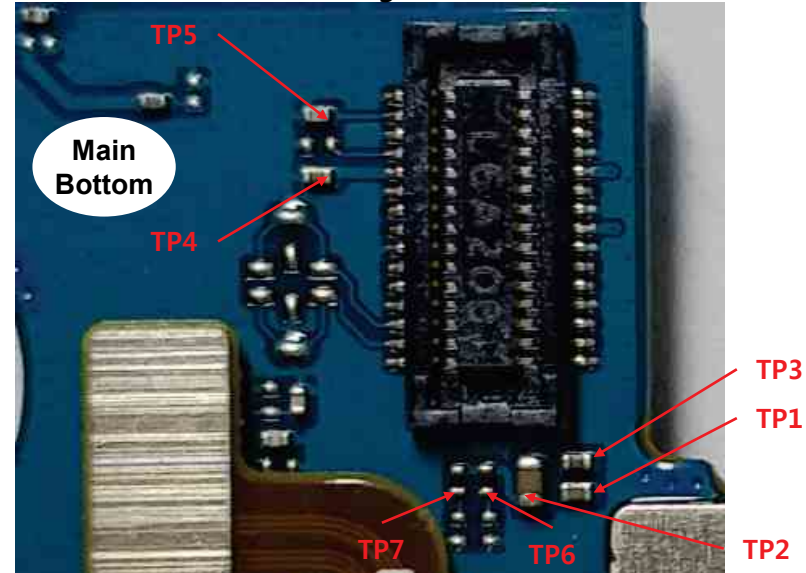
3.XX Checking Main Camera Block

13M camera control signals are generated by MT650 (U21000 : Main Chipset). And powered by MT6353 (U4100 : PMIC)

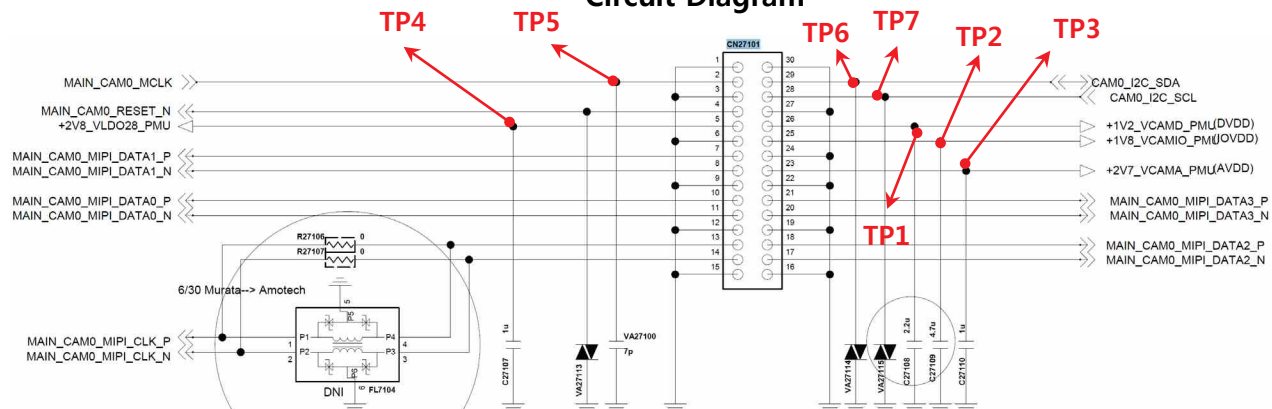
Checking Flow



Image



Circuit Diagram

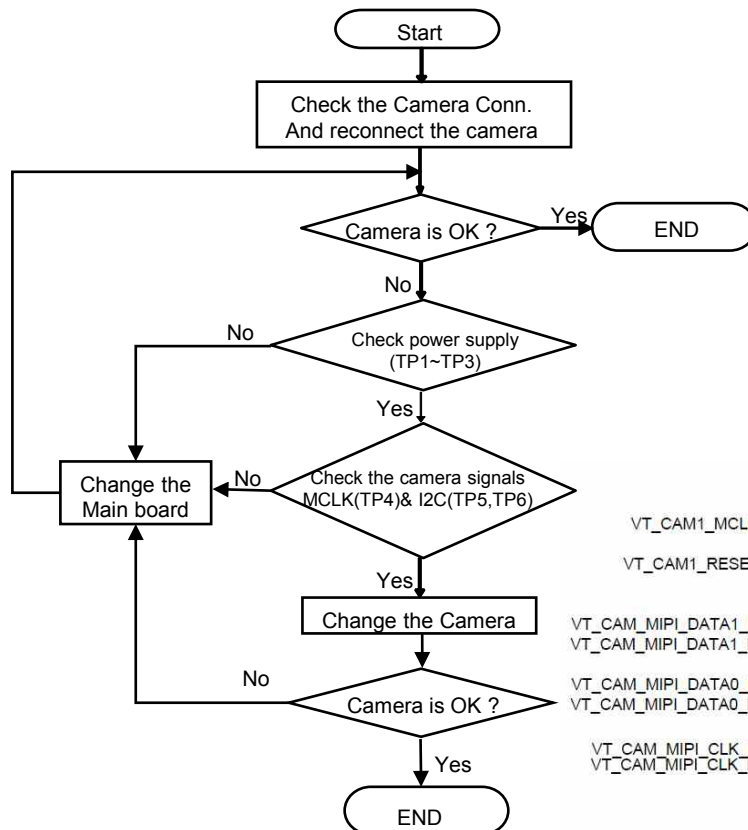


3. TROUBLE SHOOTING

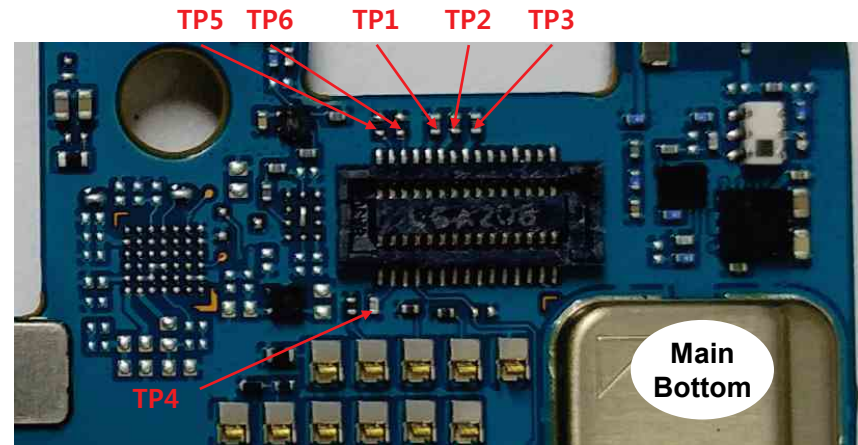
3.XX Checking Front Camera Block

5M camera control signals are generated by MT650 (U21000 : Main Chipset). And powered by MT6353 (U4100 : PMIC)

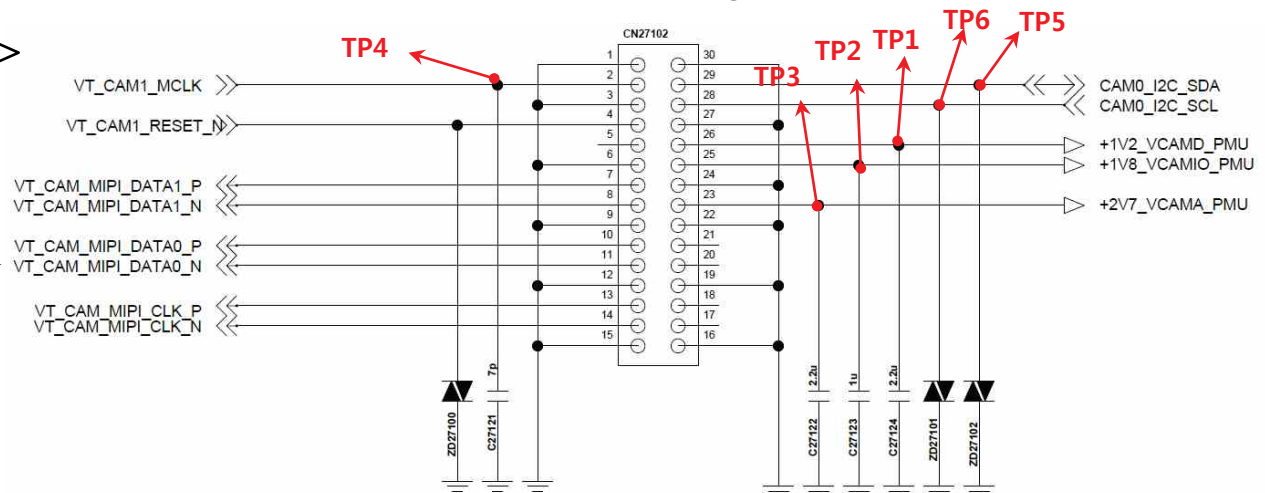
Checking Flow



Image



Circuit Diagram

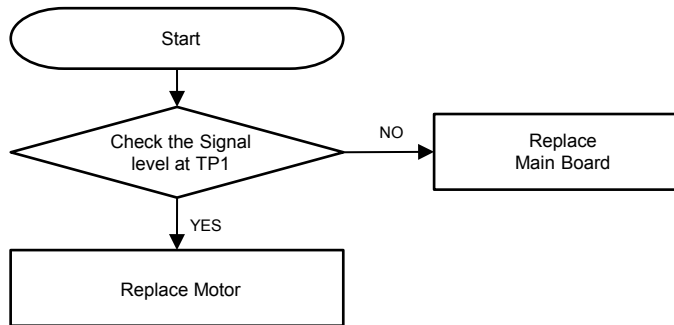


3. TROUBLE SHOOTING

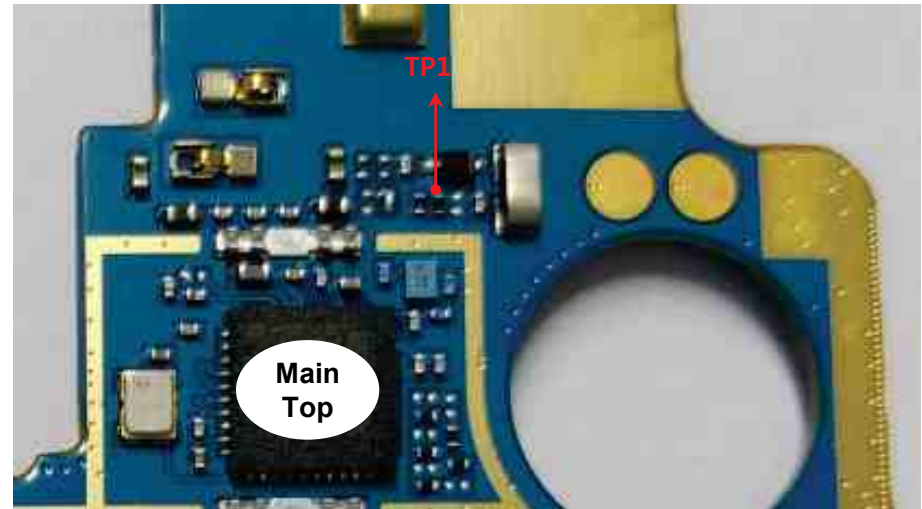
3.xx Checking Motor Block

Motor control signals are generated by MT6353 (U4100 : PMIC)

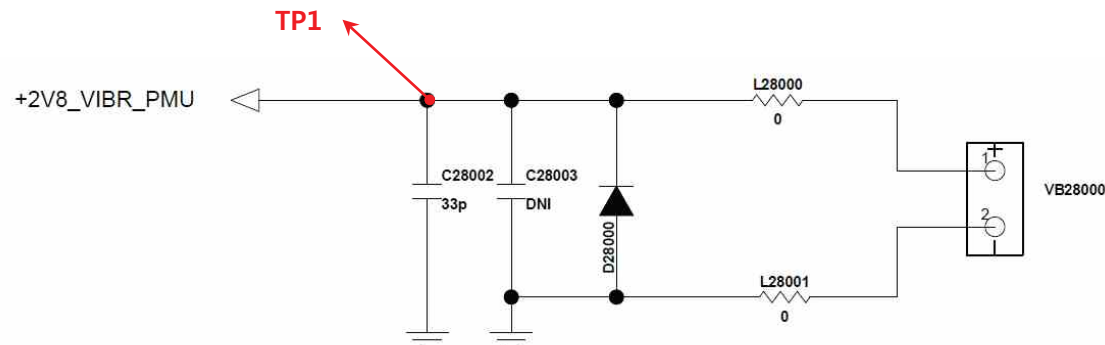
Checking Flow



Image



Circuit Diagram



3. TROUBLE SHOOTING

3.10.5 FM_Radio

Disable detecting headset insert or No FM sound from Earphone, Check the Ear Mic and MT6625L

Checking Flow

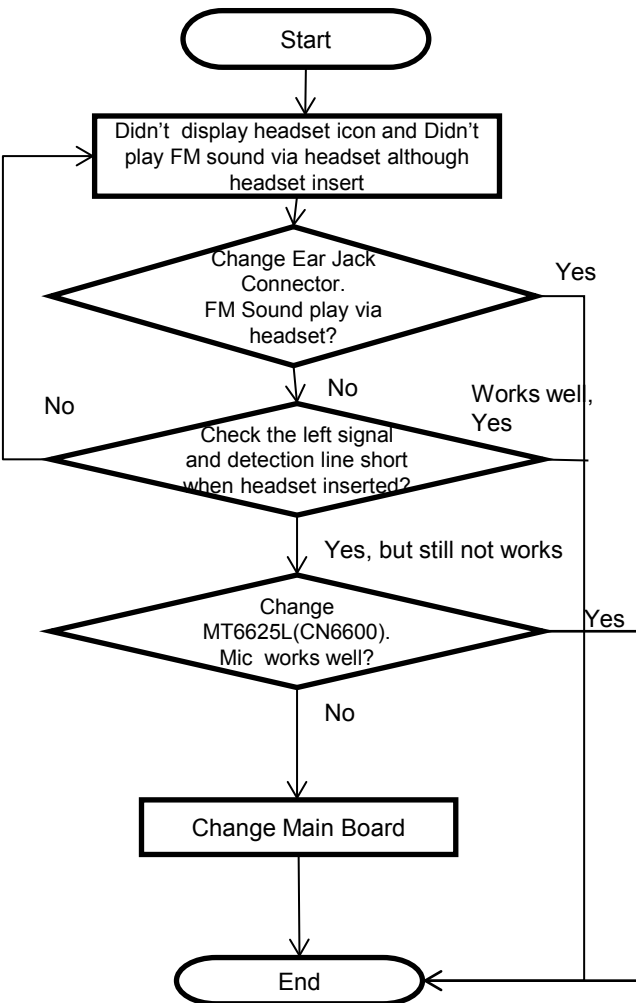
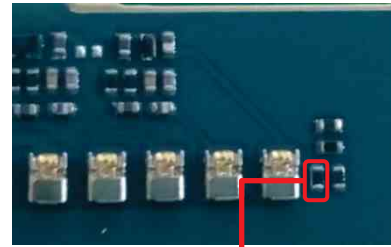
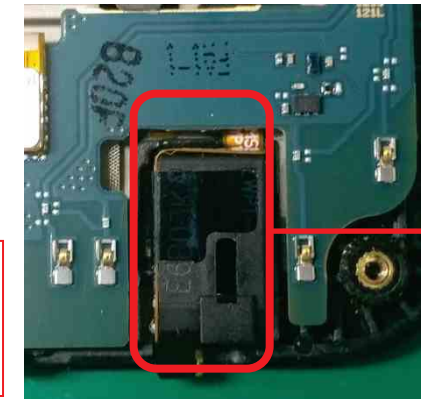


Image (Top)

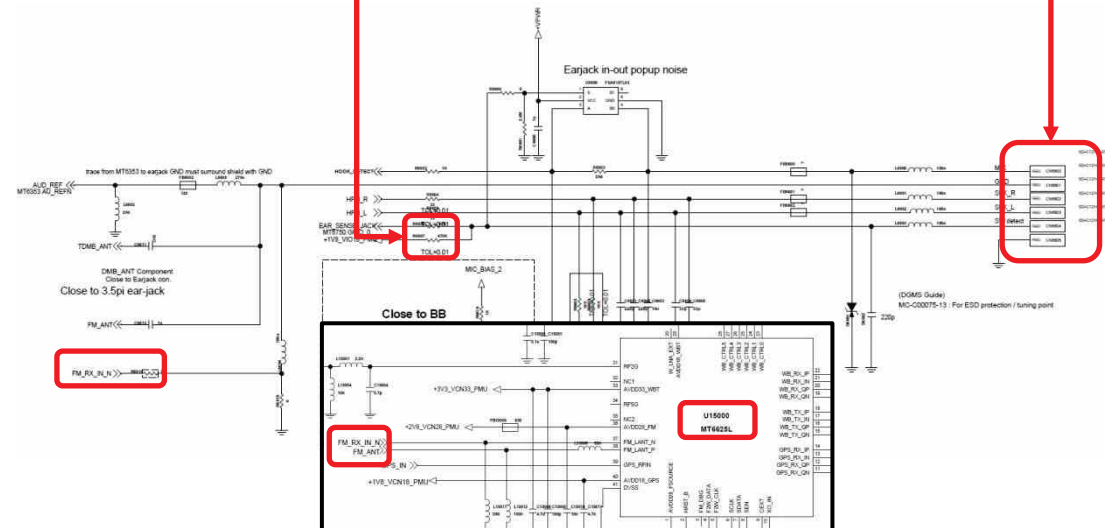


Jack detect Voltage Level Check at R6607
 - 1.8V(before Jack)
 - low(after Jack)

Image (Bottom)



Circuit Diagram



3.16 Connectivity RF PART

3.16.1 Checking RF Signal TRX path(WiFi, BT)

Checking Flow

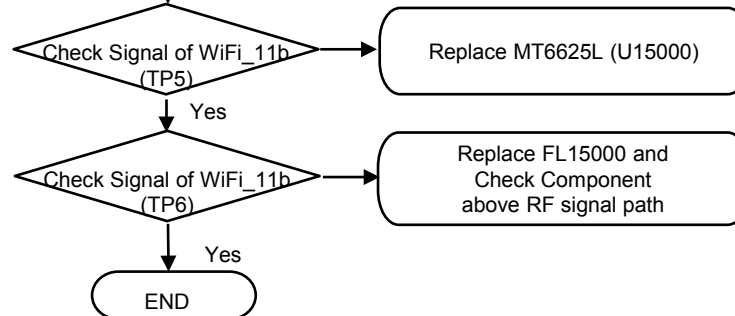
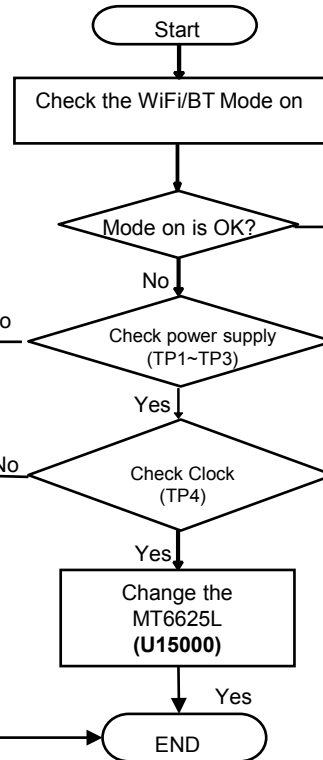
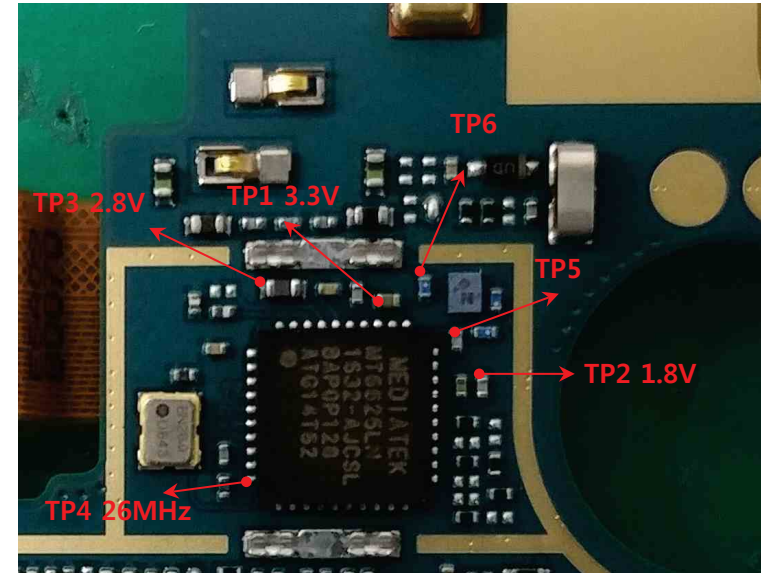


Image (Top)



3. TROUBLE SHOOTING

3.16 Connectivity RF PART

3.16.2 Checking RF Signal TRX path(GPS)

Checking Flow

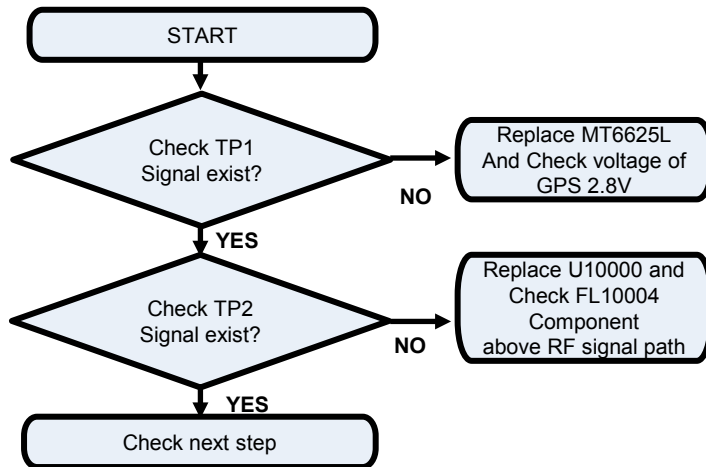
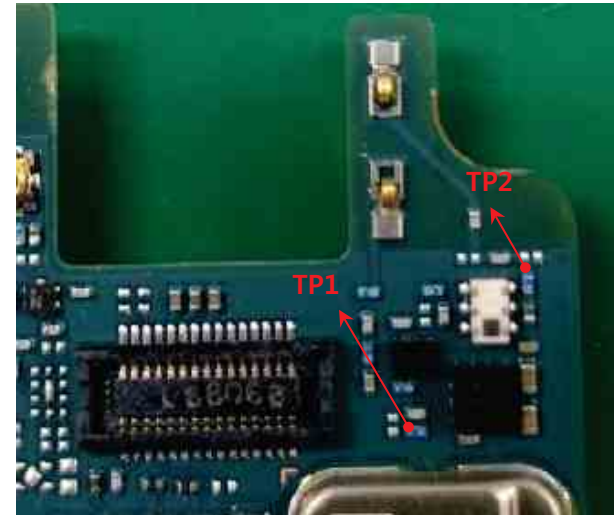
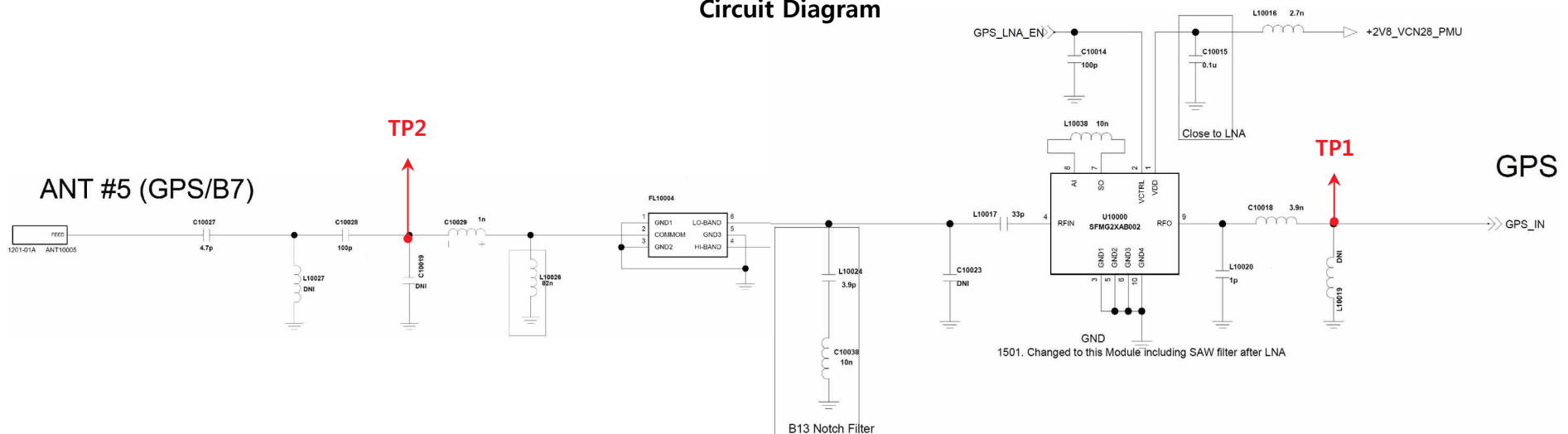


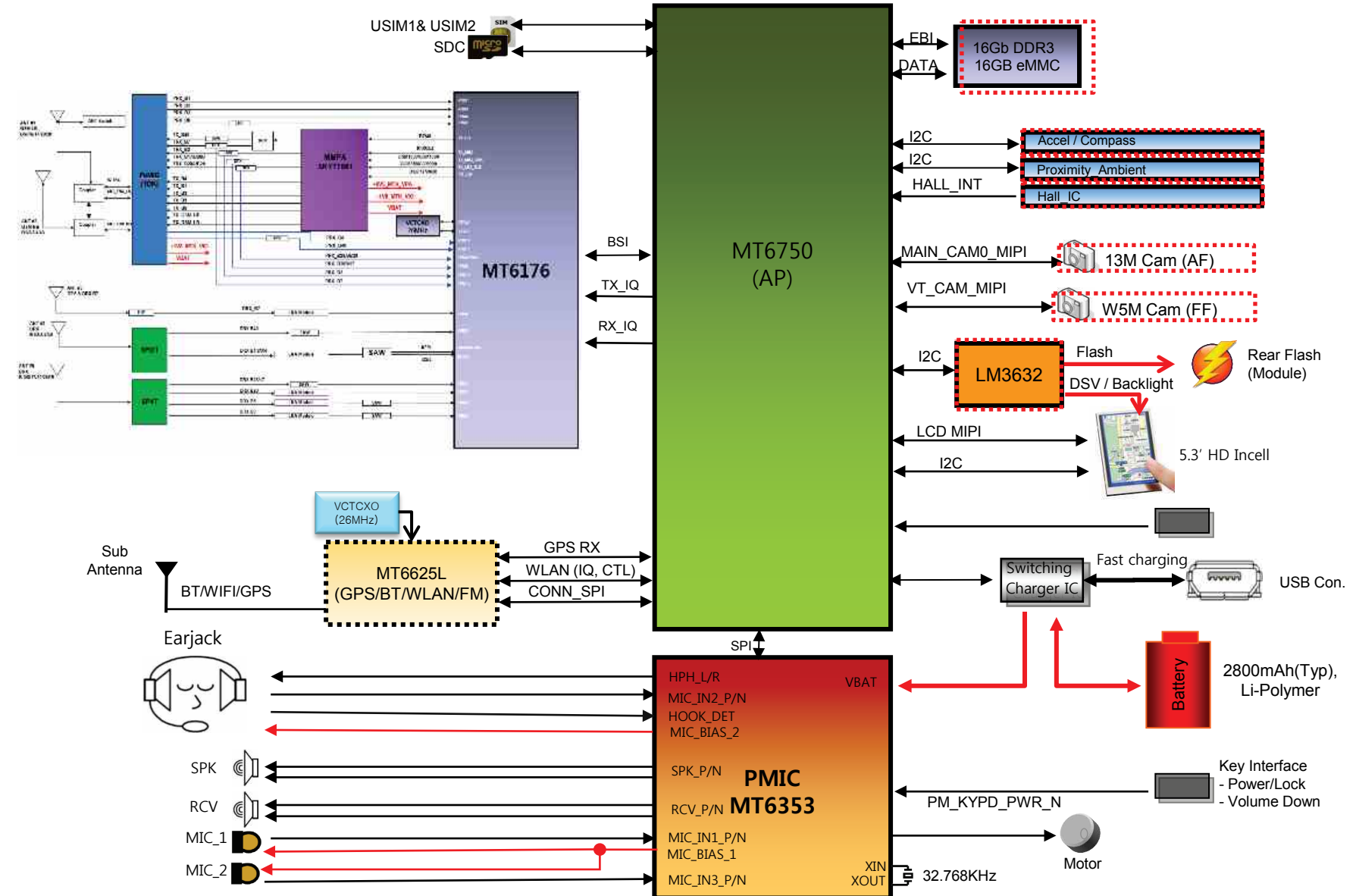
Image (Bottom)



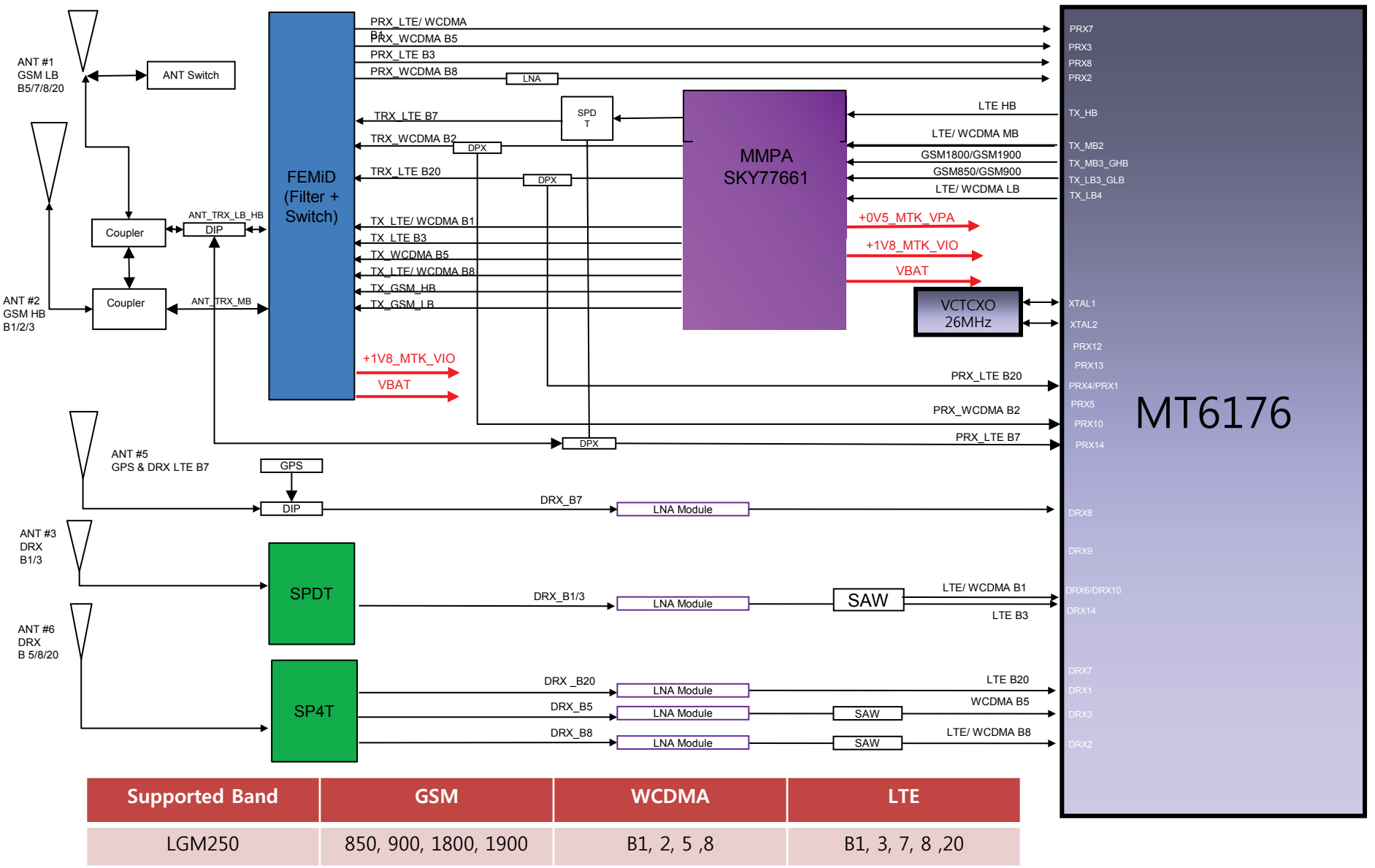
Circuit Diagram



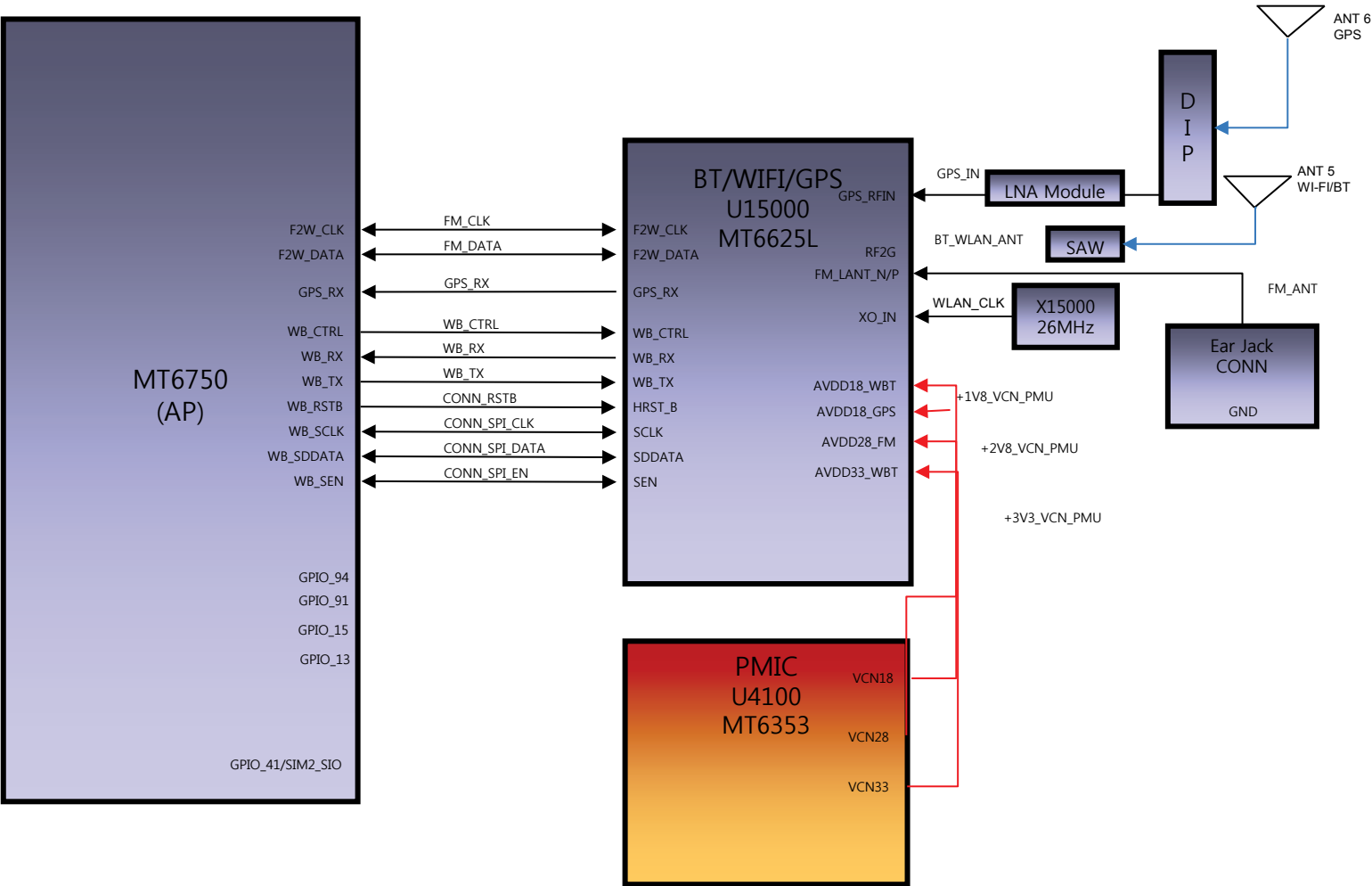
LG-M250 Block diagram : Total



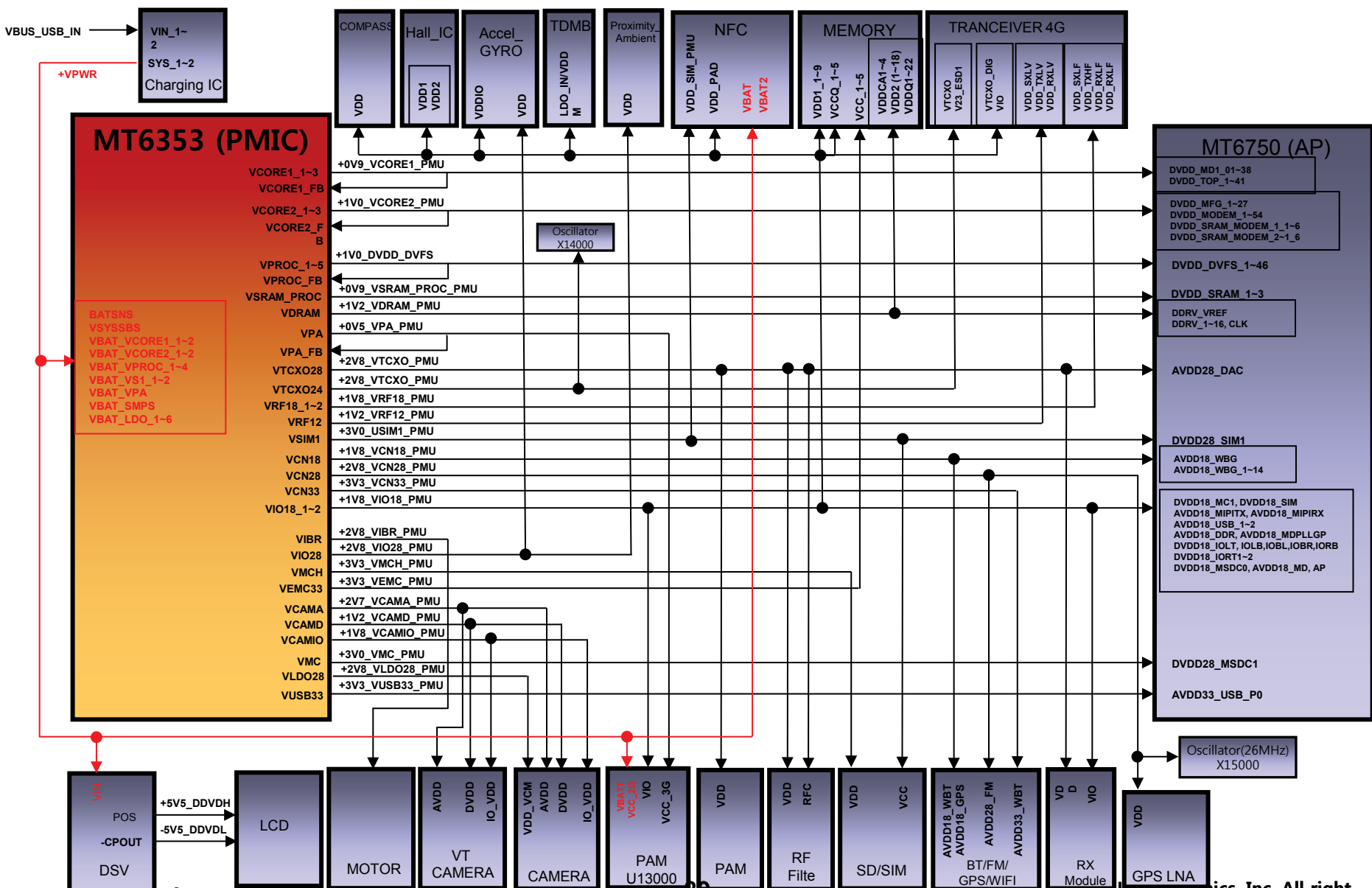
1. Block diagram : RF Part LG-M250



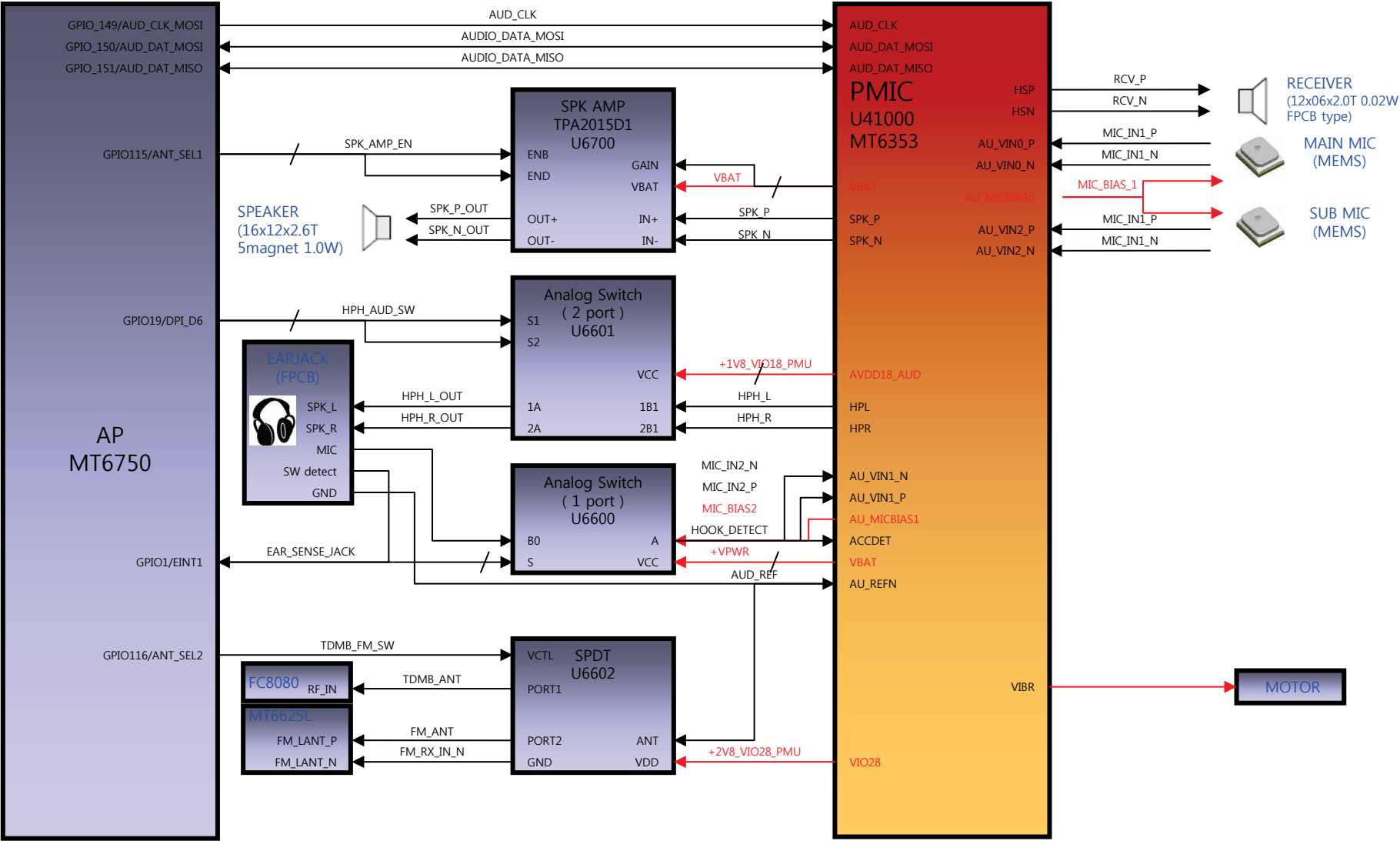
2. Block diagram : Connectivity (WIFI,BT / FM / GPS)



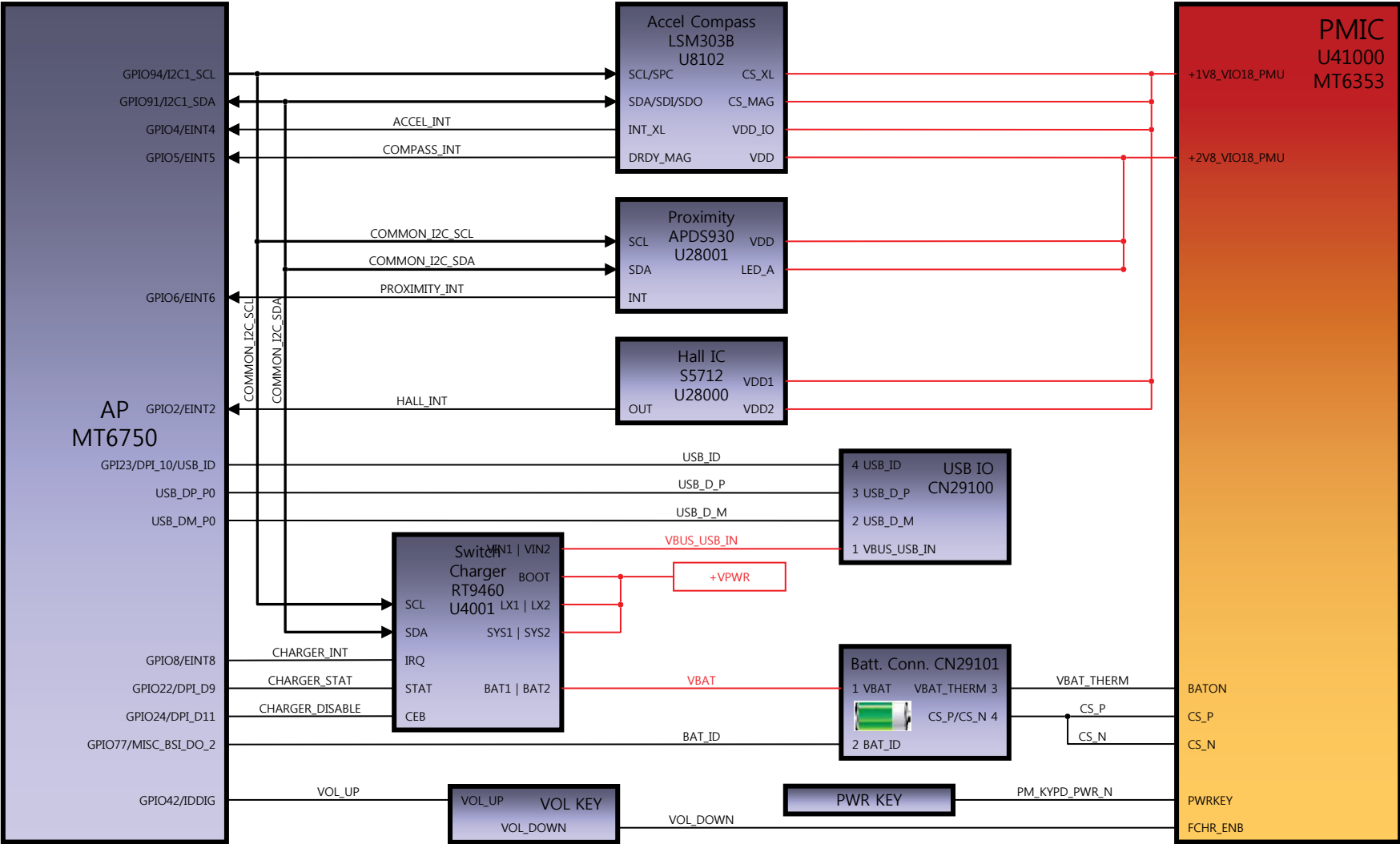
3. Block diagram : Power



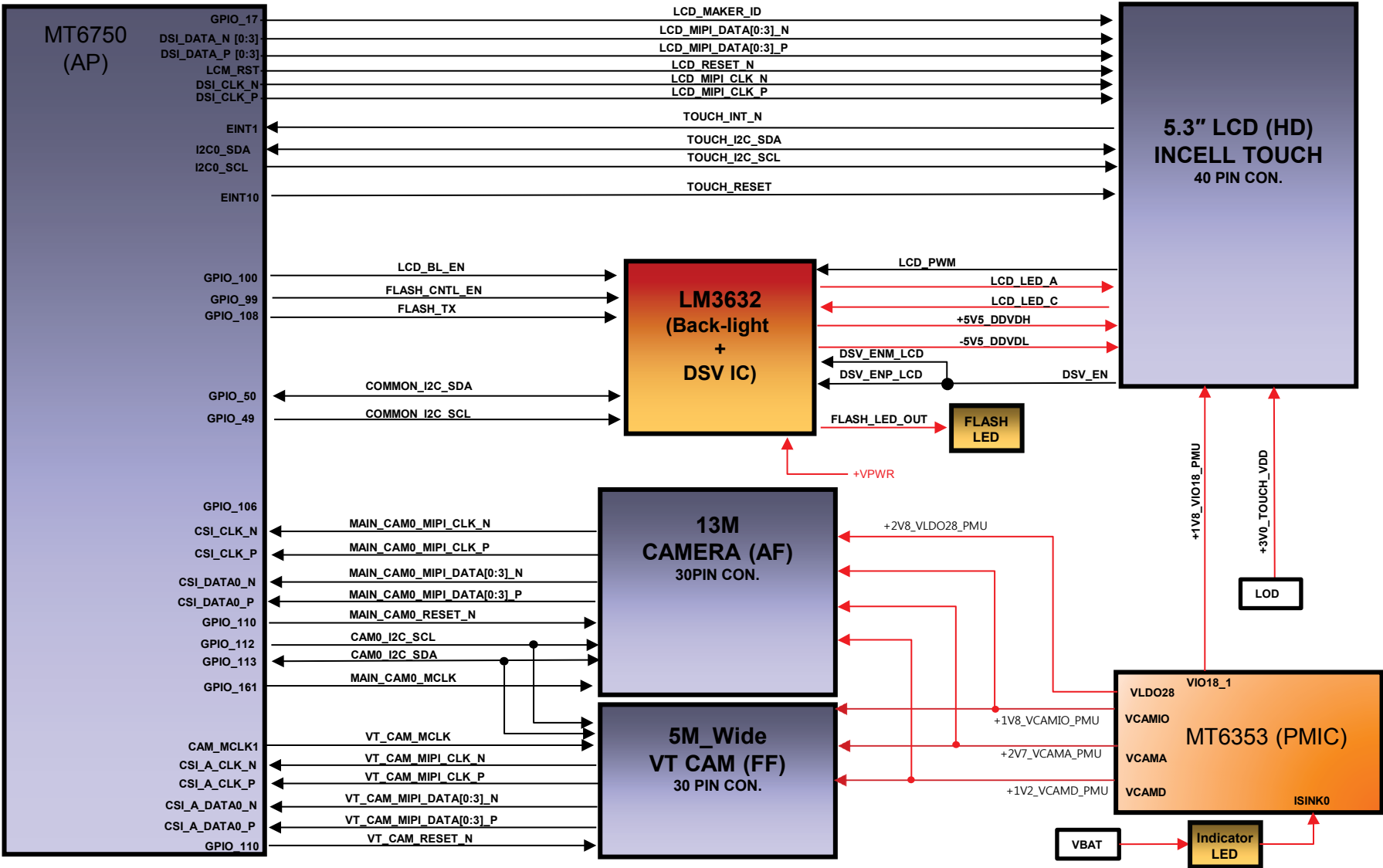
4. Block diagram : AUDIO



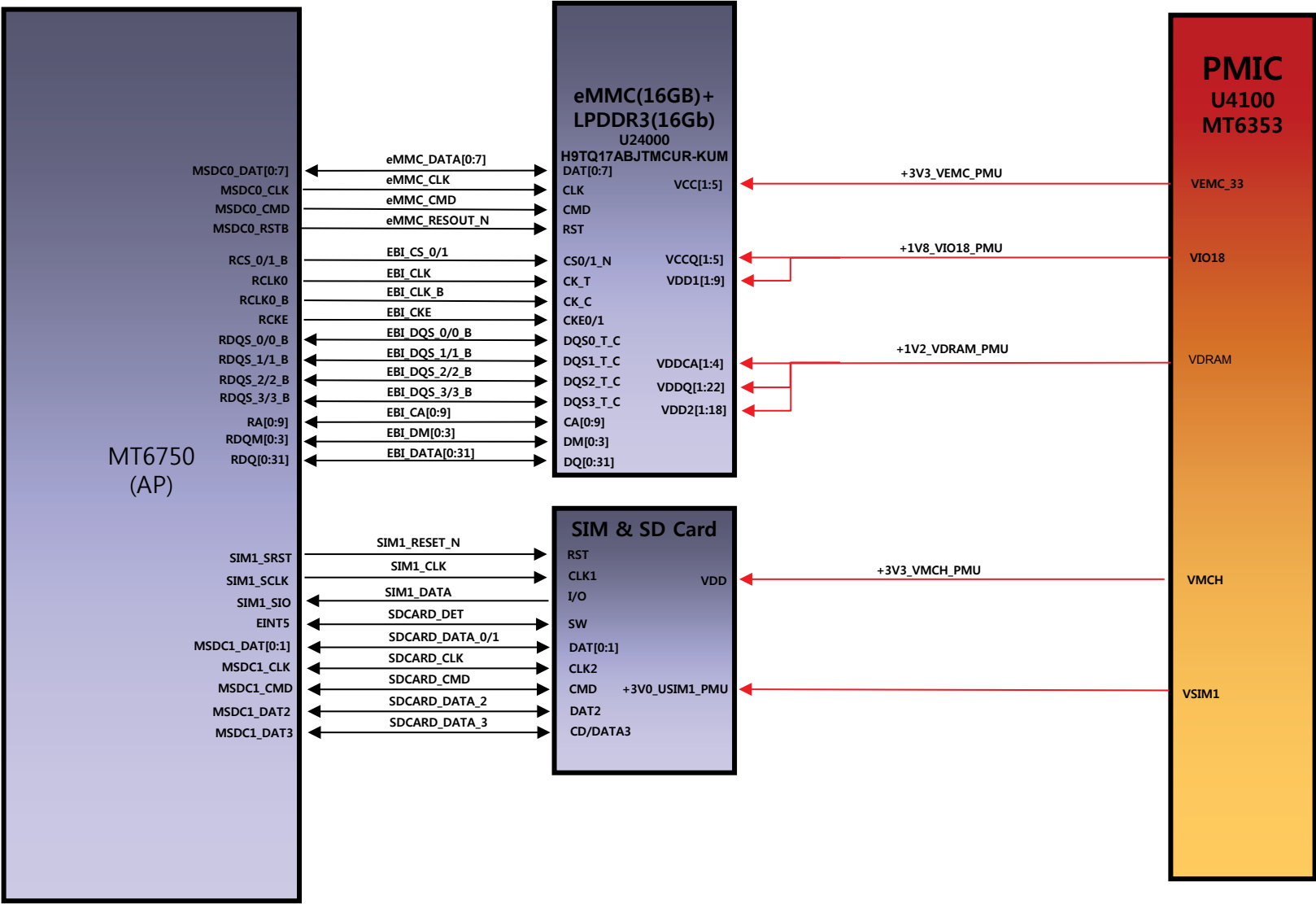
5. Block diagram : SENSOR / CHARGER



6. Block diagram : LCD(INCELL TOUCH) / Main CAM / VT CAM



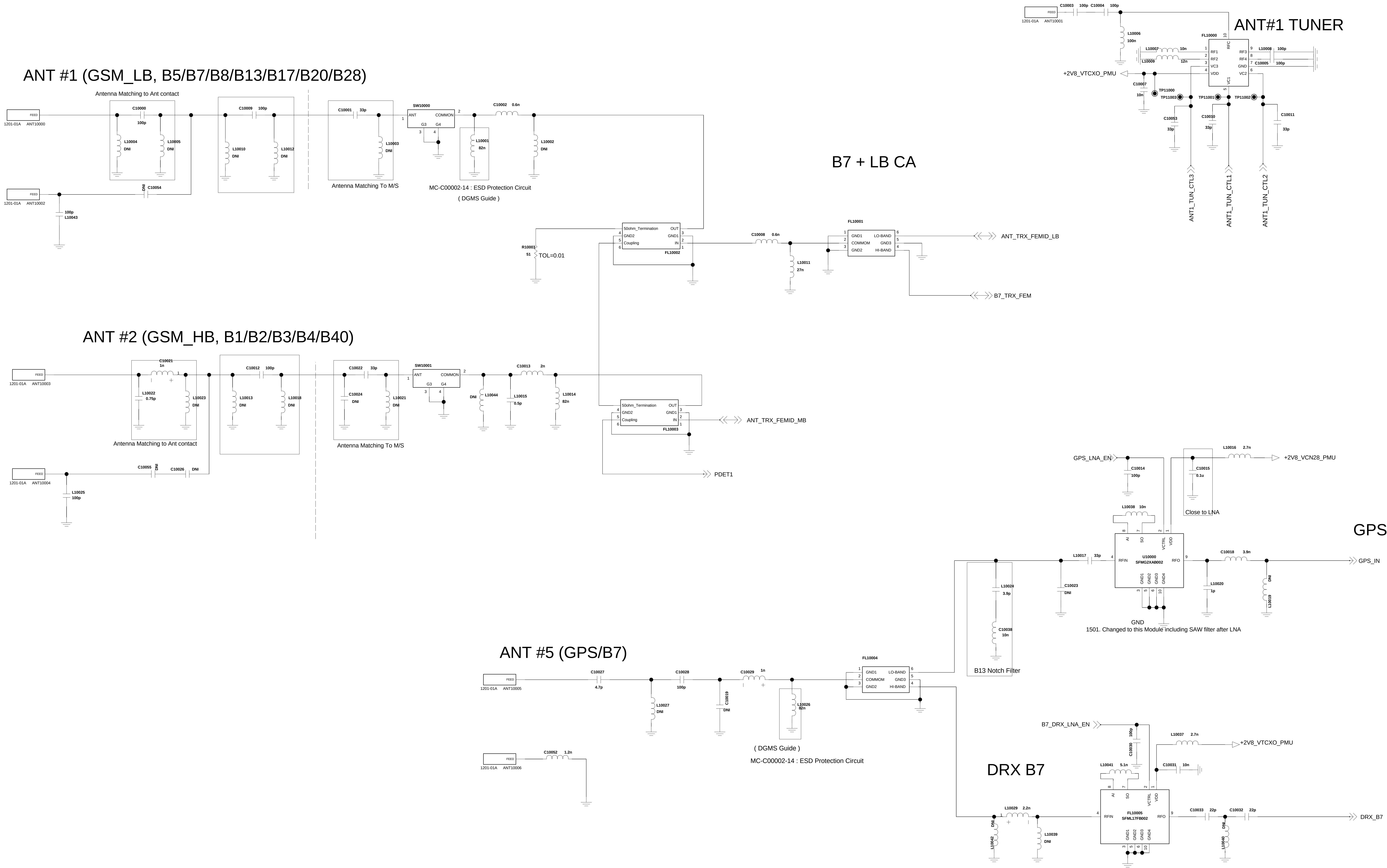
7. Block diagram : Memory / USIM / Micro SD



CIRCUIT DIAGRAM

< 1-12-1-1-2_xx_ANT > RDK

Rev_0.3



16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

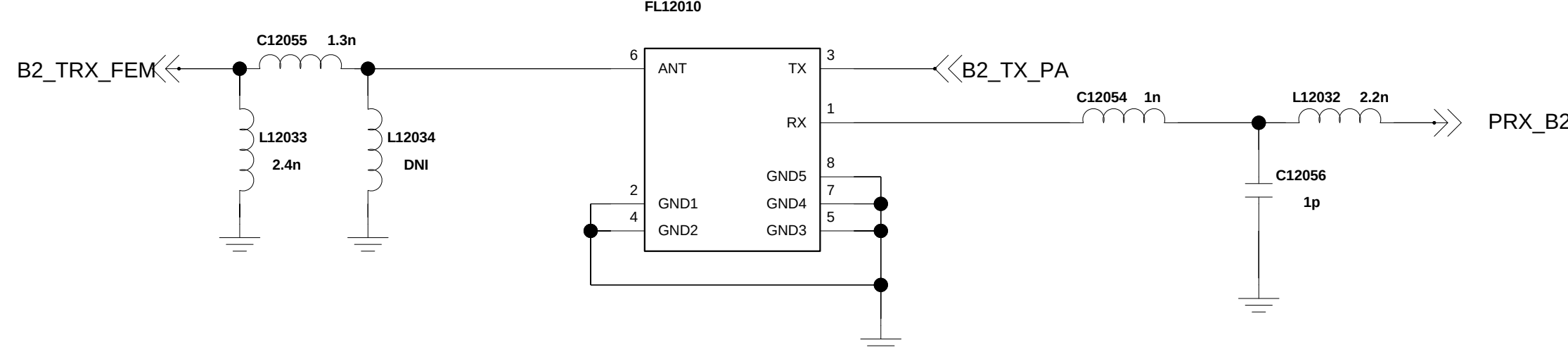
uFEMID

Global i 3 SAW Front End Module(D5322)
B4/5/8 duplexer
G1800,TX2G LPF

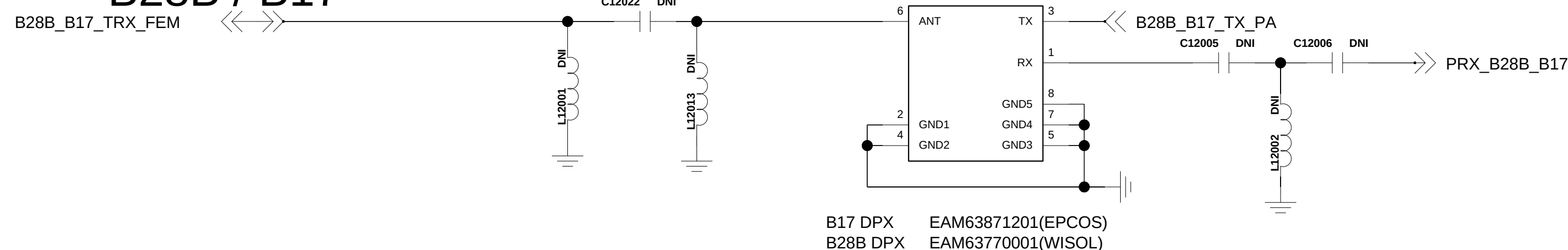
Global i 3 SAW Front End Module(D5323)
B1/3/5/8 duplexer

EAT63414801
EAT63375201

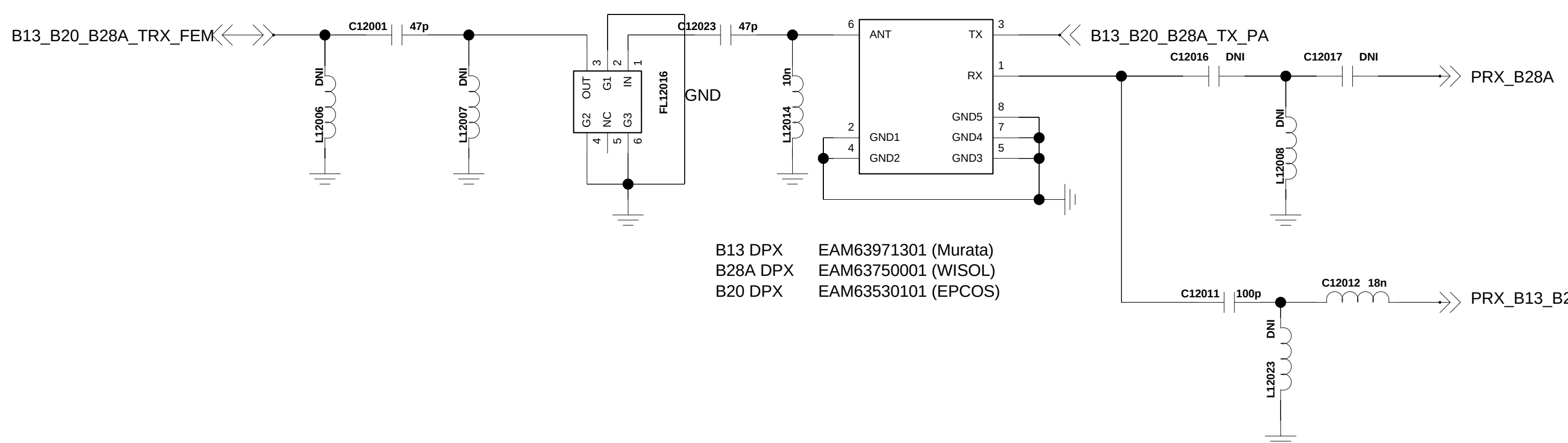
B2 / W1900 / GSM1900



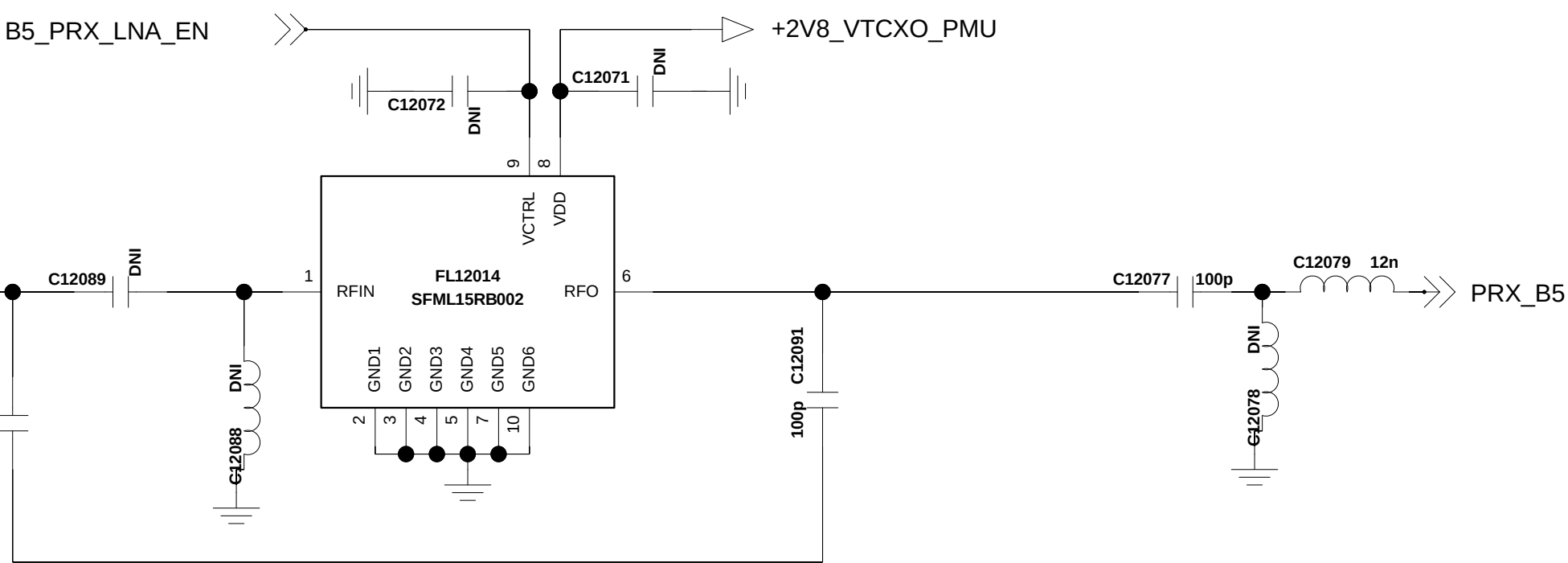
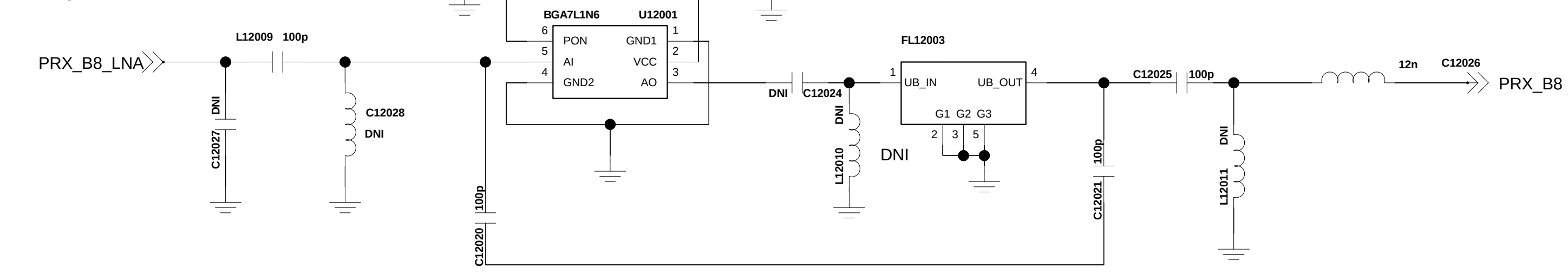
B28B / B17



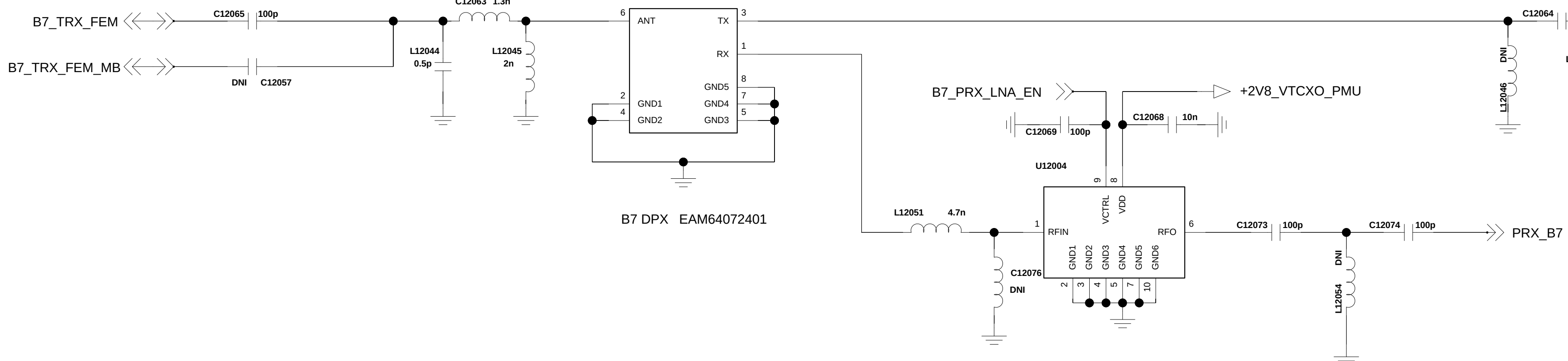
B13 / B20 / B28A



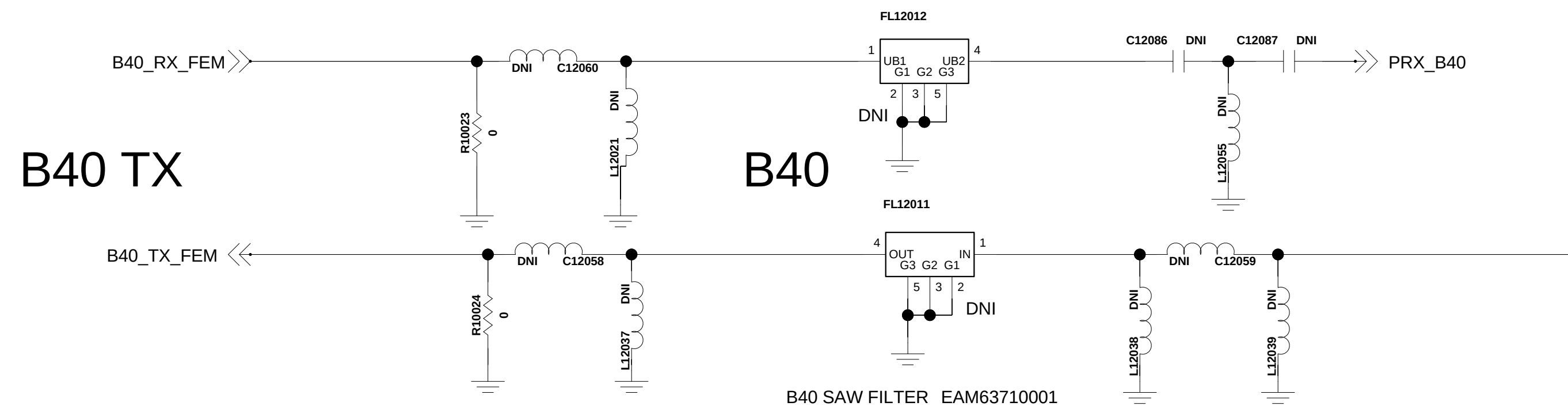
B8, G900 RX



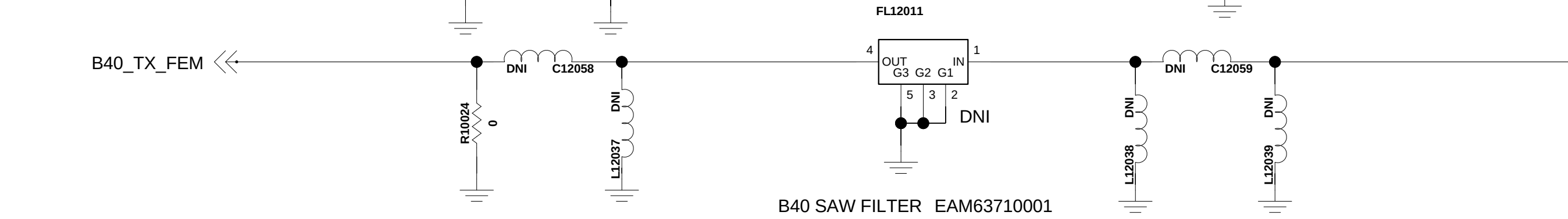
TRX B7



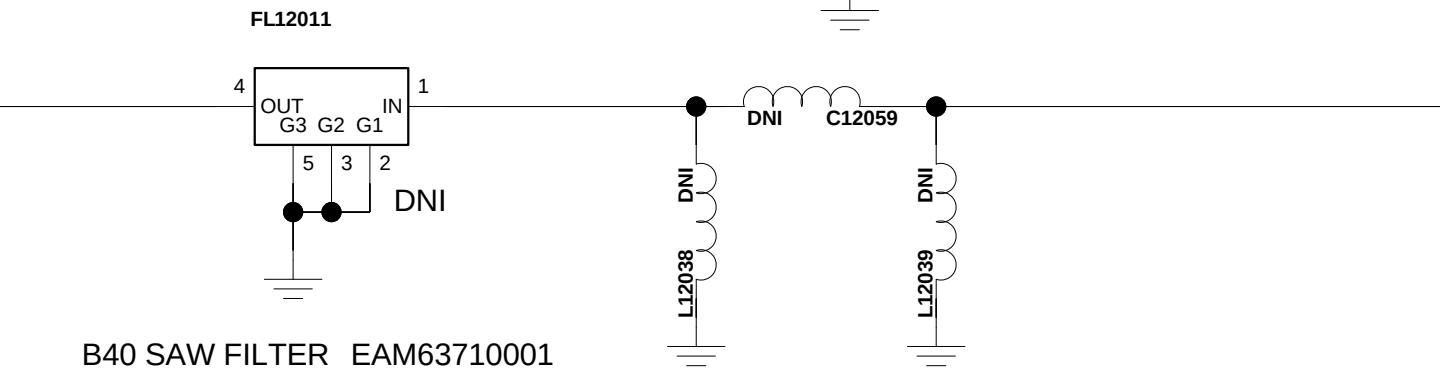
B40 RX



B40 TX

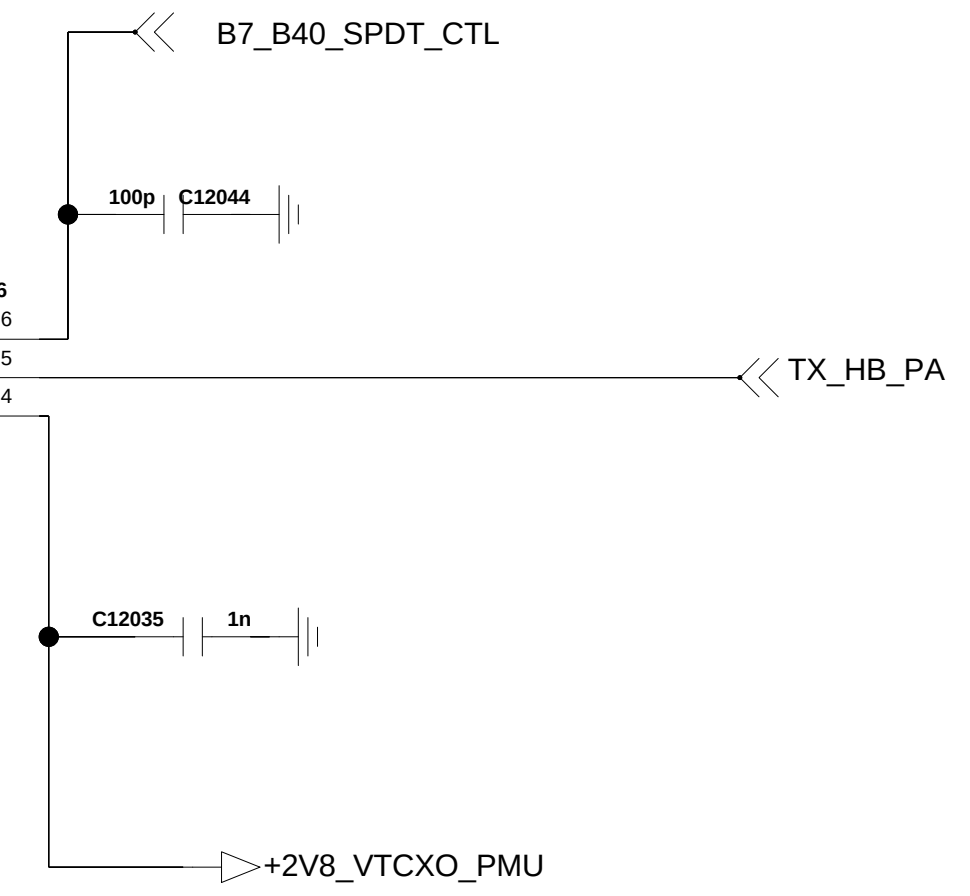


B40



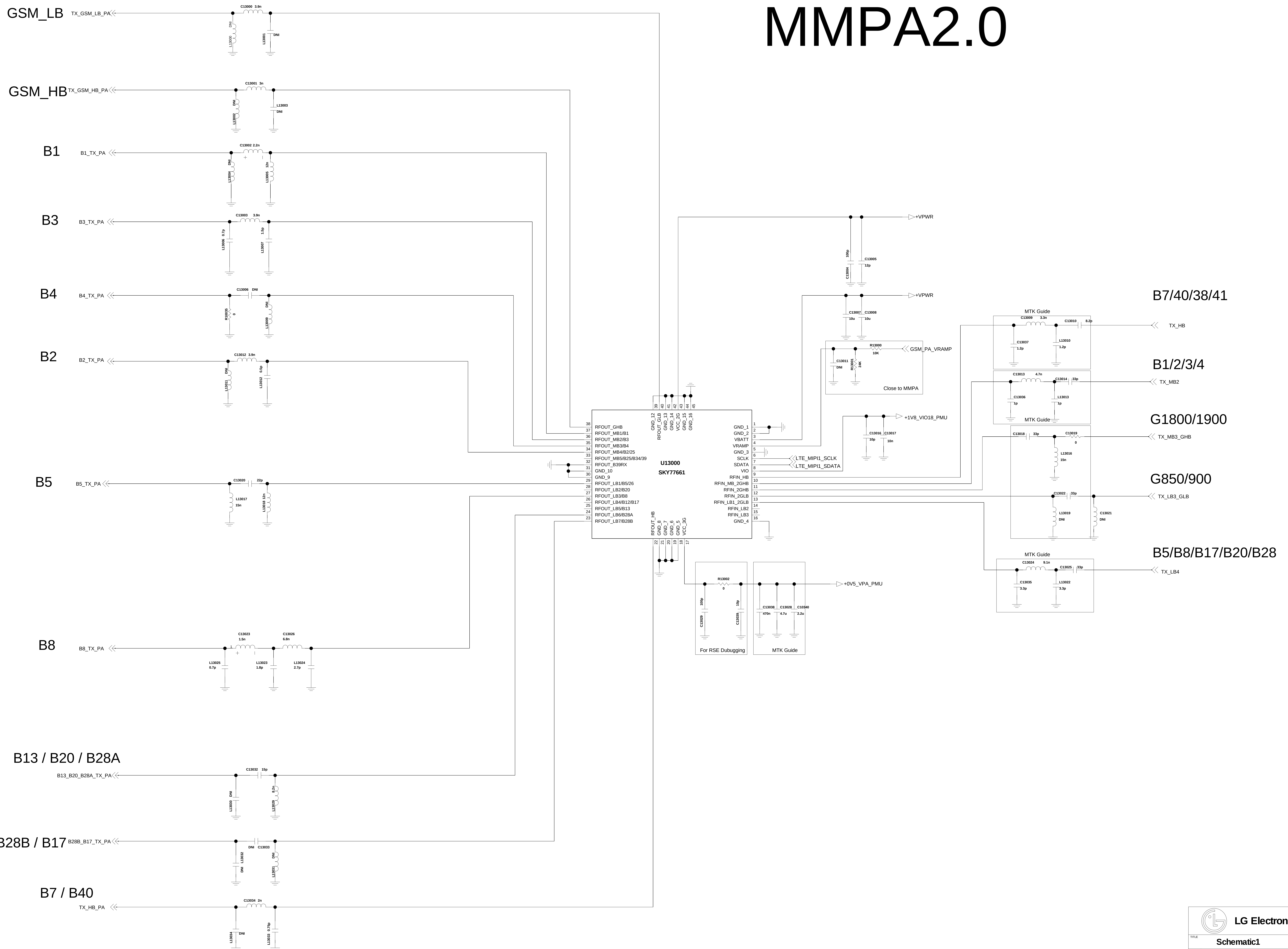
B40 SAW FILTER EAM63710001

For RSE

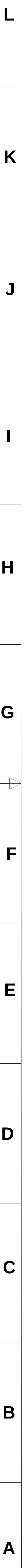


16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

MMPA2.0

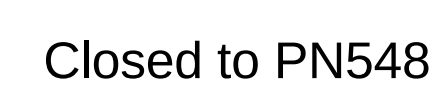


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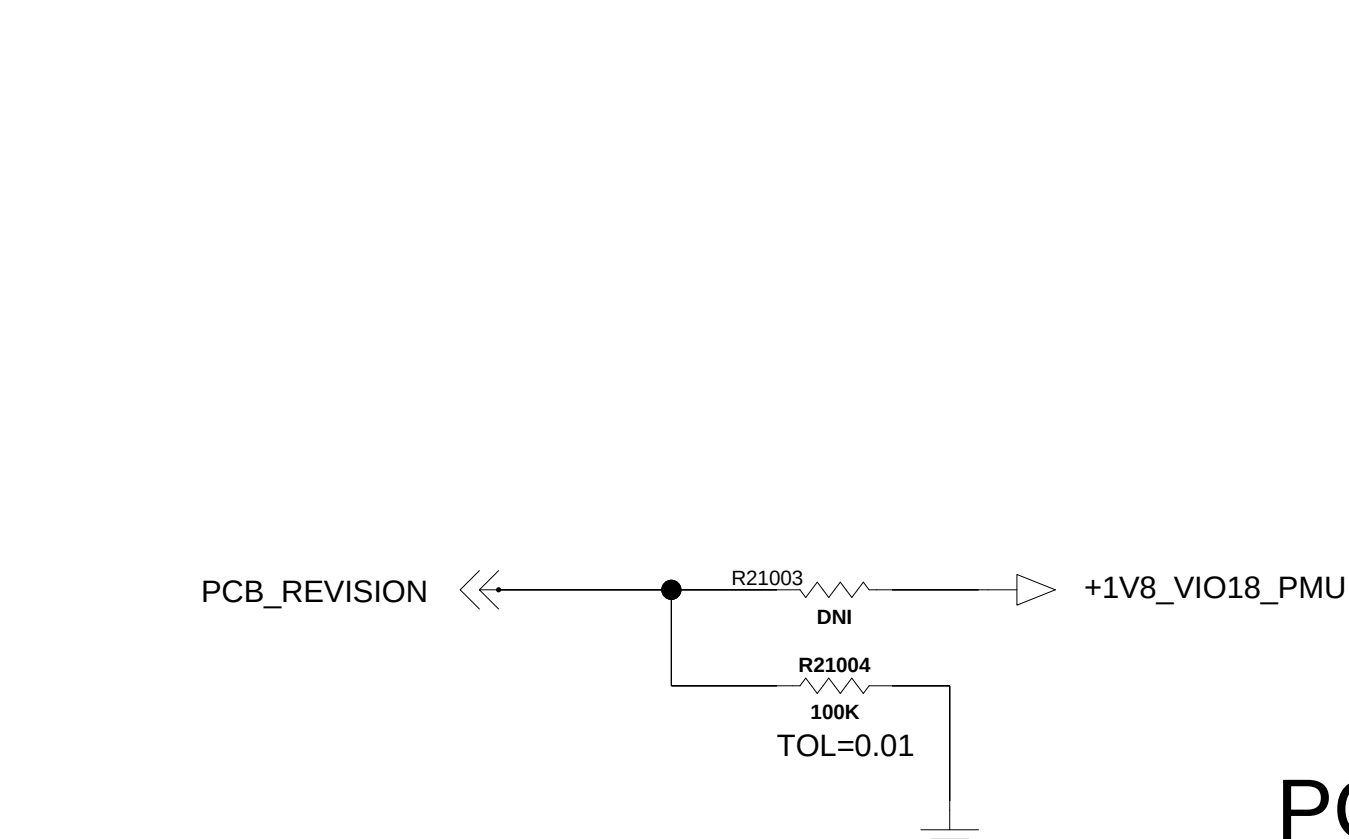
Rev_0.3

Must route with GND Shielding

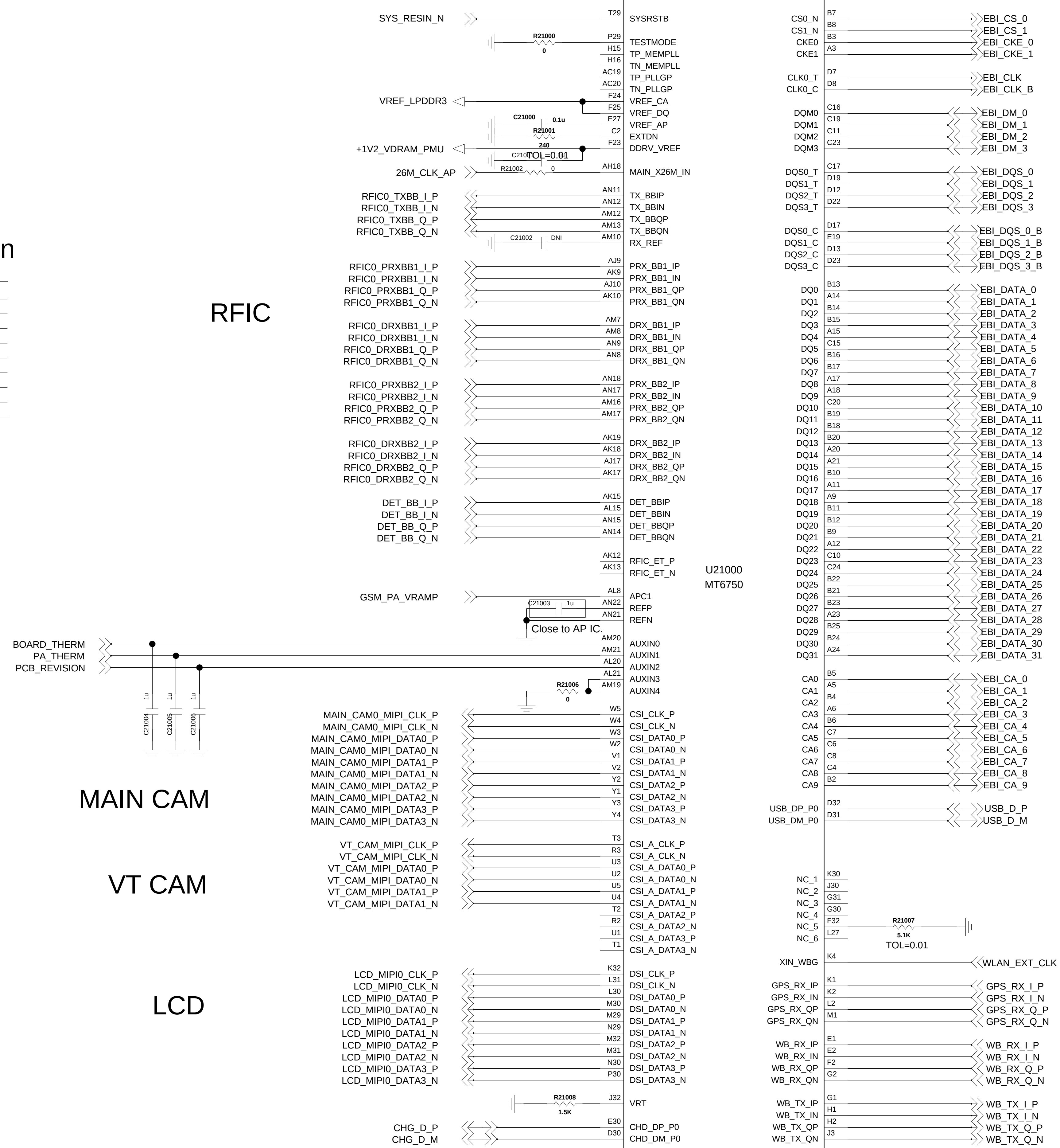
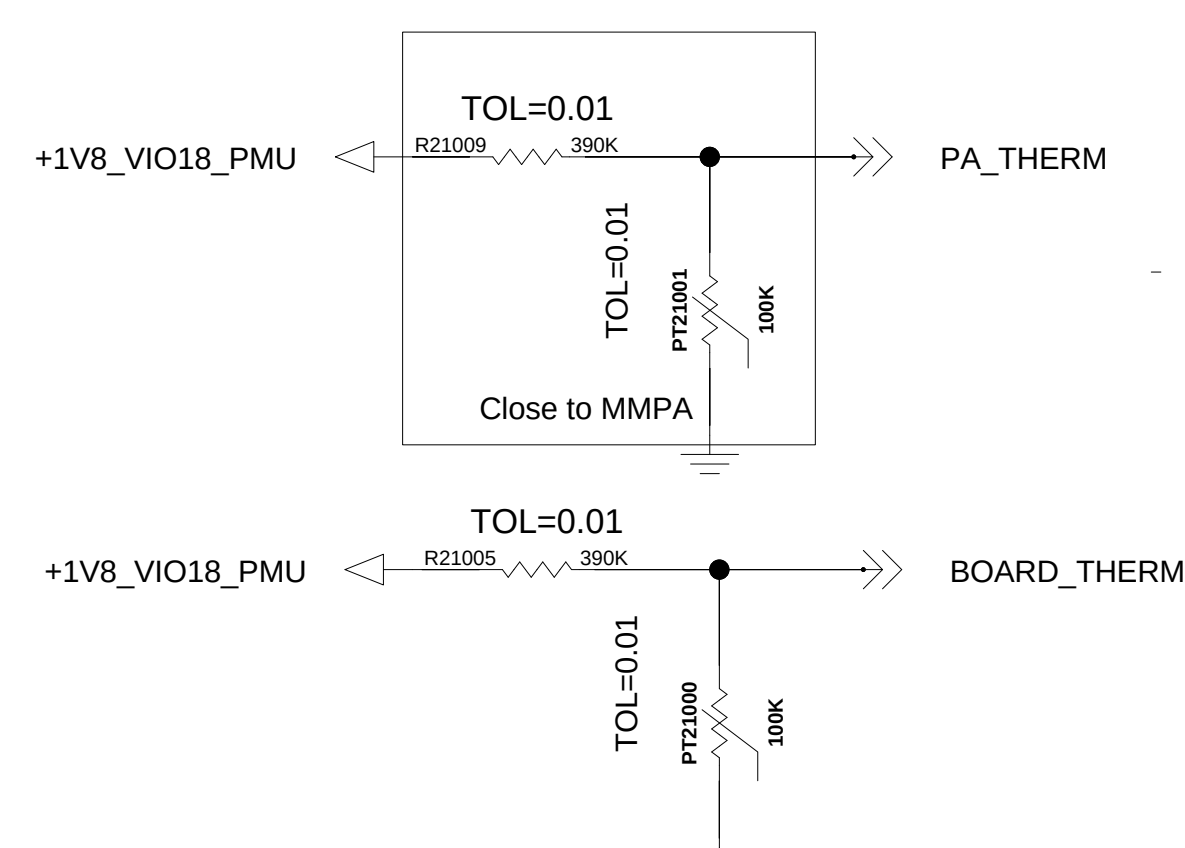


LG Electronics

< 2-1_MT6750_DATA >

PCB_Revision

0	100K	20K	0.300
0 1	100K	27K	0.383
A	100K	39K	0.505
B	100K	51K	0.608
C	100K	75K	0.771
D	100K	100K	
E	100K	DNI	
1	DNI	100K	QPPS
1.1	100K	200K	

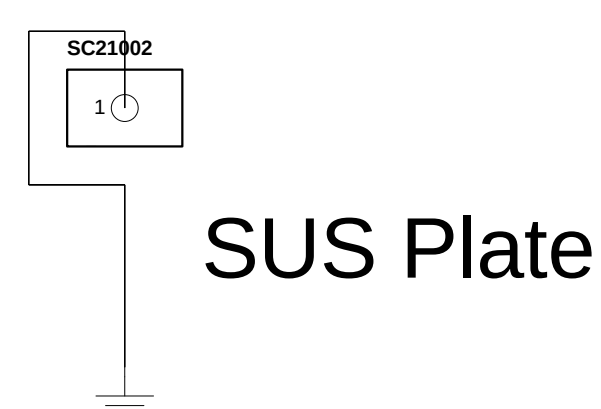
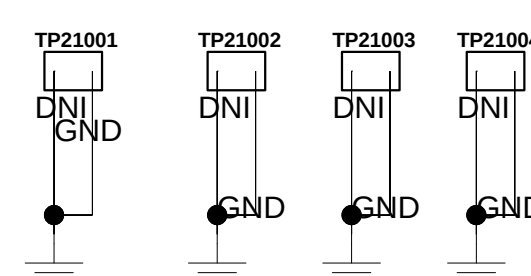


MCP(LPDDR)

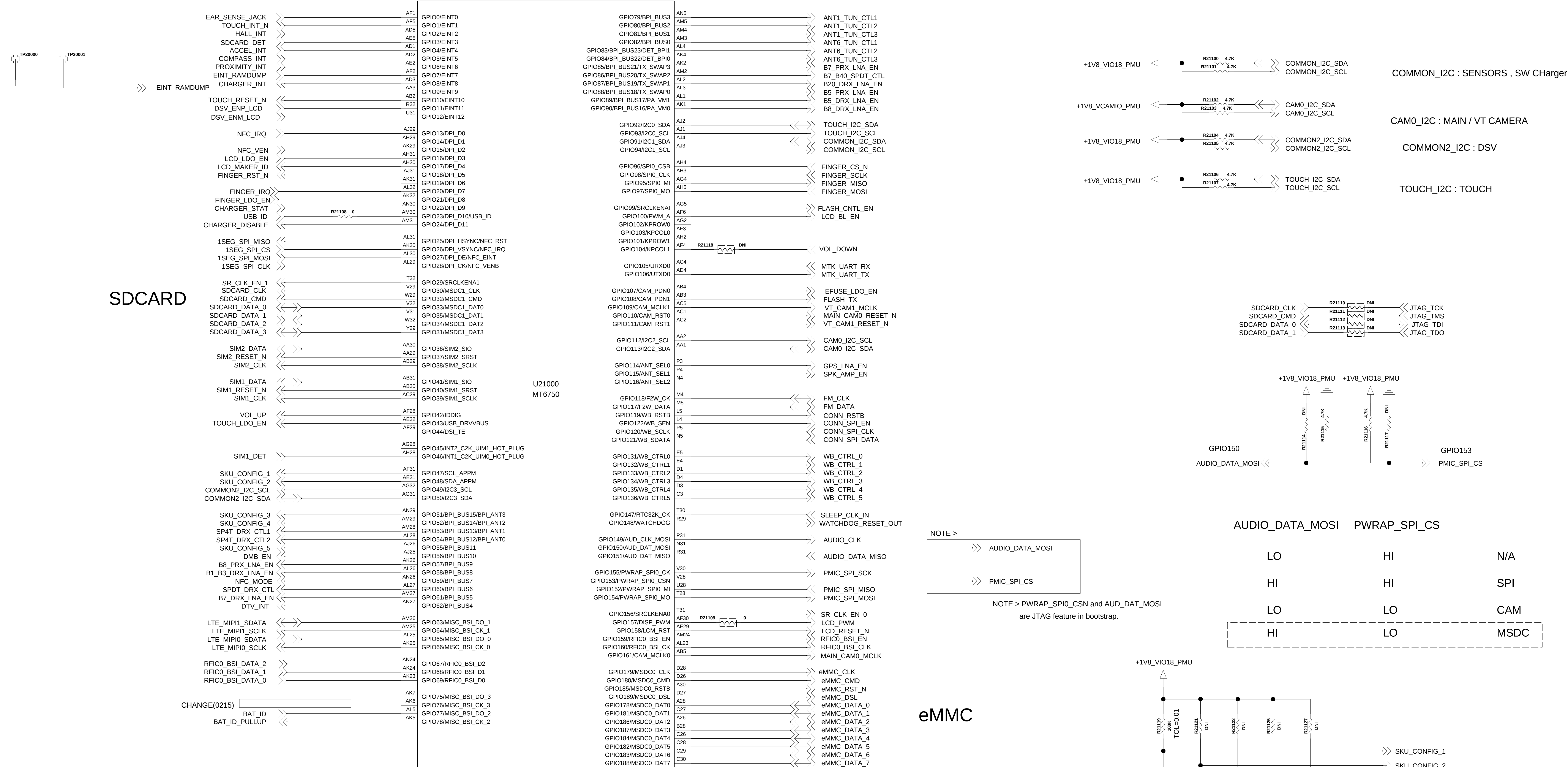
USB D

Default resistor of "SSUSB_VRT" can be NC if internal USB VRT is applied.

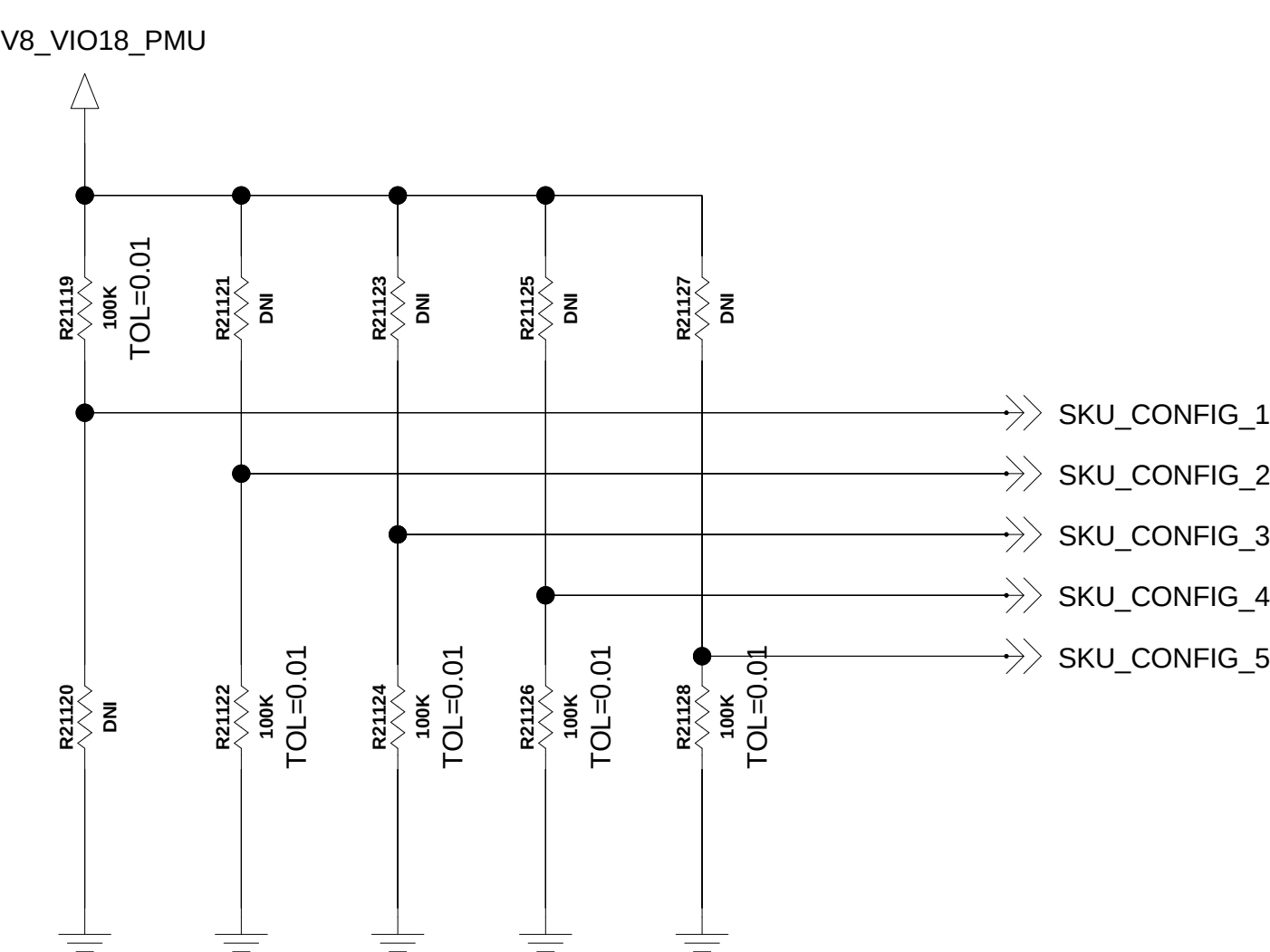
CONNECTIVITY



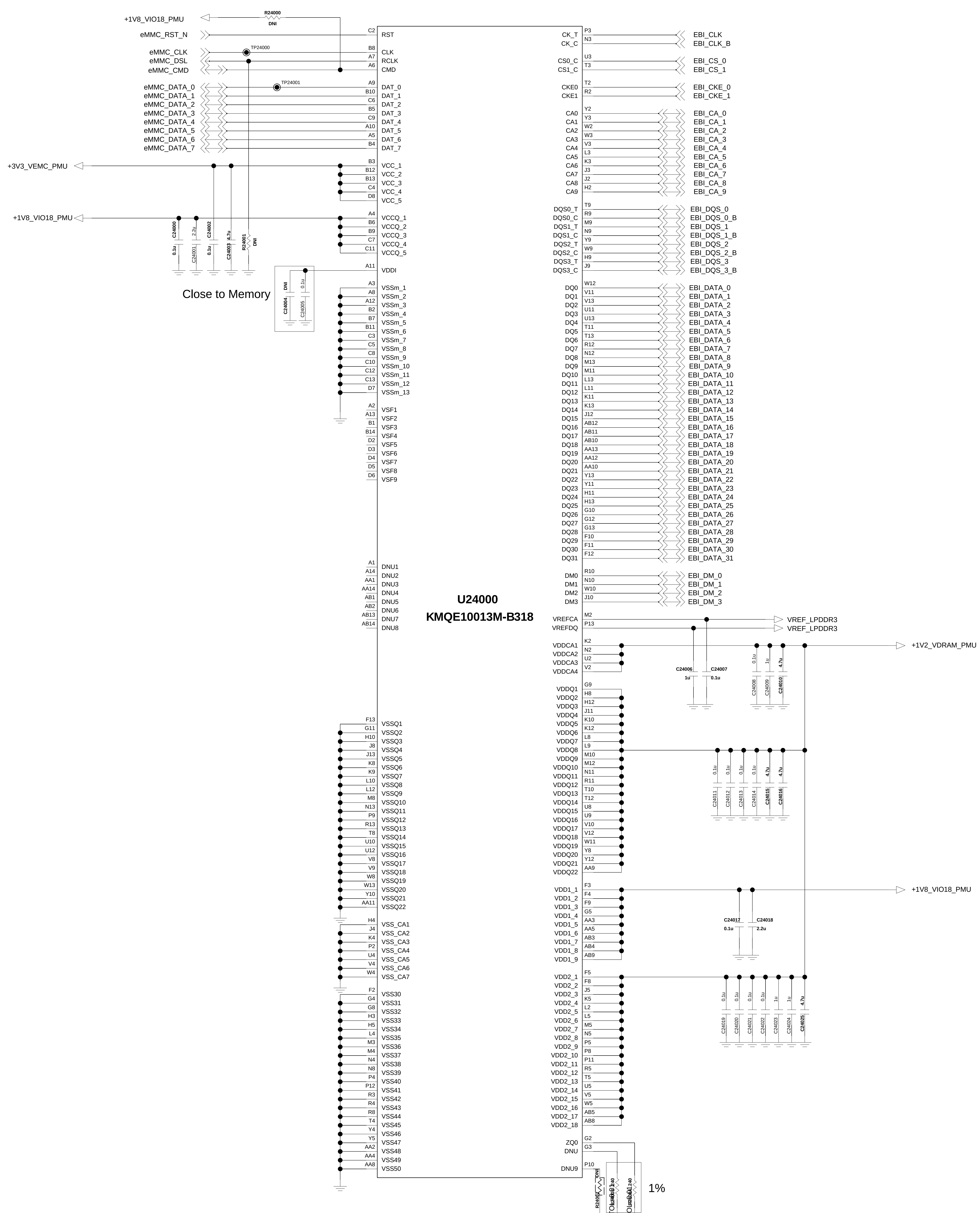
< 2-1_MT6750_GPIO >



Model	Band	NFC	SIM	GPIO_SUM	GPIO_55	GPIO_52	GPIO_51	GPIO_48	GPIO_47
M250N	1/3/7/8/20	O	SINGLE	0	0	0	0	0	0
M250	1/3/7/8/20	X	DUAL	1	0	0	0	0	1
M250E	1/3/5/7/8/20	X	DUAL	2	0	0	0	1	0
M250ds	3/7/28	X	DUAL	4	0	0	1	0	0
M250YK	1/3/5/7/8/28	X	SINGLE	5	0	0	1	0	1
M250K	1/3/5/7/8/28/40	X	DUAL	7	0	0	1	1	1
M250F	2/4/5/7/28	X	SINGLE	8	0	1	0	0	0
M250AR	2/4/5/7/28	X	SINGLE	9	0	1	0	0	1
M250H	2/4/5/7/13/17	X	SINGLE	10	0	1	0	1	0
M250I	1/3/5/8/40	X	DUAL	11	0	1	0	1	1
M250dsn	1/3/7/8/40	O	DUAL	12	0	1	1	0	0
K-M121 L	1/3/7/8/17	O	SINGLE	13	0	1	1	0	0
K-M121 S	1/3/7/8/17	O	SINGLE	14	0	1	1	0	0
K-M121 K	1/3/7/8/17	O	SINGLE	15	0	1	1	0	0
M250Y	1/3/5/7/18	X	SINGLE	18	1	0	0	1	0



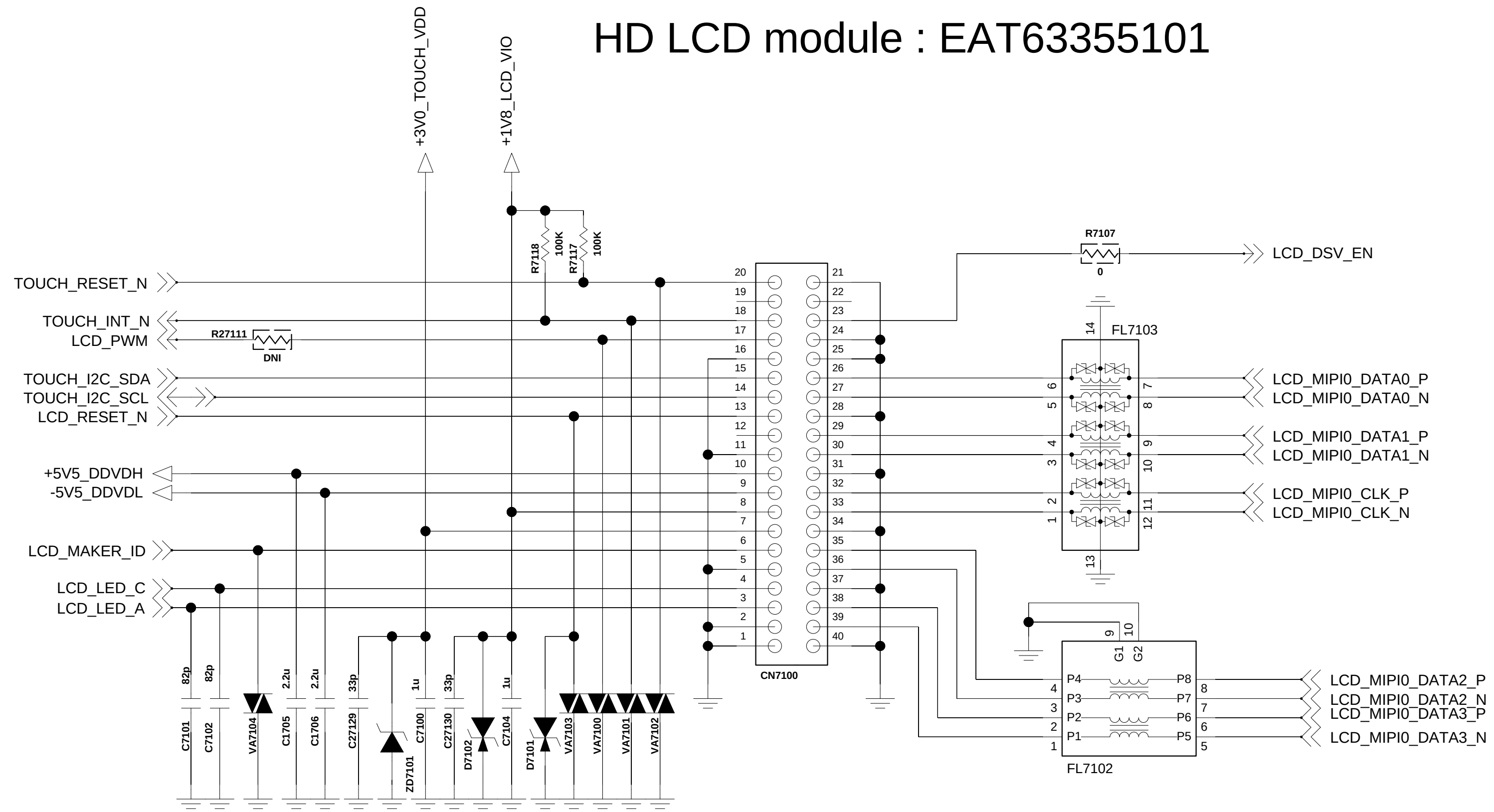
< MCP_eMMC_5_0_16Gb_DDR3_16GB_eMMC Hynix >



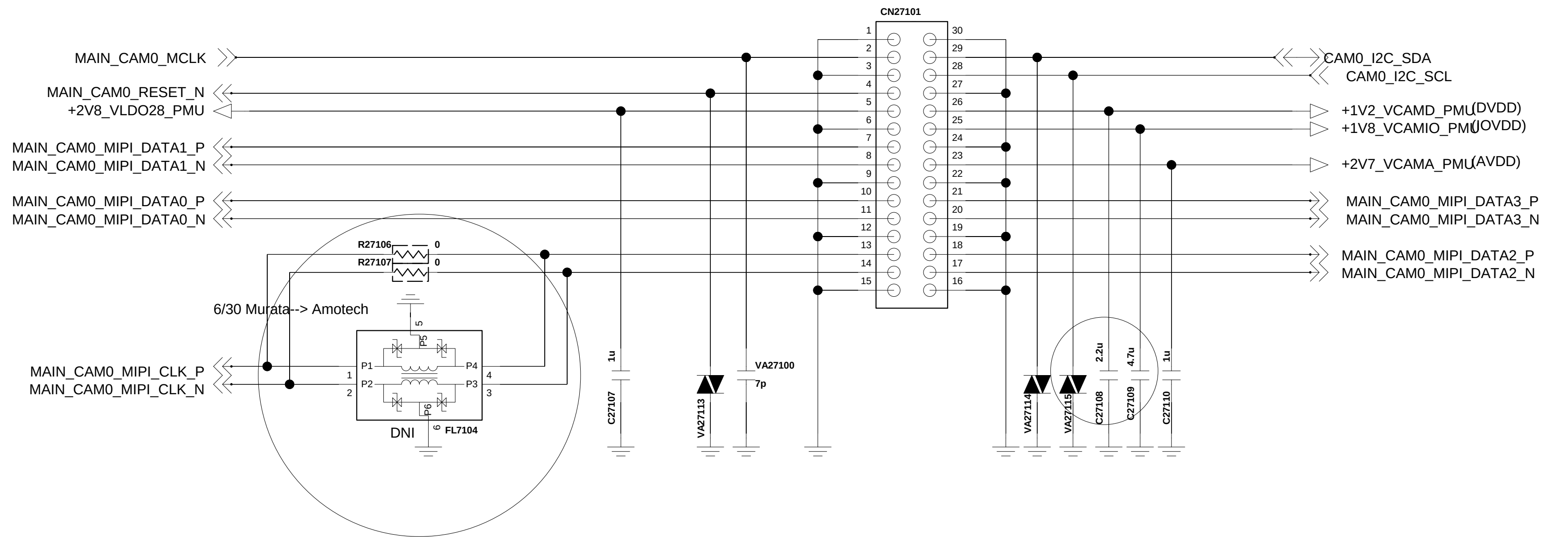
< 7-1-X_LCD_LGD_HD >
LGD 5.3 inch HD Incell Hybrid LCD

<7-3-5-1_30pin_5M_8M_13M_AF> Rev_0.3

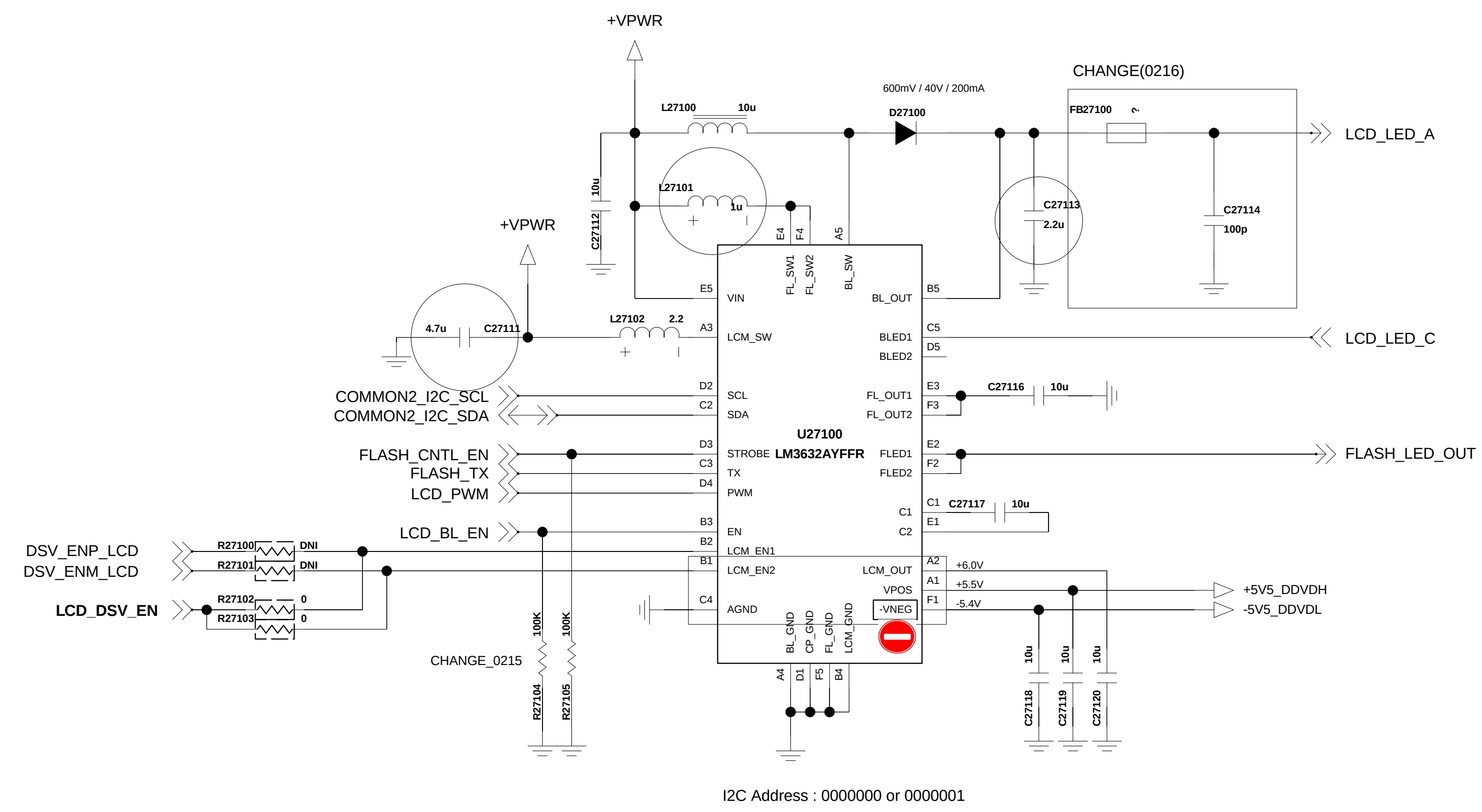
HD LCD module : EAT63355101



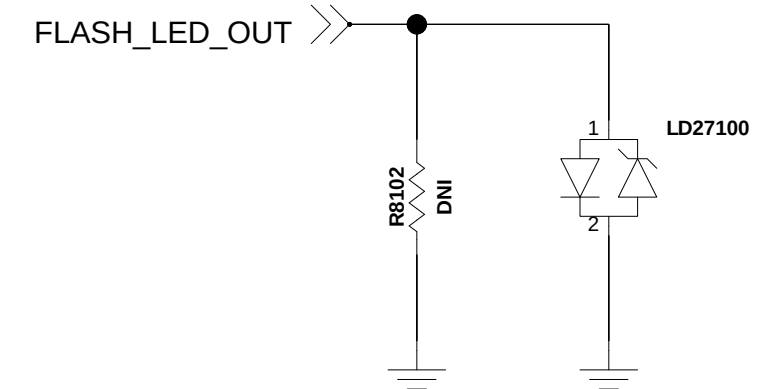
13 Mega Camera module : EBP63001801



LCD_Backlight_Flash_DSV_LM3632



<Flash_LED>

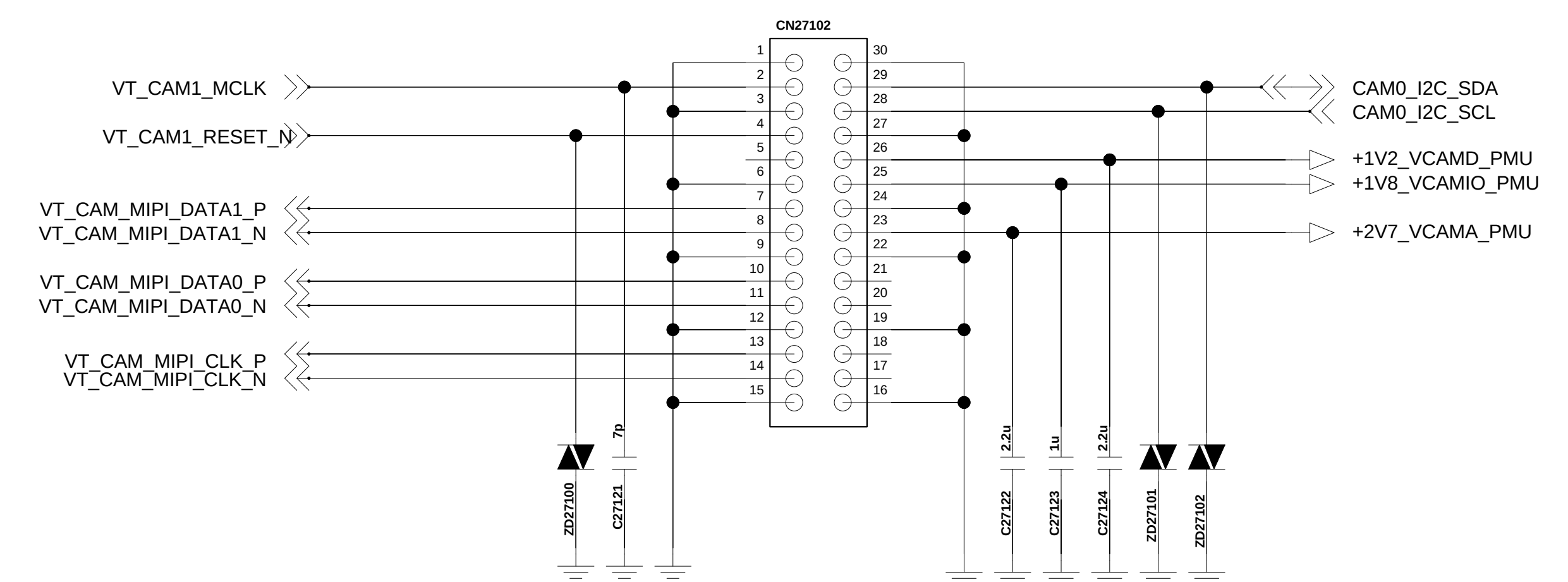


< 7-4-x_Front_Camera_30pin_5M_FF>

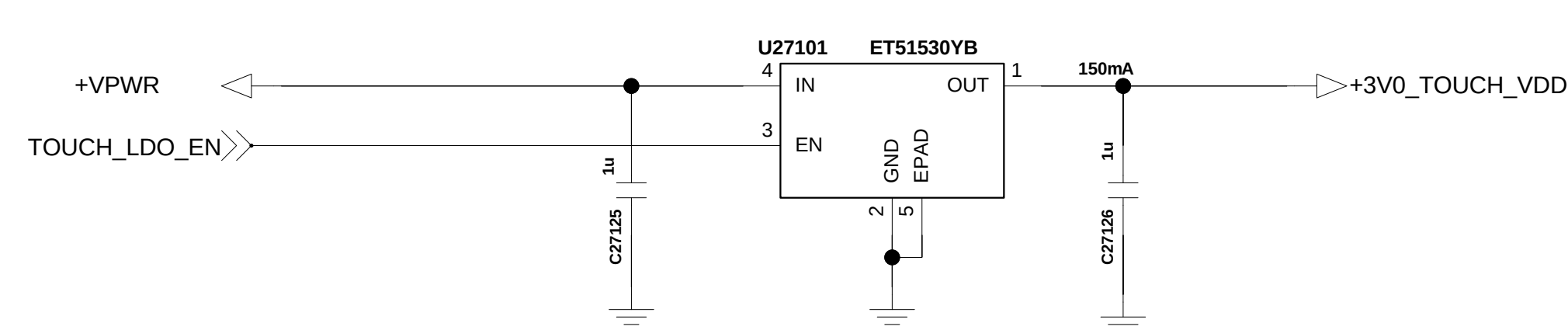
[DGMS Guide]

MC-C00081-17: 1. Add ESD Protection curcuit for Camera signal lines

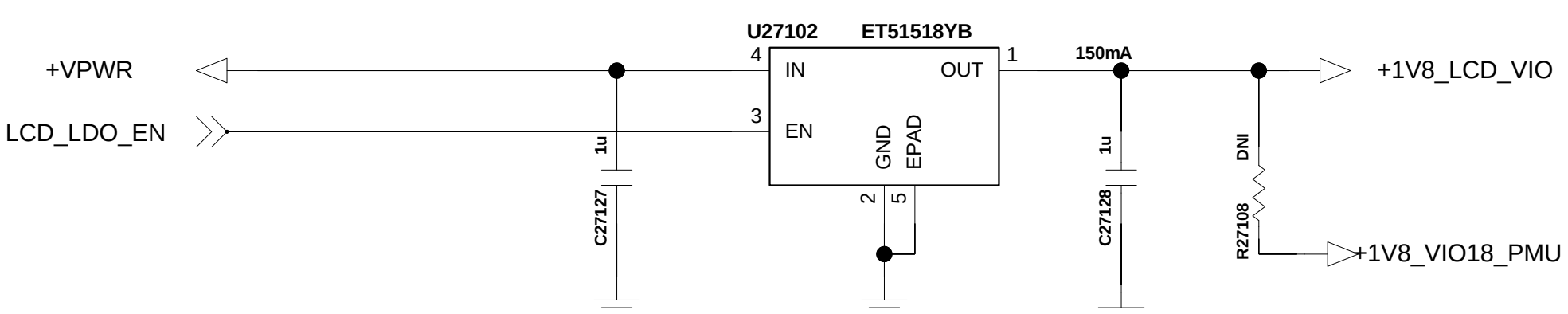
5M_Wide : EBP62982701(LGIT)



TOUCH LDO



LCD LDO



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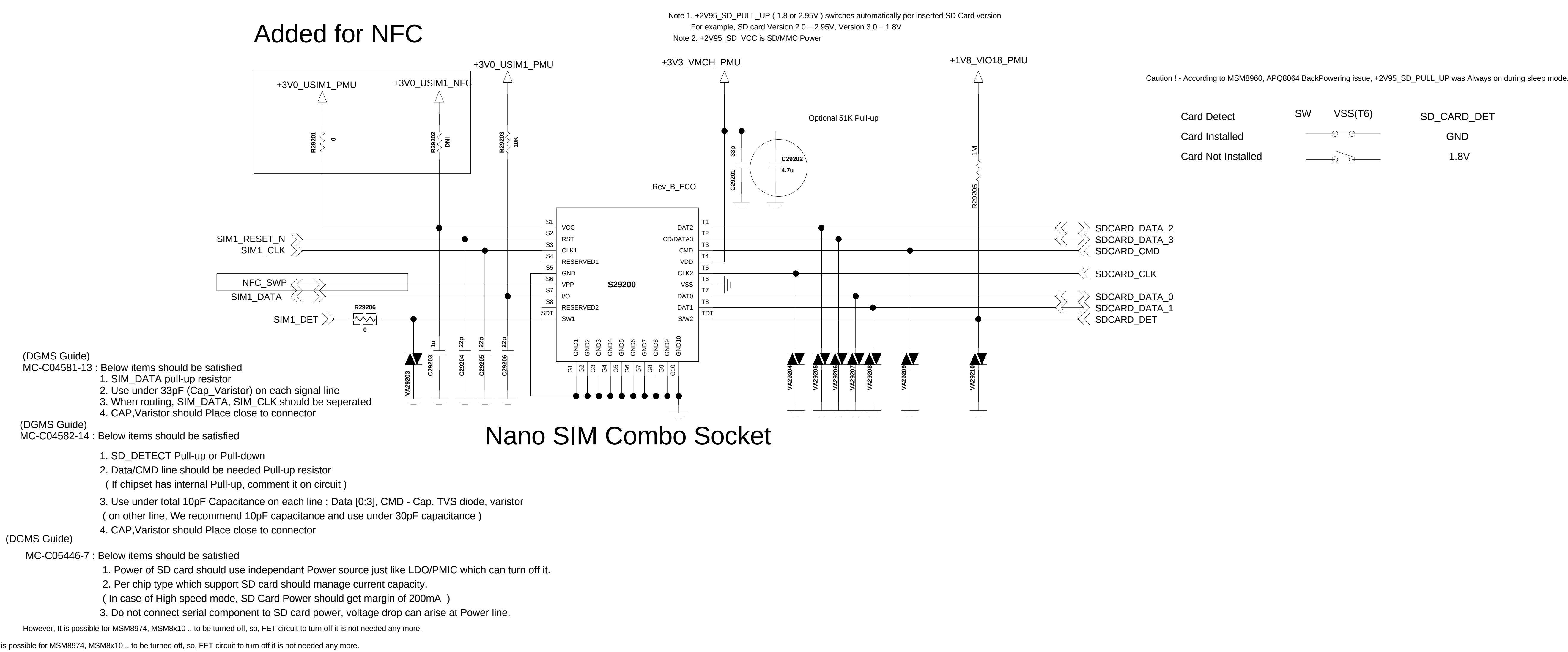
B

A

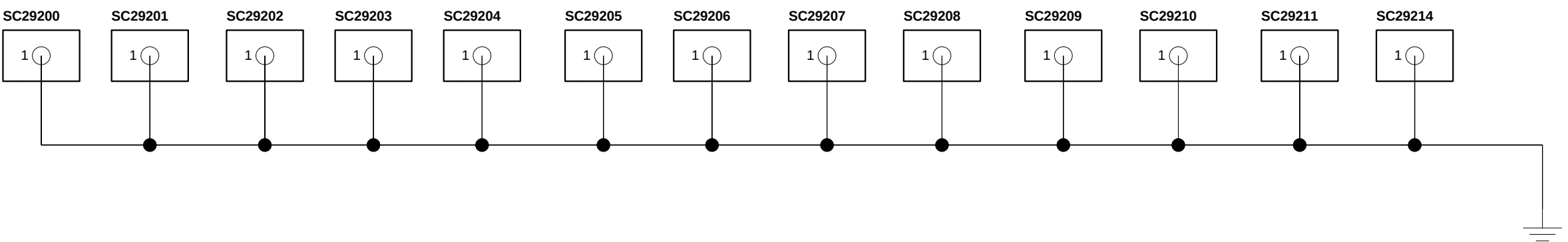
16151413121110987654321

< 9-6_u_SDCARD_Socket > Rev.1.1

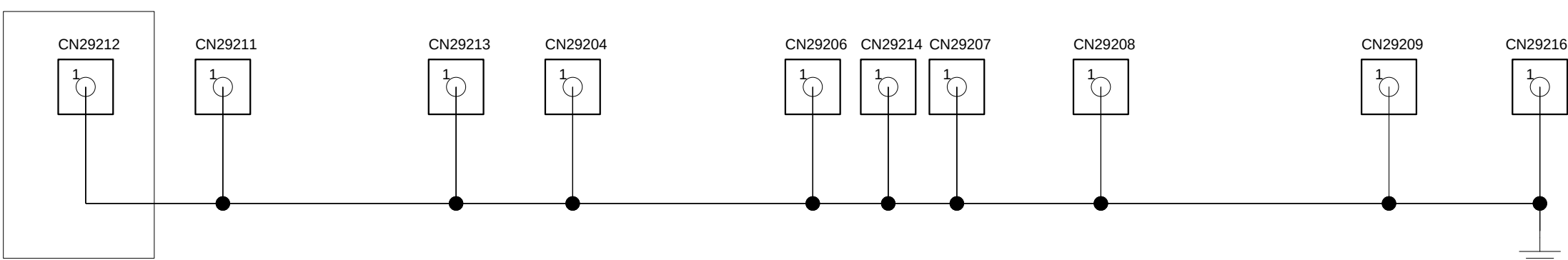
Circuit 1. u-SDCARD SIM Combo for QMC /w Low Detection



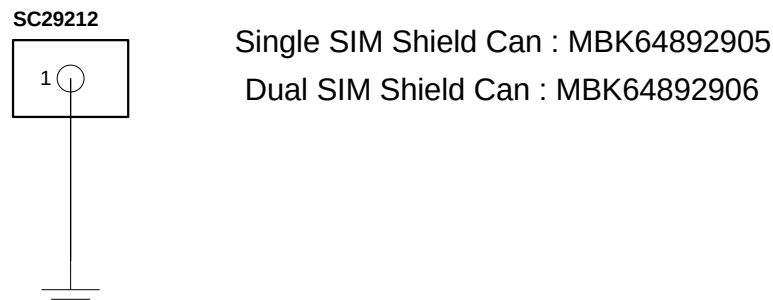
TOP SHIELD CAN CLIP



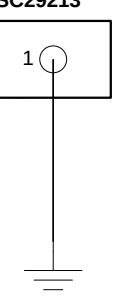
TOP_GND_CONTACT_CLIP



Bottom_Shieldcan2



Bottom_Shieldcan3



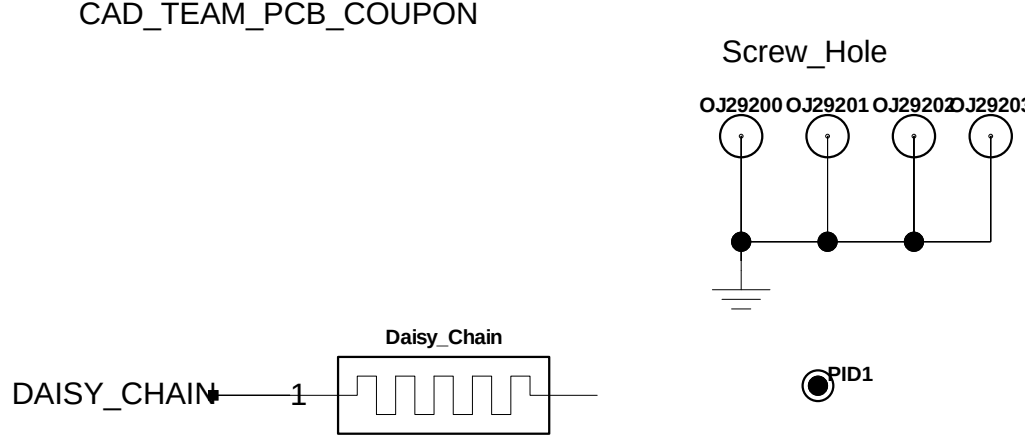
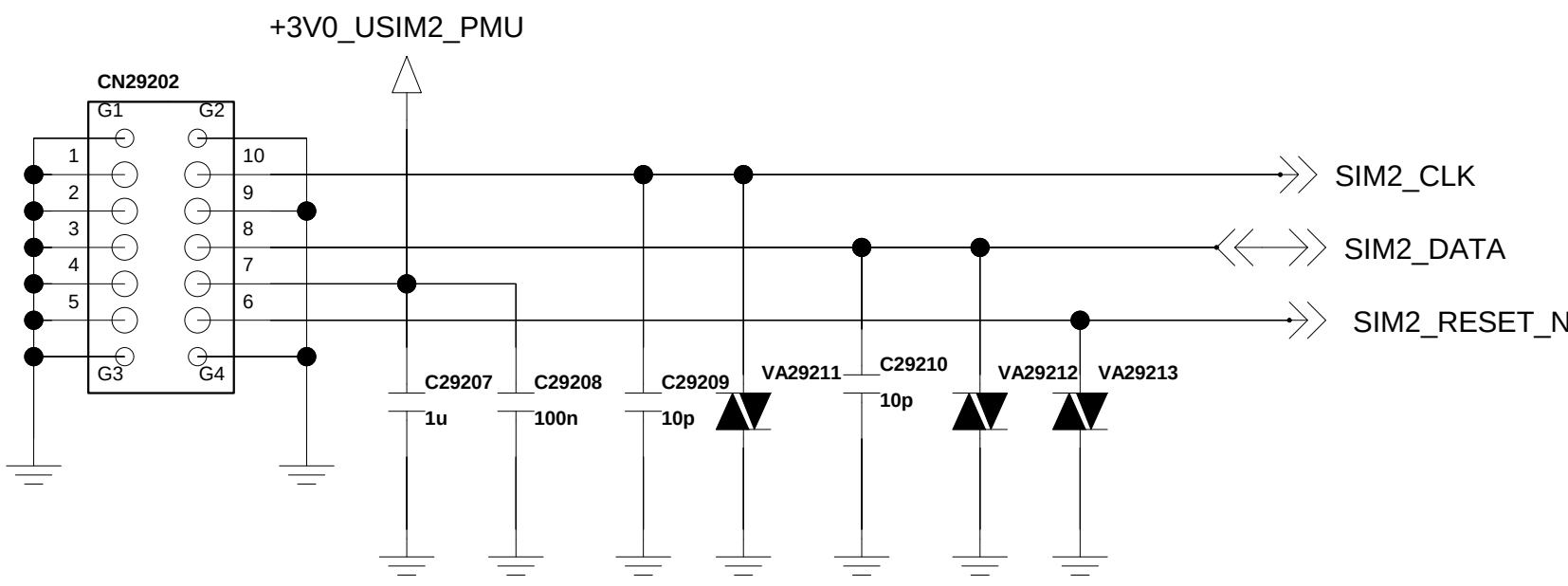
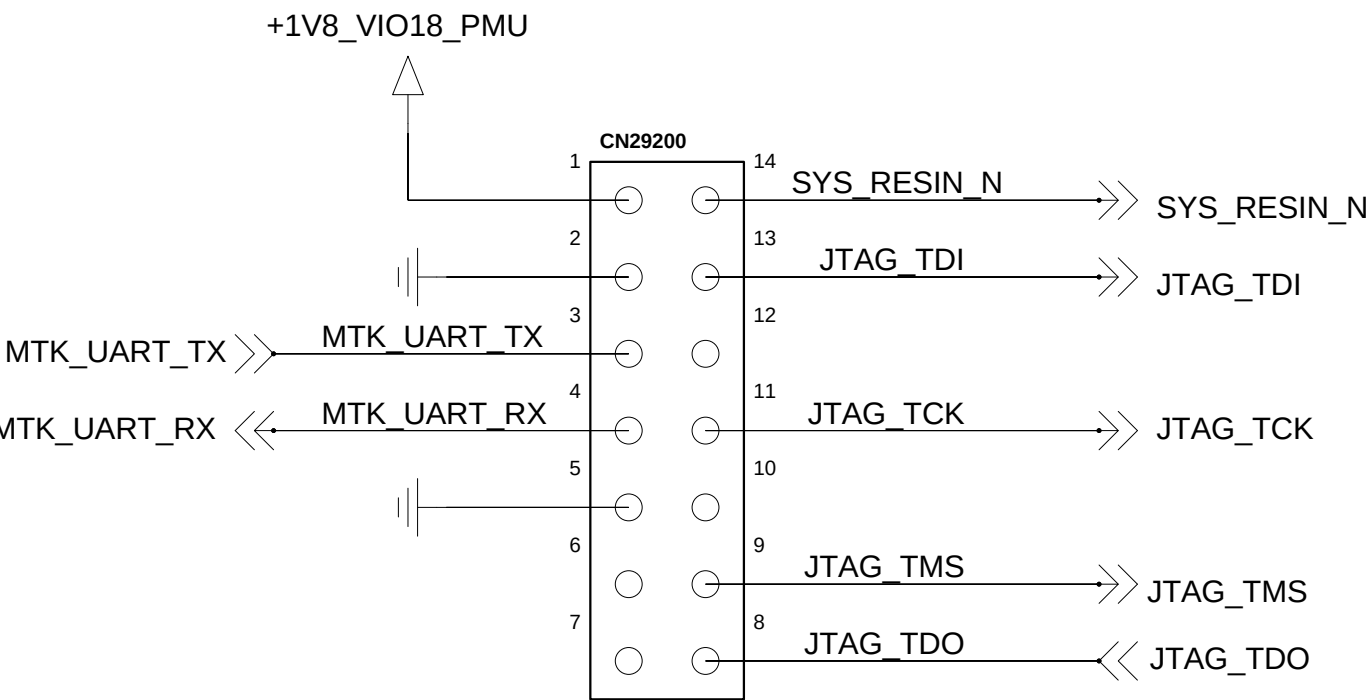
< 9-3-1_JTAG > Rev_0.4

Domestic EU_Global

Model Index

	K121L	K121S	K121K	M250n	M250E	M250	M250F	M250H	M250DS	M250AR	M250Y	M250ds	M250K	M250YK	M250I
R29207	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29208	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29209	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29210	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29211	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29212	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29213	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29214	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R29215	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI
R29216	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI
R29217	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI
R29218	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI
R29219	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI
R29220	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI
R29221	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm

Added for EU_Global Dual Nano SIM



LGE Internal Use Only

16151413121110987654321

110

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LG Electronics	
MODEL NAME	
REV: A2	DATE: 2014.02.20
PDM NUMBER	
REV: 1	Rev
DESIGN: User Name	DATE: Drawing Date

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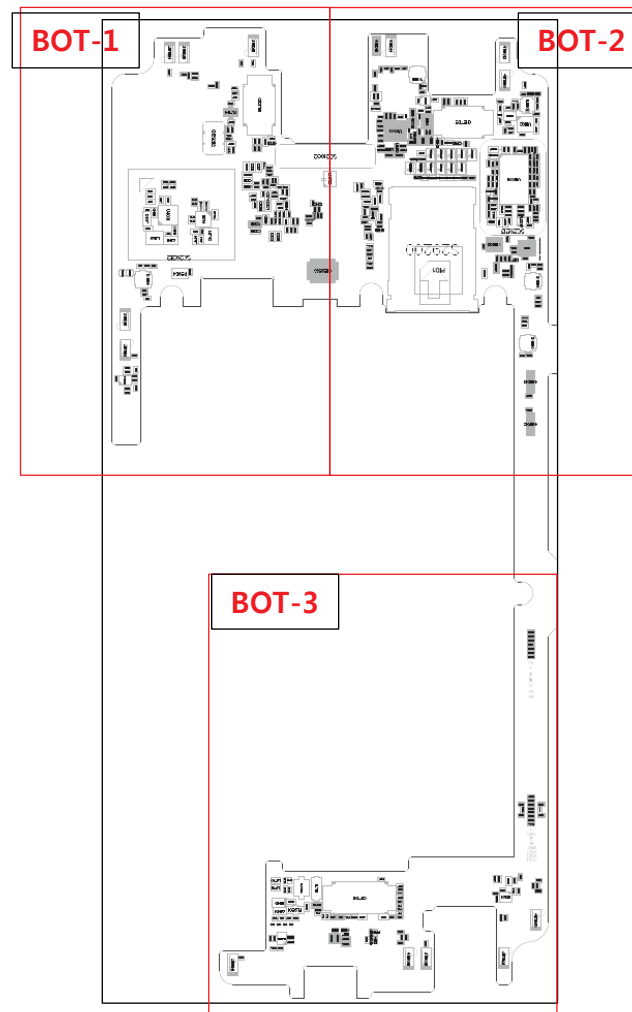
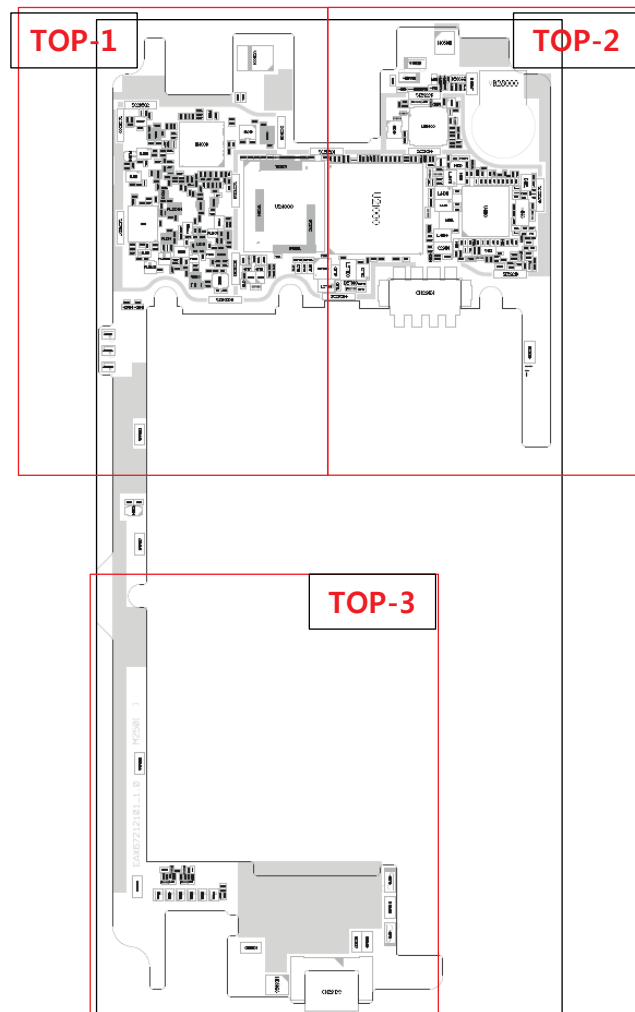
D

C

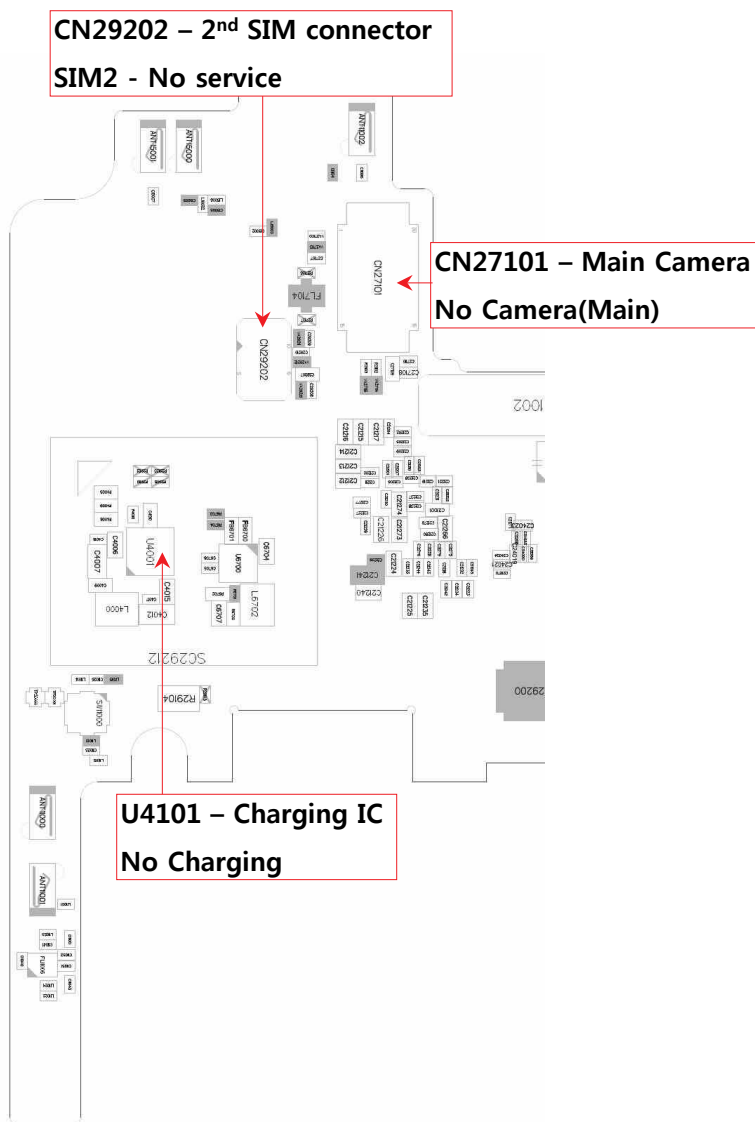
B

A

LG-M250 - MAIN_TOP & BOT

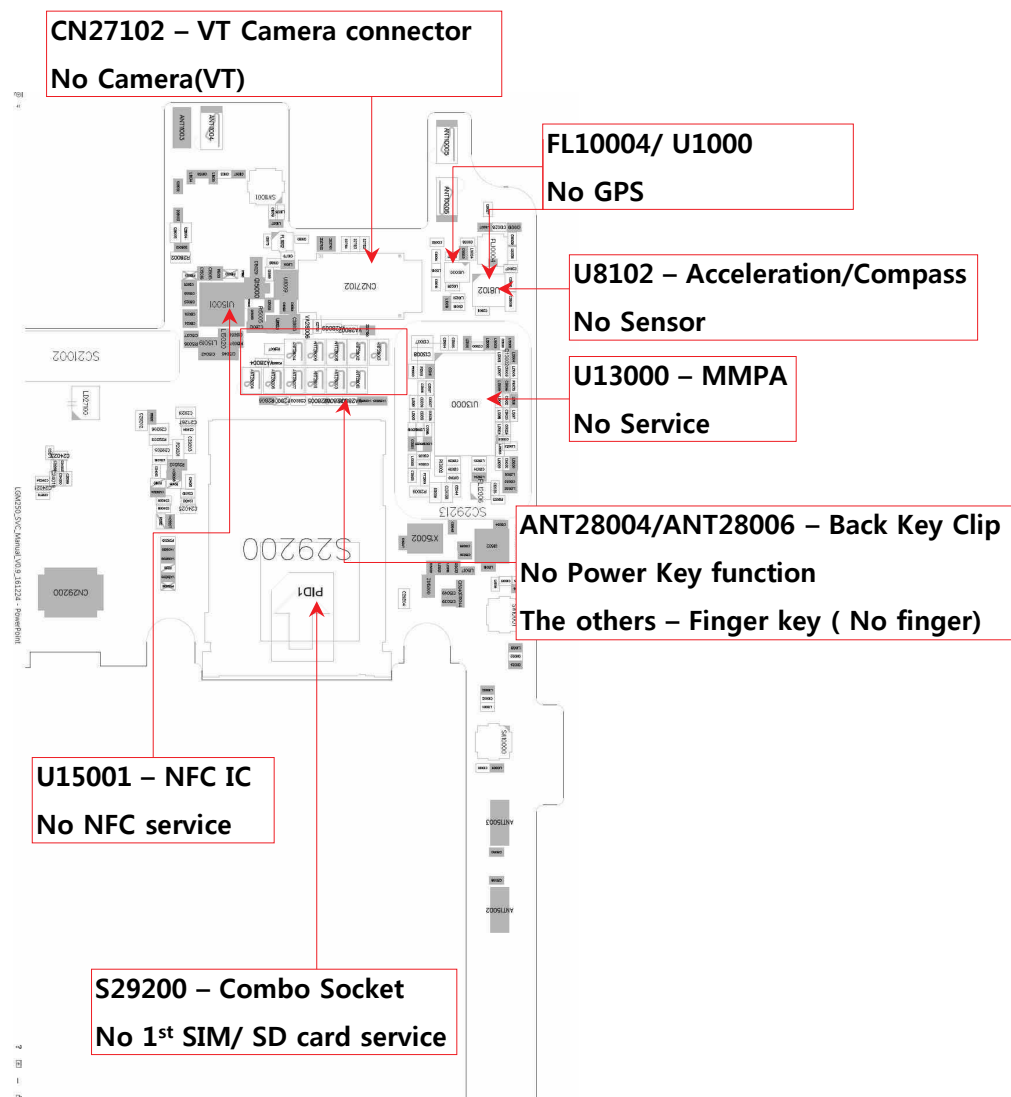


LG-M250-MAIN_BOT 1

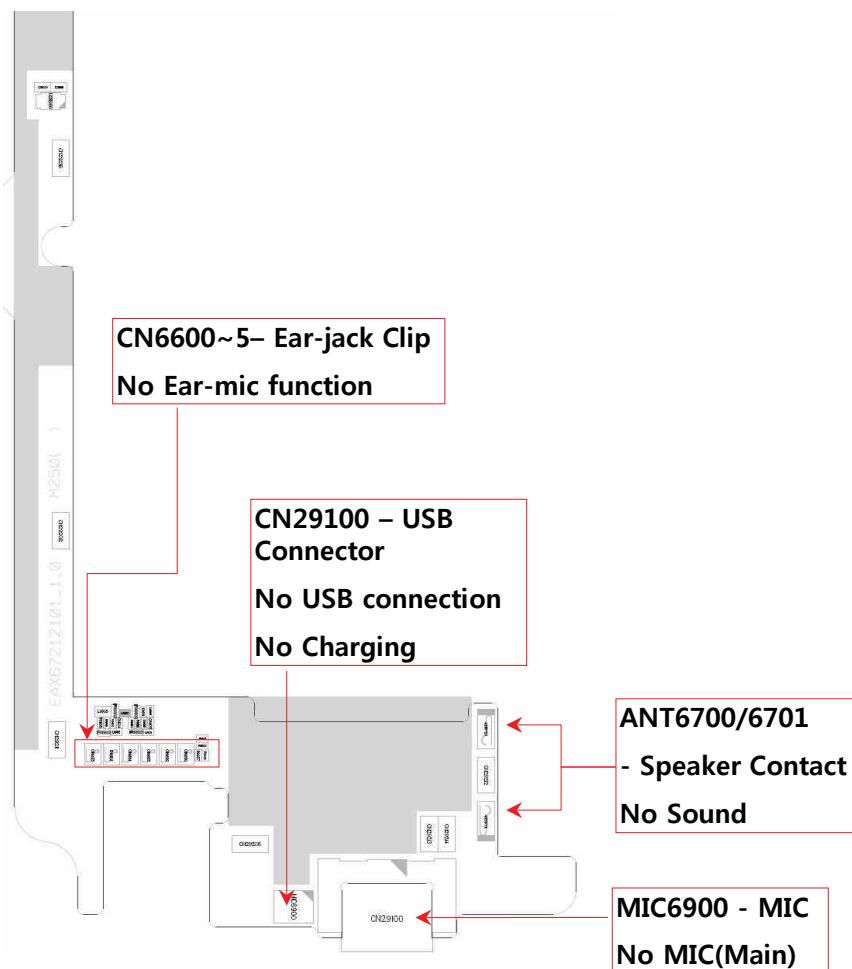


LGE Internal Use Only

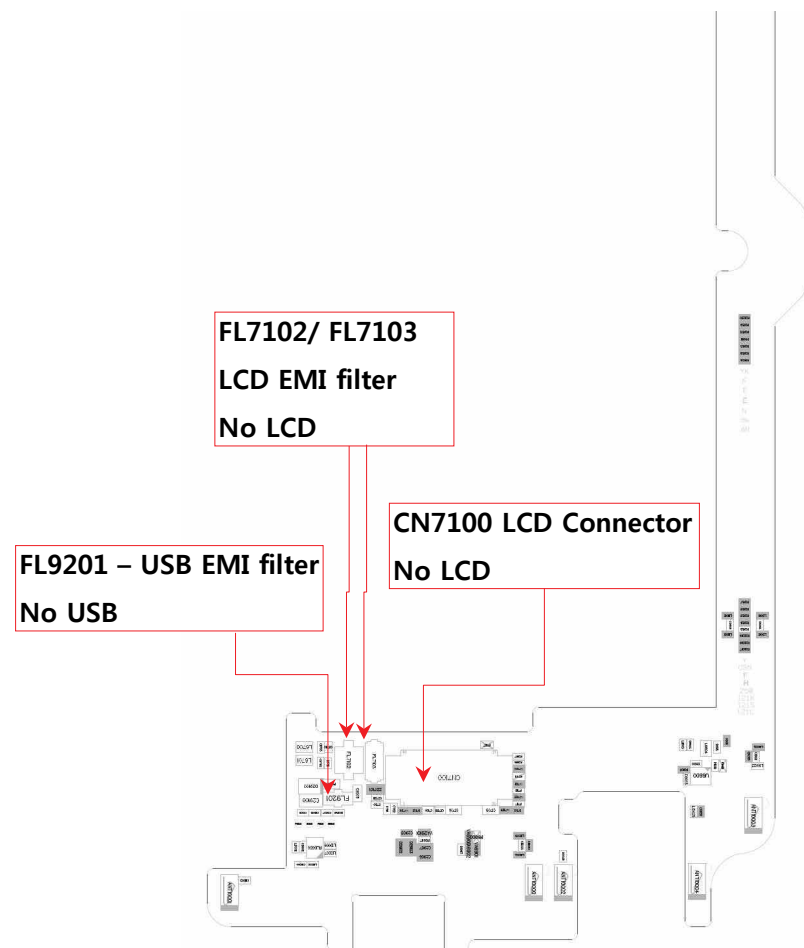
LG-M250-MAIN_BOT 2

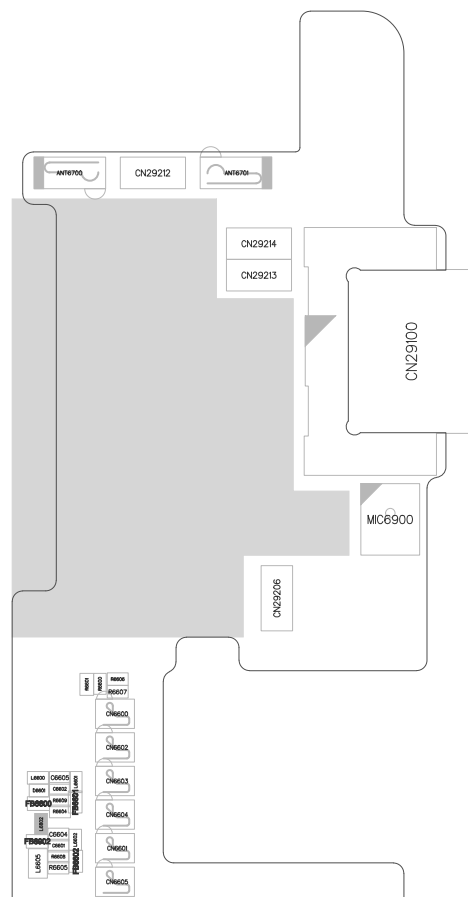
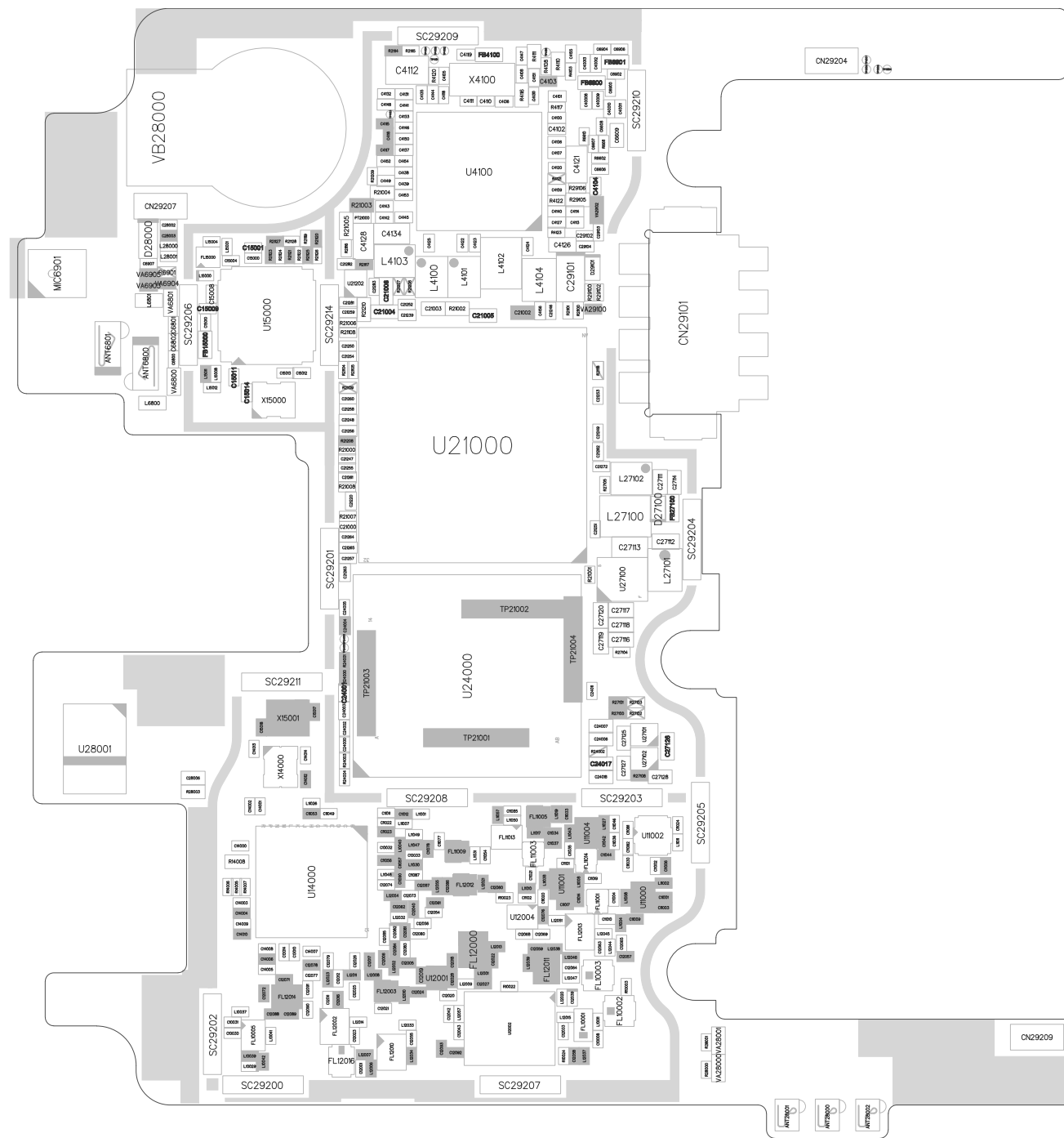


LG-M250-MAIN_TOP 3



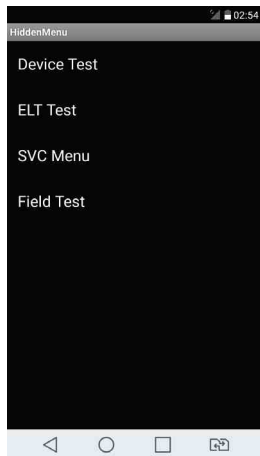
LG-M250-MAIN_BOT 3





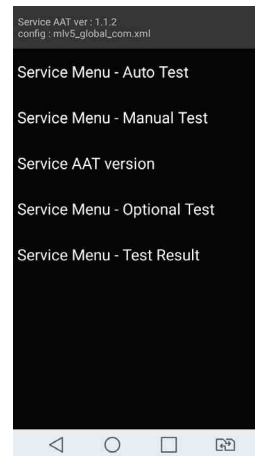


1. Hidden Menu Start



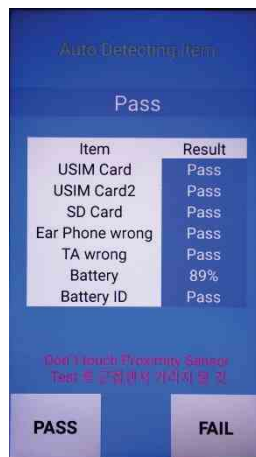
- Start shortcut key:
*#546368#*250#
- Hidden Menu List
: Start the desired menu, click

2. Device Test - SAAT



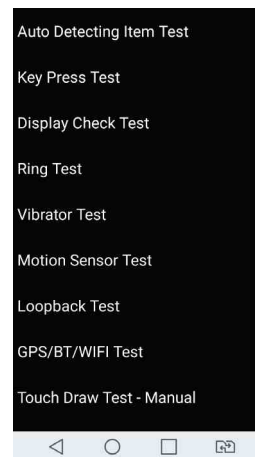
- Service Menu – Auto Test
- Service Menu – Manual Test
- Service AAT version
→ Hidden Menu version Display

2. Device Test



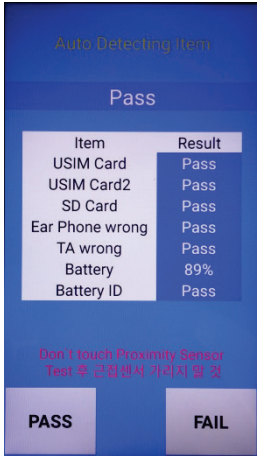
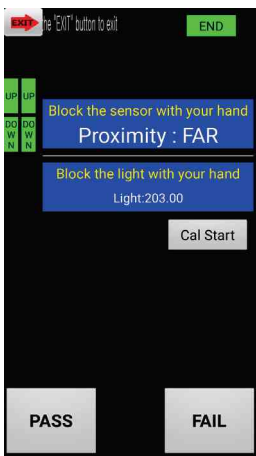
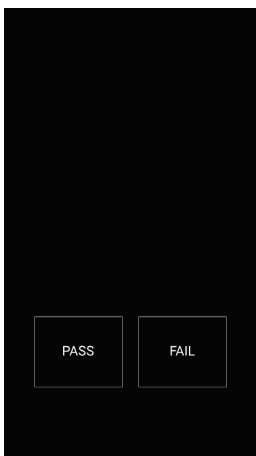
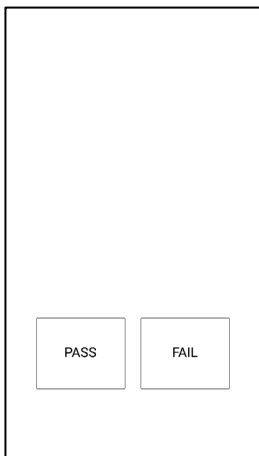
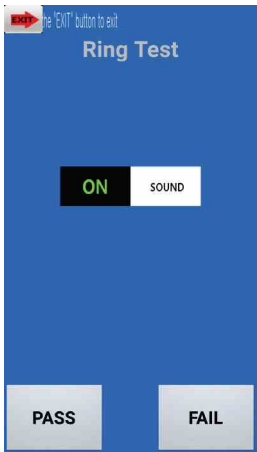
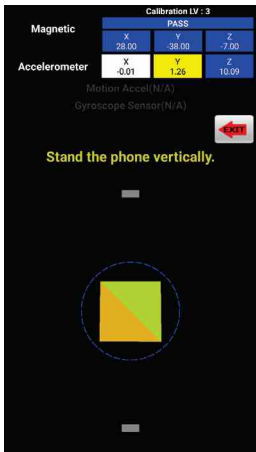
- Service Menu – Auto Test
→ All Test Items are continued one after another.
→ Continuous information on the menu, giving you ability test.

3. Device test List

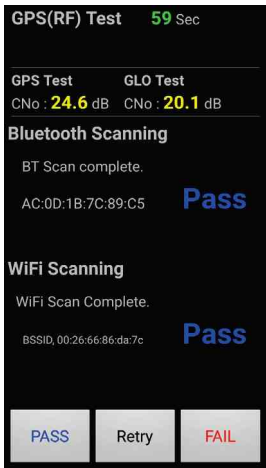
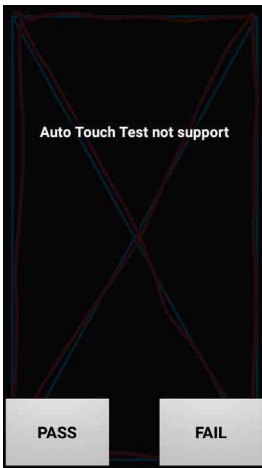
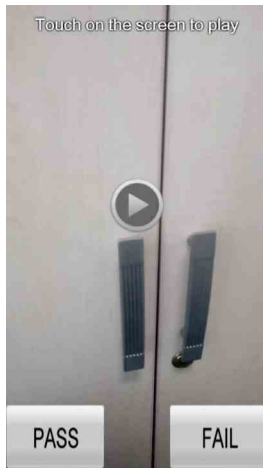
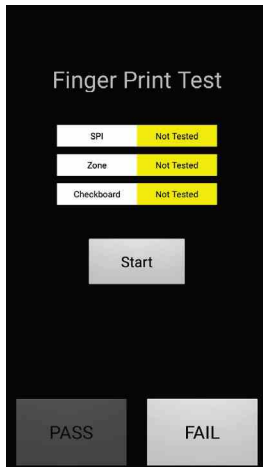


- Service Menu – Manual Test
→ Each test item can be selected and performed by user.

8. Hidden Menu

1. Auto Detecting Item Test	2. Key Press Test	3. Display Check Test (1)	4. Display Check Test (2)
			
5. Ring Test	6. Vibrator Test	7. Motion Sensor Test	8. Loop Back Test
			

8. Hidden Menu

9. GPS/BT/WiFi Test	10. Touch Draw Test	11. Camera Test (Main)	12. Camcorder Test
 GPS(BT) Test 59 Sec GPS Test GLO Test CNo : 24.6 dB CNo : 20.1 dB Bluetooth Scanning BT Scan complete. AC:0D:1B:7C:89:C5 Pass WiFi Scanning WiFi Scan Complete. BSSID: 00:26:66:86:da:7c Pass PASS Retry FAIL	 Auto Touch Test not support PASS FAIL	 Verify the picture PASS Retry FAIL	 Touch on the screen to play PASS FAIL
13. Camera Test (VT)	14. FM Radio Test	15. OTG Test	16. FingerPrint Test
 VT Camera Test EEPROM pass 2560x1920 PASS FAIL	 FM Radio Test Connected wired headset. 이어셋이 꽂혔습니다. 87.5Mhz Playing FM Radio. 소리를 확인하세요. Press down BACK key after testing. 테스트 후 back 키를 누르세요. BACK	 OTG Test Storage Mount Fail File Operation Fail PASS FAIL	 Finger Print Test SPI Not Tested Zone Not Tested Checkboard Not Tested Start PASS FAIL

1. Summary

Tool Version	DLL name	USB Driver	
LGFLASH v216	LGM250_20161216_LGFLASHv216_Download	LGUnitedMobileDriver_S52 MAN314AP22_ML_WHQL_Ver_3.14.1	
Please Check the Version to “LGST ServiceCenter Tool”			
H/W			
	Name	Part No.	SW
D/L Cable	Micro 5P (56-open-910K) USB DLC	RAD3216 7835	KDZ



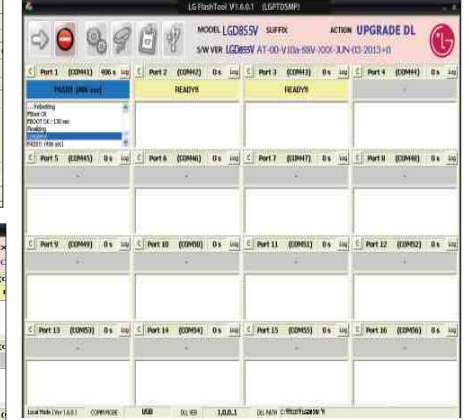
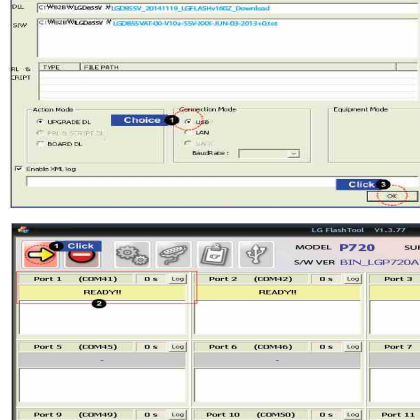
2. USB COM port Setting



3. USB D/L Cable setting



4. Flash tool D/L setting



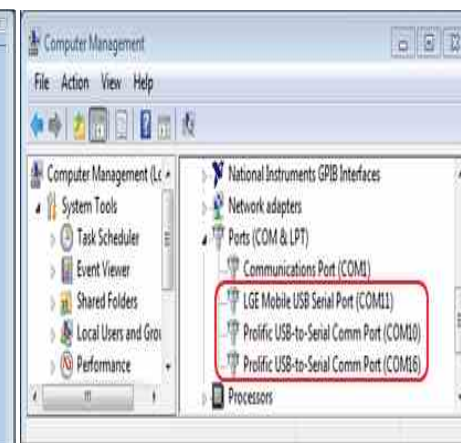
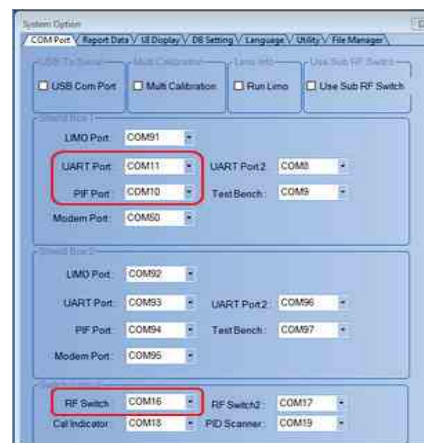
※ If you want more information, please refer LGST ServiceCenter Tool's Notification "Download User Guide".

1. Summary

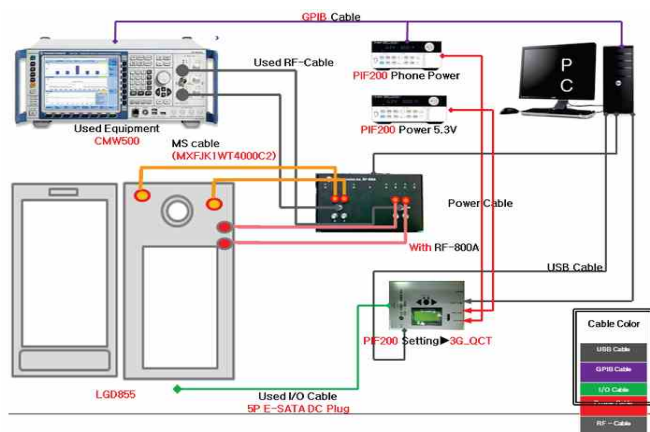
CAL INFORMATION		
S/W VERSION		
Please Check the Version to "B2B"		
H/W		
PIF	PIF200	BJAY0024021
USB Cable	Micro 5P (56-open-910K) USB DLC	RAD32167835
Power Cable	DC Power Cable	RAD32247878
I/O Cable	5P+SATA Cable(130K)	RAD32847816
RF Cable Main	MXFJM3WX6000	RAD32827895
Power Supply PIF		
Power Supply Phone	Power supply 5.0V	
PF Test Equipment		
Notice	1. Use the Battery (Refer to Attached ppt) 1) Phone states: Power off 2) If do not use the battery, TX fails. 2. Port Setting (Refer to Attached ppt) 1) Uart Port1 : Use the "LGE Mobile USB Serial Port"	
CMW 500 RF Cable connection		



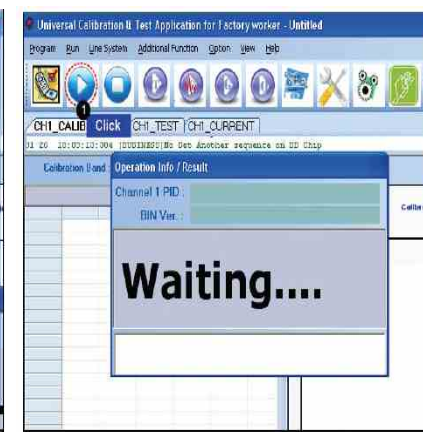
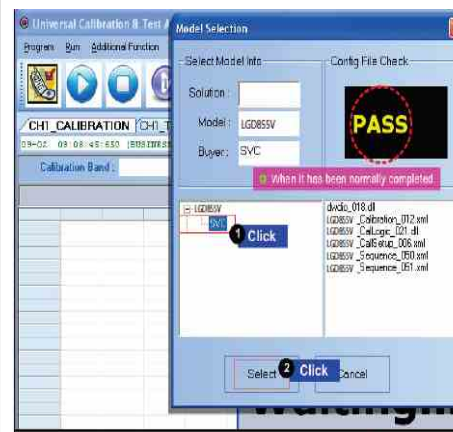
2. USB COM port Setting



3. Calibration Cable setting

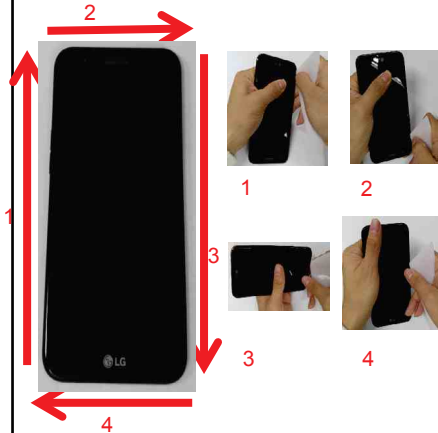
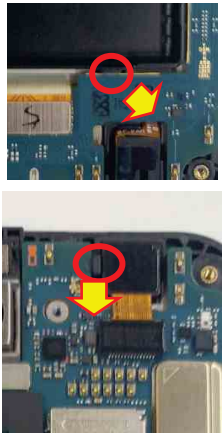
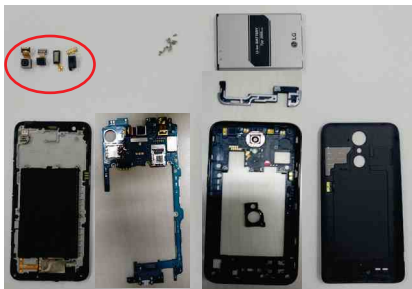
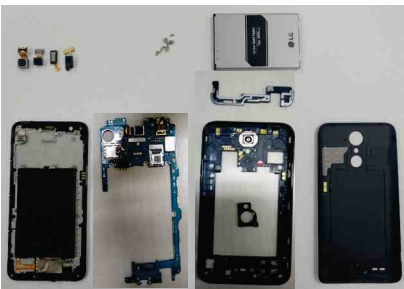


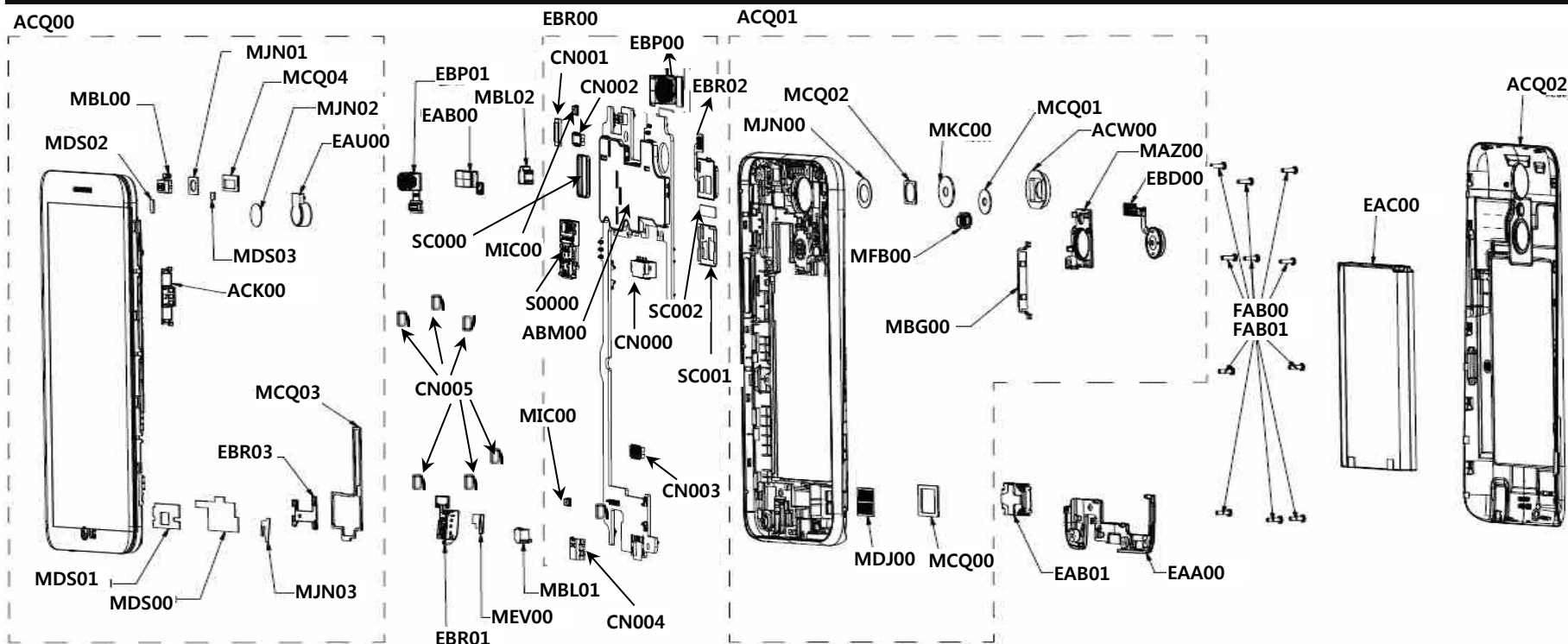
4. Tachyon setting



※ If you want more information, please refer LGST ServiceCenter Tool's Notification "RF Calibration User Guide".

11. Disassemble Guide

1. Disassemble Screw (11ea)	2. Disassemble Rear Cover	3. Disassemble Main Antenna	4. Disassemble Connector
			
5. Disassemble Main PCB	6. Disassemble HW Parts (4ea)	7. LCD Disassemble (By Heating Machine)	8. Complete Disassembling
 Disassemble Hooks			



Location no	Description	Location no	Description	Location no	Description	Location no	Description
ACQ00	Cover Assembly	MDS03	Gasket	ACQ01	Cover Assembly,Rear(SVC)	MFB00	Lens,Flash
ACK00	Contact Assembly	EAU00	Motor,DC	EBD00	Sensor,Fingerprint	MCQ02	Damper,Camera
MBL00	Cap	EBR03	PCB Assembly,Flexible	MAZ00	Bracket	EAA00	PIFA Antenna,Multiple
MJN01	Tape,Camera	MCQ03	Damper,Speaker	MBG00	Button,Side	EAB01	Speaker,Dual Mode
MDS00	Gasket	EBR02	PCB Assembly,Flexible	MCQ00	Damper,Speaker	EBR00	PCB Assembly,Main
MCQ04	Damper	EBR01	PCB Assembly,Flexible	ACW00	Decor Assembly	EBP01	Camera Module
MJN02	Tape	EAB00	Receiver	MCQ01	Damper,Motor	FAB01	Screw,Tapping
MJN03	Tape	MEV00	Insulator	MKC00	Window,Camera	FAB00	Screw,Machine
MDS01	Gasket	MBL01	Cap	MDJ00	Filter	EAC00	Rechargeable Battery,Lithium Ion
MDS02	Gasket	MBL02	Cap	MJN00	Tape,Camera	ACQ02	Cover Assembly,Battery
CN004	Connector,I/O	CN002	Connector,Terminal	S0000	Socket,DIMM/SIMM	MIC00	Microphone,Condenser
CN003	Connector,BtoB	CN001	Connector,BtoB	SC002	Plate	CN000	Connector,Terminal Block
SC000	Can,Shield	SC001	Can,Shield	CN005	Connector,Terminal	ABM00	Can Assembly,Shield
EBP00	Camera Module						

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
1	AGQ89269502	Phone Assembly	1	AGQ000000
2	ACQ89427701	Cover Assembly,EMS	1	ACQ100400
3	EBR83867801	PCB Assembly,Main	1	EBR00
4	EBR83832601	PCB Assembly,Main,SMT	1	EBR071800
5	EBR83532001	PCB Assembly,Main,SMT Bottom	1	EBR071600
6	EAE62282201	Capacitor,Ceramic,Chip	4	C12035,C4017,C6600,C6612
7	EAE62286801	Capacitor,Ceramic,Chip	1	C29208
8	EAE62502901	Capacitor,Ceramic,Chip	5	C13007,C13008,C4015,C6704,C6707
9	EAE62506501	Capacitor,Ceramic,Chip	1	C13028
10	EAE62685301	Capacitor,Ceramic,Chip	1	C4010
11	EAE62686001	Capacitor,Ceramic,Chip	1	C4006
12	EAE62726601	Capacitor,Ceramic,Chip	4	C21238,C27108,C27122,C27124
13	EAE62924001	Capacitor(High Frequency),Ceramic,Chip	2	C27121,VA27100
14	EAE62945801	Capacitor(High Frequency),Ceramic,Chip	3	C13036,L10020,L13013
15	EAE62945901	Capacitor(High Frequency),Ceramic,Chip	2	C13037,L13010
16	EAE62946001	Capacitor(High Frequency),Ceramic,Chip	1	L13023
17	EAE62946401	Capacitor(High Frequency),Ceramic,Chip	2	C13035,L13022
18	EAE62946501	Capacitor(High Frequency),Ceramic,Chip	2	C11103,L10024
19	EAE62946601	Capacitor(High Frequency),Ceramic,Chip	1	C10027
20	EAE62947001	Capacitor(High Frequency),Ceramic,Chip	1	C13039
21	EAE62962201	Capacitor(High Frequency),Ceramic,Chip	2	L10022,L13033
22	EAE63023601	Capacitor(High Frequency),Ceramic,Chip	1	L13024
23	EAE62964001	Capacitor(High Frequency),Ceramic,Chip	2	L13006,L13025
24	EAE63222701	Capacitor,Ceramic,Chip	1	C13038
25	EAE63621501	Capacitor,Ceramic,Chip	13	C21212,C21213,C21214,C21215,C21216,C21217,C21224,C21225,C21226,C21235,C21240,C21273,C21274
26	EAE64061601	Capacitor,Ceramic,Chip	5	C21267,C24010,C24015,C24016,C24025
27	EAE64061501	Capacitor,Ceramic,Chip	1	C10340
28	EAG63910001	C-Clip	13	ANT10000,ANT10001,ANT10002,ANT10003,ANT10004,ANT10005,ANT10006,ANT11000,ANT11001,ANT11002,ANT11004,ANT15000,ANT15001
29	EAE64262401	Capacitor,Ceramic,Chip	1	C4012
30	EAF63050001	Varistor	1	R28007
31	EAG63651001	Connector,BtoB	1	CN29202
32	EAG63772101	Connector,RF	4	SW10000,SW10001,SW11000,SW11001
33	EAG64651301	C-Clip	11	ANT28003,ANT28004,ANT28006,ANT28007,ANT28008,ANT28009,ANT28010,ANT28011,ANT28012,ANT28013,ANT28014
34	EAG64909901	Socket,DIMM/SIMM	1	S29200
35	EAH63353101	Diode,TVS	3	C6700,C6701,VA28004
36	EAH63092501	Diode,TVS	1	D29100
37	EAN62578201	IC,Analog Switch	1	U6600
38	EAM62071101	Filter,Bead	2	FB6700,FB6701
39	EAM62730001	Filter,Separator,Switch	1	FL11012

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
40	EAM63050001	Filter,Separator,Switch	1	FL12006
41	EAM62791701	Filter,Separator	1	FL10004
42	EAM63850001	Filter,EMI/Power	1	FL7103
43	EAM63991901	Filter,Separator,Switch	2	FL10000,FL11006
44	EAM63569201	IC,Charger	1	U4001
45	EAM62777001	IC,Speaker Amplifier	1	U6700
46	EAM63886901	IC,Acceleration Sensor	1	U8102
47	EAM63106301	IC,RF Amplifier	1	U10000
48	EAM64259501	IC,Power Amplifier	1	U13000
49	EAP61767701	Inductor,Multilayer,Chip	1	C13001
50	EAP61866601	Inductor,Multilayer,Chip	3	L10006,L11021,L6603
51	EAP61866701	Inductor,Multilayer,Chip	6	L10001,L10014,L10026,L11014,L11036,L15006
52	EAP62108301	Inductor,Multilayer,Chip	2	L11012,L13029
53	EAP61866901	Inductor,Multilayer,Chip	2	C10013,C13034
54	EAP61866401	Inductor,Multilayer,Chip	1	C13024
55	EAP62107901	Inductor,Multilayer,Chip	1	C13023
56	EAP62108101	Inductor,Multilayer,Chip	1	L10016
57	EAP62109001	Inductor,Multilayer,Chip	1	C10002
58	EAP62108401	Inductor,Multilayer,Chip	4	L10009,L11024,L13005,L13018
59	EAP62226201	Inductor,Multilayer,Chip	1	C13013
60	EAP62226101	Inductor,Multilayer,Chip	4	C10018,C13000,C13003,C13012
61	EAP62186301	Inductor,Multilayer,Chip	2	C11059,L11023
62	EAP62226001	Inductor,Multilayer,Chip	1	C10052
63	EAP62226401	Inductor,Multilayer,Chip	2	C13026,L11022
64	EAP62226501	Inductor,Multilayer,Chip	3	C10038,L10007,L10038
65	EAP62226601	Inductor,Multilayer,Chip	2	L13016,L13017
66	EAP62526401	Inductor,Multilayer,Chip	1	C13009
67	EAP63067001	Inductor,Multilayer,Chip	1	C13002
68	EAP63746801	Inductor,Multilayer,Chip	3	C10021,C10029,C11026
69	EAP63808301	Inductor,Wire Wound,Chip	3	L6604,L6700,L6701
70	EAP63666401	Inductor,Wire Wound,Chip	2	L4000,L6702
71	ECCH0002001	Capacitor,Ceramic,Chip	1	C28008
72	EAV63412401	LED,Flash	1	LD27100
73	EBG62665701	Thermistor,NTC	1	PT21001
74	ECCH0004904	Capacitor,Ceramic,Chip	5	C21001,C28004,C28010,C29203,C29207
75	ECCH0000115	Capacitor,Ceramic,Chip	4	C29204,C29205,C29206,C9203
76	ECCH0007804	Capacitor,Ceramic,Chip	2	C1705,C1706
77	ECCH0002002	Capacitor,Ceramic,Chip	1	C4009
78	ECCH0009101	Capacitor,Ceramic,Chip	18	C10015,C21268,C21269,C21270,C21271,C24008,C24012,C24013,C24014,C24019,C24020,C24021,C24022,C28007,C28011,C6705,C6706,C6903
79	ECCH0007805	Capacitor,Ceramic,Chip	1	C21266
80	ECCH0009102	Capacitor,Ceramic,Chip	1	D6602

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
81	ECCH0009103	Capacitor,Ceramic,Chip	25	C10000,C10003,C10004,C10005,C10009,C10012,C10014,C10028,C11025,C11040,C11041,C11079,C11080,C11088,C11089,C11096,C12044,C13004,C13029,C15002,C15007,L10008,L10025,L10043,L10052
82	ECCH0009104	Capacitor,Ceramic,Chip	14	C10001,C10010,C10011,C10022,C10053,C11051,C11052,C11100,C13018,C13022,C13025,C27129,C27130,L10017
83	ECCH0009105	Capacitor,Ceramic,Chip	2	C7101,C7102
84	ECCH0009106	Capacitor,Ceramic,Chip	5	C10007,C11043,C11075,C13017,C6603
85	ECCH0009109	Capacitor,Ceramic,Chip	2	C6702,C6703
86	ECCH0009208	Capacitor,Ceramic,Chip	2	L10015,L13012
87	ECCH0009216	Capacitor,Ceramic,Chip	1	C13020
88	ECCH0009502	Capacitor,Ceramic,Chip	1	L13007
89	ECCH0009514	Capacitor,Ceramic,Chip	3	C13016,C29209,C29210
90	ECCH0009520	Capacitor,Ceramic,Chip	1	C13032
91	ECCH0017301	Capacitor,Ceramic,Chip	33	C21200,C21201,C21202,C21203,C21204,C21205,C21206,C21207,C21208,C21209,C21210,C21211,C21218,C21219,C21221,C21222,C21223,C21237,C21275,C21276,C21277,C21278,C21279,C21280,C24009,C24023,C24024,C27107,C27110,C27123,C28009,C7100,C7104
92	ECCH0017601	Capacitor,Ceramic,Chip	2	C27109,C29202
93	ECCH0034801	Capacitor,Ceramic,Chip	12	C21227,C21228,C21229,C21230,C21231,C21232,C21233,C21234,C21242,C21243,C21244,C21245
94	ECCH0042301	Capacitor,Ceramic,Chip	1	C4007
95	ECZH0000813	Capacitor,Ceramic,Chip	1	C28005
96	ECZH0000830	Capacitor,Ceramic,Chip	1	C29201
97	ECZH0003503	Capacitor,Ceramic,Chip	1	C29100
98	ECZH0025911	Capacitor,Ceramic,Chip	1	C13005
99	ECZH0025917	Capacitor,Ceramic,Chip	1	C4016
100	ENBY0036001	Connector,BtoB	1	CN7100
101	ERHY0009501	Resistor,Chip	5	C13019,R10025,R10035,R29212,R6619
102	ENBY0040701	Connector,BtoB	2	CN27101,CN27102
103	ERHY0000166	Resistor,Chip	1	R21009
104	ERHY0000254	Resistor,Chip	1	R28002
105	ERHY0009507	Resistor,Chip	1	R29205
106	ERHY0009503	Resistor,Chip	1	R28004
107	ERHY0009505	Resistor,Chip	2	R13000,R4010
108	ERHY0009506	Resistor,Chip	2	R7117,R7118
109	ERHZ0000401	Resistor,Chip	3	R13002,R29201,R6700
110	ERHY0009519	Resistor,Chip	1	R13001
111	ERHY0009526	Resistor,Chip	4	R21102,R21103,R21106,R21107
112	MBK64892904	Can,Shield	1	SC29213
113	ERHY0009536	Resistor,Chip	1	R29117
114	ERHY0035601	Resistor,Chip	1	R29104
115	MBK64892906	Can,Shield	1	SC29212
116	ERHZ0000203	Resistor,Chip	3	R4005,R4006,R4009
117	MGJ65374601	Plate	1	SC21002
118	SAFO0000401	Wire Pad,Open	7	R21110,R21111,R21112,R21113,R27111,R29118,R29119

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
119	ERHZ0000405	Resistor,Chip	2	R29203,R6702
120	SAFP0000401	Wire Pad,Short	12	R15000,R15003,R27106,R27107,R28005,R29103,R29120,R29121,R29206,R6618,R6900,R7107
121	SEVY0008901	Varistor	6	VA28002,VA28005,VA28006,VA28007,VA28008,VA28009
122	ERHZ0000486	Resistor,Chip	1	R9201
123	MEZ65049701	Label	1	PID1
124	SFEY0015901	Filter,EMI/Power	1	FL7102
125	SFEY0015301	Filter,EMI/Power	1	FL9201
126	EBR83545901	PCB Assembly,Main,SMT Top	1	EBR071700
127	EAB62950801	Microphone,Condenser	2	MIC6900,MIC6901
128	EAE62282201	Capacitor,Ceramic,Chip	2	C12035,C4017,C6600,C6612
129	EAE62286801	Capacitor,Ceramic,Chip	1	C29208
130	EAE62502901	Capacitor,Ceramic,Chip	3	C13007,C13008,C4015,C6704,C6707
131	EAE62506501	Capacitor,Ceramic,Chip	2	C13028
132	EAE62685301	Capacitor,Ceramic,Chip	4	C4010
133	EAE62726601	Capacitor,Ceramic,Chip	3	C21238,C27108,C27122,C27124
134	EAE627262301	Capacitor,Ceramic,Chip	11	C29104,C4100,C4102,C4127,C4135,C4136,C4137,C4138,C4139,C4140,C4155
135	EAE62767801	Capacitor,Ceramic,Chip	1	C4112
136	EAE62946901	Capacitor(High Frequency),Ceramic,Chip	1	C13010
137	EAE62945801	Capacitor(High Frequency),Ceramic,Chip	1	C13036,L10020,L13013
138	EAE63286601	Capacitor,Ceramic,Chip	4	C4122,C4123,C4124,C4125
139	EAE62964001	Capacitor(High Frequency),Ceramic,Chip	1	L13006,L13025
140	EAE62963801	Capacitor(High Frequency),Ceramic,Chip	1	L12047
141	EAE63541901	Capacitor,Ceramic,Chip	1	C27113
142	EAE64022001	Acoustic Noise MLCC	1	C4121
143	EAE64061601	Capacitor,Ceramic,Chip	8	C21267,C24010,C24015,C24016,C24025
144	EAE64281701	Capacitor,TA,Polymer	1	C29101
145	EAG63652301	C-Clip	2	ANT6700,ANT6701
146	EAG63749801	Connector,Terminal Block	1	CN29101
147	EAE64383601	Capacitor,Ceramic,Chip	3	C27118,C27119,C27120
148	EAG63910001	C-Clip	2	ANT10000,ANT10001,ANT10002,ANT10003,ANT10004,ANT10005,ANT10006,ANT11000,ANT11001,ANT11002,ANT11004,ANT15000,ANT15001
149	EAG64149901	Connector,I/O	1	CN29100
150	EAG64651301	C-Clip	9	ANT28003,ANT28004,ANT28006,ANT28007,ANT28008,ANT28009,ANT28010,ANT28011,ANT28012,ANT28013,ANT28014
151	EAH61992801	Diode,Schottky	1	D27100
152	EAH62213401	Diode,TVS	1	D29101
153	EAG64451301	Connector,Terminal	10	CN29204,CN29206,CN29207,CN29208,CN29209,CN29211,CN29212,CN29213,CN29214,CN29216
154	ELCH0003842	Inductor,Multilayer,Chip	2	L6800,L6801
155	EAM62691501	Filter,Ceramic	1	FL12016
156	EAH63353101	Diode,TVS	1	C6700,C6701,VA28004
157	EAM63031101	Filter,Separator,Switch	1	U11002
158	EAM62230101	Filter,Bead	1	FB6902

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
159	EAM63450501	Filter,Saw,Dual	1	FL11013
160	EAM62252001	Filter,Bead	3	FB6600,FB6601,FB6602
161	EAM63670001	Filter,Saw	1	FL15000
162	EAM63530101	Filter,Duplexer	1	FL12002
163	EAM63930301	Filter,Duplexer	1	FL12010
164	EAM64030601	Filter,Separator	1	FL10001
165	EAM63410201	Filter,Saw	1	FL11003
166	EAM63090201	Filter,Saw	1	FL11001
167	EAM63730401	Filter,Bead	1	FB27100
168	EAM64072401	Filter,Duplexer	1	FL12013
169	EAM62357301	IC,Hall Effect Switch	1	U28000
170	EAM63390101	Filter,Saw	1	FL11014
171	EAM63486501	IC,WiFi	1	U15000
172	EAM63606501	IC,LDO Voltage Regulator	1	U21202
173	EAM63648001	IC,Proximity	1	U28001
174	EAM63788301	IC,RF Amplifier	1	FL10005
175	EAM63807801	IC,RF Amplifier	1	U12004
176	EAM64086201	IC,RF Transceiver,4G	1	U14000
177	EAM64254101	IC,MCP,eMMC	1	U24000
178	EAM64294701	IC,PMIC	1	U4100
179	EAM64345901	IC,Digital Baseband Processor,4G	1	U21000
180	EAP61866301	Inductor,Multilayer,Chip	1	L10041
181	EAM64086901	IC,Sub PMIC	1	U27100
182	EAM64407901	IC,Voltage Regulator	1	U27102
183	EAM64428001	IC,LDO Voltage Regulator	1	U27101
184	EAP61747501	Inductor,Multilayer,Chip	1	L11050
185	EAP61767801	Inductor,Multilayer,Chip	1	L12033
186	EAP61866601	Inductor,Multilayer,Chip	4	L10006,L11021,L6603
187	EAP63106701	Inductor,Wire Wound,Chip	1	L27100
188	EAP63185901	Inductor,Multilayer,Chip	1	L27101
189	EAP61866701	Inductor,Multilayer,Chip	1	L10001,L10014,L10026,L11014,L11036,L15006
190	EAP61866901	Inductor,Multilayer,Chip	1	C10013,C13034
191	EAP61946101	Inductor,Multilayer,Chip	2	C12055,C12063
192	EAP63666401	Inductor,Wire Wound,Chip	1	L4000,L6702
193	EAP62107901	Inductor,Multilayer,Chip	2	C13023
194	EAP62108101	Inductor,Multilayer,Chip	1	L10016
195	EAP62108001	Inductor,Multilayer,Chip	3	C11019,L12032,L15001
196	EAP63786501	Inductor,Wire Wound,Chip	3	L4100,L4101,L4103
197	EAP62108401	Inductor,Multilayer,Chip	2	L10009,L11024,L13005,L13018
198	EAT63375201	Module,Rx Module	1	U12002
199	EAP62108201	Inductor,Multilayer,Chip	2	C11002,L11049
200	EAP62108301	Inductor,Multilayer,Chip	1	L11012,L13029

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
201	EAP62108701	Inductor,Multilayer,Chip	2	L10011,L11011
202	EAP62108501	Inductor,Multilayer,Chip	3	C11022,C11049,C12012
203	EAP63826001	Inductor,Wire Wound,Chip	1	L4102
204	EAP62225901	Inductor,Multilayer,Chip	2	C12054,C12085
205	EBC61856201	Resistor,Chip	3	R21001,R24003,R24004
206	EAP62109001	Inductor,Multilayer,Chip	1	C10002
207	EBC61997701	Resistor,Chip	1	R6601
208	EAW63023701	Oscillator,TCXO	1	X15000
209	EAW63243401	Oscillator,VCTCXO	1	X14000
210	EAP62226201	Inductor,Multilayer,Chip	2	C13013
211	EAP62226501	Inductor,Multilayer,Chip	2	C10038,L10007,L10038
212	EBC62575901	Resistor,Chip	2	R14006,R14007
213	EBC62581901	Resistor,Chip	2	R6604,R6605
214	EAP62226601	Inductor,Multilayer,Chip	1	L13016,L13017
215	EBC62035901	Resistor,Chip	2	R4122,R4123
216	EBC62595901	Resistor,Chip	1	R14005
217	EAP62226701	Inductor,Multilayer,Chip	1	L12015
218	EAP63067001	Inductor,Multilayer,Chip	2	C13002
219	EBG62665701	Thermistor,NTC	1	PT21001
220	ERHY0009547	Resistor,Chip	1	R4117
221	ECA30360101	Coupler,RF Directional	2	FL10002,FL10003
222	ECCH0000198	Capacitor,Ceramic,Chip	2	C24018,C6609
223	EAP63729001	Inductor,Wire Wound,Chip	1	L6605
224	EAP63748801	Inductor,Multilayer,Chip	1	L11048
225	EAP63769801	Inductor,Wire Wound,Chip	1	L27102
226	ECCH0002001	Capacitor,Ceramic,Chip	2	C28008
227	ECCH0009216	Capacitor,Ceramic,Chip	6	C13020
228	ECCH0009228	Capacitor,Ceramic,Chip	1	C15011
229	ECCH0017501	Capacitor,Ceramic,Chip	2	C4128,C4134
230	ECZH0025920	Capacitor,Ceramic,Chip	1	C14001
231	ECCH0000182	Capacitor,Ceramic,Chip	2	C24007,C24017
232	EDSY0010501	Diode,Switching	1	D28000
233	ECCH0017301	Capacitor,Ceramic,Chip	36	C21200,C21201,C21202,C21203,C21204,C21205,C21206,C21207,C21208,C21209,C21210,C21211,C21218,C21219,C21221,C21222,C21223,C21237,C21275,C21276,C21277,C21278,C21279,C21280,C24009,C24023,C24024,C27107,C27110,C27123,C28009,C7100,C7104
234	ECCH0004904	Capacitor,Ceramic,Chip	6	C21001,C28004,C28010,C29203,C29207
235	ERHY0000166	Resistor,Chip	1	R21009
236	EAX67212101	PCB,Main	1	EAX010001
237	ERHY0009301	Resistor,Chip	2	R28000,R28001
238	ERHY0009302	Resistor,Chip	1	R29106
239	ECZH0000813	Capacitor,Ceramic,Chip	1	C28005

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
240	ECCH0009101	Capacitor,Ceramic,Chip	32	C10015,C21268,C21269,C21270,C21271,C24008,C24012,C24013,C24014,C24019,C24020,C24021,C24022,C28007,C28011,C6705,C6706,C6903
241	ERHY0009303	Resistor,Chip	2	R6608,R6609
242	ECCH0009102	Capacitor,Ceramic,Chip	2	D6602
243	ERHY0009307	Resistor,Chip	1	R6607
244	ECCH0009103	Capacitor,Ceramic,Chip	41	C10000,C10003,C10004,C10005,C10009,C10012,C10014,C10028,C11025,C11040,C11041,C11079,C11080,C11088,C11089,C11096,C12044,C13004,C13029,C15002,C15007,L10008,L10025,L10043,L10052
245	ERHY0009311	Resistor,Chip	1	R10003
246	ECCH0009104	Capacitor,Ceramic,Chip	3	C10001,C10010,C10011,C10022,C10053,C11051,C11052,C11100,C13018,C13022,C13025,C27129,C27130,L10017
247	ERHY0009312	Resistor,Chip	1	R21007
248	ECCH0009106	Capacitor,Ceramic,Chip	6	C10007,C11043,C11075,C13017,C6603
249	ERHY0009504	Resistor,Chip	3	R4103,R6602,R6610
250	ECCH0009208	Capacitor,Ceramic,Chip	2	L10015,L13012
251	ERHY0009506	Resistor,Chip	2	R7117,R7118
252	ECCH0009226	Capacitor,Ceramic,Chip	1	C12033
253	ERHY0009527	Resistor,Chip	1	R6606
254	ERHY0009511	Resistor,Chip	3	R21008,R4120,R6611
255	ECZH0025911	Capacitor,Ceramic,Chip	1	C13005
256	ERHY0009543	Resistor,Chip	1	R29102
257	ECZH0025916	Capacitor,Ceramic,Chip	7	C28002,C6604,C6605,C6801,C6802,C6902,C6906
258	ERHY0009526	Resistor,Chip	7	R21102,R21103,R21106,R21107
259	ECZH0025917	Capacitor,Ceramic,Chip	2	C4016
260	ERHZ0000236	Resistor,Chip	1	R14008
261	ERHY0009559	Resistor,Chip	1	R4108
262	ERHZ0000401	Resistor,Chip	1	R13002,R29201,R6700
263	ERHZ0000404	Resistor,Chip	1	R4116
264	ERHZ0000249	Resistor,Chip	1	R28003
265	ERHZ0000267	Resistor,Chip	1	R4110
266	EXXY0018701	Crystal	1	X4100
267	ERHZ0000308	Resistor,Chip	1	R29105
268	ERHY0009501	Resistor,Chip	12	C13019,R10025,R10035,R29212,R6619
269	SFBH0008101	Filter,Bead	1	FB15000
270	SFBH0008102	Filter,Bead	2	FB6900,FB6901
271	SFBH0008106	Filter,Bead	1	FB4100
272	ERHY0009536	Resistor,Chip	5	R29117
273	ERHZ0000204	Resistor,Chip	1	R21004
274	ERHZ0000295	Resistor,Chip	1	R4111
275	ERHZ0000320	Resistor,Chip	1	R21210
276	SAFO0000401	Wire Pad,Open	2	R21110,R21111,R21112,R21113,R27111,R29118,R29119
277	MBV62321701	Clip	13	SC29200,SC29201,SC29202,SC29203,SC29204,SC29205,SC29206,SC29207,SC29208,SC29209,SC29210,SC29211,SC29214

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
278	SAFP0000401	Wire Pad,Short	7	R15000,R15003,R27106,R27107,R28005,R29103,R29120,R29121,R29206,R6618,R6900,R7107
279	SEVY0005201	Varistor	4	VA28000,VA28001,VA6800,VA6801
280	MEZ65049703	Label	0	PID2
281	RAA34548901	Resin,PC	0	RAA050100
282	EBR83847801	PCB Assembly,Main,Insert	1	EBR071500
283	ABM75396801	Can Assembly,Shield	1	ABM070300
284	MCQ68970101	Damper	1	MCQ000000
285	MCQ68970102	Damper	1	MCQ000001
286	MDS65756301	Gasket	1	MDS000000
287	MEV65994901	Insulator	1	MEV000000
288	MBK64892901	Can,Shield	1	MBK070300
289	MBK64892907	Can,Shield	1	MBK070301
290	MCQ68970106	Damper	1	MCQ000002
291	RAA34549103	Resin,PC	0	RAA050100
292	MEV66133101	Insulator	1	MEV000000
293	EBR83532201	PCB Assembly,Flexible	1	EBR02
294	EBR83570601	PCB Assembly,Flexible,Insert	1	EBR070100
295	MJN70247901	Tape	1	MJN000000
296	EBR83570301	PCB Assembly,Flexible,SMT	1	EBR070400
297	EBR83532301	PCB Assembly,Flexible,SMT Top	1	EBR070300
298	EAG63650901	Connector,BtoB	1	CN10001
299	EAX67192801	PCB,Flexible	1	EAX010700
300	EBR83570401	PCB Assembly,Flexible,SMT Bottom	1	EBR070200
301	EAG64909801	Socket,DIMM/SIMM	1	J10001
302	EBP63001801	Camera Module	1	EBP000000
303	ACQ89286301	Cover Assembly,Rear	1	ACQ063300
304	ACQ89346601	Cover Assembly,Rear(SVC)	1	ACQ105800
305	EAA64508001	PIFA Antenna,RF	1	EAA030106
306	EAA64529701	PIFA Antenna,WiFi	1	EAA030105
307	EAA64549601	PIFA Antenna,GPS	1	EAA030107
308	EAA64568901	PIFA Antenna,RF	1	EAA030104
309	EBD62925601	Sensor,Fingerprint	1	EBD00
310	MAZ65468801	Bracket	1	MAZ00
311	MBG66263701	Button,Side	1	MBG00
312	MCK69524801	Cover,Rear	1	MCK063301
313	MCQ68948001	Damper,Speaker	1	MCQ00
314	MCQ68986401	Damper,Motor	1	MCQ01
315	MJN70091402	Tape,Protect	1	MJN061102
316	MKC66159601	Window,Camera	1	MKC00
317	MDJ65084601	Filter	1	MDJ00
318	MJN70267701	Tape,Protect	1	MJN061106
319	MJN70113901	Tape,Camera	1	MJN00

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
320	MJN70187401	Tape,Protect	1	MJN061105
321	MJN70091404	Tape,Protect	1	MJN061104
322	MEZ64319901	Label,After Service	1	MEZ000900
323	MCQ68986301	Damper,Camera	1	MCQ02
324	MFB64393601	Lens,Flash	1	MFB00
325	MJN70091403	Tape,Protect	1	MJN061103
326	ACW75437001	Decor Assembly	1	ACW00
327	MCR66687701	Decor	1	MCR000000
328	MJN70148201	Tape,Camera	1	MJN009400
329	MJN70091401	Tape,Protect	1	MJN061101
330	EAA64587601	PIFA Antenna,Multiple	1	EAA00
331	EAB64370201	Speaker,Dual Mode	1	EAB01
332	ACQ89345801	Cover Assembly,Bar	1	ACQ003400
333	ACQ89404801	Cover Assembly	1	ACQ00
334	MBL67097601	Cap	1	MBL00
335	ACK73469001	Contact Assembly	1	ACK00
336	ACW75377401	Decor Assembly	1	ACW000000
337	MCR66687801	Decor	1	MCR000000
338	MJN70110401	Tape,Decor	1	MJN020800
339	MJN70091201	Tape,Camera	1	MJN01
340	MDS65792801	Gasket	1	MDS00
341	ACQ89287201	Cover Assembly,Front	1	ACQ032700
342	ACQ89317601	Cover Assembly,Front(Sub)	1	ACQ033200
343	MCK69394001	Cover,Front	1	MCK032700
344	MDQ65056601	Frame	1	MDQ000000
345	MICE0016905	Insert,Nut	4	MET099500
346	MHK65669701	Sheet	1	MHK000000
347	MCQ68986201	Damper	1	MCQ04
348	MJN70148101	Tape	1	MJN02
349	MJN70091101	Tape	1	MJN03
350	MDS65754201	Gasket	1	MDS01
351	MDS65850401	Gasket	1	MDS02
352	MDS65850402	Gasket	1	MDS03
353	EAU63423201	Motor,DC	1	EAU00
354	EAT63355101	Module,Hybrid Touch LCD	1	EAT130000
355	EBR83532202	PCB Assembly,Flexible	1	EBR03
356	EBR83570302	PCB Assembly,Flexible,SMT	1	EBR070400
357	EBR83532302	PCB Assembly,Flexible,SMT Top	1	EBR070300
358	EAX67192802	PCB,Flexible	1	EAX010700
359	EBR83570402	PCB Assembly,Flexible,SMT Bottom	1	EBR070200
360	MBL67097602	Cap	1	MBL01
361	MBL67097603	Cap	1	MBL02

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
362	MCQ68927001	Damper,Speaker	1	MCQ03
363	MJN70112001	Tape,Protect	1	MJN061100
364	EBP62982701	Camera Module	1	EBP01
365	EBR83507801	PCB Assembly,Flexible	1	EBR01
366	EBR83525701	PCB Assembly,Flexible,SMT	1	EBR070400
367	EBR83563201	PCB Assembly,Flexible,SMT Top	1	EBR070300
368	EAX67116301	PCB,Flexible	1	EAX010700
369	EAG64991101	Jack,Phone	1	EAG140301
370	EBR83541401	PCB Assembly,Flexible,SMT Bottom	1	EBR070200
371	EBR83507901	PCB Assembly,Flexible,Insert	1	EBR070100
372	MJN70227701	Tape	1	MJN000000
373	EAB64468201	Receiver	1	EAB00
374	MEV66170801	Insulator	1	MEV00
375	MLAZ0038301	Label	1	MEZ000000
376	GMEY0014301	Screw,Machine	11	FAB00
377	MEZ66680401	Label,Approval	1	MEZ002101
378	FAB32218901	Screw,Tapping	0	FAB01
379	SAD36243201	Software,Mobile	1	SAD010000
380	AAD88270701	Addition Assembly	1	AAD000000
381	ACQ89393101	Cover Assembly,Battery	1	ACQ02
382	MCK69434201	Cover,Battery	1	MCK004100
383	MHK65765401	Sheet	1	MHK000000
384	MHK65866601	Sheet	1	MHK000001
385	MHK65866611	Sheet	1	MHK000002
386	MHK65866621	Sheet	1	MHK000003
387	EAC63360001	Rechargeable Battery,Lithium Ion	1	EAC00
388	EAD62377921	Cable,Assembly	1	EAD010000
389	EAY62709914	Adapters	1	EAY060000
390	MEZ66451301	Label,Approval	1	MEZ002100
391	MJN70111901	Tape,USP Film	1	MJN107400
392	AFN77632601	Manual Assembly,Operation	1	AFN053800
393	MBM65696601	Card,Quick Reference	1	MBM062600
394	MBM64995926	Card,Warranty	1	MBM087200
395	MFL69741501	Manual,Operation	0	MFL053800
396	AGF78416403	Package Assembly	1	AGF000000
397	AGJ74578401	Pallet Assembly	1	AGJ000000
398	MAY67888902	Box,Pallet Sleeve	0	MAY115200
399	MAY67829102	Box,Carton	0	MAY010800
400	MBL67256901	Cap,Box	0	MBL007000
401	MLAJ0004402	Label,Master Box	0	MEZ047200
402	MLAZ0050901	Label	0	MEZ000000
403	MGA64059201	Pallet	0	MGA000000

13. REPLACEMENT PART LIST

No	Part No	Description	Qty	Location No.
404	MAF63267402	Bag,Vinyl	1	MAF086500
405	MAY67889601	Box,Master	0	MAY084000
406	MAY67868101	Box,Unit	1	MAY047100
407	MEZ64571201	Label,Unit Box	1	MEZ084100
408	MEZ64686921	Label,Unit Box	1	MEZ084101
409	MLAC0004541	Label,Barcode	1	MEZ084102
410	MLAJ0004402	Label,Master Box	0	MEZ047200
411	MFZ65055805	Packing,Blister	1	MFZ005500

